

Expertise-Leveling Technology and the Division of Creative Labor

Giuseppe Musillo*

September 8, 2026

Abstract

What happens when an expertise-leveling technology diffuses through an industry? I study digital sampling, which makes recorded performances reusable and reduces the expertise required for music production. I assemble novel data on production credits, sample use, and content for 70,712 US records released between 1980 and 2000. A difference-in-differences design identifies the causal effects of geographic exposure to a new sampler's 1988 promotional campaign attached to Michael Jackson's tour. Exposure increased sampling. Main artists on sampled records were more likely to be self-taught and less likely to report formal musical training. Participation in music production, measured as participants per 100,000 working-age residents, showed no detectable change. On exposed records, the number of people working exclusively in writing, composition, arrangement, and production grew by 0.5, roughly 15 percent of the pre-period mean, with no detectable broadening of individual task bundles. Records became more acoustically distinct from recent music before the organizational response emerged, while lyrics showed no significant change. These findings suggest that making existing performances easier to reuse can support specialized creative contributions and music that departs from the recent past.

JEL: J21, J24, O33, L82, Z11

Keywords: technology and labor, task reallocation, creative industries, sampling, credited work

*Department of Economics, Tilburg University. Email: g.musillo@tilburguniversity.edu. I am especially grateful to my supervisors, Mery Ferrando, Anna Salomons, and David Schindler, for their guidance, comments, and support throughout this project. I thank Carmela Accettura, Lorenzo Alderighi, David Autor, Patricio Dalton, Ruben Enikolopov, Aurélie Gillen, Matteo Giugovaz, Hugo Reichardt, Marta Reynal-Querol, Milica Vranic, Till Wicker, and Niccolò Zaccaria for helpful comments and suggestions. I am grateful to Mike McRoberts, AKAI's US product manager at the time of the MPC60's introduction, for insightful conversations and for generously sharing his knowledge of AKAI and the MPC60. I also owe special thanks to Matteo Pocchiarri for tremendous help with data collection, and to Paul van Geldrop for fundamental help with data processing. I also thank seminar participants at LMU, Tilburg University, and UPF, as well as participants at the 4th Milan PhD Economics Workshop, the 11th Monash-Paris-Warwick-Zurich-CEPR Text-As-Data Workshop, the 19th Workshop on Social Economy for Young Economists, the Warwick 14th PhD Economics Conference, the 31st ENTER Jamboree, the Venice Summer Institute Workshop on Workforce Adjustments to Technological Change, and the AYEWE Labor Economics Workshop. All errors are my own.

1 Introduction

Much of the debate over new technologies asks how many jobs they will replace. Yet employment is only one margin of adjustment. Technological change can also reshape who can perform a task, how many people participate in an industry, the backgrounds of those who participate, how work is divided among them, and what is ultimately produced. A long line of work shows that computerization and digital technologies substituted for routine and rules-based tasks while raising the returns to expertise, education, and judgment (Tinbergen, 1974; Autor et al., 2003; Acemoglu and Autor, 2011). Technology can also reduce the knowledge, training, or experience required to perform a task, narrowing the gap between what experts and non-experts can do. Such expertise-leveling technological change can expand access to production and change which tasks workers are relatively well suited to perform, with consequences for the division of labor and output. Generative AI provides a contemporary example. It raises productivity in several settings, with the largest gains among less-experienced or lower-performing workers, compressing differences associated with prior expertise (Noy and Zhang, 2023; Brynjolfsson et al., 2025). Recent theoretical and empirical work has begun to examine how such technologies reshape the expertise required within occupations and workers’ comparative advantage across tasks (Autor and Thompson, 2025; Althoff and Reichardt, 2026). Evidence on broader labor-market adjustment, however, remains limited: studies of AI document changes in tasks and work practices alongside limited detectable effects on aggregate employment, earnings, or hours (Acemoglu et al., 2022; Humlum and Vestergaard, 2026). For recent generative AI technologies, the available evidence necessarily captures only the early stages of adjustment.

I study the longer-term labor-market and production consequences of expertise-leveling technological change through the diffusion of digital sampling in US recorded music, a historical episode that can be followed for more than a decade. Sampling allows producers to reuse recorded sound as an input into new music, reducing the need to recreate every musical element through a new performance and opening creative possibilities through the combination and manipulation of existing recordings. I focus on the AKAI MPC60, introduced in 1988, which combined sampling with rhythm programming, sequencing, and arrangement within a single workstation and made several expertise-intensive production tasks easier to perform. I assemble novel data linking production credits, documented sample use, and measures of acoustic and lyrical content for 70,712 records released between 1980 and 2000 and recorded in 7,183 US studios.

The question is especially relevant today as expertise-leveling technologies begin to reshape creative and technical tasks across industries. In music, hip-hop artist and producer Dr. Dre

recently compared resistance to AI to earlier resistance to drum machines and synthesizers and described it as “a new tool for creativity” ([The New York Times, 2026](#)). Creative production is itself economically substantial: arts and cultural production accounted for \$1.17 trillion in value added, 4.2 percent of US GDP, and 5.4 million jobs in 2023 ([Bureau of Economic Analysis, 2025](#)). Beyond artistic production, the broader question concerns how technology changes the expertise required to develop and implement ideas. The U.S. Department of Labor’s O*NET identifies “Thinking Creatively” as an important activity across scientific, technical, managerial, and professional occupations ([National Center for O*NET Development, 2026](#)). In professional services, concerns that AI may absorb tasks traditionally assigned to junior workers have prompted debate about how work will change and which skills will become more valuable, including judgment, communication, and client-facing skills ([The Financial Times, 2026a,b](#)).¹

This historical episode is particularly well suited to studying these adjustments for four reasons. First, the sampler is a clear example of an expertise-leveling technology. Reusing recorded performances reduced the instrumental expertise needed to incorporate musical elements into a record, while integrating sampling, sequencing, rhythm programming, and arrangement made several production tasks easier to perform. This episode forms part of a broader transformation in which digital technologies lowered barriers to music production and enabled musicians to take on tasks previously outside their roles ([Askin and Mol, 2018](#)). Whether easier access to these tasks led musicians to perform more of them themselves or changed how they worked together is an empirical question.

Second, the allocation of tasks is observable. Production credits identify both the musicians associated with a record and the broad tasks attributed to each of them, so a record is simultaneously an output and a production unit. I can therefore distinguish musicians taking on broader task bundles from changes in team composition. Third, the output is measurable. Acoustic characteristics and lyrics allow me to ask whether changes in production are accompanied by changes in the records produced. Fourth, the campaign promoting the sampler generated geographic variation in exposure, allowing these margins of adjustment to be studied within a causal design.

I use this setting to study how an industry adjusts when a technology makes expertise-intensive tasks more accessible, drawing on production records and artists’ musical biographies. To identify causal effects, I exploit geographic variation in exposure to the sampler’s promotional campaign. In 1988, AKAI organized hands-on demonstrations attached to Michael Jackson’s US *Bad* Tour. I compare the eighteen commuting zones reached by the campaign

¹For related theoretical work on how automating junior workers’ tasks affects training and career viability, see [Garicano and Rayo \(2025\)](#).

with commuting zones that hosted other major US tours that year, including tours by Bruce Springsteen and Pink Floyd, but were not exposed to the campaign. Using a difference-in-differences design, I compare changes between the pre-campaign period, 1980–1988, and the post-campaign period, 1989–2000, in exposed and comparison commuting zones.

I document five findings. First, campaign exposure increased sampling. After 1988, the share of records that include a documented sample rises by 2.4 percentage points in exposed relative to comparison commuting zones, against a pre-period mean of roughly 1 percent, and the share of studios using sampling rises as well. The increase begins soon after the campaign, establishing that campaign exposure accelerated the diffusion of sampling.

Second, I find no detectable change in participation in music production, measured as participants per 100,000 residents aged 20–64 in each commuting zone and year.

Third, sampling is disproportionately associated with artists outside traditional musical training pathways. Main artists on sampled records are substantially more likely to be self-taught and less likely to report formal or conservatory musical training than artists on other records. They are also younger on average. These differences are consistent with sampling lowering expertise-related barriers to music production.

Fourth, records in exposed commuting zones include more people working exclusively in writing, composition, arrangement, and production. Over 1993–2000, their number increases by 0.52 per record, relative to a pre-period mean of 3.48, an increase of roughly 15 percent. I do not detect an offsetting decline in another observed contributor category.² Holding the recording studio fixed, the share of production credits attributed to writing, composition, arrangement, and production also rises, while performance and technical tasks lose relative weight. Despite the sampler integrating several production tasks within a common workflow, individual task bundles do not measurably broaden: musicians become no more likely to combine performance or technical tasks with writing, composition, arrangement, and production, nor to accumulate more distinct tasks within the latter group. The organizational adjustment therefore occurs through team composition.

Fifth, campaign exposure also changes the records produced. Following work on differentiation in cultural products ([Askin and Mauskopf, 2017](#)), I measure each record’s acoustic distance from music released in the preceding year. Campaign exposure increases this distance by 0.17 pre-period standard deviations, indicating greater acoustic distinctiveness relative to recent music. Applying the same measure to lyrics, whose writing the sampler does not directly simplify, yields no significant change.

²Removing potentially attribution-related writing and composition credits from sampled records after 1991, to account for credits given to creators of the sampled material, explains only 17 percent of the increase in people working exclusively in these creative tasks.

I find that documented sampling increases soon after the campaign, acoustic differentiation emerges around 1991, and the change in team composition appears only around 1993. The division of labor therefore adjusts several years after sampling begins to diffuse and after changes in output become apparent. [Demirer et al. \(2026\)](#) distinguish short-run deployment of technology within fixed jobs from longer-run adjustment of job boundaries and skill requirements. The timing of the results illustrates why focusing only on the first years after a technology diffuses can miss changes in the organization of work.

This paper contributes to three literatures. First, it contributes to research on the labor-market impacts of skill-biased and routine-task-displacing technological change ([Autor et al., 2003](#); [Acemoglu and Autor, 2011](#); [Goos et al., 2014](#); [Acemoglu and Restrepo, 2019](#)).³ An emerging literature studies expertise-leveling technological change. [Autor and Thompson \(2025\)](#) examine how automation changes the tasks that remain for workers and therefore the expertise required within occupations, while [Althoff and Reichardt \(2026\)](#) show how task simplification reshapes comparative advantage, occupational sorting, and workers' allocation of time across tasks. I provide causal evidence on how an expertise-leveling technology changes production across an industry over more than a decade. This complements evidence on AI's effects in specific tasks, firms, and occupations by examining diffusion, participation, task allocation, and output together, alongside evidence on artists' musical training.⁴ Observing who performs which tasks within production teams also allows me to identify how technological integration changes the allocation of work, including adjustments that occupational employment measures would miss. The results show that work and output can change without a detectable participation response, with output responding before the organization of work adjusts. This offers a perspective on AI's potential effects across sectors where it makes expertise-intensive tasks easier to perform and combine: gains in individual task performance can be accompanied by a wider reorganization of production that unfolds over several years.

Second, the paper advances research on specialization and the division of labor by showing how technological integration can be accompanied by more specialized human contributions. Comparative advantage, coordination costs, and the accumulation of knowledge shape how

³Related work examines changes in task content and skill requirements, the emergence of new work, and worker adjustment within firms and occupations. See, among others, [Spitz-Oener \(2006\)](#); [Lin \(2011\)](#); [Atalay et al. \(2020\)](#); [Deming and Noray \(2020\)](#); [Autor et al. \(2024\)](#); [Feigenbaum and Gross \(2024\)](#); [Bessen et al. \(2025\)](#); [Salomons et al. \(2025\)](#).

⁴Recent studies examine effects of generative AI on productivity, worker performance, labor-market outcomes, and creative output ([Noy and Zhang, 2023](#); [Brynjolfsson et al., 2025](#); [Humlum and Vestergaard, 2026](#); [Abbas Nejad et al., 2025](#); [Doshi and Hauser, 2024](#); [Peng et al., 2023](#); [Lin, 2025](#); [Kim et al., 2026](#); [Reimers and Waldfogel, 2026](#)). Historical technological transformations can reorganize production over much longer horizons ([Baily et al., 2026](#)).

tasks are allocated across workers and teams (Becker and Murphy, 1992; Garicano, 2000; Jones, 2009). Dessein and Santos (2006) emphasize the tradeoff between specialization and adapting interdependent tasks to local information, while Autor and Handel (2013) document substantial variation in task content within occupations and connect task assignment to worker characteristics and wages. Observing the allocation of tasks across musicians working on the same record makes it possible to distinguish broader individual task bundles from changes in team composition. The sampler made several tasks easier for one person to perform, but the organizational response involved more people concentrated in creative tasks, with no detectable broadening of individual task bundles. What a technology enables one person to do therefore does not determine how people organize production: greater individual technological capability can support teams that combine more specialized human contributions.

Third, the paper contributes to the economics of creative production by bringing the organization of creative work into the analysis of technological change and output. Existing research studies how technological change affects the quantity, quality, and value of creative output (Waldfogel, 2018; Aguiar and Waldfogel, 2018), differentiation and exploration in music (Askin and Mauskopf, 2017; Kim et al., 2024), and the organization and returns to creative labor (Rosen, 1981; Caves, 2000; Caoui and Galasso, 2025; Borowiecki, 2022). Qualitative accounts of recorded music also document how digital technologies transformed work inside the recording studio (Katz, 2004; Moorefield, 2010). I connect these questions by observing who performs which tasks on the same records whose content I measure, identifying the effects of a production technology on both the division of creative labor and the music produced. The evidence shows that creative output can change before the organization of work adjusts, and that making existing performances easier to reuse can support music that is more acoustically distinct from the recent catalogue.

The remainder of the paper proceeds as follows. Section 2 describes recorded-music production, digital sampling, and the promotional campaign. Section 3 introduces the data and measures, and Section 4 develops the empirical strategy. Section 5 studies the diffusion of sampling and participation in recorded-music production, Section 6 examines the division of labor within records, and Section 7 studies changes in acoustic and lyrical content. Section 8 discusses the interpretation and broader implications of the results, and Section 9 concludes.

2 Background

2.1 Teams in recorded-music production

Recorded music is produced by project teams whose members perform different tasks at different stages of production. Songwriters and composers create musical material and lyricists the words. Arrangers translate that material into parts for the instruments and performers used on a recording, while producers shape and coordinate the recording process, including choices over material, personnel, arrangement, and sound. During the session, vocalists and instrumentalists perform those parts and recording engineers capture them. The resulting multitrack recording is then mixed and mastered into a finished master. Other contributors perform tasks surrounding the release rather than the recording itself, including photography, design, management, and artists-and-repertoire.

Recording technology has repeatedly changed how these tasks can be combined. Multitrack recording shifted the studio from a place primarily used to document performances toward one in which recordings could be constructed from separately captured and manipulated parts (Moorefield, 2010; Katz, 2004). Performance, engineering, arrangement, and production consequently became increasingly separable parts of the production process, often associated with distinct forms of expertise. This organization is characteristic of project-based creative production, where specialized contributors are assembled around individual goods rather than employed along a fixed production line (Caves, 2000).

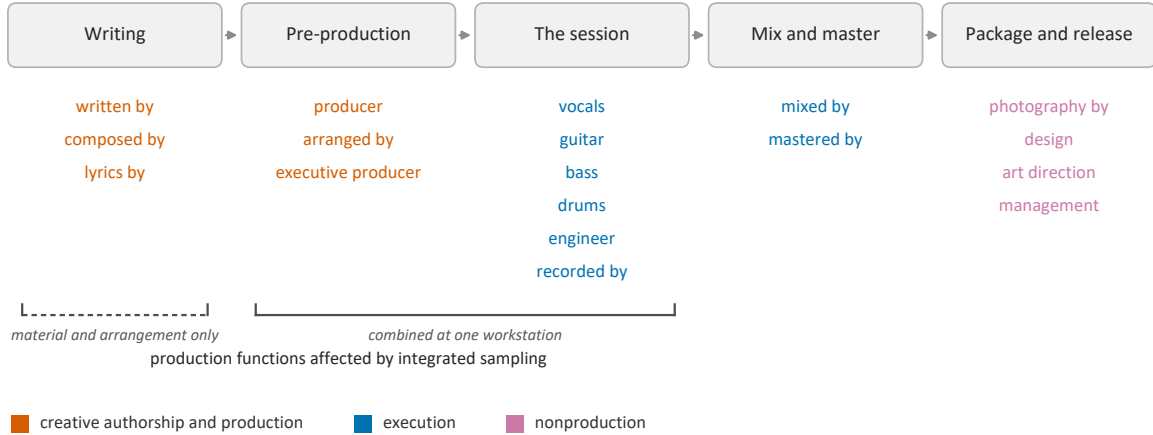
Recorded music makes this division of labor unusually observable because released records conventionally identify both the people associated with production and the tasks attributed to them. Figure 1 places common credit labels along the production sequence and shows where the technology studied in this paper could alter that sequence. Sampling and sequencing brought several tasks involved in constructing a recording into the same workstation, while mixing, mastering, and tasks surrounding the release remained outside the technology’s direct reach. Section 3.2 formalizes how these tasks are classified in the empirical analysis.

2.2 Digital sampling and the sampler

Digital sampling turns existing recordings into inputs that can be cut, repeated, rearranged, and combined into new music. Unlike musical notation, a sample preserves a particular recorded performance (Katz, 2004). A producer can therefore reuse a performance already embodied in a recording rather than recreate the same musical element through a new studio session. Sampling consequently changes both the source of musical material and the expertise required to turn an idea into a finished recording.

Figure 1

Music recording pipeline



How a record is made, and who is credited for what. *Note.* Stages of record production, with common credit labels placed at the stage where the work occurs and colored by the three categories used in the main analysis. The categories partition every classified credit; a record in the sample carries about twenty-four of them. The annotation beneath indicates production tasks affected by integrated sampling, not tasks performed by the machine itself. The solid segment marks tasks that could be combined at one workstation, while the dashed segment over writing indicates the weaker link between sampling and the construction of musical material. Integrated sampling does not itself write or compose. Mixing and mastering, together with sleeve and business tasks, lie outside the technology’s direct reach.

Early digital samplers made this possible but remained expensive and cumbersome. The Fairlight CMI, introduced in 1979, cost around \$25,000.⁵ The E-mu SP-1200, introduced in 1987 at roughly \$3,000, allowed recorded sounds to be sampled and organized into songs, but with more limited sampling capacity (E-mu Systems, 1987; Vintage Synth Explorer, 2026). The AKAI MPC60, designed by Roger Linn and released in 1988 at roughly \$5,000, combined sampling, rhythm programming, sequencing, and arrangement within a single workstation (Trask, 1988). From the same device, a user could select and reshape recorded material, construct rhythms, sequence musical events, and organize the resulting parts into a track.

The economic change was therefore broader than cheaper reuse of recorded sound. Existing performances could substitute for some newly performed parts, while sequencing and looping reduced the instrumental, technical, and coordination requirements involved in assembling musical elements (Arditi, 2014; Moorefield, 2010). Tasks that had previously relied on distinct forms of expertise could increasingly be carried out within a common digital workflow. Thomas Bangalter describes this margin directly when discussing Daft Punk’s production process. Confronted with “the limitation of our own playing capabilities,” they instead

⁵Price documentation at <https://www.perfectcircuit.com/signal/fairlight-cmi-history>; a period review of the MPC60, including its launch price, is Trask (1988), with specifications at <https://www.vintagesynth.com/Akai/mpc60>.

used technology to “cut and paste and loop stuff.”⁶ Some production tasks could therefore be performed with less of the instrumental, technical, or coordination expertise previously required.

Selecting source material, deciding how musical elements should fit together, and shaping the resulting recording remained human tasks. By making several production tasks easier to perform, the sampler could encourage musicians to take on broader task bundles and reduce their reliance on specialists. It could also make specialized inputs easier to combine: recorded parts could be rearranged within the same workstation, potentially reducing the coordination required to assemble contributions from different musicians. Production could then draw on musicians with distinct creative capabilities while each remained concentrated in particular tasks. The organizational question is whether the technology’s diffusion led individuals to perform broader task bundles or changed the composition of the teams producing each record.

2.3 Copyright clearance and attribution

Sampling also interacted with copyright law because it reused creative work produced by others. Through much of the 1980s, sampling often proceeded without formal licenses (Demers, 2006). That environment changed with *Grand Upright Music, Ltd. v. Warner Bros. Records Inc.*, 780 F. Supp. 182 (S.D.N.Y. 1991). The December 1991 ruling treated unlicensed sampling as copyright infringement and contributed to formal sample clearance becoming standard industry practice.

Sample clearance is the process of obtaining permission to use sampled material. It matters for the credit data because licensing can result in writers or composers of the underlying work receiving credit on the new record even when they did not participate in producing it. A writing or composition credit can therefore represent either a task performed in producing the new record or attribution to the creator of an earlier work used as an input.

Because this legal change occurred during the post-campaign period, changes in authorship credits may reflect both tasks performed in producing new records and changes in attribution. Section 8.2 assesses this channel directly and quantifies how much potentially attribution-related crediting can account for the organizational result.

⁶Thomas Bangalter, interview with Andre Torres, “Quantum Leap,” *Wax Poetics*, Issue 55, Summer 2013. The interview is reproduced at <https://magazine.waxpoetics.com/article/daft-punk-go-from-sampling-disco-records-to-creating-a-dance-masterpiece/>.

2.4 The *Bad* Tour and the AKAI road clinics

In 1988, Michael Jackson’s *Bad* World Tour crossed the United States, playing major arenas and stadiums across the country (Halstead and Cadman, 2003; Billboard, 1988). AKAI used the US leg of the tour to promote the newly introduced MPC60. Mike McRoberts, then AKAI’s US product manager, recalls that members of Jackson’s touring band proposed a series of hands-on demonstrations at the January 1988 NAMM trade show. The resulting road clinics followed the tour and introduced the sampler to local musicians and recording professionals.⁷

The campaign combined hands-on product demonstration with the geographic reach of one of the most prominent touring acts of the period. Rather than encountering the sampler only through conventional advertising, local musicians and recording professionals could see touring musicians demonstrate how it could be incorporated into a production workflow.

Because the clinics followed the tour itinerary, exposure varied geographically. The historical record identifies realized tour stops much more completely than it identifies the occurrence of, or attendance at, every individual clinic. I therefore observe which commuting zones were reached by the campaign more reliably than I observe which particular musicians, producers, or studios attended a demonstration. The empirical treatment is defined accordingly at the commuting-zone level. Section 4.1 develops the exposure design, while Appendix B.4 documents the itinerary and surviving clinic evidence.

Figure 2 documents the campaign using the MPC60 itself, an AKAI promotional photograph from a clinic, and a contemporaneous road-clinic advertisement.

3 Data

I assemble data on records produced in US recording studios, the people credited on those records and the tasks attributed to them, documented sampling, and the commuting zones in which production took place. I then link records to measures of acoustic and lyrical content and to biographical information on their main artists. The central advantage of the data is that each record is observed both as an output and as a production unit: I observe who is associated with it and which tasks are attributed to each person.

⁷The account of the NAMM approach, the weekly clinic format, and the clinic rig is McRoberts’s own, in an interview at <https://www.bboytechreport.com/2013/01/11/interview-with-mike-mcroberts/>. NAMM is the National Association of Music Merchants, whose annual show is the industry’s principal trade fair for musical instruments and music technology. Appendix B.4 gives the campaign’s documentary record in full.

Figure 2



The campaign, in primary sources. *Note.* Panels (a) and (b) are AKAI promotional photographs from the 1988 campaign, kindly shared by Mike McRoberts; panel (c) is a road-clinic advertisement from *The Valley Advocate* (Hatfield, MA), March 14, 1988.

3.1 Data sources

Records and credits. The primary source is Discogs, a user-maintained discographic database containing release dates, tracks, recording studios, genres, and the credit lists printed with released music. Discogs records individual releases rather than underlying records, so multiple pressings or formats associated with the same master are collapsed to a single record observation, as described in Appendix A.1. I retain records released between 1980 and 2000 whose recording location can be assigned to the continental United States.

The resulting unit is a *record*, not necessarily a full-length album. Singles, EPs, and twelve-inch releases are retained because they are produced through the same recording process and carry the same types of production credits. Short-form releases account for 37.5 percent of the estimation sample. They contain fewer documented credits on average than long-form releases, a difference that matters particularly for outcomes expressed as shares and that I examine in Section 6 and Appendix A.8.

For each record, the credit list identifies the people or entities named on the release and the tasks attributed to them. These include writing, composition, performance, arrangement, production, recording, engineering, mixing, mastering, and tasks surrounding the release. These credits provide the basis for measuring how production tasks are allocated across the people associated with the same record.

Sampling. Documented sampling comes from WhoSampled, which records links between a recording and earlier recordings whose material it reuses. I classify a record as documenting sampling when at least one of its tracks is linked to a sampled source. Because these relationships are assembled retrospectively and coverage is incomplete, the measure captures documented rather than exhaustive sample use. Appendix A.2 describes the matching procedure and measurement checks.

Studios and geography. Recording studios named in Discogs are standardized and geocoded to counties and then to 1990 commuting zones (Tolbert and Sizer, 1996). Commuting zones aggregate counties according to commuting flows, approximate local labor markets, and provide the geographic unit used throughout the paper. Tour dates and venues are assembled from contemporaneous sources, principally *Billboard*’s Boxscore reports, and matched to the commuting zone containing each venue.

Studio equipment. I digitize the regional studio directories published by *Mix* magazine from 1980 through 1993. The directories contain studio addresses and staff and, in the more detailed listing format, information on consoles, recorders, monitoring equipment, and instruments. I recover 18,869 listings covering 8,328 studios, of which 96.6 percent can be assigned to a commuting zone.

These directories are the only source in the paper that observes studio equipment directly rather than inferring production practices from record credits. I do not use them to estimate adoption effects because equipment is reported only in the more detailed listing format, and the probability of appearing in that format changes with campaign exposure. The set of studios for which equipment is observed is therefore not stable across treatment status and time. Appendix A.5 describes the source and Appendix D.6 reports the descriptive equipment evidence.

Population and local characteristics. Working-age population, aged 20–64, is constructed from annual SEER county population estimates and aggregated to commuting zones. It provides the denominator for the participation measure in Section 5.2. Predetermined local characteristics, including population, population density, urban share, education, and income, come from the 1980 Census. County employment comes from the County Business Patterns series as imputed by Eckert et al. (2021), and local personal income from the BEA. Appendix A.4 documents construction and validation.

Recording content. I link records to Spotify Echo Nest acoustic characteristics, MusicFlamingo audio embeddings, and Genius lyrics. These sources provide complementary representations of the finished record: acoustic characteristics summarize dimensions of recorded sound, audio embeddings provide a higher-dimensional representation of the recording, and lyrics capture its verbal content. Section 7 introduces the resulting outcomes, while Appendix H documents their construction and measurement.

Biographies. For the descriptive analysis of artist backgrounds, I link main artists to Wikidata and corresponding Wikipedia biographies and collect information on musical training, education, birthplace, birth year, and related characteristics. Missing biographical characteristics remain missing rather than being coded as absent. Because documentation varies systematically across artists, these comparisons are interpreted conditional on the observed biographical sample. Appendix A.7 describes construction and coverage.

3.2 Measuring the division of labor

The central measurement problem is to observe both how credited work is distributed across production tasks and how those tasks are allocated across people within the same record. I begin by assigning every classified credit to one of three mutually exclusive task families.

Execution contains tasks involved in performing and technically realizing the recording, including vocals, instrumental performance, recording, engineering, mixing, and mastering. *Creative authorship and production* contains writing, composition, arrangement, and production, tasks involved in originating the musical material or shaping how it is realized in the recording. *Nonproduction* contains credited tasks surrounding the release but outside production of the recording itself, including photography, sleeve design, management, and artists-and-repertoire. Figure 1 places the main credit labels along the production process and shows how they map into the three families.

The classification follows the task attributed by the credit rather than the technology used to perform it. A synthesizer performance, for example, is classified as execution for the same reason as a guitar performance: both identify instrumental performance. Reclassifying work because it was performed electronically would mechanically build technological change into the outcome. The credit catalogue and its family assignments were therefore fixed before estimating the organizational outcomes. Appendix A.6 documents the underlying labels and classification rules.

I construct three sets of measures from this taxonomy.

The first captures *team composition*. Every contributor on a record falls into exactly one of four mutually exclusive cells: credited only in execution, only in creative authorship and

production, only in nonproduction, or in two or more of the three families. I observe both the number of contributors in each cell and their share of the credited team. A contributor credited only as a writer belongs to the creative-authorship-and-production-only cell, while a contributor credited as both writer and guitarist belongs to the multi-family cell. The four counts sum exactly to total credited team size.

The second captures the *allocation of credited work*. For each record, I calculate the share of classified credit lines assigned to execution, creative authorship and production, and nonproduction. Because the three families partition classified credits, these shares sum to one. These measures describe the relative weight of different tasks in the credit list, while the team-composition measures describe how those tasks are distributed across people.

The third captures *individual breadth*. I first calculate the share of contributors credited in two or more of the three broad families. I then construct a narrower measure within creative authorship and production: the share of contributors holding two or more distinct tasks within that family. The first asks whether contributors span different parts of production; the second asks whether they accumulate multiple tasks within the part of production toward which the organizational response is concentrated.

For interpretation, I further divide creative authorship and production into *authorship*, consisting of writing and composition, and *production and arrangement*, consisting of production, arrangement, and conducting. This decomposition allows the analysis to distinguish changes concentrated in authorship from changes in the broader set of production tasks.

All of these measures capture *credited responsibility*. A credit establishes that a task was attributed to a person on the released record. It does not measure hours worked, contractual authority, decision rights, effort, or uncredited contributions. The analysis therefore studies how documented production tasks are allocated across people rather than the amount of labor embodied in each credit.

3.3 The estimation sample

The underlying continental-US data contain 75,589 records released between 1980 and 2000 whose recording location can be assigned to a commuting zone. The estimation sample restricts this universe to the commuting zones used in the exposure design developed in Section 4.1.

Eighteen commuting zones hosted a realized US stop of Michael Jackson’s *Bad* Tour in 1988 and are classified as exposed to the AKAI campaign. The comparison group consists of 106 commuting zones that hosted one of the other major US tours included in the design during the same year but were not reached by the AKAI promotion. Section 4.1 explains the

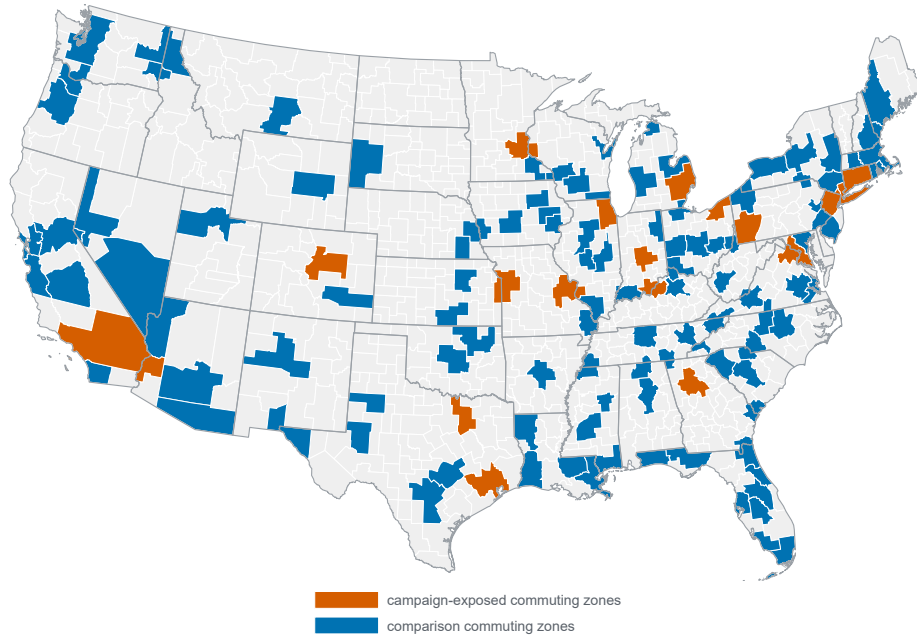
identification rationale for this comparison.

The resulting sample contains 124 commuting zones observed from 1980 through 2000, or 2,604 commuting-zone-years, together with 70,712 records produced in 7,183 recording studios. Eight of these commuting zones contain no records in the record-level sample and enter only through commuting-zone outcomes; the record-level sample therefore covers 116 commuting zones. Of these records, 70,707 contain at least one classified production credit. The eighteen exposed commuting zones account for 44,430 records, reflecting the geographic concentration of US recorded-music production. As discussed in Section 4.2, record-level estimates therefore weight commuting zones according to the amount of music they produce rather than giving each commuting zone equal weight.

Of the 70,712 records, 20,847 were released in 1980–1987, 3,197 in 1988, and 46,668 in 1989–2000. The campaign year is retained in the estimation sample and treated as part of the pre-period in the pooled specifications, as described in Section 4.3. A total of 1,648 studios are observed producing records both before and after the campaign and therefore contribute to identification in specifications with studio fixed effects.

Figure 3 shows the geographic distribution of exposed and comparison commuting zones.

Figure 3



Commuting zones in the exposure design. *Note.* Exposed commuting zones are the 18 commuting zones containing a realized 1988 *Bad* Tour stop during the AKAI campaign. Comparison commuting zones hosted another major US tour in 1988 without the AKAI promotional campaign. Commuting zones outside the estimation sample are shown in the background.

3.4 Descriptive statistics

Table 1 reports pre-campaign descriptive statistics for the main commuting-zone- and record-level measures by treatment arm. Exposed commuting zones are substantially larger, denser, more urban, and more active in recorded-music production than comparison commuting zones before the campaign. At the record level, however, the two groups are more similar in the allocation of credited tasks. During the pre-period, 1980–1988, creative authorship and production accounts for 36.4 percent of classified credits in exposed commuting zones and 37.0 percent in comparison commuting zones, while execution accounts for 54.9 and 54.7 percent, respectively.

These statistics characterize the commuting zones represented in the design. The identifying comparison, developed in Section 4.4, comes from changes over time rather than from equality in pre-campaign levels.

Table 1: The measures used in the analysis

	campaign-exposed			comparison		
	<i>N</i>	mean	s.d.	<i>N</i>	mean	s.d.
<i>Panel A. Commuting zone, market-year</i>						
Share of releases crediting a sample	162	0.017	0.026	606	0.008	0.048
Share of active studios with such a release	162	0.037	0.051	606	0.018	0.080
Credited people per 100,000 working-age residents	162	26.4	26.9	954	12.5	38.5
Releases	162	97.1	176.0	954	8.718	24.5
Active studios	162	32.9	54.0	954	3.560	8.324
Credited people	162	1,070	2,065	954	95.3	288.8
<i>Panel B. Record: the organization of credited work</i>						
Share of credits: performance	15,726	0.381	0.257	8,317	0.392	0.274
Share of credits: technical	15,726	0.168	0.151	8,317	0.155	0.163
Share of credits: creative authorship and production	15,726	0.364	0.236	8,317	0.370	0.264
Share of credits: nonproduction	15,726	0.088	0.098	8,317	0.083	0.105
Share of people working only in performance	15,726	0.288	0.258	8,317	0.294	0.274
Share of people working only in technical	15,726	0.181	0.159	8,317	0.162	0.171
Share of people working only in creative authorship and production	15,726	0.256	0.263	8,317	0.276	0.294
Share of people working only in nonproduction	15,726	0.104	0.113	8,317	0.098	0.120
Record credits a sample	15,727	0.037	0.188	8,317	0.019	0.137
<i>Panel C. Record: content</i>						
Acoustic atypicality: distance from the genre’s previous year	6,128	0.263	0.055	2,627	0.256	0.060
Lyrical novelty: distance from the two years behind	3,025	0.038	0.015	1,297	0.037	0.014

Note. The measures the paper estimates on, over 1980–1988, for the 124 commuting zones of the estimation sample. 1988 is in the pre-treatment period here as it is in every specification. Panel A is the commuting-zone panel; a share is undefined in a zone-year with no release, which is why the counts differ down the panel. Panel B reports two different partitions of the same record and they are not eight separate measures: the first four are shares of the record’s credit lines and the next four are shares of its people, counting a person once however often they are credited and only where every credit they hold on that record falls in one category. Panel C covers the records carrying a content measure, which is about half the sample for the acoustic measure and a fifth for the lyrical one; Appendix H reports the coverage tests. Predetermined market and record characteristics, the post-campaign window and normalized differences are in Appendix C. Descriptive throughout; no coefficient is estimated in this table.

4 Empirical strategy

The empirical challenge is to identify the effect of sampling technology when adoption was chosen by studios and artists. I address this challenge using AKAI’s 1988 promotion of the MPC60 alongside Michael Jackson’s US *Bad* Tour. The campaign exposed some commuting zones to demonstrations of the newly introduced sampler but not others. I compare commuting zones reached by the campaign with commuting zones that hosted other major US tours in the same year but were not exposed to the promotion.

The design follows the same sequence as the empirical analysis. I first ask whether campaign exposure accelerated documented sampling and changed participation in local recorded-music production. I then ask whether the division of labor within records changed and whether campaign exposure is reflected in the records produced. Throughout, the treatment is local campaign exposure. Realized sampling is an outcome of that exposure and is not used as an assigned treatment.

4.1 Exposure design

An exposed commuting zone is one that hosted a realized US stop of Michael Jackson’s *Bad* Tour in 1988.⁸ AKAI organized demonstrations of the sampler around these tour stops, as described in Section 2.4. This produces eighteen exposed commuting zones.

I choose as the comparison group commuting zones that hosted one of the other major US tours described in Section 3.3 during the same year but were not reached by the AKAI campaign. Major tours systematically visited large urban areas with the venues and demand needed to support arena and stadium concerts. Restricting the comparison group to other major-tour destinations therefore compares exposed commuting zones with places selected along the same broad margin of national touring activity.

Two scheduled *Bad* Tour stops provide an additional falsification exercise. Cincinnati and Tacoma had announced tour dates that were subsequently cancelled, for routing and illness respectively. These destinations passed much of the tour’s selection process but did not receive a realized tour stop and therefore did not receive campaign exposure as defined in the analysis. I use them to ask whether commuting zones selected for the tour but not ultimately exposed display the same subsequent changes as exposed commuting zones.

Treatment is defined as local campaign exposure at the commuting-zone level. The estimates are therefore reduced-form effects of exposure to the campaign rather than effects of individual clinic attendance or MPC60 adoption. I first establish whether the campaign

⁸Commuting zones group counties linked by commuting flows and approximate local labor markets. Section 3.1 describes the geographic assignment of studios and tour venues.

increased documented sampling. Subsequent estimates continue to capture the effect of campaign exposure rather than the causal effect of realized sampling itself.

4.2 Units of analysis

Treatment varies across commuting zones, while the outcomes are observed at two levels.

Diffusion and participation are measured at the commuting-zone-year level. These outcomes ask whether documented sampling became more prevalent within a commuting zone and whether participation changed relative to the local working-age population.

The division of labor and product characteristics are measured at the record level. A record is both an output and a production unit: its credits identify the musicians associated with it and the tasks attributed to them. Record-level regressions therefore ask how records produced in exposed commuting zones changed following campaign exposure.

This distinction also determines the weighting of the estimand. A commuting-zone-year regression gives each commuting zone one observation in a year, whereas a record-level regression gives greater weight to commuting zones producing more records. The organizational estimates should therefore be interpreted as effects on recorded-music production where that production is concentrated, rather than as effects on the average US commuting zone. Section 6 reports the corresponding equal-weighted commuting-zone exercise.

For the organizational outcomes, my preferred record-level specification also holds the recording studio fixed. It therefore compares records produced by the same studios before and after campaign exposure. A complementary specification replaces studio fixed effects with commuting-zone fixed effects, allowing changes in the composition of active studios to contribute to the response.

4.3 Event-study and pooled specifications

I use event studies as the main specification because they show both differential movements before the campaign and the timing of subsequent adjustment.

For commuting-zone outcomes, I estimate

$$Y_{mt} = \sum_{\tau \neq 1988} \beta_{\tau} (\text{Exposed}_m \times \mathbf{1}\{t = \tau\}) + \mu_m + \rho_{d,t} + \varepsilon_{mt}, \quad (1)$$

where Y_{mt} is an outcome in commuting zone m and year t , Exposed_m indicates campaign exposure, μ_m are commuting-zone fixed effects, and $\rho_{d,t}$ are census-division-by-year fixed effects. The omitted year is 1988. Each β_{τ} therefore measures the difference between exposed

and comparison commuting zones in year τ relative to their difference in 1988, comparing commuting zones within the same census division.

For record-level outcomes, I index each record a by its recording studio $s(a)$, commuting zone $m(a)$, census division $d(a)$, and release year $t(a)$. The preferred event-study specification is

$$Y_a = \sum_{\tau \neq 1988} \beta_\tau \left(\text{Exposed}_{m(a)} \times \mathbf{1}\{t(a) = \tau\} \right) + \mu_{s(a)} + \rho_{d(a),t(a)} + \varepsilon_a. \quad (2)$$

The studio fixed effects $\mu_{s(a)}$ absorb time-invariant differences across recording studios. The coefficients therefore compare changes in records produced by the same studios in exposed commuting zones with contemporaneous changes in records produced in comparison commuting zones within the same census division.

I use pooled difference-in-differences specifications to summarize the average post-campaign contrast. At the commuting-zone level, I estimate

$$Y_{mt} = \beta \left(\text{Exposed}_m \times \text{Post}_t \right) + \mu_m + \rho_{d,t} + \varepsilon_{mt}, \quad (3)$$

where $\text{Post}_t = 1$ from 1989 onward. The coefficient β summarizes the average post-campaign change in exposed commuting zones relative to comparison commuting zones.

For record-level organizational outcomes, the corresponding specification is

$$Y_a = \beta \left(\text{Exposed}_{m(a)} \times \text{Post}_{t(a)} \right) + \mu_{s(a)} + \rho_{d(a),t(a)} + \varepsilon_a. \quad (4)$$

I complement this with

$$Y_a = \beta \left(\text{Exposed}_{m(a)} \times \text{Post}_{t(a)} \right) + \mu_{m(a)} + \rho_{d(a),t(a)} + \varepsilon_a, \quad (5)$$

which is still estimated at the record level but replaces studio fixed effects with commuting-zone fixed effects. Equation (4) captures changes within continuing studios. Equation (5) additionally allows changes in which studios are active to contribute to the estimated response.

The campaign took place during 1988. Because the data record release year rather than the date at which production occurred relative to the local demonstration, I retain 1988 in the sample and treat it as part of the pre-period in the pooled specifications. The pooled estimates therefore compare 1980–1988 with 1989–2000, while the event studies use 1988 as the omitted reference year. Appendix C.6 reports the sensitivity of the results to alternative treatments of the campaign year.

The two specifications serve different purposes. The event studies show the timing of adjustment and whether exposed and comparison commuting zones were already moving

differently before the campaign. The pooled coefficient provides a single average contrast between the pre- and post-campaign periods.

4.4 Identification

The identifying assumption is that, absent the campaign, exposed and comparison commuting zones would have followed parallel trends after conditioning on commuting-zone fixed effects and census-division-by-year effects.

The two groups differ substantially in levels before the campaign. As Table 1 shows, exposed commuting zones are larger, denser, more urban, and more active in recorded-music production. The design does not require these levels to be equal. Identification requires their outcomes to have followed similar trends in the absence of campaign exposure.

The comparison-tour design makes this assumption more plausible by restricting the estimation sample to commuting zones reached by major national tours in the same year. I then examine its validity in several ways. First, the event studies show whether exposed and comparison commuting zones were already moving differently before 1988. Second, I allow predetermined 1980 commuting-zone characteristics, including density, urban share, and income, to generate separate year-specific trajectories. Third, I assign placebo exposure using other major tours that were not accompanied by the AKAI campaign. Fourth, I compare exposed commuting zones with those containing the documented cancelled *Bad* Tour destinations described above. I also apply the design to local population, employment, and income, outcomes for which the sampler provides no direct mechanism for a discrete response to campaign exposure.

For the main organizational outcome, I additionally use reassignment exercises that compare the observed estimate with estimates obtained after assigning exposure to alternative sets of commuting zones matched on census division and pre-campaign recording activity. Appendix E.13 also applies the approach of [Rambachan and Roth \(2023\)](#) to assess sensitivity to departures from parallel trends.

Because musicians, producers, and production practices can move across commuting zones, campaign effects may also diffuse beyond the exposed commuting zones. Spillovers into comparison commuting zones would reduce the contrast between the two groups, so the estimates should be interpreted as effects of local campaign exposure relative to the comparison commuting zones in the design.

Analyses based on realized sampling, including the comparison of artist backgrounds, are descriptive because sampling is chosen rather than assigned. I identify these analyses explicitly when reporting them.

4.5 Inference

Treatment varies at the commuting-zone level, so standard errors are clustered by commuting zone throughout. Because only eighteen commuting zones are exposed, the main confidence intervals use the bias-reduced cluster-robust CR2 estimator with coefficient-specific Satterthwaite degrees of freedom. Conventional cluster-robust inference, CRV1, is reported as a benchmark in the appendix. Where CR2 inference is unavailable, I state the alternative inference procedure explicitly.

5 Results: diffusion and participation

I begin with two commuting-zone-level margins of adjustment. First, did campaign exposure accelerate the diffusion of sampling? Second, did participation in recorded-music production change relative to local population? These results establish whether the campaign changed use of the technology it promoted and whether that diffusion was accompanied by a change in how many people participated before turning to the division of labor within records.

5.1 Campaign exposure and the diffusion of sampling

Figure 4 reports the event-study estimates for documented sampling. Before the campaign, exposed and comparison commuting zones show little differential movement: all pre-period coefficients lie within 1.6 percentage points of the 1988 reference year and their confidence intervals include zero. After 1988, the series turns positive, reaching 2.5 percentage points in 1990, 4.7 points in 1992, and 5.0 points in 1993 before declining later in the decade.

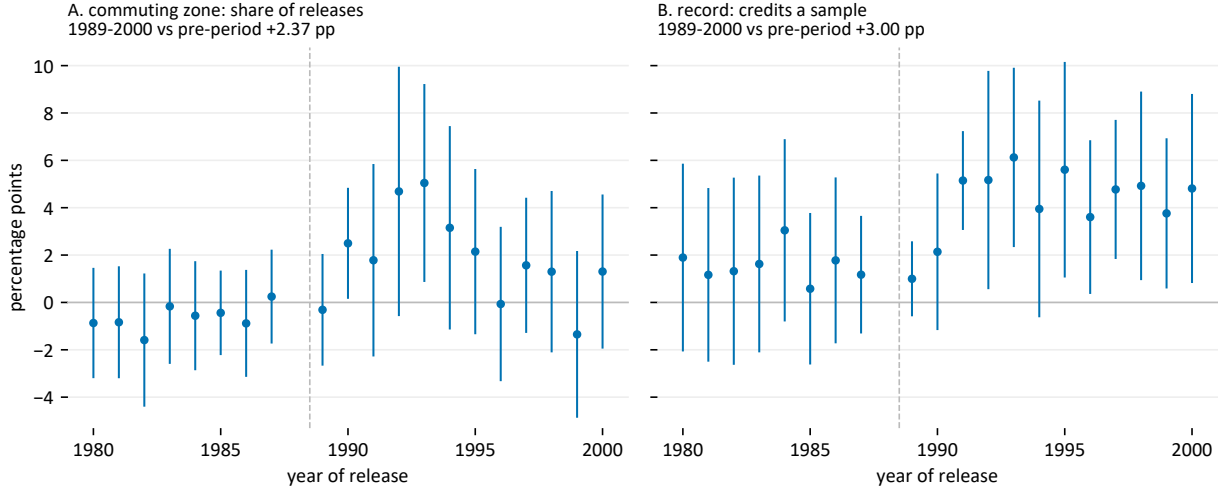
Pooling 1989–2000, campaign exposure is associated with a 2.37 percentage point increase in the share of records documenting a sample, relative to a pre-campaign mean of 1.01 percent. The corresponding studio-level measure moves in the same direction. Campaign exposure therefore increased documented sampling.

The timing and direction of the result are similar across alternative comparison samples and leave-one-out exercises. Reassigning exposure using comparable major tours that were not accompanied by the AKAI campaign produces a substantially smaller estimate, while commuting zones associated with unrealized *Bad* Tour dates show no corresponding increase. Appendix D.4 reports these exercises and the underlying itinerary distinctions.

The pooled magnitude is more sensitive to allowing predetermined commuting-zone characteristics to generate separate trajectories. Interacting 1980 density, urban share, and income with year attenuates the commuting-zone estimate. The annual path nevertheless continues to show a positive post-campaign differential, with much of the attenuation arising

Figure 4

Documented sampling, by year, at two units



Documented sampling rises in commuting zones exposed to the AKAI campaign. *Note.* Annual coefficients on campaign exposure interacted with year indicators; 1988 is the omitted reference year. The left panel reports the share of a commuting zone's records documenting a sample. The right panel reports the corresponding record-level indicator. Bars are 95% confidence intervals under the CR2 estimator with Satterthwaite degrees of freedom. Neither panel includes predetermined commuting-zone characteristics.

from changes in the estimated pre-period contrast. I therefore interpret the diffusion evidence primarily as establishing that campaign exposure shifted documented sampling rather than attaching particular weight to a single pooled magnitude. Realized sampling remains an endogenous outcome in all subsequent analyses.

5.2 Participation shows no detectable change

I next ask whether campaign exposure changed participation in recorded-music production. I measure participation as the number of unique credited contributors per 100,000 residents aged 20–64 in each commuting zone and year.

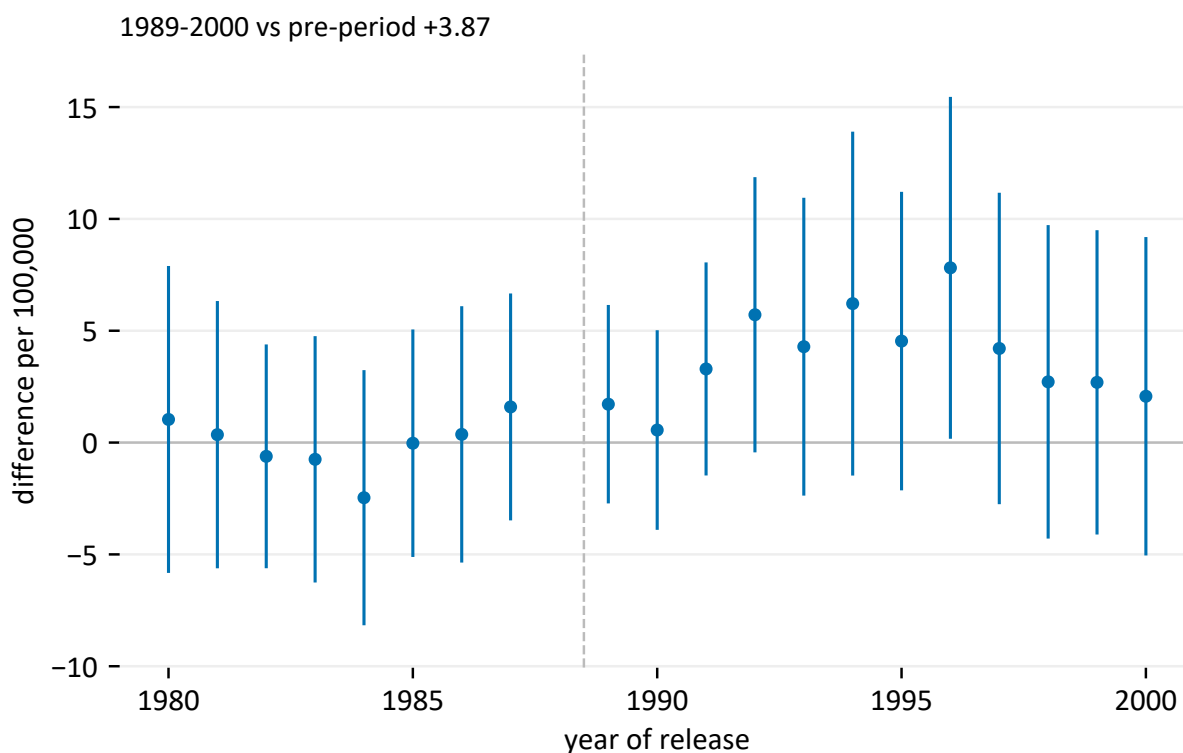
Figure 5 reports the annual estimates. The post-campaign coefficients are generally positive in the baseline specification but imprecisely estimated. Pooling 1989–2000, campaign exposure is associated with 3.88 additional credited contributors per 100,000, relative to a pre-period mean of 14.5. Allowing predetermined 1980 density, urban share, and income to generate their own trajectories changes the estimate to -3.32 . A Poisson specification for contributor counts with working-age population as an offset leads to the same qualitative conclusion.

The data therefore show no detectable change in participation relative to local population.

This is not a mechanical consequence of the population denominator: working-age population itself shows little differential movement after the campaign. At the same time, the result does not imply that the absolute scale of recorded-music production was unchanged. Credited contributor counts rise in exposed commuting zones in both the baseline specification and the specification allowing predetermined characteristics to generate separate trajectories. The evidence therefore distinguishes growth in the absolute number of credited participants from a change in participation relative to the size of the local population.

Figure 5

Credited people per 100,000 working-age residents, by year



Participation shows no detectable change after campaign exposure. *Note.* Annual coefficients from regressions of unique credited contributors per 100,000 residents aged 20–64 on campaign exposure interacted with year indicators; 1988 is the omitted reference year. Bars are 95% confidence intervals under the CR2 estimator with Satterthwaite degrees of freedom.

Appendix Figure D.4 decomposes participation according to whether contributors are associated with records documenting sampling. The positive movement in the baseline specification is concentrated in the smaller sampling-linked component, with little corresponding movement among other contributors.

Together, these commuting-zone-level results locate the margin of adjustment studied

next. Campaign exposure increased documented sampling, while participation relative to local population showed no detectable change. Section 6 therefore turns inside the production unit and asks how tasks were reallocated among the musicians associated with each record.

6 Results: the organization of production

Campaign exposure is followed by a reorganization of production beginning around 1993. The relative allocation of credited work shifts away from execution and toward creative authorship and production. Looking inside that reallocation, records come to contain more musicians whose tasks are confined to creative authorship and production, while individual task bundles do not measurably broaden. The organizational response therefore operates primarily through team composition rather than through broader individual task bundles.

Throughout this section, the preferred specification is Equation (4), which compares records produced by the same studio before and after local campaign exposure with contemporaneous changes in comparison commuting zones. The complementary commuting-zone specification in Equation (5) allows changes in the composition of active studios to contribute to the response and is reported in Appendix E. Standard errors are clustered by commuting zone throughout.

6.1 The allocation of credited work shifts toward creative authorship and production

I begin with the broad allocation of credited work across the three task families. Figure 6 reports annual estimates for the share of classified credits assigned to creative authorship and production, execution, and nonproduction. Because the three families partition every classified credit, the coefficients sum to zero within each year.

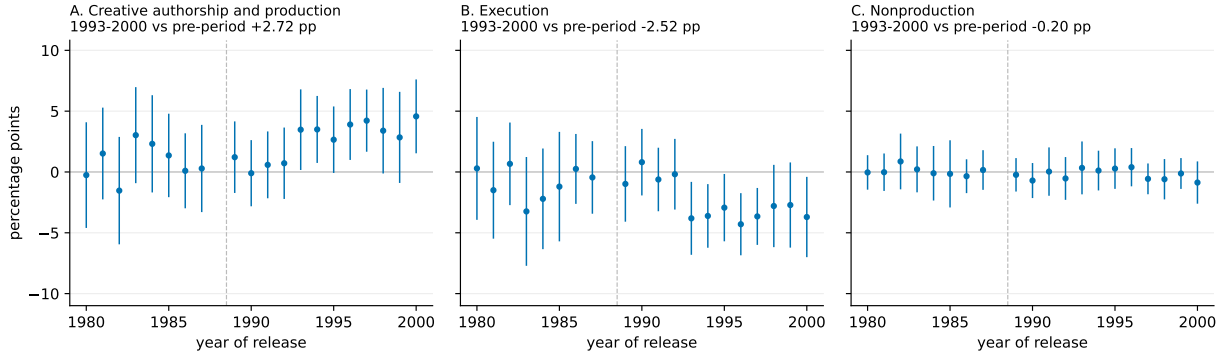
Before the campaign, the three shares show little differential movement. For creative authorship and production, the fitted pre-period slope is close to zero and no pre-campaign annual coefficient separates from zero. Little movement is visible during 1989–1992 either.

Beginning around 1993, the allocation changes. Over 1993–2000, the share of credits assigned to creative authorship and production is 2.72 percentage points higher relative to the pre-period, from a baseline of 34.3 percent, while execution is 2.52 points lower from a baseline of 55.9 percent. Five of the eight annual creative-authorship-and-production coefficients and six of the eight execution coefficients separate from zero. Nonproduction changes by only -0.20 percentage points and remains approximately flat.

These movements describe a single reallocation rather than three independent results.

Figure 6

The allocation of credited work across the three families



Credited work shifts in relative weight from execution toward creative authorship and production, beginning around 1993. *Note.* Annual coefficients from record-level regressions of each family’s share of classified credits on campaign exposure interacted with year indicators; 1988 is the omitted reference year. Studio and division-year effects, following Equation (2). The three families partition every classified credit, so the annual coefficients sum to zero at each year and the panels share one vertical scale. Bars are 95% confidence intervals under the bias-reduced estimator with coefficient-specific Satterthwaite degrees of freedom. Standard errors are clustered by commuting zone. 70,707 records, 116 commuting zones, 18 exposed.

With nonproduction largely unchanged, the differential movement occurs between execution and creative authorship and production. The timing is also concentrated in the later post-campaign years. Pooling 1989–2000 gives a smaller increase of 1.57 percentage points in creative authorship and production because the estimate averages the relatively flat 1989–1992 period with the later movement. I therefore use the annual path to describe when the reallocation emerges and the pooled specification to summarize the average post-campaign contrast.

The movement is weighted toward authorship. Creative authorship and production contains two blocks of almost equal pre-period weight. *Authorship*, comprising writing and composition, accounts for 16.8 percent of classified credits, while *production and arrangement* accounts for 17.5 percent. Over 1993–2000, authorship rises by 1.71 percentage points and production and arrangement by 1.01 points. The reallocation is therefore weighted toward authorship but is not confined to it. Appendix E.6 reports the finer decomposition.

The concentration in writing and composition raises an attribution concern. Following changes in sample-clearance practices after 1991, writers and composers of sampled material could receive credit on a new record without participating in producing it. Section 8.2 addresses this issue directly. I therefore state the main organizational result in terms of the broader creative-authorship-and-production category and interpret the particular concentration in authorship more cautiously.

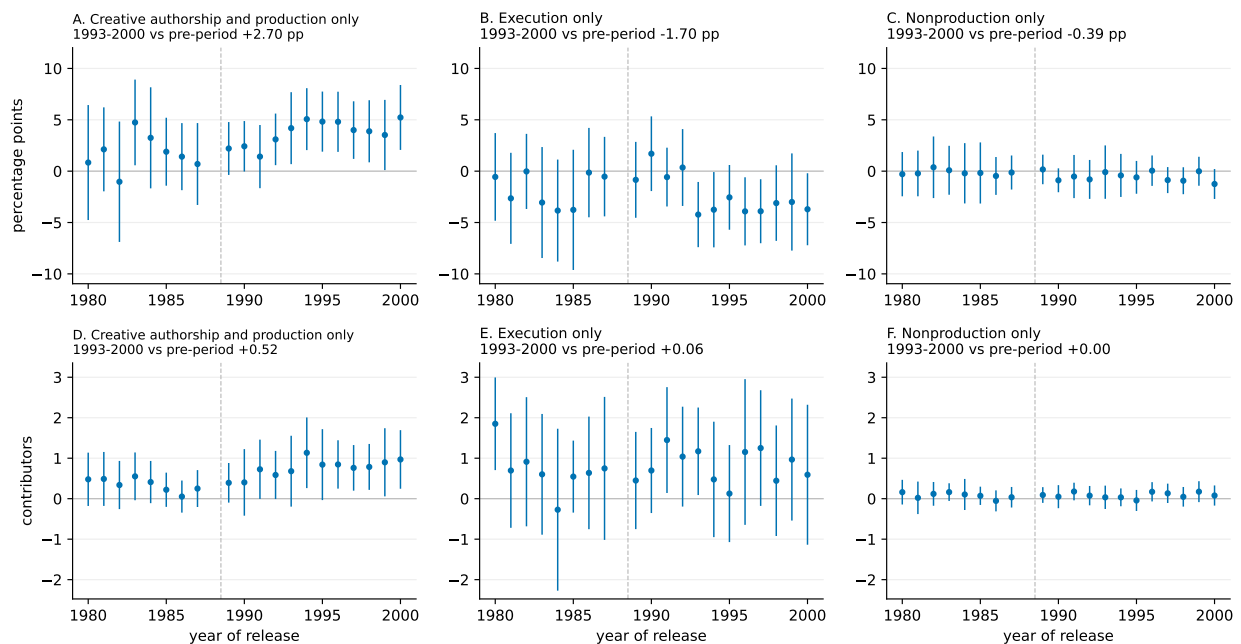
Credit shares establish the direction and timing of the reallocation, but they depend on the number of credits elsewhere on the record and are more sensitive to alternative control sets and release formats. I therefore next examine how the same tasks are distributed across the musicians associated with each record.

6.2 The reallocation occurs through team composition

As defined in Section 3.2, every contributor on a record falls into exactly one of four mutually exclusive cells: credited only in creative authorship and production, only in execution, only in nonproduction, or in two or more of the three families. These cells exhaust the credited team and can be expressed either as shares of contributors or as counts. Figure 7 reports both.

Figure 7

The same partition of the credited team, as a share of contributors and as a count



Records gain contributors credited exclusively in creative authorship and production beginning around 1993.

Note. Annual coefficients; 1988 is the omitted reference year. Studio and division-year effects, following Equation (2). A contributor is assigned to a family only if every credit they hold on the record falls within it. Panels A–C report shares of the credited team and Panels D–F counts of contributors per record. Contributors credited in two or more families and total team size complete the partition and are reported in Appendix E.10. Bars are 95% confidence intervals under the bias-reduced estimator with coefficient-specific Satterthwaite degrees of freedom. Standard errors are clustered by commuting zone. 70,707 records, 116 commuting zones, 18 exposed.

The share measures reproduce the reallocation observed in credit lines. Over 1993–2000, the share of contributors credited exclusively in creative authorship and production rises by 2.70 percentage points from a pre-period mean of 23.5 percent. The execution-only share

falls by 1.70 points, while the nonproduction-only share changes little. The same reallocation is therefore visible when the unit is the person rather than the credit line.

The count measures show where that change comes from. Records gain 0.52 contributors whose credited tasks fall entirely within creative authorship and production over 1993–2000, relative to a pre-period mean of 3.48, an increase of about 15 percent. Six of the eight annual coefficients from 1993 onward separate from zero, while none of the eight pre-campaign coefficients does. As with the credit shares, little movement is visible immediately after the campaign and the increase becomes sustained beginning around 1993.

The result does not depend on defining the post-period from that later path. Under the baseline pooled specification, which sets $\text{Post} = 1[t \geq 1989]$, the estimate is +0.44 contributors and remains statistically distinguishable from zero. Averaging the relatively flat 1989–1992 period into the post-treatment window therefore reduces the magnitude, as implied by the annual estimates, but does not eliminate the increase.

The remaining cells show no corresponding offset in their point estimates. Contributors credited only in execution move by +0.06 from a pre-period mean of 7.72, contributors credited only in nonproduction are essentially unchanged, and contributors spanning two or more families move by −0.12. Total credited team size increases by 0.45. Some of these estimates are too imprecise, or their pre-period paths too uneven, to support strong null claims. The narrower conclusion is that the increase in contributors concentrated in creative authorship and production is not mechanically generated by a decline in the observed number of contributors in another cell.

The absolute count is the most reliable measure of this organizational change. Unlike a share, it does not increase mechanically when the denominator falls, and it is substantially more stable across release formats and alternative specifications. Allowing predetermined 1980 commuting-zone characteristics to generate separate year-specific trajectories leaves the estimate essentially unchanged at +0.54 contributors. Reassigning exposure to alternative sets of eighteen commuting zones matched on census division and pre-campaign recording activity places the observed estimate above roughly 96 percent of reassignments. The result also survives bounded departures from parallel trends using the approach of [Rambachan and Roth \(2023\)](#). Extending the sample to 1965–2005, which the raw source supports but the main estimation frame does not use, yields an estimate of +0.39 contributors with an interval that still excludes zero. [Appendix E.7](#), [Appendix E.12](#), [Appendix E.13](#), and [Appendix C.9](#) report these exercises.

The count also clarifies the role of sample clearance. [Section 8.2](#) removes potentially attribution-related writing and composition credits from sampled records after 1991 and reconstructs the contributor measure. Removing all eligible authorship credits accounts for

about 17 percent of the increase in contributors concentrated in creative authorship and production. Copyright attribution can therefore explain part of the measured increase, but not most of it.

Where the increase is located. The record-level estimand weights commuting zones in proportion to the amount of recorded music they produce. This matters because recorded-music production is highly concentrated geographically. Collapsing the same observations to commuting-zone-by-year averages and weighting each zone-year by its number of records gives an estimate of +0.59 contributors, close to the record-level result. Giving every commuting-zone-year equal weight instead produces essentially no change. As discussed in Section 4.2, the organizational result therefore describes the records produced in commuting zones where recorded music is concentrated rather than the average US commuting zone. Appendix C.8 reports the full weighting comparison.

These measures remain measures of credited responsibility. A contributor count identifies the people to whom tasks are attributed on the released record; it does not measure hours worked, contractual authority, decision rights, or uncredited labor.

6.3 No generalist response

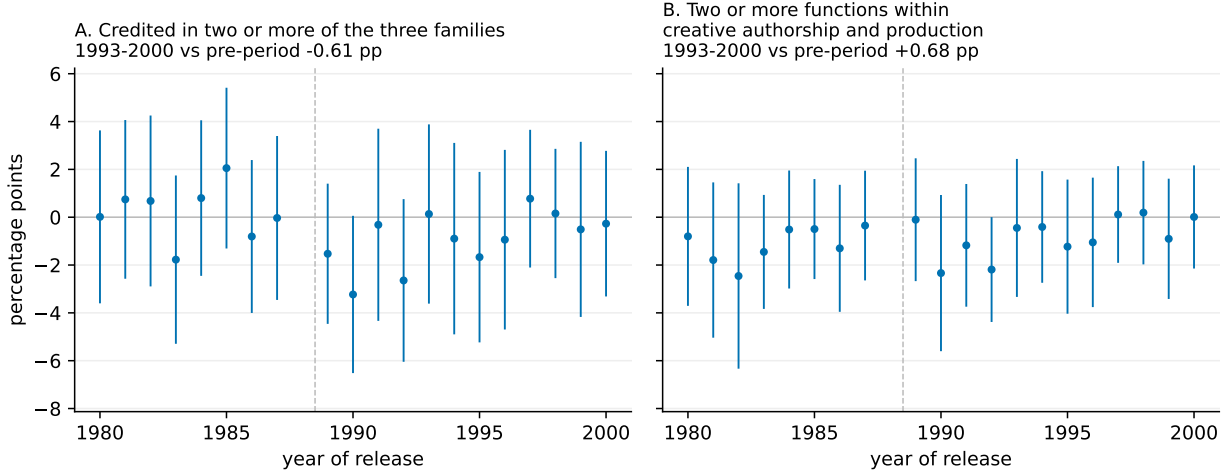
Whether technological change leads individuals to span a broader set of tasks or instead reinforces specialization is a central question in theories of knowledge accumulation and team production (Jones, 2009). The increase in contributors concentrated in particular tasks could coexist with a second response in which individuals themselves take on broader task bundles. Because the sampler combined several production tasks within one workstation, technological integration could have allowed the same person to accumulate tasks that had previously been distributed across different people. I test for such a generalist response on two margins.

The first measure captures breadth across the three broad task families: the share of contributors credited in two or more of them. Campaign exposure changes this share by -1.24 percentage points in the pooled post-period, relative to a pre-period mean of 18.6 percent. The second measure asks whether contributors accumulate multiple distinct tasks within creative authorship and production itself. This share changes by -0.04 percentage points from a pre-period mean of 10.3 percent. Neither measure separates from zero in the pooled specification, over 1993–2000, or in any individual year.

These estimates do not establish that contributors’ underlying work was literally unchanged, and they allow for modest changes in breadth. They do show that measurable broadening is absent on both margins. Technological integration therefore did not translate into broader credited task bundles within individuals.

Figure 8

Contributors do not measurably broaden their credited functions



Contributors do not measurably broaden their credited task bundles. *Note.* Annual coefficients; 1988 is the omitted reference year. Studio and division-year effects, following Equation (2). Panel A reports the share of a record’s contributors credited in two or more of the three broad families. Panel B reports the share of contributors holding two or more distinct tasks within creative authorship and production. Panel A is mechanically related to the exclusivity decomposition in Figure 7; Panel B is not. Bars are 95% confidence intervals under the bias-reduced estimator with coefficient-specific Satterthwaite degrees of freedom. Standard errors are clustered by commuting zone. 70,707 records, 116 commuting zones, 18 exposed.

Taken together, the organizational evidence has a consistent structure. Credited work shifts in relative terms from execution toward creative authorship and production; records increasingly contain contributors whose credited tasks are confined to the latter category; and individual task bundles do not measurably broaden. The reorganization therefore occurs primarily through team composition rather than through the emergence of broader generalists.

6.4 Expertise leveling

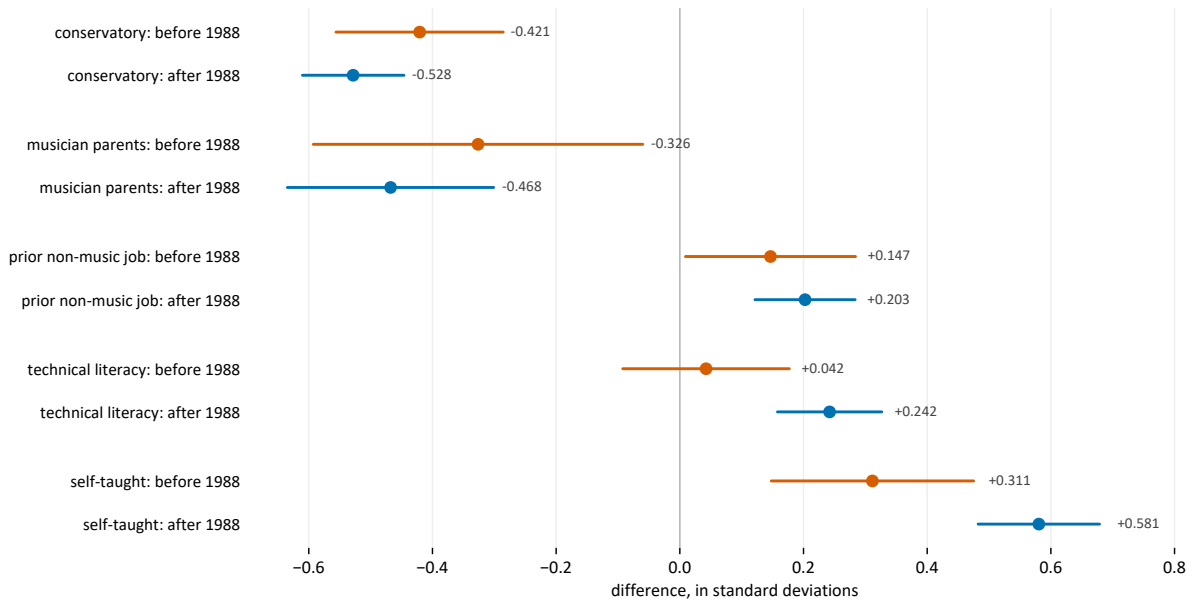
The characteristics of artists associated with sampled records provide complementary evidence on the expertise margin.

The largest differences concern musical training. In 1989–2000, 65 percent of main artists associated with sampled records are coded as self-taught, compared with 37 percent among other artists. Formal musical training is reported for 15 percent of artists associated with sampled records and 40 percent of other artists; conservatory training for 10 and 30 percent, respectively; and higher education for 16 and 36 percent. Artists associated with sampled records are also younger and somewhat more likely to have technically oriented backgrounds.

Sampled production is therefore disproportionately associated with artists outside traditional channels of specialist musical training. This comparison is descriptive rather than

Figure 9

Adopters against other main artists, on five predetermined characteristics



Artists associated with sampled records are less formally trained. *Note.* Differences between main artists associated with sampled and non-sampled records, standardized by the pooled dispersion of each characteristic and reported separately before and after 1988. The comparison is descriptive: realized sampling is chosen rather than assigned. Coverage varies across characteristics and is reported in Appendix Table F.4.

causal, but it provides complementary evidence that the technology was used disproportionately by artists for whom traditional forms of musical expertise were less prevalent, consistent with the expertise-leveling characterization of the technological episode.

7 Results: a different sound

The preceding sections show that campaign exposure increased documented sampling and changed the division of labor within records. I now ask whether campaign exposure is also reflected in the records produced. The outcome measures how far a record lies from what the industry had recently produced.

Following [Askin and Mauskopf \(2017\)](#), I locate each record relative to music released in the preceding year, weighting each comparison record by the proximity of its genre to the record’s own, and measure one minus that weighted average similarity. Higher values therefore indicate greater distance from the industry’s recent output.⁹ I construct this distance using two representations of the same output: the acoustic characteristics of the recording and an embedding of its lyrics.¹⁰

The content analysis uses commuting-zone rather than studio fixed effects. The object of interest is the position of records produced within a commuting zone relative to the industry’s recent output, rather than the output of a particular recording room. Musical influence can travel across studios through artists, producers, performers, and local scenes, making the commuting zone the natural unit to hold fixed. Appendix [H.3](#) reports the corresponding studio-fixed-effect estimates and documents the specification history.

7.1 Sound moves away from its recent past

Figure [10](#) reports the annual estimates for the acoustic and lyrical representations. In acoustic characteristics, the coefficients from 1980 through 1990 are small and statistically indistinguishable from zero. Beginning in 1991, the estimates turn positive and remain so through 2000, with eight of the twelve post-campaign coefficients statistically significant. Pooling the post-campaign period, records produced in exposed commuting zones are 0.167

⁹The comparison set is the released catalogue rather than the chart, so the measure captures supply-side differentiation rather than reproducing the original construct exactly. The eleven acoustic characteristics are the same as those used by [Askin and Mauskopf \(2017\)](#). The genre weighting is close to inert here and the measure is therefore effectively a catalogue-wide comparison rather than a within-genre one: weights between distinct genres average 0.94, own-genre records account for 18.5 percent of the comparison mass and 5.9 percent for hip hop, and the measure correlates 0.97 with a completely genre-blind version. Appendix [H.5](#) reports the corresponding within-genre measure.

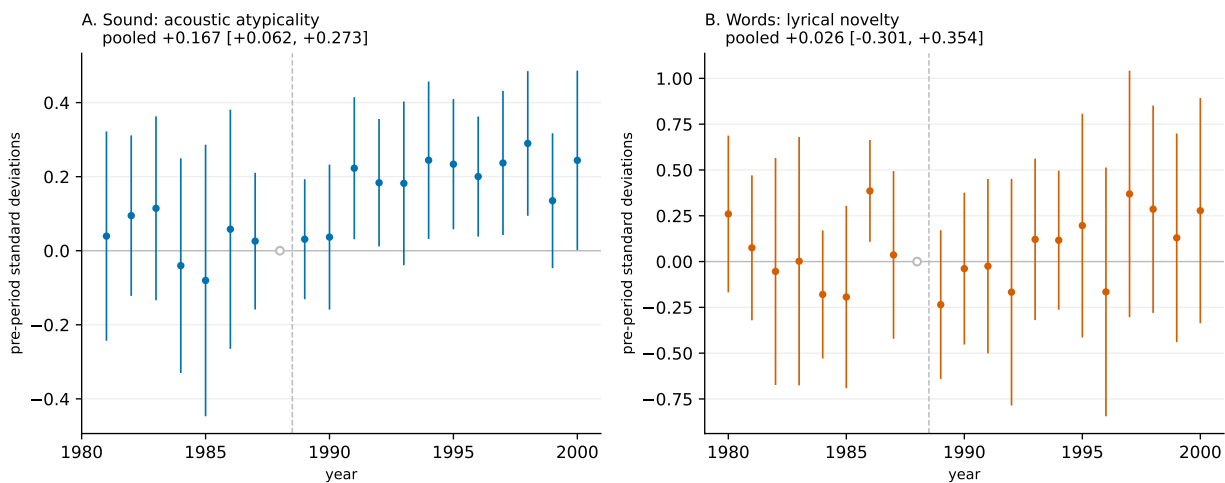
¹⁰Lyrical distance is corrected for the mechanical relationship between lyric length and measured similarity using within-song truncation experiments. Appendix [H.2](#) describes the construction and its assumptions.

pre-period standard deviations farther from the recent catalogue than records produced in comparison commuting zones. The annual path therefore indicates a persistent acoustic repositioning rather than a short-lived movement.

The corresponding lyrical estimate is $+0.026$ pre-period standard deviations and is not statistically significant in the pooled specification or in any post-campaign year. I therefore detect a change in the acoustic position of records but no comparable detectable movement in lyrics. This does not establish that the acoustic and lyrical responses differ from one another, since the lyrical estimates are substantially less precise and permit changes of economically meaningful size in either direction.¹¹

Figure 10

How far a record sits from what came before it



Records in exposed commuting zones move acoustically away from the recent catalogue. *Note.* Annual coefficients on campaign exposure; 1988 is the omitted reference year and is retained in the sample as a pre-period observation. Panel A reports one minus a record's genre-weighted mean acoustic similarity to the previous year's releases; Panel B reports the analogous lyrical-distance measure. Outcomes are standardized by their pre-period dispersion and the panels use independent vertical scales. Commuting-zone and census-division-by-year fixed effects. Bars are 95% confidence intervals under the bias-reduced estimator with coefficient-specific Satterthwaite degrees of freedom, clustered by commuting zone. Neither outcome was pre-specified.

The result adds a product margin to the adjustment documented in the preceding sections. Campaign exposure changes not only documented sampling and the organization of production, but also the acoustic characteristics of the records produced in exposed commuting zones. Its direction is notable. By lowering the cost of copying and recombining existing recordings, the sampler made imitation technologically easier. Yet records produced in exposed commuting zones become more acoustically differentiated from the recent catalogue rather than moving

¹¹A second, independently constructed representation of the audio yields an estimate of the same sign and comparable magnitude on the records for which both measures are available, although with much wider confidence intervals. The headline acoustic result is also present when the sample is restricted to those records. Appendix H.6 reports these exercises, and Appendix H.2 reports the lyrical measurement exercises.

closer to it. The technology is therefore associated with acoustic repositioning away from recent output, not convergence toward it.

The distance measure does not identify exactly what form that repositioning takes. It cannot distinguish a commuting zone whose records spread out across acoustic space from one whose records move together toward a different region of that space. The reduced-form result is therefore a change in the position of output from exposed commuting zones relative to the recent catalogue, rather than a claim about greater dispersion within exposed commuting zones. I return to this interpretation in Section 8.

The timing also differs across margins. Documented sampling begins to increase soon after the campaign, the acoustic position of records shifts beginning in 1991, and the change in the division of labor becomes visible around 1993. Product characteristics and production organization therefore do not adjust simultaneously.

7.2 The content signature of sampled records

The reduced-form estimates above ask how campaign exposure changed the records produced in a commuting zone. A different question is what records that actually document sampling look like. Because realized sampling is chosen rather than assigned, these comparisons are descriptive and are not interpreted as causal effects of sampling.

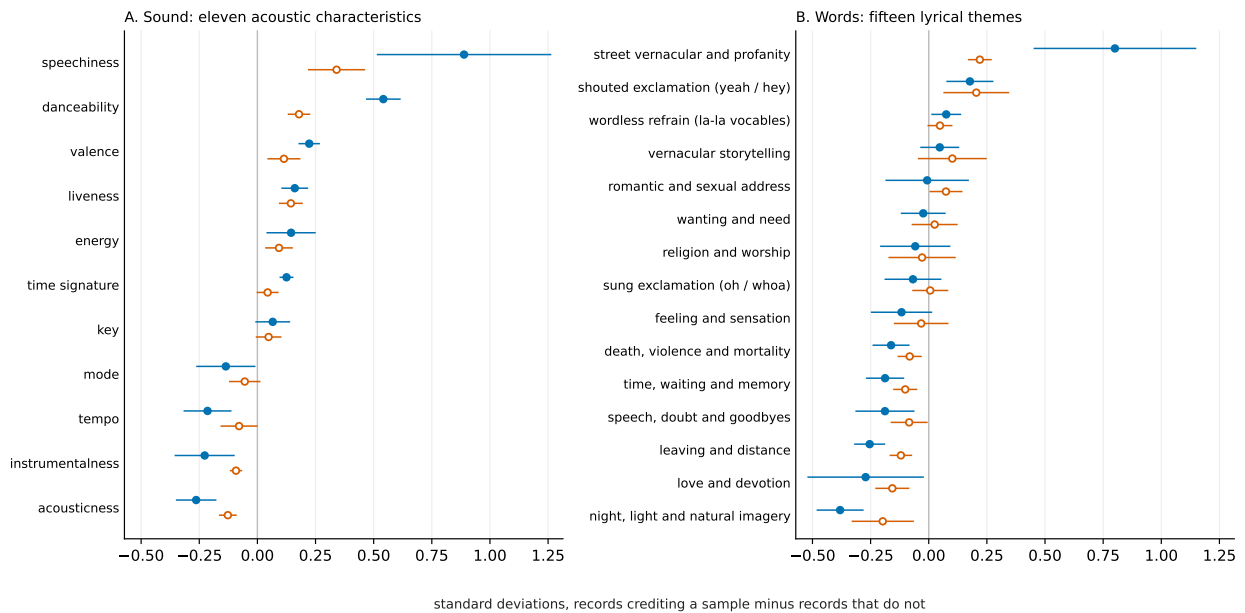
Figure 11 compares sampled and non-sampled records in both representations. Sampled records have a distinctive content profile, much of which reflects genre composition. In sound, ten of the eleven acoustic characteristics differ between sampled and non-sampled records. Speechiness is 0.89 standard deviations higher and danceability 0.54 standard deviations higher among sampled records. Holding broad genre fixed absorbs a median of about 59 percent of the acoustic differences. In lyrics, the largest difference is in street vernacular and profanity, at about 0.80 standard deviations, of which genre absorbs roughly three quarters. Broad genre similarly absorbs about half of the differences across lyrical themes more generally.

The descriptive signature differs from the reduced-form product response. Campaign exposure shifts a record's overall acoustic position relative to the recent catalogue, while no single acoustic characteristic reproduces the broad cross-sectional profile associated with documented sampling. The exposure effect is therefore not simply a scaled-down version of the observable characteristics of sampled records.

The magnitudes point in the same direction. Campaign exposure raises documented sampling by about 2.4 percentage points, while the reduced-form acoustic effect is 0.167 pre-period standard deviations averaged across the records produced in exposed commuting

Figure 11

What a record that credits a sample looks like, before and after genre



Sampled records have a distinctive content profile, much of which reflects genre. *Note.* Differences between records documenting a sample and records without a documented sample, in standard deviations of each outcome. Filled markers include studio and census-division-by-year fixed effects; hollow markers additionally hold broad genre fixed on the same observations. Realized sampling is chosen rather than assigned, so these differences are descriptive. Bars are 95% confidence intervals under the bias-reduced estimator with coefficient-specific Satterthwaite degrees of freedom, clustered by commuting zone.

zones. An average movement of that size is difficult to attribute solely to the addition of a small number of records that document sampling. Together with the absence of a single acoustic characteristic reproducing the sampled-record profile, this pattern is consistent with campaign exposure affecting the broader composition or characteristics of local production rather than operating only through the content of documented sampled records.

8 Discussion

The results distinguish technological integration from the organization of human work. The sampler combined several production tasks within a single workstation, yet musicians do not measurably broaden their credited task bundles. Instead, records in exposed commuting zones come to contain more musicians whose credited tasks are concentrated in creative authorship and production, while the relative allocation of credits shifts away from execution and toward those tasks. The adjustment therefore occurs primarily through team composition rather than through broader individual task bundles.

This section develops that interpretation, examines how sample clearance affects the authorship component of the result, and considers which features of the episode travel beyond recorded music.

8.1 Technological integration and organizational specialization

Campaign exposure is followed by records containing more musicians working exclusively in creative authorship and production, while individual task bundles show no detectable broadening. The organizational response therefore takes the form of specialization through team composition: production combines more musicians concentrated in particular tasks, even though the technology makes several tasks easier for one person to perform.

The sampler could affect task allocation through several channels. Sampling allowed existing recorded performances to substitute for some newly performed parts, while sequencing and rhythm programming reduced the instrumental, technical, and coordination requirements involved in assembling musical elements. Lowering these requirements could encourage musicians to perform more tasks themselves, reducing their reliance on specialists. At the same time, selecting material, deciding how elements fit together, and shaping the resulting recording remained human tasks. Reflecting on electronic music production, Thomas Bangalter described “part of the creative process that is very much human-based and not algorithmic of any sort.”¹² Making some production tasks easier could consequently change

¹²Thomas Bangalter, quoted from a 2023 BBC interview in Liberty Dunworth, “Daft Punk’s Thomas

the relative value of the expertise and creative judgment required for others.

One interpretation emphasizes this change in comparative advantage (Becker and Murphy, 1992; Garicano, 2000; Jones, 2009). As some execution tasks become easier to perform, production may place greater weight on musicians' capabilities in selecting, developing, and combining musical material. Musicians with different creative strengths could contribute to the same record while remaining concentrated in particular tasks. The relative decline in execution credits and the increase in musicians working exclusively in creative authorship and production are consistent with this reallocation.

The tradeoff between specialization and adaptation provides a second perspective. In Dessein and Santos (2006), allowing workers to tailor tasks to local information creates coordination demands when their decisions are interdependent. Broader task bundles can help maintain coordination, at the cost of specialization. Better communication can therefore reduce specialization if it encourages sufficiently greater adaptation. Applied to music production, this framework suggests that easier experimentation could favor broader task bundles when changes to rhythm, arrangement, and sound require closely coordinated decisions. A musician able to revise these elements together could respond more readily to an emerging musical idea.

A separate force could favor specialist collaboration. By making recorded parts easier to combine and rearrange within a common workflow, the sampler could reduce the cost of assembling specialized inputs (Becker and Murphy, 1992; Garicano, 2000; Agrawal and Goldfarb, 2008). If integrating contributions from different musicians became easier, production could draw on their distinct capabilities without requiring each musician to span several production tasks. The relevant tradeoff is therefore between the value of coordinating adjustments within one person's task bundle and the gains from combining specialized contributions across people.

The observed change in team composition is consistent with gains from specialized contributions or lower costs of combining them. The credit data do not directly measure comparative advantage, coordination costs, or how production decisions were adapted, so they cannot establish which force generated the response. Acoustic differentiation likewise measures a property of the resulting output rather than the process of adaptation inside the studio. The evidence establishes that technological integration was accompanied by more musicians concentrated in creative tasks, without detectable broadening of individual task bundles.

Musicians' backgrounds provide complementary evidence on expertise requirements. Main

Bangalter explains the reason for their split," *NME*, April 4, 2023, <https://www.nme.com/news/music/daft-punks-thomas-bangalter-explains-the-reason-for-their-split-3425585>.

artists on sampled records are more likely to be self-taught and less likely to report formal or conservatory musical training. Sampling was therefore used disproportionately by artists outside traditional channels of specialist musical training. This association is consistent with lower expertise-related barriers to production, although it does not establish that the technology caused these artists to participate.

The broader implication is that expertise-leveling technology can change both the tasks an individual can perform and the gains from collaborating with others. Its organizational consequences depend on how those changes affect the benefits of specialization and the costs of coordinating interdependent tasks. In the sampling episode, the observable adjustment occurred through the composition of production teams.

Which capabilities of the technology matter. Two appendix exercises help distinguish sample reuse from the broader production system embodied in the sampler. The organizational change is not concentrated in genres or records that themselves document sampling, and much of the increase in musicians concentrated in creative authorship and production appears on records with no documented sample. By contrast, predetermined measures of programming, sequencing, and drum-machine activity are positively associated with the differential organizational response in some specifications.

These patterns are consistent with the response reflecting the broader bundle of machine-based production capabilities associated with the sampler. The device also functioned as a drum machine and sequencer, so documented sampling captures only part of the production practices that its diffusion could alter. The evidence therefore supports considering these capabilities together, although it does not identify which individual capability generated the organizational response.

Timing. The organizational response emerges gradually. Documented sampling increases soon after the campaign, whereas the change in team composition becomes visible around 1993. This delay is consistent with diffusion, learning, complementary investment, or the gradual development of new production practices. The design does not distinguish among these mechanisms. The pooled post-campaign estimate for the number of musicians working exclusively in creative authorship and production remains positive, so the organizational result does not depend on defining 1993 as the beginning of the treatment period.

The product results clarify the sequence. Acoustic differentiation emerges around 1991, before the change in team composition becomes clearly visible, while lyrics show no comparable detectable change. The division of labor therefore adjusts after sampling begins to diffuse and after changes in output become apparent. This timing connects to [Demirer et al. \(2026\)](#), who

distinguish short-run deployment of technology within fixed jobs from longer-run adjustment of job boundaries and skill requirements. Following the episode over a longer horizon reveals organizational changes that would be missed by observing only the initial years of diffusion.

8.2 Property rights and authorship attribution

The authorship component of the result requires separate interpretation because sampling operates on creative work produced by other people. A writing or composition credit on a sampled record can therefore represent two different economic objects: participation in creating the new record or attribution to the creator of an earlier work used as an input. Credits do not necessarily map one-for-one into contemporaneous production.

This distinction became more important during the sample period. Through much of the 1980s, sampling frequently proceeded without formal licenses. Following *Grand Upright Music, Ltd. v. Warner Bros. Records Inc.* in December 1991, formal sample clearance became standard industry practice. Clearing sampled material could generate writing or composition credits for creators of the underlying work even when those individuals did not participate in producing the new recording.

The timing makes clearance an important alternative explanation. The organizational movement becomes visible shortly after the legal environment changed, and the increase within creative authorship and production is weighted toward writing and composition. Neither fact identifies an effect of clearance: the ruling applied nationally, while the empirical design compares changes across exposed and comparison commuting zones, and gradual technological diffusion could generate similar timing.

I account for this issue directly in Appendix E.11. I remove potentially attribution-related writing and composition credits from sampled records after 1991 and reconstruct the main contributor outcome. Removing all eligible authorship credits explains only about 17 percent of the increase in musicians credited exclusively in creative authorship and production. Copyright attribution can therefore account for part of the measured increase, but not most of it.

This adjustment does not estimate how much attribution actually occurred. The data cannot link every credited person on a new record to the creators of each sampled source, and documentation of sampled material is incomplete. Its purpose is narrower: to assess whether sample-related attribution is mechanically large enough to account for the organizational result. Most of the increase remains after potentially attribution-related credits are removed.

Several additional patterns point in the same direction but are less decisive. Production and arrangement also move positively even though clearance directly generates writing and

composition credits rather than production credits. Most of the measured increase appears on records without a documented sample, although incomplete documentation means that some of these records may nevertheless use sampled material. Conversely, the increase is disproportionately associated with musicians making their first appearance in the credit panel, which is compatible both with new participation in production and with clearance bringing previously unobserved writers onto later records. None of these decompositions alone identifies the contribution of clearance.

The credit-share outcomes cannot resolve the issue either. Adding writing credits through attribution can simultaneously raise the creative-authorship-and-production share and reduce the execution share even if the number of execution credits remains unchanged. This denominator logic is one reason the absolute contributor count carries more interpretive weight than the movements in credit shares.

Taken together, the evidence does not imply that sample clearance was irrelevant or that credits measure contemporaneous production perfectly. Clearance changed attribution on at least some sampled records. The adjustment for potentially attribution-related credits nevertheless indicates that direct mechanical attribution is too small to explain most of the increase in musicians concentrated in creative authorship and production. The broader organizational result therefore survives the attribution concern, while the particular concentration of the movement in authorship should be interpreted more cautiously.

The distinction matters beyond music. When a technology uses prior human-created work as an input, observed authorship can reflect both current production and the institutional allocation of rights over earlier work. Changes in property rights can therefore alter measured credit without producing an equivalent change in contemporaneous labor.

8.3 What travels beyond recorded music

The episode provides a longer-run example of how an expertise-leveling technology can propagate through production. Its value for other settings is not that digital sampling and newer technologies are technologically equivalent, or that the magnitudes estimated here should transport directly. The relevant similarities are narrower: technology can reduce the expertise required for particular tasks, alter the relative scarcity of human capabilities, and interact with institutions governing the prior work on which production builds.

The first lesson concerns the margin of adjustment. Participation relative to local population shows no detectable change, while the division of labor changes substantially. A headcount alone would therefore miss the clearest response. More generally, technologies can reorganize which people perform which tasks without producing a commensurate change in

the number of people participating.

The second concerns horizon. The adjustment is sequential. Documented sampling increases first, while the division of labor changes only several years later. Historical technological transformations often required complementary changes in organization before their consequences became visible (Baily et al., 2026). The sampling episode makes that reorganization observable within a production unit. A study ending shortly after the introduction of the technology would have captured diffusion while missing much of the later organizational response.

This point is relevant to contemporary generative artificial intelligence, where current evidence necessarily covers relatively short horizons. Generative AI and digital sampling operate differently, but both can make tasks that previously required scarce expertise easier to perform. The historical episode therefore illustrates why short-run effects on employment or participation may not reveal the eventual division-of-labor response.

A third lesson concerns institutions. When a technology operates on prior human-created inputs, technological substitution and the property-rights regime governing those inputs are distinct economic margins. In recorded music, sample clearance changed who received authorship credit at the same time that the technology was changing current production. Similar issues arise whenever access to knowledge or creative inputs depends on licensing, attribution, or ownership (Heller and Eisenberg, 1998; Scotchmer, 1991; Demers, 2006; Caoui and Galasso, 2025). Changes in measured authorship can therefore reflect both technological change and institutional rules governing the inputs to that technology.

Finally, organizational change and product change can occur at different times. Campaign exposure produces a clear change in the division of labor and a detectable acoustic repositioning of the resulting records, while the lyrical representation does not show a comparable response. Observing the finished product alone would therefore provide an incomplete account of how production adjusted.

The broader lesson is consequently about margins rather than about a particular technology. Expertise-leveling technological change can affect participation, the allocation of tasks across people, and the products they create in different ways and at different speeds. Understanding its consequences requires observing these margins separately rather than inferring one from another.

8.4 Limits

Several limits bound the interpretation of the results. First, credits measure attributed tasks rather than the complete allocation of labor. They do not reveal hours, wages, effort,

contractual authority, uncredited contributions, or who had final decision rights. The results therefore concern the organization of credited work.

Second, treatment is geographic exposure to the campaign rather than verified clinic attendance or observed adoption of the sampler. The estimated coefficients are reduced-form effects of local campaign exposure. Analyses conditional on realized sampling, including the comparison of artist backgrounds, describe groups selected into sampling rather than randomly assigned adopters.

Third, the causal design relies on parallel counterfactual evolution between exposed and comparison commuting zones. Pre-treatment event studies, alternative comparison tours, predetermined-characteristic adjustments, and sensitivity exercises provide evidence on this assumption but cannot establish it. Residual differential shocks to exposed commuting zones remain the principal threat to the causal interpretation.

Fourth, the product analysis is exploratory. The acoustic evidence is detectable in the primary representation and has the same sign in an independently constructed representation, but the latter is substantially less precise, while lyrics show no comparable detectable movement. I therefore place greater weight on the organizational results than on any particular interpretation of the product response.

Fifth, the organizational estimates describe the records produced in commuting zones where recorded music is concentrated rather than the average American commuting zone. Record-level regressions weight commuting zones in proportion to their production, and the contributor-count result disappears when every commuting zone receives equal weight. This is a scope condition: the two weighting schemes answer different questions, and the paper’s organizational estimand concerns production where recorded music is actually made.

Finally, the setting is one creative industry and one technological episode. The results do not establish a general tendency for expertise-leveling technologies to increase specialization. They document one economically meaningful possibility: technological integration can be accompanied by specialization through team composition even when individual task bundles do not measurably broaden. Whether the same response emerges elsewhere will depend on the tasks affected, the distribution of expertise, coordination costs, and the institutions governing production inputs.

9 Conclusion

What happens when an expertise-leveling technology diffuses through an industry? I study digital sampling in recorded music using geographic variation in exposure to AKAI’s 1988 promotion of the MPC60 and production credits from 70,712 US records released between 1980

and 2000. Campaign exposure increased documented sampling, while participation relative to local population showed no detectable change. The clearest adjustment instead occurred inside the production unit, in how tasks were allocated across the musicians associated with each record.

Beginning around 1993, records in exposed commuting zones contain about 0.52 additional contributors whose tasks are confined to creative authorship and production, relative to a pre-period mean of 3.48, an increase of roughly 15 percent. I do not detect a corresponding decline in another observed contributor category, and individual task bundles do not measurably broaden. Credit shares move in the same direction, away from execution and toward creative authorship and production. Removing potentially attribution-related writing and composition credits from sampled records after 1991 explains only about 17 percent of the contributor-count increase, indicating that changes in sample clearance cannot account for most of the organizational result. Campaign exposure is also reflected in output: records produced in exposed commuting zones become more acoustically differentiated from the recent catalogue, while lyrics show no comparable detectable change.

The episode therefore separates technological integration from the organization of human work. The sampler expanded the set of production tasks one person could technically perform, but production did not reorganize through broader individual task bundles. Instead, the adjustment occurred through team composition, with records combining more human contributions concentrated in particular tasks. These margins also moved at different speeds: documented sampling increased first, acoustic characteristics began to shift next, and the division of labor changed only several years later. Expertise-leveling technological change can therefore reshape participation, the division of labor, and output differently and at different speeds. Understanding its consequences requires observing these margins separately rather than inferring one from another.

References

- Abbas Nejad, Kian, Giuseppe Musillo, Till Wicker, and Niccolò Zaccaria**, “Labor Market Signals: The Role of Large Language Models,” *Available at SSRN 5185035*, 2025.
- Acemoglu, Daron and David Autor**, “Skills, tasks and technologies: Implications for employment and earnings,” in “Handbook of labor economics,” Vol. 4, Elsevier, 2011, pp. 1043–1171.
- **and Pascual Restrepo**, “Automation and New Tasks: How Technology Displaces and Reinstates Labor,” *Journal of Economic Perspectives*, 2019, *33* (2), 3–30.
- **, David Autor, Jonathon Hazell, and Pascual Restrepo**, “Artificial Intelligence and Jobs: Evidence from Online Vacancies,” *Journal of Labor Economics*, 2022, *40* (S1), S293–S340.
- Agrawal, Ajay and Avi Goldfarb**, “Restructuring Research: Communication Costs and the Democratization of University Innovation,” *American Economic Review*, 2008, *98* (4), 1578–1590.
- Aguiar, Luis and Joel Waldfogel**, “Quality predictability and the welfare benefits from new products: Evidence from the digitization of recorded music,” *Journal of Political Economy*, 2018, *126* (2), 492–524.
- Althoff, Lukas and Hugo Reichardt**, “Task-Specific Technical Change and Comparative Advantage,” NBER Working Paper 35353, National Bureau of Economic Research 2026.
- Arditi, David**, “Digital downsizing: The effects of digital music production on labor,” *Journal of Popular Music Studies*, 2014, *26* (4), 503.
- Askin, Noah and Joeri Mol**, “Institutionalizing Authenticity in the Digitized World of Music,” *Research in the Sociology of Organizations*, 2018, *55*, 159–202.
- **and Michael Mauskopf**, “What makes popular culture popular? Product features and optimal differentiation in music,” *American Sociological Review*, 2017, *82* (5), 910–944.
- Atalay, Enghin, Phai Phongthientham, Sebastian Sotelo, and Daniel Tannenbaum**, “The Evolution of Work in the United States,” *American Economic Journal: Applied Economics*, 2020, *12* (2), 1–34.
- Autor, David and Neil Thompson**, “Expertise,” *Journal of the European Economic Association*, 2025, *23* (4), 1203–1271.

- , **Caroline Chin, Anna Salomons, and Bryan Seegmiller**, “New frontiers: The origins and content of new work, 1940–2018,” *The Quarterly Journal of Economics*, 2024, *139* (3), 1399–1465.
- Autor, David H. and Michael J. Handel**, “Putting Tasks to the Test: Human Capital, Job Tasks, and Wages,” *Journal of Labor Economics*, 2013, *31* (S1), S59–S96.
- Autor, David H, Frank Levy, and Richard J Murnane**, “The skill content of recent technological change: An empirical exploration,” *The Quarterly journal of economics*, 2003, *118* (4), 1279–1333.
- Baily, Martin Neil, David M. Byrne, Aidan T. Kane, and Paul E. Soto**, “Generative AI: What Type of Technology?,” *AEA Papers and Proceedings*, May 2026, *116*, 26–30.
- Baker, Andrew, Brantly Callaway, Scott Cunningham, Andrew Goodman-Bacon, and Pedro H. C. Sant’Anna**, “Difference-in-Differences Designs: A Practitioner’s Guide,” *Journal of Economic Literature*, 2026, *64* (2), 498–557.
- Barnes, Brooks**, “Michael Jackson: A Thrilling Pop Icon, a Troubled Soul,” *The Seattle Times*, June 2009. Reports that the Tacoma Dome Bad Tour dates were cancelled before opening night.
- Becker, Gary S. and Kevin M. Murphy**, “The Division of Labor, Coordination Costs, and Knowledge,” *Quarterly Journal of Economics*, November 1992, *107* (4), 1137–1160.
- Bessen, James, Maarten Goos, Anna Salomons, and Wiljan van den Berge**, “What Happens to Workers at Firms that Automate?,” *The Review of Economics and Statistics*, 2025, *107* (1), 125–141.
- Billboard**, “Billboard Boxscore,” *Billboard*, 1988, *100* (11, 12, 15, 17, 22, 42, 43, 45). Boxscore listings for Michael Jackson Bad Tour dates, March 12–November 5, 1988.
- , “Billboard Boxscore,” *Billboard*, 1989, *101* (6), 32. Boxscore listing for Michael Jackson Los Angeles Sports Arena dates.
- Borowiecki, Karol Jan**, “Good Reverberations? Teacher Influence in Music Composition since 1450,” *Journal of Political Economy*, April 2022, *130* (4), 991–1090.
- Brynjolfsson, Erik, Danielle Li, and Lindsey R. Raymond**, “Generative AI at Work,” *The Quarterly Journal of Economics*, 2025, *140* (2), 889–942.

- Bureau of Economic Analysis**, “Arts and Cultural Production Satellite Account, U.S. and States, 2023,” Statistical Release, U.S. Department of Commerce April 2025.
- Caoui, El Hadi and Alberto Galasso**, “Fractional Ownership and Copyright Licensing: Evidence from the Music Industry,” NBER Working Paper 34336, National Bureau of Economic Research October 2025. Also CEPR Discussion Paper DP19686.
- Caves, Richard E.**, *Creative Industries: Contracts Between Art and Commerce*, Harvard University Press, 2000.
- Cullen, Zoë B., Ester Faia, Elisa Guglielminetti, Ricardo Perez-Truglia, and Concetta Rondinelli**, “The Innovation Race: Experimental Evidence on Advanced Technologies,” NBER Working Paper 34532, National Bureau of Economic Research 2026.
- Demers, Joanna**, *Steal This Music: How Intellectual Property Law Affects Musical Creativity*, University of Georgia Press, 2006.
- Deming, David J. and Kadeem Noray**, “Earnings Dynamics, Changing Job Skills, and STEM Careers,” *The Quarterly Journal of Economics*, 2020, 135 (4), 1965–2005.
- Demirer, Mert, John J. Horton, Nicole Immorlica, Brendan Lucier, and Peyman Shahidi**, “Chaining Tasks, Redefining Work: A Theory of AI Automation,” Working Paper 34859, National Bureau of Economic Research 2026.
- Dessein, Wouter and Tano Santos**, “Adaptive Organizations,” *Journal of Political Economy*, 2006, 114 (5), 956–995.
- Doshi, Anil R. and Oliver P. Hauser**, “Generative AI Enhances Individual Creativity but Reduces the Collective Diversity of Novel Content,” *Science Advances*, 2024, 10 (28), eadn5290.
- E-mu Systems**, *SP1200 Sampling Percussion System Service Manual* E-mu Systems 1987.
- Eckert, Fabian, Teresa C. Fort, Peter K. Schott, and Natalie J. Yang**, “Imputing Missing Values in the US Census Bureau’s County Business Patterns,” Working Paper 26632, National Bureau of Economic Research 2021.
- Feigenbaum, James and Daniel P Gross**, “Answering the call of automation: How the labor market adjusted to mechanizing telephone operation,” *The Quarterly Journal of Economics*, 2024, 139 (3), 1879–1939.

- Garicano, Luis**, “Hierarchies and the Organization of Knowledge in Production,” *Journal of political economy*, 2000, 108 (5), 874–904.
- **and Luis Rayo**, “Training in the Age of AI: A Theory of Career Viability,” Discussion Paper 20634, CEPR 2025. Revised March 2026.
- Goos, Maarten, Alan Manning, and Anna Salomons**, “Explaining Job Polarization: Routine-Biased Technological Change and Offshoring,” *American Economic Review*, 2014, 104 (8), 2509–2526.
- Griliches, Zvi**, “Hybrid Corn: An Exploration in the Economics of Technological Change,” *Econometrica*, 1957, 25 (4), 501–522.
- Halstead, Craig and Chris Cadman**, *Michael Jackson: The Solo Years*, Authors OnLine, 2003.
- Heller, Michael A. and Rebecca S. Eisenberg**, “Can Patents Deter Innovation? The Anticommons in Biomedical Research,” *Science*, 1998, 280 (5364), 698–701.
- Humlum, Anders and Emilie Vestergaard**, “Still Waters, Rapid Currents: Early Labor Market Transformation under Generative AI,” NBER Working Paper 33777, National Bureau of Economic Research March 2026. Originally issued May 2025; revised March 2026.
- Jones, Benjamin F.**, “The Burden of Knowledge and the “Death of the Renaissance Man”: Is Innovation Getting Harder?,” *The Review of Economic Studies*, 2009, 76 (1), 283–317.
- Katz, Mark**, *Capturing Sound: How Technology Has Changed Music*, University of California Press, 2004.
- Kim, Khwan, Noah Askin, and James A Evans**, “Disrupted routines anticipate musical exploration,” *Proceedings of the National Academy of Sciences*, 2024, 121 (6), e2306549121.
- Kim, Sueyoul, Ginger Zhe Jin, and Eungik Lee**, “Does Generative AI Crowd Out Human Creators? Evidence from Pixiv,” NBER Working Paper, National Bureau of Economic Research 2026.
- Lin, Jeffrey**, “Technological Adaptation, Cities, and New Work,” *The Review of Economics and Statistics*, 2011, 93 (2), 554–574.

- Lin, Sijie**, “Learning to Prompt: Human Adaptation in Production with Generative AI,” 2025. Job market paper, Rotman School of Management, University of Toronto. Version December 2025. SSRN 5788402.
- Moorefield, Virgil**, *The producer as composer: Shaping the sounds of popular music*, Mit Press, 2010.
- National Center for O*NET Development**, “4.A.2.b.2 – Thinking Creatively,” 2026.
- Noy, Shakked and Whitney Zhang**, “Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence,” *Science*, 2023, *381* (6654), 187–192.
- Peng, Sida, Eirini Kalliamvakou, Peter Cihon, and Mert Demirer**, “The Impact of AI on Developer Productivity: Evidence from GitHub Copilot,” Working Paper, arXiv preprint 2023. arXiv:2302.06590.
- Rambachan, Ashesh and Jonathan Roth**, “A More Credible Approach to Parallel Trends,” *Review of Economic Studies*, 2023, *90* (5), 2555–2591.
- Reimers, Imke and Joel Waldfogel**, “AI and the Quantity and Quality of Creative Products: Have LLMs Boosted Creation of Valuable Books?,” NBER Working Paper 34777, National Bureau of Economic Research January 2026. Revised May 2026.
- Rosen, Sherwin**, “The Economics of Superstars,” *The American Economic Review*, 1981, *71* (5), 845–858.
- Salomons, Anna, Cäcilia vom Baur, and Ulrich Zierahn-Weilage**, “Expertise at Work: New Technologies, New Skills, and Worker Impacts,” 2025. Working paper.
- Scotchmer, Suzanne**, “Standing on the Shoulders of Giants: Cumulative Research and the Patent Law,” *Journal of Economic Perspectives*, 1991, *5* (1), 29–41.
- Spitz-Oener, Alexandra**, “Technical Change, Job Tasks, and Rising Educational Demands: Looking Outside the Wage Structure,” *Journal of Labor Economics*, 2006, *24* (2), 235–270.
- The Financial Times**, “AI Isn’t Destroying Entry-Level Jobs. It’s Changing Them,” *Financial Times*, July 2026. By Andrew Hill.
- , “Junior Consultants Called Back to Office as AI Increases Need for Human Skills,” *Financial Times*, August 2026.
- The New York Times**, “Michael Jackson Silent on Ending His Tours,” *The New York Times*, December 1988.

—, “Dr. Dre and Jimmy Iovine Think A.I. Is Good for Music,” *The New York Times*, August 23 2026. By Jordyn Holman.

The Vindicator, “Jackson Cancels Cincinnati Shows,” *The Vindicator*, February 1988. Contemporaneous notice of cancelled Riverfront Coliseum dates.

Tinbergen, Jan, “Substitution of graduate by other labour,” *Kyklos*, 1974, 27 (2), 217–226.

Tolbert, Charles M. and Molly Sizer, “U.S. Commuting Zones and Labor Market Areas: A 1990 Update,” Technical Report, U.S. Department of Agriculture, Economic Research Service 1996. Staff Paper AGES-9614.

Trask, Simon, “Akai MPC60 MIDI Production Center,” *Music Technology*, April 1988, pp. 40–44. Contemporary review of the Akai MPC60, including its launch price.

Vintage Synth Explorer, “E-mu SP-1200,” 2026. Accessed May 9, 2026.

Waldfoegel, Joel, *Digital Renaissance: What Data and Economics Tell Us about the Future of the Creative Industries*, Princeton University Press, 2018.

A Data construction

This appendix documents the construction of the datasets and measures used in the paper.

A.1 From Discogs releases to the record sample

Discogs records individual *releases*, so the same underlying record can appear multiple times across pressings, countries, and formats. I collapse releases sharing a master identifier to a single observation. I retain the earliest release year and, where several releases remain tied, select the observation with the largest number of credited contributors, then the largest track count, and finally the lowest release identifier.

I exclude genres outside recorded-music production proper (Stage & Screen, Children’s, Non-Music, and Brass & Military). I then restrict the data to records released between 1980 and 2000 whose recording location can be assigned to the continental United States and to the commuting zones included in the exposure design described in Section 3.3. The resulting estimation sample contains 70,712 records.

Figure A.1 shows how the units of the estimation sample nest and the coverage of the main measures.

A.2 Matching to WhoSampled

Sample relationships come from WhoSampled and are matched to Discogs tracks in two stages. I first match exactly on lowercased and punctuation-stripped track title, artist, and year. Remaining observations are matched using Jaro–Winkler similarity with a threshold of 0.90. The two sets of matches are then combined and deduplicated.

A record is classified as documenting sampling when at least one of its tracks is linked to a sampled source. I also construct a record-level sampling rate equal to the share of tracks with a documented sampled source. Because WhoSampled is assembled retrospectively and does not provide exhaustive coverage of all sampling, these variables measure documented rather than complete sample use.

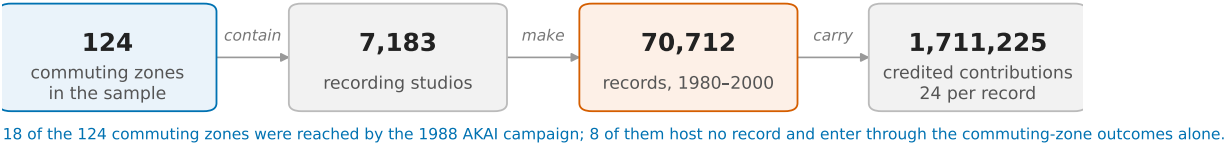
A.3 Studios and geography

Studio names are extracted from the recording credits, standardized, and geocoded using manual lookup together with the Google Maps Geocoding API. Coordinates are assigned to counties and then to 1990 commuting zones through the Autor–Dorn crosswalk.

When a record names multiple studios, I assign it to the studio identified as the main recording location. If no studio is explicitly designated as primary, I use the first listed

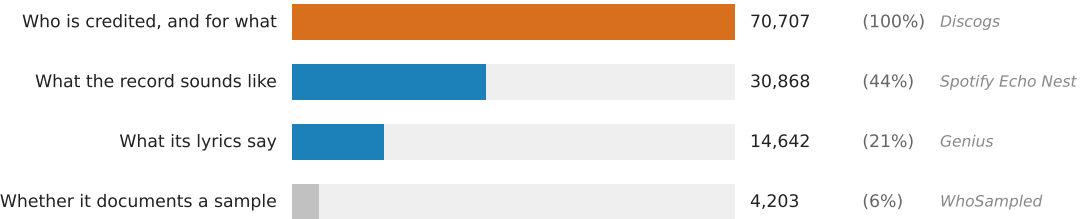
A. Where a record sits

The campaign reached commuting zones; the outcomes are measured on the records made in them. All counts are the estimation sample.



B. What is observed about it

Share of the 70,712 records carrying each measure. One denominator throughout, so the bars are comparable.



Commuting-zone population, income and employment are from the 1980 Census, BEA and SEER; the campaign's reach is from the 1988 tour itineraries.

Figure A.1: Where a record sits, and what is observed about it

Note. All counts refer to the 70,712-record estimation sample rather than the wider continental-US universe described in Section 3.4. Panel A shows the nesting of the empirical design: campaign exposure varies across commuting zones, while record-level outcomes are observed on records produced in studios located within those markets. Standard errors are therefore clustered by commuting zone. Eight of the 124 commuting zones contain no record in the record-level sample and enter only through commuting-zone outcomes. Of the 70,712 records, 70,707 contain at least one classified credit. Panel B reports coverage of the principal record-level measures. Credit lists are observed for essentially the full sample, while acoustic and lyrical measures are available only for subsets of records. Appendix H examines whether campaign exposure predicts coverage of these content measures.

recording studio. Records naming studios in more than one commuting zone are therefore assigned to the commuting zone of their primary studio. Alaska, Hawaii, and US territories are excluded throughout.

A.4 Population and economic series

The denominator for the participation measure is constructed from the SEER county population estimates, which report annual population by single year of age from 1969 onward. I aggregate residents aged 20–64 from counties to commuting zones.

SEER uses a modified county geography, so county codes that do not map directly into the county-to-commuting-zone crosswalk are resolved individually. The only code spanning two commuting zones is the pre-1983 Yuma–La Paz combination. I split this observation using the corresponding 1994 population shares separately by age band and flag the affected commuting-zone-years.

I validate the resulting series against the published 1980 continental-US Census total and against tabulated 1980 Census population at the commuting-zone level. County employment comes from the County Business Patterns series imputed by [Eckert et al. \(2021\)](#), while personal income comes from the BEA local-area accounts. Both are aggregated to commuting zones and used in the negative-control and robustness exercises reported in Section 5.1.

A.5 The *Mix* directories

Mix magazine published regional directories of recording studios during the sample period. Listings report a studio’s name, address, telephone number, owner, and manager. In the more detailed of the magazine’s two listing formats, they also report consoles, recorders, monitoring equipment, and instruments. These directories are the only source used in the paper that directly observes studio equipment.

I identify 80 issues containing studio directories between 1980 and 1993. Because the magazine changed the names of the directory sections over time and did not consistently label issues as directories, I identify relevant pages using the fields printed within the listings. The resulting data contain 18,869 listings covering 8,328 distinct studios. Of these, 96.6 percent can be assigned to a commuting zone using their postal information, and 1,256 studios are observed both before and after 1988.

Listings are segmented from page images and read into a fixed schema. Telephone number, state, and postal code are also extracted deterministically and used as internal checks on the extracted listing. Ninety-nine percent of extracted values remain traceable to the underlying

source text. Equipment indicators used in the analysis are identified directly from the raw listing text rather than from the structured extraction.

The directories provide useful descriptive evidence but do not support a causal estimate of equipment adoption. Inclusion in the directory itself is relatively stable across campaign exposure among studios observed on both sides of 1988. The detailed listing format that contains equipment information is not: its probability changes differentially with campaign exposure. Equipment ownership is therefore observed for a changing subset of studios. Appendix D.6 reports these data descriptively rather than interpreting them as an adoption outcome.

Matching the directories to studios in the estimation sample using studio name and commuting zone links 434 studios, or 7.6 percent of the record-sample studios. This overlap is too small to support a useful within-studio equipment analysis.

A.6 Classifying credits into function families

The unit of classification is a contributor-function pair at the record level. A person credited as both a writer and guitarist therefore contributes one classified credit to writing and one to instrumental performance.

Credit labels are normalized for case, punctuation, and aliases and mapped to a catalogue of 814 distinct functions covering 1,808,567 contributor-function observations in the record sample. Classification follows an ordered set of rules at the function level. The 238 functions accounting for 99 percent of observations are individually adjudicated, with each assignment retaining the rule used to classify it. Less common functions inherit the family associated with their underlying Discogs credit category and are flagged as not individually reviewed. The catalogue and its assignments were fixed before estimating the organizational outcomes.

The underlying catalogue separates performance, technical work, creative authorship and production, and nonproduction. The main analysis combines performance and technical work into the broader *execution* family. The largest underlying functions are:

- *Performance*: vocals (76,225), guitar (69,147), backing vocals (67,995), and drums (60,790);
- *Technical*: engineer (114,653), mixed by (62,788), recorded by (39,237), and mastered by (32,131);
- *Creative authorship and production*: written by (224,312), producer (139,686), arranged by (47,186), and executive producer (23,799);

- *Nonproduction*: photography by (45,338), design (31,337), art direction (18,692), and management (14,411).

The resulting three analysis families are therefore execution, creative authorship and production, and nonproduction, as defined in Section 3.2.

Producer and arrangement credits. Producer and arrangement credits are assigned to creative authorship and production. The classification follows the function attributed by the credit. Producers make decisions over material, arrangements, performances, and the shape of the recording, while engineers are separately credited for technically realizing those decisions. Arrangement translates musical material into the parts and structure used in the recording and is therefore classified with the functions that shape the musical material rather than with its execution.

The alternative classifications would not mechanically favor the paper’s result. Assigning producers to technical work or arrangers to execution would move two large categories away from creative authorship and production. More generally, the taxonomy classifies functions according to what is attributed to the contributor rather than according to the technology through which the function was performed.

A.7 Biographical characteristics

The descriptive artist analysis in Section 6.4 combines Wikidata and Wikipedia information for billed main artists.

Construction proceeds in three stages. First, artists are matched to Wikidata using catalogue identifiers, yielding information including birth date, occupations, and instruments. Second, the corresponding Wikipedia biography is classified into a fixed set of categories covering musical training, socioeconomic background, previous occupations, and technical literacy. Musical training distinguishes self-taught artists, structured individual instruction, and formal training, with conservatory training recorded as a subset of formal training. Self-taught and formally trained are therefore mutually exclusive by construction.

Third, characteristics that are not stated in the underlying sources remain missing rather than being inferred. The two exceptions are technical literacy and prior non-music career, which are mention-based indicators and therefore take a value of zero when the biography contains no corresponding evidence. These measures should consequently be interpreted as lower bounds. Birthplace is retained as free text rather than geocoded.

Biographical coverage is incomplete and is itself related to how extensively an artist is documented. For this reason, all analyses using these characteristics are descriptive and

interpreted conditional on the observed biographical sample.

A.8 The measures before the exposure design

This subsection examines the credit measures descriptively before applying the empirical design. All objects are unconditional means or shares computed over records. No treatment coefficient is estimated.

National trends in the three function families. Figure A.2 reports the three credit-share measures nationally and separately by treatment arm. Between 1980–1987 and 1993–2000, the record-weighted share assigned to creative authorship and production falls from 36.3 to 31.7 percent, while execution rises from 55.2 to 57.6 percent and nonproduction from 8.5 to 10.7 percent.

Because the three families partition classified credits, their movements are mechanically related. These national changes are descriptive and should not be confused with the differential changes across exposed and comparison markets estimated in Section 6.

Genre profiles. Figure A.4 and Table A.1 report pre-campaign credit patterns by genre. The taxonomy reproduces familiar differences in how genres organize and document production without being calibrated to generate them.

Hip hop has the largest creative-authorship-and-production share and relatively small credited teams. Classical records carry an unusually large composition and conducting component. Jazz is the most execution-intensive genre in the pre-period. Funk and soul have among the largest credited teams, while rock has the lowest contributor exclusivity, consistent with a convention in which musicians frequently both write and perform.

Validation exercises. Five diagnostic predictions were specified before computing these genre profiles. Three are borne out directly. Hip hop’s writer share exceeds rock’s by more than the specified margin (26.1 versus 14.5 percent); producer credits increase over the sample period; and records produced in larger commuting zones contain larger credited teams (15.3 versus 10.5 contributors).

Two predictions do not meet their specified thresholds. Classical has the largest conducting share, but at 4.7 percent it is 2.3 times rather than three times the next-highest genre. Classical and jazz were also predicted to rank among the three most execution-intensive genres when pooling the sample period, which they do not.

Two additional descriptive checks help explain these failures. Classical has by far the largest composer share, 14.9 percent compared with a next-highest value of 4.0 percent, while

jazz is the most execution-intensive genre when the comparison is restricted to the pre-period, at 59.0 percent. Because these checks were added after observing the original diagnostics, I treat them as corroborating evidence rather than as part of the original validation exercise.

Studio size and market size. Credited team size rises monotonically with studio activity, from 11.8 contributors among studios producing one or two sample records to 18.1 among studios producing more than one hundred. The composition of credited functions changes much less, with the creative-authorship-and-production share moving from 37.3 to 35.7 percent across the same groups.

Market-size comparisons are less informative at the record level because recording activity is geographically concentrated. The largest third of recording commuting zones contains 97.4 percent of all records, leaving only 461 and 1,629 post-period records in the bottom and middle thirds, compared with 33,552 in the largest third. Within rock, contributor exclusivity varies little across either market size or studio size, remaining between 71 and 79 percent.

Ensemble credits. Some execution credits refer to groups rather than individuals. An orchestra, choir, horn section, or named house band may therefore appear as a single credit even though several musicians performed. This differs from uncredited labor: the contribution is documented, but several people may be compressed into one credited entity.

I identify potential ensemble credits using a rule based on both the credit label and the credited entity name. Labels indicating collectives include *orchestra*, *choir*, *chorus*, *ensemble*, *strings*, *horns*, *brass*, *brass band*, and *rhythm section*. Entity-name matching is restricted to non-main-artist execution credits so that a band performing on its own record is not treated as a collapsed ensemble. Credits describing an individual who directs or writes for an ensemble, such as *orchestrated by* or *chorus master*, are excluded.

The rule identifies 34,300 of 1,798,606 classified credits, or 1.9 percent overall and 3.2 percent of execution credits. Because the data do not reveal how many musicians a collective credit represents, the rule identifies ensemble crediting but cannot recover the underlying number of performers.

Ensemble compression is concentrated in classical music, where 12.1 percent of execution credits are ensemble credits, compared with 1.1 percent in rock and 0.4 percent in hip hop. A sensitivity exercise treats each ensemble credit as representing k performers. As k increases from one to twenty, classical moves from eighth of eleven genres in execution share (51.8 percent) to first (64.6 percent), while the next-largest movement for another genre is only 3.3 percentage points. This exercise is illustrative because the data provide no basis for choosing a particular value of k .

For the causal analysis, the relevant concern is differential change in ensemble crediting across treatment arms. The share of records containing an ensemble credit falls from 11.3 to 9.9 percent in exposed commuting zones and from 12.8 to 10.8 percent in comparison commuting zones, a difference-in-differences of +0.60 percentage points. Excluding ensemble credits entirely changes the national function shares by less than 0.1 percentage points. Ensemble compression therefore affects levels in particular genres but does not appear to generate the differential organizational result.

Record weighting and credit weighting. The national trend in function shares depends importantly on the unit of weighting. Weighting each record equally, the creative-authorship-and-production share falls by 4.6 percentage points between 1980–1987 and 1993–2000. Weighting each individual credit equally, it instead rises by 0.7 percentage points.

Figure A.3 shows that the difference is concentrated among records with short credit lists. Records carrying one to three classified credits lose 20.4 percentage points in their creative-authorship-and-production share, while records carrying four to six lose 22.3 points. Records with thirteen or more classified credits instead show changes between +0.6 and +3.7 points. When a record contains only a small number of credits, its category shares are mechanically coarse and sensitive to changes in documentation.

Release format explains much of this pattern. Short-form releases account for 37.5 percent of the sample and contain an average of 15.5 classified credits, compared with 28.8 for long-form releases. Their share of the record sample rises from roughly 30 to 41 percent over the period. A shift-share decomposition attributes about 57 percent of the record-weighted decline in creative authorship and production to changes in release-format composition. The movement is concentrated in vinyl, whose mean documented credit count falls from 22.8 to 14.0, while compact-disc credit lists remain close to 28.

This is primarily a weighting and documentation issue rather than evidence of declining authorship activity. Absolute counts move in the opposite direction: authorship credit lines per record rise by 8.4 percent, contributors holding an authorship credit by 7.3 percent, and total credited team size by 1.8 percent. Section 6.2 therefore places greater weight on contributor counts and explicitly examines sensitivity to release format.

Decomposing the national trend. Between 1980–1987 and 1993–2000, the record-weighted creative-authorship-and-production share falls by 4.61 percentage points, compared with a 0.72-point increase when credits are weighted equally. Execution changes by +2.42 and –2.15 points under the two weighting schemes, respectively, while nonproduction changes by +2.19 and +1.43 points.

A shift-share decomposition across genres attributes -4.40 percentage points of the record-weighted change to within-genre movements, $+0.27$ points to changing genre composition, and -0.48 points to their interaction. Thus roughly 95 percent of the total change comes from within-genre movements. A corresponding decomposition by credit-list length attributes 92 percent to within-bin changes. Across release formats, changing format composition accounts for -2.63 percentage points, or approximately 57 percent of the total.

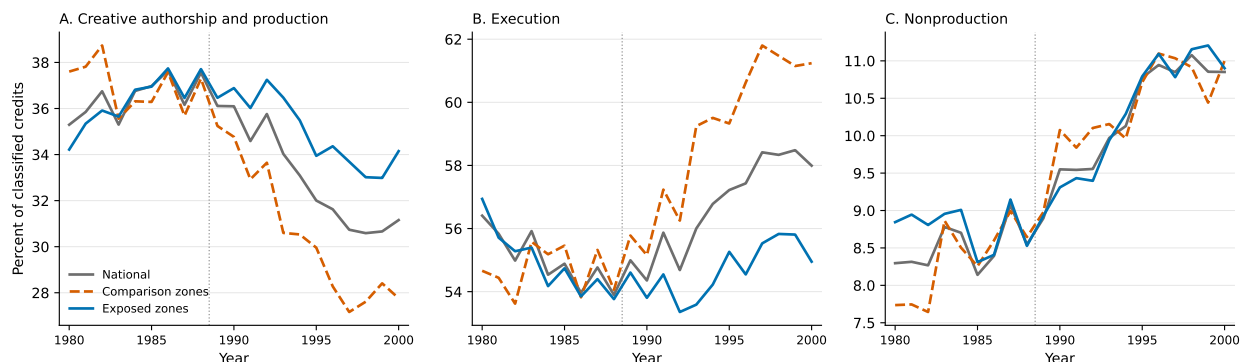


Figure A.2: **National credit composition changes over the sample period in both treatment arms.** Mean record-weighted composition of classified credits by year, 1980–2000, for the continental United States and separately for exposed and comparison commuting zones. The three function families partition classified credits and sum to 100 percent in every year. The vertical rule marks the 1988 campaign. Descriptive throughout; no treatment coefficient is estimated.

The 1988 inflection. Several national credit measures turn around 1988. The creative-authorship-and-production share rises before 1988 and falls afterward, while both multi-family breadth and the producer-credit share also peak near 1988. Because 1988 is the omitted year in the event-study specifications, I examine whether this national movement differs across treatment arms.

The change in the annual slope of the creative-authorship-and-production share is -0.75 percentage points per year in exposed commuting zones and -0.60 in comparison commuting zones, a differential of -0.14 points. Across the five series examined, the treatment-arm differential is at most one third of the corresponding common component. The pattern is therefore predominantly national rather than treatment-specific.

There is also no discrete break in record coverage at 1988. The number of records grows by 9.4 percent in 1987 and 7.4 percent in 1988, while the largest annual increase in the sample is 12.4 percent in 1992. The same inflection remains when restricting the data to studios observed both before and after the campaign. These descriptive comparisons do not substitute for the pre-campaign coefficients in the event studies, but they show that

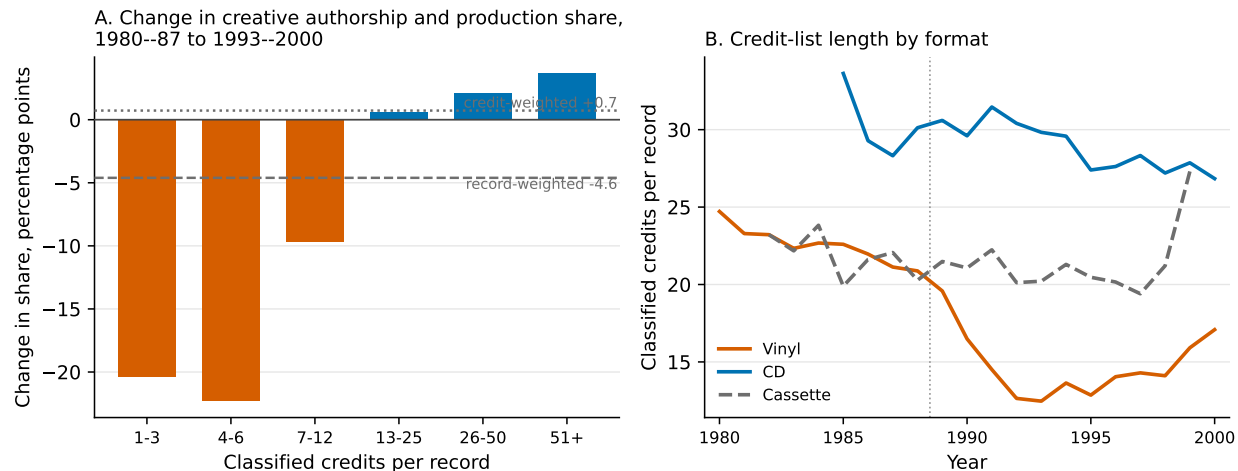


Figure A.3: **The national composition trend depends on the unit of weighting, with the difference concentrated among thinly credited records.** Panel A reports the change in the creative-authorship-and-production share between 1980–1987 and 1993–2000 within bins of classified credits per record. The dashed line reports the change when each record receives equal weight, corresponding to the unit of the record-level regressions; the dotted line reports the change when each classified credit receives equal weight. Panel B reports mean classified credits per record by release format. Format-years containing fewer than 100 records are suppressed. Descriptive throughout; no treatment coefficient is estimated.

the national turn around 1988 is shared across treatment arms rather than mechanically generated by the exposure definition.

B Additional background

This appendix provides additional detail on recorded-music production, the technological change introduced by sampling, and the 1988 AKAI campaign.

B.1 The classical division of labor

Michael Jackson’s *Thriller* (1982) illustrates the specialized organization of recorded-music production before the diffusion of integrated sampling technology. Its credits include track-specific writers, arrangers, a large roster of session musicians, centralized production, and separately credited recording engineers.¹³ This organization reflects the broader structure of project-based creative production, in which specialized contributors are assembled around

¹³The full credit list is documented at <https://www.discogs.com/release/152946-Michael-Jackson-Thriller>.

Table A.1: Composition of credited work by genre, 1980–1987

Genre	Records	Team	Function-family share (%)			Excl. (%)	Function share (%)		
			Auth.	Exec.	Nonpr.		Writ.	Comp.	Prod.
Hip Hop	513	7.4	59.4	36.9	3.7	85.8	26.1	1.1	26.8
Funk/Soul	3,230	19.0	42.8	51.6	5.6	87.3	18.6	1.1	15.1
Latin	392	13.4	42.0	49.4	8.6	89.4	21.3	1.4	10.6
Reggae	332	10.7	41.8	52.5	5.7	87.3	12.7	1.5	18.8
Electronic	2,677	14.4	39.9	52.8	7.2	81.6	15.7	1.6	17.0
Classical	824	12.4	36.8	51.8	11.4	92.0	3.9	14.9	10.2
Folk/World/Country	3,899	15.2	36.4	56.5	7.1	89.7	18.8	1.2	12.7
Blues	764	17.6	34.6	56.2	9.2	86.9	19.8	0.8	10.7
Pop	1,050	17.3	34.0	57.5	8.5	84.6	15.8	1.5	11.5
Rock	5,946	12.5	31.8	57.5	10.7	73.9	14.5	1.2	14.0
Jazz	2,829	17.5	30.5	59.0	10.5	87.5	16.2	3.1	8.4

Notes. Records released from 1980 through 1987 in the continental United States. *Team* is the number of credited contributors per record. The three function-family shares partition classified credits and sum to 100 percent. *Excl.* is the share of contributors whose credits all fall within a single function family. The three individual-function shares are shares of classified credits and do not sum to the creative-authorship-and-production family because that family also contains arrangement and conducting. Descriptive throughout.

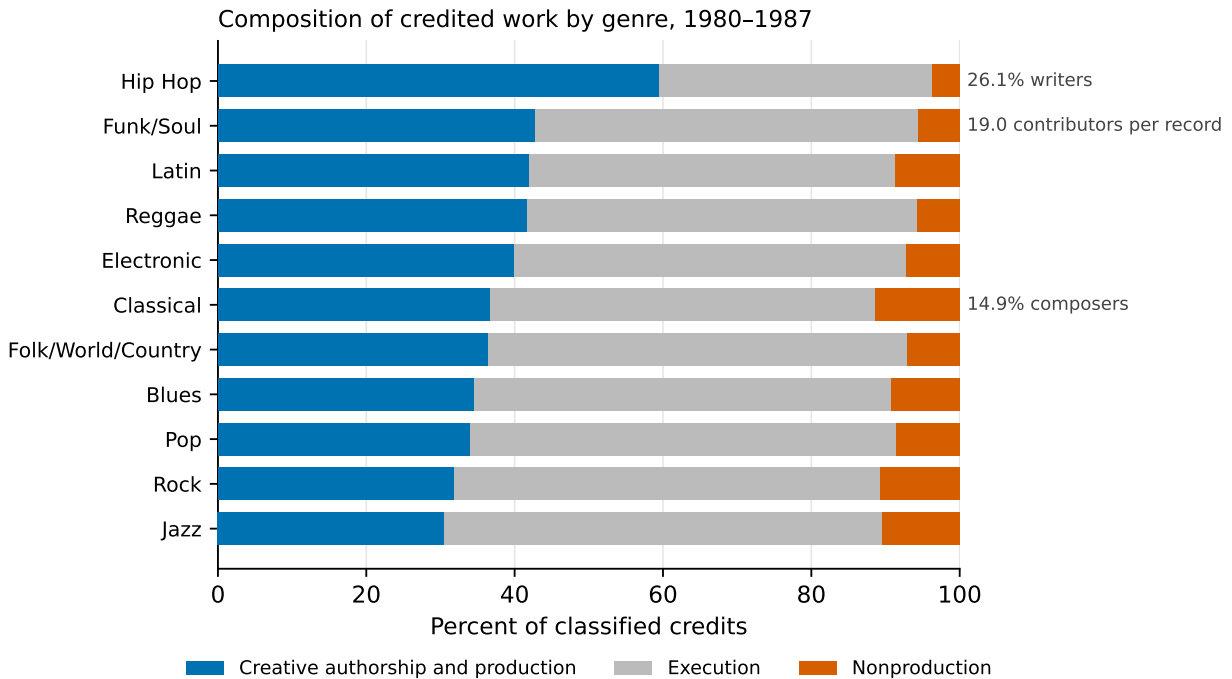


Figure A.4: **The credit measures reproduce familiar differences in production and crediting across genres.** Mean composition of classified credits by genre over 1980–1987 for records produced in the continental United States. The three function families partition classified credits and sum to 100 percent within each genre. Annotations report selected individual-function shares. Genres are ordered by their creative-authorship-and-production share. Descriptive throughout.

individual goods whose commercial value is uncertain at the time of production (Caves, 2000).

Recording technology had already changed this division of labor before the sampler arrived. In particular, multitrack recording shifted the studio from a place primarily used to document a performance toward a place in which the recording itself could be constructed from separately captured and manipulated parts (Moorefield, 2010; Katz, 2004). Production, arrangement, engineering, and performance consequently became increasingly separable functions. Producers also became associated with distinctive sonic choices that were difficult to separate from the musical content itself, blurring the boundary between production and composition well before the technological episode studied here.

B.2 From ensemble performance to digital construction

A simple example illustrates how sampling and sequencing can change the production process. Stevie Wonder’s *Pastime Paradise* (1976) was produced using a conventional studio ensemble, with separately credited instrumental performers alongside the writing and production functions. Coolio featuring L.V.’s *Gangsta’s Paradise* (1995) builds on the same musical idea but organizes it through a different production process. Core elements are re-recorded, looped, and sequenced into the new record by producer Doug Rasheed, together with a smaller set of performance inputs and a layer of engineering and production credits.

Gangsta’s Paradise uses an interpolation rather than a direct sample: the underlying musical material is recreated rather than copied from the original recording. The distinction matters legally, but the production contrast is similar. Digital sequencing and repetition reduce the need to coordinate a full ensemble in real time and allow more of the recording to be constructed through the selection, repetition, and arrangement of musical elements.

A second pair illustrates what this change looks like in the credit data. Table B.1 compares two records produced in the same city on opposite sides of the campaign: Stevie Wonder’s *Characters* (1987) and 2Pac’s *All Eyez On Me* (1996). The two records contain almost the same number of classified credits, 111 and 112, but their composition differs sharply. Performance credits fall from 78 to 15, while creative-authorship-and-production credits rise from 10 to 69. The corresponding share of credited work assigned to creative authorship and production rises from 0.09 to 0.62.

Both records belong to the estimation sample, so the table illustrates the same objects used in the record-level analysis. The comparison is purely descriptive: the records differ in genre, period, personnel, and many other dimensions. Its purpose is to show what the credit data make observable, rather than to identify an effect of the technology.

Table B.1: Two records, thirteen years apart

	<i>Characters</i> Stevie Wonder, 1987	<i>All Eyez On Me</i> 2Pac, 1996
<i>Panel A. The eight credit labels that differ most</i>		
Written by	1	29
Backing vocals	33	8
Featuring	1	23
Choir	12	0
Producer	3	14
Kazoo	8	0
Mixed by	1	9
Engineer	4	11
<i>Panel B. Credits by family</i>		
Creative authorship and production	10	69
Performance	78	15
Technical	11	21
Nonproduction	12	7
<i>Performance and technical together</i>	<i>89</i>	<i>36</i>
<i>Panel C. The same, as shares of the record's credits</i>		
Creative authorship and production	0.090	0.616
Performance	0.703	0.134
Technical	0.099	0.188
Nonproduction	0.108	0.062
Classified credits	111	112
Distinct contributors	62	76

Notes. Credited roles from the two records' Discogs credit lists (releases 12718274 and 226430), classified with the same role taxonomy as every album estimate in the paper; a contributor may hold several roles, so credits exceed the headcount. **Both records are observations in the estimation sample**—made in Los Angeles studios, in the primary sample, thirteen years apart on either side of the campaign—and Panel C reproduces their stored family shares exactly. **The team is a similar size; its composition is not:** 111 credits against 112, and 62 contributors against 76, while the share going to authorship and production runs from 0.090 to 0.616 and the share going to performance and technical work runs the other way. **The pair differs in genre as well as in period, and two records cannot separate the two.** Nothing here is an estimate: it illustrates what the credit list makes visible, and the comparison that identifies anything is Equation (4), fitted on 70,707 records with studio fixed effects.

B.3 How sampling could change the division of labor

Sampling changes the object of production by making recorded sound itself reusable. Producers can select, transform, repeat, and combine existing material rather than generating every musical element through a new performance (Katz, 2004). This technological change can affect several parts of the production process.

First, recorded performances can substitute for some newly performed parts. Second, sequencing and looping can reduce the instrumental and coordination requirements involved in assembling musical elements. Third, the relative importance of functions involved in selection, arrangement, writing, and production can change as more of the recording is constructed through recombination.

These forces do not imply a unique organizational response. The sampler expands the set of functions that one person can technically perform, which could allow contributors to take on broader function bundles. But it can also change comparative advantage and reduce coordination costs between specialized inputs, allowing production to combine contributors concentrated in different functions.

This ambiguity motivates the three organizational margins measured in the paper. I examine the allocation of credited work across execution, creative authorship and production, and nonproduction; the composition of the people carrying those functions; and the breadth of the function bundles attributed to individual contributors. The credit taxonomy classifies functions according to the activity attributed to the contributor rather than the technology used to perform it, so technological change is not mechanically incorporated into the outcomes.

B.4 The campaign in detail

The *Bad World Tour* was one of the largest live-music events of the late 1980s and toured extensively through the United States in 1988 (Halstead and Cadman, 2003; Billboard, 1988, 1989). During the US leg, AKAI used the tour to promote the newly introduced MPC60.

Mike McRoberts, then AKAI's US product manager, describes the origin of the campaign at the January 1988 NAMM show.¹⁴ Members of Jackson's touring band proposed holding hands-on demonstrations as the tour moved across the country. The resulting road clinics were organized around tour stops, with touring musicians demonstrating an integrated production setup centered on the MPC60 alongside other AKAI equipment.¹⁵

¹⁴NAMM is the National Association of Music Merchants; the NAMM Show is the industry's principal trade show for musical instruments and music technology.

¹⁵Interview with Mike McRoberts describing the NAMM approach, the weekly clinics, and the clinic setup: <https://www.bboytechreport.com/2013/01/11/interview-with-mike-mcroberts/>.

The historical record is substantially more complete for the tour itinerary than for individual clinic attendance. Concert dates and venues can therefore be established systematically, while surviving documentation does not identify every clinic or every person who attended one. For this reason, the empirical design defines exposure at the commuting-zone level using realized *Bad* Tour stops during the campaign.

The itinerary also contains several useful non-realizations. Cincinnati and Tacoma had scheduled and announced *Bad* Tour dates that were subsequently cancelled, for tour restructuring and illness respectively ([The Vindicator, 1988](#); [Barnes, 2009](#)). Neither market hosted a realized stop and neither is classified as exposed. These destinations had nevertheless progressed substantially through the tour-selection process, making them useful for the falsification exercise described in Section 4.1 and reported in Appendix 5.1.

St. Louis illustrates why treatment is defined using realized exposure rather than whether every scheduled performance occurred. Jackson performed there on March 13, 1988, while a subsequent scheduled performance was cancelled because of illness ([The New York Times, 1988](#)). Because the market received a realized tour stop during the campaign, St. Louis remains classified as exposed.

Two additional cities, Birmingham and Chapel Hill, appear in historical accounts as possible destinations but do not have the same documentary status as Cincinnati and Tacoma. I therefore do not use them as documented cancelled stops in the main falsification exercise. They are reported separately in the itinerary checks.

The resulting treatment definition follows the historical object that can be measured consistently: whether a local market was reached by a realized *Bad* Tour stop during the AKAI campaign. It does not require observing individual clinic attendance or subsequent ownership of the sampler.

The use of local demonstrations as a source of technology diffusion has parallels with a broader literature on geographic adoption and information transmission. [Griliches \(1957\)](#) provides the classic example of geographically uneven technology diffusion, while [Cullen et al. \(2026\)](#) documents strategic interdependence in the adoption of advanced technologies. These mechanisms provide a rationale for why exposure to a high-profile local demonstration could affect adoption beyond the individuals who directly attended it.

C The sample, specification choices, and inference

This appendix provides additional detail on the estimation sample, the relationship between the paper’s alternative specifications, the treatment of the campaign year, and several sensitivity exercises that clarify the interpretation of the main estimates.

C.1 The estimation sample

C.2 From the commuting zone to the record

The diffusion result can be estimated at either the commuting-zone-year or record level. These specifications differ not only in the unit of observation but also in the weighting of local markets and in the fixed effects they absorb. Table C.5 changes these elements one at a time.

Moving from commuting-zone-years to records changes the weighting of the estimand. At the commuting-zone-year level, each active local market receives one observation in a given year. At the record level, commuting zones receive weight in proportion to the amount of recorded music they produce. The latter therefore places greater weight on the main recording centers.

Moving from year effects to census-division-by-year effects also narrows the comparison. The latter specification compares exposed and comparison commuting zones within the same census division in each year and reduces the estimated diffusion coefficient by roughly one percentage point in this specification ladder.

The final difference concerns the campaign year. The baseline convention retains 1988 as part of the pre-period. Dropping 1988 changes the record-level estimates more than the commuting-zone estimates, reflecting the greater weight placed on records released during the campaign year.

The headline diffusion result in Section 5.1 uses commuting-zone-years, commuting-zone and census-division-by-year fixed effects, and retains 1988 in the pre-period. The preferred organizational specification in Section 6 instead operates at the record level and holds the recording studio fixed. Applying that organizational specification to documented sampling produces a smaller and less precise diffusion estimate. The difference illustrates why the paper estimates diffusion and organization at the units appropriate to the corresponding economic questions.

C.3 Conventional cluster-robust inference

The main text uses the bias-reduced CR2 estimator because only 18 commuting zones are exposed. For comparison, the figures below reproduce the principal commuting-zone results using conventional cluster-robust CRV1 inference.

Table C.1: Summary statistics

Panel A. Records

	campaign-exposed commuting zones					comparison commuting zones				
	<i>N</i>	mean	s.d.	median	share zero	<i>N</i>	mean	s.d.	median	share zero
<i>Panel B. Record characteristics</i>										
classified credits per album	44,430	25.1	25.3	18.0	0.000	26,282	22.5	21.8	17.0	0.000
distinct people credited	44,426	16.2	16.8	11.0	0.000	26,281	14.0	14.3	10.0	0.000
tracks	44,430	9.006	7.263	9.000	0.000	26,282	9.204	6.137	10.0	0.000
recording studios named	44,430	1.777	1.789	1.000	0.000	26,282	1.589	1.427	1.000	0.000
released on a major label	44,430	0.411	0.492	0.000	0.589	26,282	0.317	0.465	0.000	0.683
studio also active on the other side of 1988	44,430	0.675	0.468	1.000	0.325	26,282	0.566	0.496	1.000	0.434
<i>Panel C1. Outcomes, record, 1980–1988</i>										
share of credits for performance	15,726	0.381	0.257	0.412	0.192	8,317	0.392	0.274	0.423	0.215
share of credits for technical	15,726	0.168	0.151	0.130	0.115	8,317	0.155	0.163	0.113	0.166
share of credits for creative authorship and production	15,726	0.364	0.236	0.316	0.037	8,317	0.370	0.264	0.315	0.059
share of credits for nonproduction	15,726	0.088	0.098	0.069	0.343	8,317	0.083	0.105	0.057	0.392
share of people working only in performance	15,726	0.288	0.258	0.263	0.290	8,317	0.294	0.274	0.269	0.323
share of people working only in technical	15,726	0.181	0.159	0.151	0.162	8,317	0.162	0.171	0.125	0.242
share of people working only in creative authorship and production	15,726	0.256	0.263	0.167	0.207	8,317	0.276	0.294	0.176	0.256
share of people working only in nonproduction	15,726	0.104	0.113	0.083	0.364	8,317	0.098	0.120	0.066	0.419
record credits a sample	15,727	0.037	0.188	0.000	0.963	8,317	0.019	0.137	0.000	0.981
<i>Panel C2. Outcomes, record, 1989–2000</i>										
share of credits for performance	28,700	0.357	0.255	0.355	0.170	17,964	0.409	0.261	0.421	0.146
share of credits for technical	28,700	0.190	0.160	0.158	0.100	17,964	0.185	0.169	0.143	0.120
share of credits for creative authorship and production	28,700	0.350	0.229	0.327	0.076	17,964	0.301	0.229	0.273	0.127
share of credits for nonproduction	28,700	0.103	0.117	0.077	0.338	17,964	0.104	0.124	0.075	0.352
share of people working only in performance	28,700	0.266	0.258	0.214	0.294	17,964	0.292	0.277	0.250	0.304
share of people working only in technical	28,700	0.183	0.163	0.154	0.172	17,964	0.174	0.171	0.143	0.219
share of people working only in creative authorship and production	28,700	0.238	0.247	0.167	0.270	17,964	0.192	0.241	0.105	0.386
share of people working only in nonproduction	28,700	0.120	0.132	0.091	0.362	17,964	0.121	0.139	0.091	0.386
record credits a sample	28,703	0.097	0.296	0.000	0.903	17,965	0.038	0.191	0.000	0.962

Panel B. Commuting zones

	campaign-exposed commuting zones					comparison commuting zones				
	<i>N</i>	mean	s.d.	median	share zero	<i>N</i>	mean	s.d.	median	share zero
<i>Panel B. Predetermined market characteristics</i>										
population, 1980	18	3,792,387	3,081,396	2,722,558	0.000	106	772,513	808,735	541,055	0.000
employment, 1980	18	1,446,617	1,181,723	964,631	0.000	106	257,782	299,994	176,253	0.000
log population density, 1980	18	5.320	0.819	5.011	0.000	106	3.804	0.948	3.844	0.000
urban share, 1980	18	0.877	0.069	0.885	0.000	106	0.693	0.161	0.706	0.009
college share, 1980	18	0.191	0.044	0.186	0.000	106	0.161	0.039	0.156	0.000
unemployment rate, 1980	18	0.060	0.021	0.060	0.000	106	0.065	0.021	0.064	0.000
log median income, 1979	18	9.021	0.127	9.011	0.000	106	8.716	0.218	8.718	0.000
residents aged 16 and over, 1980	18	2,882,061	2,365,324	2,104,839	0.000	106	586,793	626,320	400,028	0.000
records released before 1988	18	3,675	7,571	776.5	0.000	106	286.2	947.4	42.0	0.085
studios active before 1988	18	131.1	193.9	64.5	0.000	106	15.9	30.7	7.000	0.085
people credited before 1988	18	5,377	9,428	2,128	0.000	106	550.9	1,465	112.5	0.170
other 1988 tours hosted	18	5.778	1.987	6.500	0.056	106	2.594	1.632	2.000	0.000
<i>Panel C1. Outcomes, market-year, 1980–1988</i>										
share of releases crediting a sample	162	0.017	0.026	0.000	0.568	606	0.008	0.048	0.000	0.894
share of active studios with a release crediting a sample	162	0.037	0.051	0.000	0.568	606	0.018	0.080	0.000	0.894
credited people per 100,000 working-age residents	162	26.4	26.9	18.3	0.000	954	12.5	38.5	3.129	0.365
releases	162	97.1	176.0	31.0	0.000	954	8.718	24.5	1.500	0.365
active studios	162	32.9	54.0	14.0	0.000	954	3.560	8.324	1.000	0.365
credited people	162	1,070	2,065	300.0	0.000	954	95.3	288.8	11.0	0.365
<i>Panel C2. Outcomes, market-year, 1989–2000</i>										
share of releases crediting a sample	216	0.054	0.062	0.040	0.352	847	0.022	0.073	0.000	0.779
share of active studios with a release crediting a sample	216	0.085	0.084	0.071	0.352	847	0.041	0.118	0.000	0.779
credited people per 100,000 working-age residents	216	36.0	34.9	23.6	0.000	1,272	17.5	45.9	5.011	0.334
releases	216	132.9	226.3	43.0	0.000	1,272	14.1	36.1	2.000	0.334
active studios	216	48.9	76.7	21.0	0.000	1,272	5.937	14.2	1.000	0.334
credited people	216	1,595	2,915	491.5	0.000	1,272	158.2	450.6	19.0	0.334

Note. The estimation sample by treatment arm. Panel A contains one observation per record and Panel B describes comparison commuting zones. Exposed commuting zones are the 18 that hosted a realized 1988 *Bad* Tour stop during the AKAI campaign. Comparison commuting zones are 106 in 1988 without the AKAI promotional campaign. Predetermined commuting-zone characteristics are measured before treatment.

Table C.2: Summary statistics: commuting zones

	campaign-exposed commuting zones					comparison commuting zones				
	<i>N</i>	mean	s.d.	median	share zero	<i>N</i>	mean	s.d.	median	share zero
<i>Panel B. Predetermined market characteristics</i>										
population, 1980	18	3,792,387	3,081,396	2,722,558	0.000	106	772,513	808,735	541,055	0.000
employment, 1980	18	1,446,617	1,181,723	964,631	0.000	106	257,782	299,994	176,253	0.000
log population density, 1980	18	5.320	0.819	5.011	0.000	106	3.804	0.948	3.844	0.000
urban share, 1980	18	0.877	0.069	0.885	0.000	106	0.693	0.161	0.706	0.009
college share, 1980	18	0.191	0.044	0.186	0.000	106	0.161	0.039	0.156	0.000
unemployment rate, 1980	18	0.060	0.021	0.060	0.000	106	0.065	0.021	0.064	0.000
log median income, 1979	18	9.021	0.127	9.011	0.000	106	8.716	0.218	8.718	0.000
residents aged 16 and over, 1980	18	2,882,061	2,365,324	2,104,839	0.000	106	586,793	626,320	400,028	0.000
records released before 1988	18	3,675	7,571	776.5	0.000	106	286.2	947.4	42.0	0.085
studios active before 1988	18	131.1	193.9	64.5	0.000	106	15.9	30.7	7.000	0.085
people credited before 1988	18	5,377	9,428	2,128	0.000	106	550.9	1,465	112.5	0.170
other 1988 tours hosted	18	5.778	1.987	6.500	0.056	106	2.594	1.632	2.000	0.000
<i>Panel C1. Outcomes, market-year, 1980–1988</i>										
share of releases crediting a sample	162	0.017	0.026	0.000	0.568	606	0.008	0.048	0.000	0.894
share of active studios with a release crediting a sample	162	0.037	0.051	0.000	0.568	606	0.018	0.080	0.000	0.894
credited people per 100,000 working-age residents	162	26.4	26.9	18.3	0.000	954	12.5	38.5	3.129	0.365
releases	162	97.1	176.0	31.0	0.000	954	8.718	24.5	1.500	0.365
active studios	162	32.9	54.0	14.0	0.000	954	3.560	8.324	1.000	0.365
credited people	162	1,070	2,065	300.0	0.000	954	95.3	288.8	11.0	0.365
<i>Panel C2. Outcomes, market-year, 1989–2000</i>										
share of releases crediting a sample	216	0.054	0.062	0.040	0.352	847	0.022	0.073	0.000	0.779
share of active studios with a release crediting a sample	216	0.085	0.084	0.071	0.352	847	0.041	0.118	0.000	0.779
credited people per 100,000 working-age residents	216	36.0	34.9	23.6	0.000	1,272	17.5	45.9	5.011	0.334
releases	216	132.9	226.3	43.0	0.000	1,272	14.1	36.1	2.000	0.334
active studios	216	48.9	76.7	21.0	0.000	1,272	5.937	14.2	1.000	0.334
credited people	216	1,595	2,915	491.5	0.000	1,272	158.2	450.6	19.0	0.334

Table C.3: Summary statistics: records

	campaign-exposed commuting zones					comparison commuting zones				
	<i>N</i>	mean	s.d.	median	share zero	<i>N</i>	mean	s.d.	median	share zero
<i>Panel B. Record characteristics</i>										
classified credits per album	44,430	25.1	25.3	18.0	0.000	26,282	22.5	21.8	17.0	0.000
distinct people credited	44,426	16.2	16.8	11.0	0.000	26,281	14.0	14.3	10.0	0.000
tracks	44,430	9.006	7.263	9.000	0.000	26,282	9.204	6.137	10.0	0.000
recording studios named	44,430	1.777	1.789	1.000	0.000	26,282	1.589	1.427	1.000	0.000
released on a major label	44,430	0.411	0.492	0.000	0.589	26,282	0.317	0.465	0.000	0.683
studio also active on the other side of 1988	44,430	0.675	0.468	1.000	0.325	26,282	0.566	0.496	1.000	0.434
<i>Panel C1. Outcomes, record, 1980–1988</i>										
share of credits for performance	15,726	0.381	0.257	0.412	0.192	8,317	0.392	0.274	0.423	0.215
share of credits for technical	15,726	0.168	0.151	0.130	0.115	8,317	0.155	0.163	0.113	0.166
share of credits for creative authorship and production	15,726	0.364	0.236	0.316	0.037	8,317	0.370	0.264	0.315	0.059
share of credits for nonproduction	15,726	0.088	0.098	0.069	0.343	8,317	0.083	0.105	0.057	0.392
share of people working only in performance	15,726	0.288	0.258	0.263	0.290	8,317	0.294	0.274	0.269	0.323
share of people working only in technical	15,726	0.181	0.159	0.151	0.162	8,317	0.162	0.171	0.125	0.242
share of people working only in creative authorship and production	15,726	0.256	0.263	0.167	0.207	8,317	0.276	0.294	0.176	0.256
share of people working only in nonproduction	15,726	0.104	0.113	0.083	0.364	8,317	0.098	0.120	0.066	0.419
record credits a sample	15,727	0.037	0.188	0.000	0.963	8,317	0.019	0.137	0.000	0.981
<i>Panel C2. Outcomes, record, 1989–2000</i>										
share of credits for performance	28,700	0.357	0.255	0.355	0.170	17,964	0.409	0.261	0.421	0.146
share of credits for technical	28,700	0.190	0.160	0.158	0.100	17,964	0.185	0.169	0.143	0.120
share of credits for creative authorship and production	28,700	0.350	0.229	0.327	0.076	17,964	0.301	0.229	0.273	0.127
share of credits for nonproduction	28,700	0.103	0.117	0.077	0.338	17,964	0.104	0.124	0.075	0.352
share of people working only in performance	28,700	0.266	0.258	0.214	0.294	17,964	0.292	0.277	0.250	0.304
share of people working only in technical	28,700	0.183	0.163	0.154	0.172	17,964	0.174	0.171	0.143	0.219
share of people working only in creative authorship and production	28,700	0.238	0.247	0.167	0.270	17,964	0.192	0.241	0.105	0.386
share of people working only in nonproduction	28,700	0.120	0.132	0.091	0.362	17,964	0.121	0.139	0.091	0.386
record credits a sample	28,703	0.097	0.296	0.000	0.903	17,965	0.038	0.191	0.000	0.962

Table C.4: Share of records documenting a sample, by year and treatment arm

campaign-exposed commuting zones				comparison commuting zones		
	releases	share crediting a sample	market average	releases	share crediting a sample	market average
<i>The share of releases crediting a sample, by year</i>						
1980	1,670	0.020	0.010	1,061	0.008	0.005
1981	1,608	0.020	0.009	958	0.009	0.003
1982	1,613	0.024	0.013	906	0.012	0.021
1983	1,599	0.021	0.017	940	0.007	0.001
1984	1,545	0.033	0.010	800	0.007	0.004
1985	1,656	0.033	0.018	826	0.021	0.007
1986	1,804	0.042	0.015	896	0.017	0.009
1987	2,038	0.044	0.025	927	0.032	0.009
1988	2,194	0.077	0.031	1,003	0.056	0.018
1989	2,143	0.090	0.038	1,041	0.056	0.035
1990	2,141	0.113	0.059	1,092	0.055	0.021
1991	2,179	0.117	0.062	1,162	0.044	0.038
1992	2,420	0.116	0.083	1,311	0.054	0.023
1993	2,440	0.121	0.084	1,589	0.043	0.022
1994	2,546	0.107	0.078	1,615	0.050	0.026
1995	2,607	0.094	0.053	1,789	0.032	0.015
1996	2,634	0.091	0.038	1,712	0.036	0.025
1997	2,467	0.082	0.041	1,731	0.021	0.009
1998	2,482	0.084	0.040	1,663	0.027	0.014
1999	2,451	0.073	0.030	1,674	0.027	0.027
2000	2,193	0.079	0.036	1,586	0.028	0.011

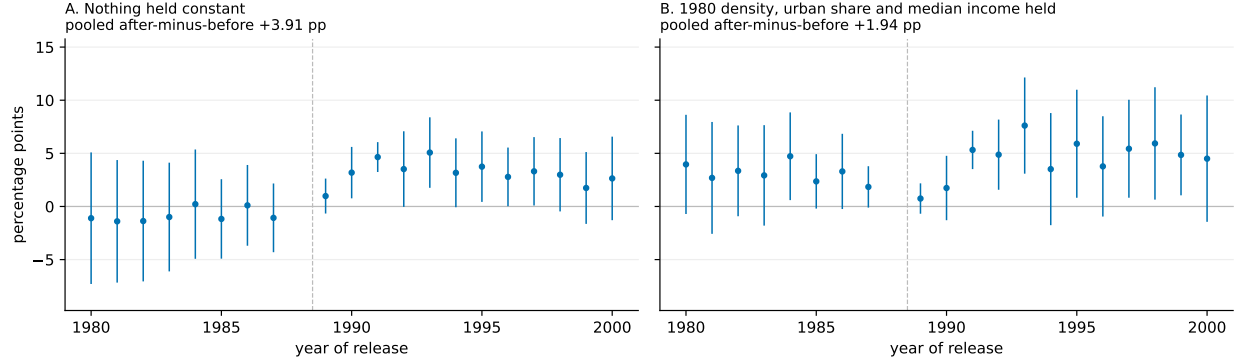
Table C.5: The unit and campaign-year bridge for documented sampling

Specification	Unit	Fixed effects	1988	$\hat{\beta}$ (pp)	95% interval	Separates
Record level, commuting zone and year	record	commuting zone + year	kept as pre	3.91	[+0.70, +7.13]	yes
Record level, studio and year	record	studio + year	kept as pre	2.85	[+0.13, +5.57]	yes
Record level, studio and year	record	studio + year	dropped	2.44	[−0.52, +5.39]	no
Commuting-zone level, division-year	CZ-year	commuting zone + division-year	dropped	2.43	[+0.50, +4.37]	yes
Section 5.1 headline	CZ-year	commuting zone + division-year	kept as pre	2.37	[+0.52, +4.22]	yes
Record specification of Section 6	record	studio + division-year	kept as pre	1.62	[−0.12, +3.35]	no
Record level, studio and division-year	record	studio + division-year	dropped	0.96	[−1.08, +3.00]	no

Note. The outcome is documented sampling: the share of records documenting a sample in commuting-zone-year specifications and a record-level indicator in record specifications. Coefficients are in percentage points. Intervals use the bias-reduced CR2 estimator with coefficient-specific Satterthwaite degrees of freedom and are clustered by commuting zone. “1988 kept as pre” retains the campaign year in the pre-period; “dropped” removes those observations from the estimation sample.

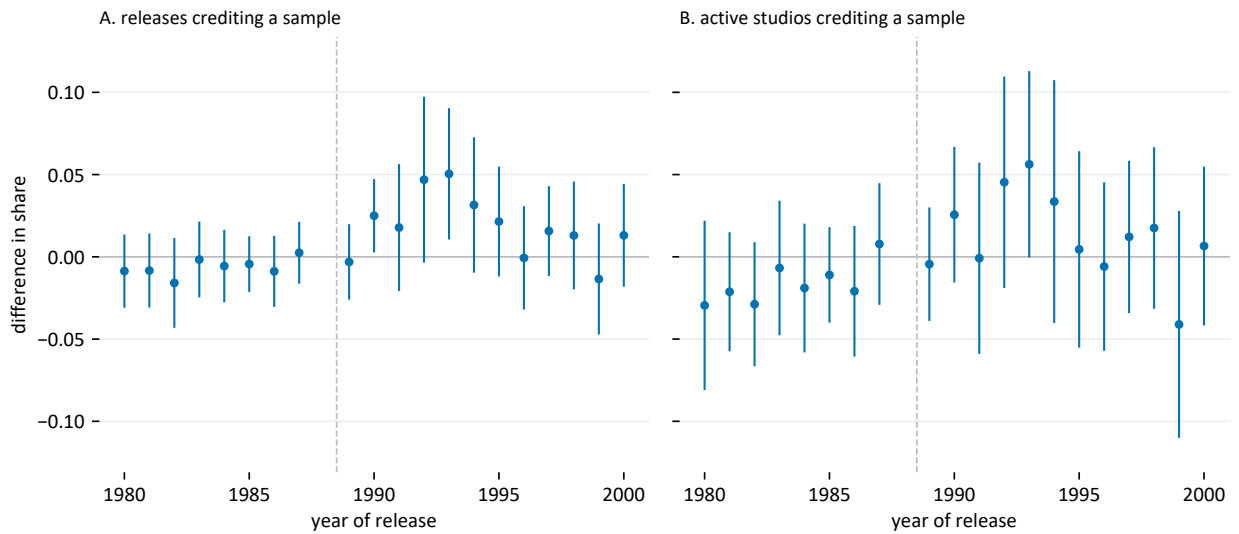
Figure C.1: Documented sampling at the record level, conventional intervals

Whether a record credits a sample, by the year it was released



Note. Record-level counterparts to Figure 4, with conventional cluster-robust intervals. The figure is shown on the same vertical scale as the corresponding CR2 estimates.

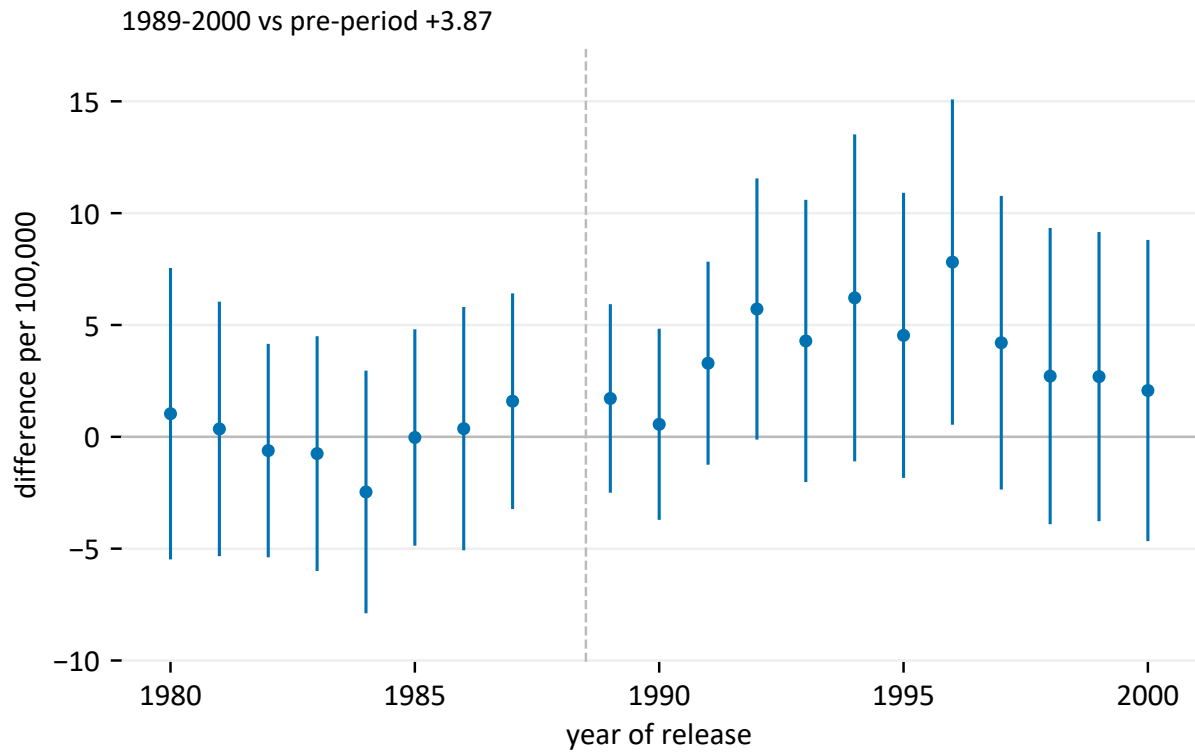
Figure C.2: Documented sampling, conventional cluster-robust inference



Note. Figure 4 using conventional CRV1 inference with $G - 1$ degrees of freedom. The vertical scale matches the corresponding figure in the main text.

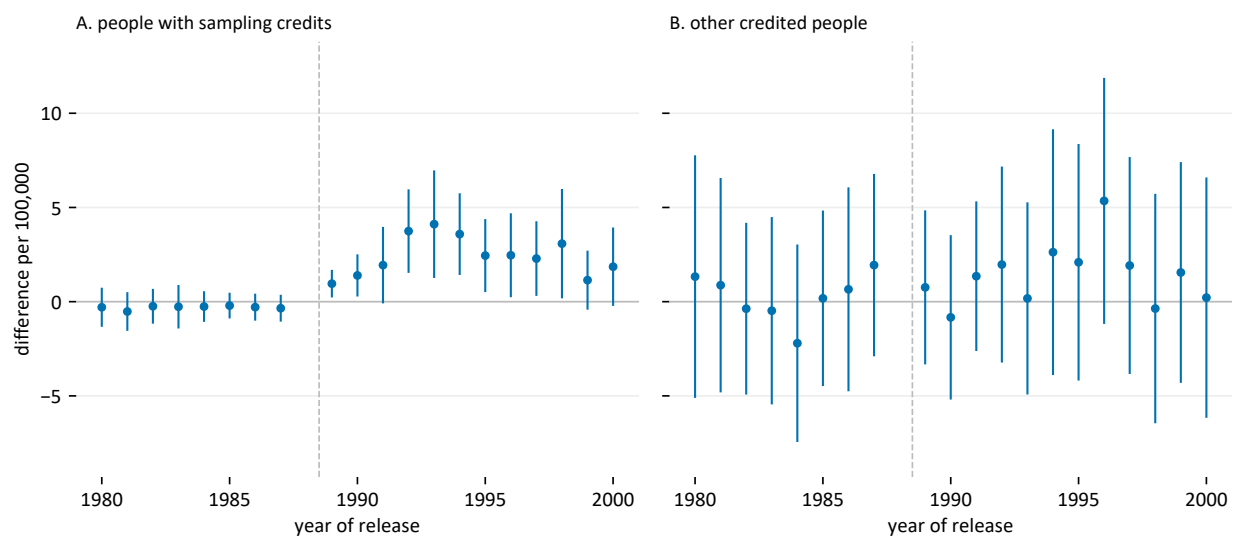
Figure C.3: Participation, conventional cluster-robust inference

Credited people per 100,000 working-age residents, by year



Note. Figure 5 using conventional CRV1 inference, on the same vertical scale.

Figure C.4: Participation by sampling-linked status, conventional cluster-robust inference



Note. Figure D.4 using conventional CRV1 inference, on the same vertical scale.

C.4 Multiple hypotheses

Table C.6 reports Benjamini–Hochberg and Holm adjustments for the family of record-level function-share and contributor-exclusivity outcomes. The adjusted values are based on conventional CRV1 p -values. They are included as a supplementary multiple-testing diagnostic rather than as the basis for the paper’s inference, which relies on the CR2 confidence intervals reported in the main text.

Table C.6: Multiplicity-adjusted p -values, record-level outcomes

Outcome	Coef.	p	p (BH)	p (Holm)
Performance	−0.0107	0.207	0.553	1.000
Technical	−0.0027	0.654	0.748	1.000
Creative authorship and production	+0.0157	0.101	0.404	0.707
Nonproduction	−0.0023	0.544	0.748	1.000
Performance (specialist)	−0.0041	0.563	0.748	1.000
Technical (specialist)	−0.0008	0.893	0.893	1.000
Creative authorship and production (specialist)	+0.0211	0.027	0.218	0.218
Nonproduction (specialist)	−0.0035	0.430	0.748	1.000

Notes. Unadjusted CRV1 p -values for the eight four-family companion outcomes—performance, technical, creative authorship and production, and nonproduction, each in credit-share and exclusivity form—with Benjamini–Hochberg and Holm adjustments applied within that family. **They are not the rows of Table E.3**, which merges performance and technical into execution and adds the two-or-more-categories residual. The family is defined post hoc, which is the conservative direction here. The paper’s reporting rule remains the confidence interval; this table exists so the reader can apply a threshold convention if they hold one.

C.5 The pooled coefficient and averages of the annual estimates

The pooled difference-in-differences coefficient and the average of the post-campaign event-study coefficients summarize different contrasts (Baker et al., 2026). The pooled coefficient compares the average post-campaign period with the average pre-campaign period. By contrast, each annual event-study coefficient is normalized to 1988, so averaging the post-campaign coefficients produces an average post-period difference relative to 1988.

The two quantities coincide when the average pre-campaign event-study coefficient relative to 1988 is zero. More generally, their difference reflects the average position of the pre-campaign coefficients relative to the reference year. Table C.7 reports the pooled coefficient, the average post-campaign event-study coefficient, and the corresponding average pre-campaign coefficient for the principal outcomes.

The distinction is most relevant for the count of contributors credited exclusively in creative authorship and production. The average post-campaign coefficient relative to 1988

Table C.7: Pooled coefficients and averages of annual estimates

	Pooled Post $\times$ Exposed	Post-period average vs 1988	Pre-period average vs 1988
<i>Panel A. Records, studio and division-year effects</i>			
Creative authorship and production, share of credits	+1.57 [-0.88, 4.01]	+2.58 [0.28, 4.89]	+0.85 [-1.82, 3.53]
Execution, share of credits	-1.34 [-3.46, 0.79]	-2.38 [-4.50, -0.25]	-0.92 [-3.53, 1.69]
Contributors exclusive to creative authorship, share	+2.11 [-0.33, 4.55]	+3.72 [1.66, 5.78]	+1.74 [-1.46, 4.94]
Contributors exclusive to creative authorship, count	+0.44 [0.06, 0.82]	+0.75 [0.19, 1.31]	+0.35 [-0.03, 0.73]
Credited team size	+0.59 [-0.36, 1.53]	+1.69 [0.20, 3.17]	+1.31 [-0.11, 2.73]
<i>Panel B. Records, commuting-zone and year effects</i>			
Creative authorship and production, share of credits	+4.14 [-0.08, 8.35]	+2.75 [-0.72, 6.22]	-1.44 [-4.52, 1.65]
Execution, share of credits	-4.16 [-7.86, -0.46]	-3.16 [-6.29, -0.03]	+0.96 [-1.80, 3.72]
Contributors exclusive to creative authorship, share	+4.56 [-0.26, 9.38]	+3.87 [0.22, 7.52]	-0.60 [-3.93, 2.72]
Contributors exclusive to creative authorship, count	+0.68 [0.13, 1.24]	+1.03 [0.47, 1.60]	+0.44 [0.05, 0.83]
Credited team size	-0.21 [-2.51, 2.09]	+1.85 [0.04, 3.66]	+2.39 [0.46, 4.31]
<i>Panel C. Commuting zones</i>			
Records documenting a sample, share	+2.37 [0.52, 4.22]	+1.81 [-0.30, 3.93]	-0.64 [-2.62, 1.34]
Credited contributors per 100,000 aged 20–64	+3.87 [-1.48, 9.23]	+3.82 [-1.48, 9.12]	-0.06 [-4.86, 4.73]

Note. Two summary parameters formed from the same annual estimates. The pooled column is the coefficient on Exposed $\times$ Post reported in the body, which compares the post-period average with the *pre-period* average. The second column averages the annual coefficients over 1989–2000, each measured against 1988 alone. The third column is the average of the eight pre-period annual coefficients, also measured against 1988, and is the difference between the first two columns. The two summary parameters coincide only when the pre-period coefficients average to zero. Shares are in percentage points and counts in contributors. Brackets are 95% confidence intervals under the bias-reduced estimator with per-coefficient Satterthwaite degrees of freedom, clustered by commuting zone. On the count of contributors exclusive to creative authorship and production the pre-period average is 68% of the 1993–2000 step reported in Section 6.2; none of the eight pre-period years separates from zero individually, which is a different statement and is why both columns are shown. The estimates the paper reports are the pooled ones, which difference this level away rather than carrying it.

is +0.752 contributors, while the pooled 1989–2000 coefficient is +0.441. The corresponding average pre-campaign coefficient relative to 1988 is +0.349.

The event-study coefficients therefore do not describe a perfectly flat pre-period relative to 1988, even though no individual pre-campaign coefficient separates from zero. This is one reason the main text places greater weight on the pooled pre-versus-post coefficient and on the later 1993–2000 contrast than on the level of the post-campaign coefficients relative to a single reference year. Both contrasts remove the average pre-period difference rather than treating 1988 alone as the counterfactual level.

C.6 Treatment of the campaign year

The AKAI campaign took place during 1988. Because release dates do not reveal whether production occurred before or after the local demonstration, the baseline specifications retain records released in 1988 and assign the year to the pre-period. Event studies use 1988 as the omitted reference year.

One way to examine this convention is to omit 1987 instead and estimate a coefficient for 1988. Table C.8 reports this coefficient for the main outcomes under both fixed-effect specifications. None of the twelve outcome-by-specification estimates separates from zero.

The estimates remain imprecise enough that this exercise cannot rule out economically meaningful movement during 1988. For the count of contributors concentrated in creative authorship and production, for example, the 1988 coefficient is -0.250 , with an interval from -0.706 to $+0.205$. The evidence therefore supports the narrower conclusion that I do not detect a differential movement during the campaign year.

Table C.9 reports the pooled estimates under three alternative codings of 1988: retaining the year as pre-treatment, dropping it from the estimation sample, or assigning it to the post-period.

For the main contributor-count outcome, the corresponding coefficients are +0.441, +0.385, and +0.291. Dropping 1988 reduces the estimate by about 13 percent and leaves its interval above zero, at $[+0.029, +0.741]$. Assigning the campaign year to the post-period reduces the coefficient by about 34 percent, to +0.291, with an interval of $[-0.005, +0.587]$.

The first two codings are the most natural given that the campaign occurred during 1988 and records released that year may have been produced before local campaign exposure. The third provides a more demanding alternative in which all 1988 releases are treated as post-campaign observations.

Table C.8: The campaign year relative to 1987

	1988 against 1987 (zero under no anticipation)	Share of the 1993–2000 step
<i>Panel A. Records, studio and division-year effects</i>		
Creative authorship and production, share of credits	−0.29 [−3.87, 3.30]	−0.11
Execution, share of credits	+0.45 [−2.54, 3.43]	−0.18
Contributors exclusive to creative authorship, share	−0.70 [−4.68, 3.29]	−0.26
Contributors exclusive to creative authorship, count	−0.25 [−0.71, 0.21]	−0.49
Credited team size	−1.21 [−3.49, 1.07]	−2.68
<i>Panel B. Records, commuting-zone and year effects</i>		
Creative authorship and production, share of credits	−0.02 [−2.96, 2.92]	−0.00
Execution, share of credits	+0.38 [−2.16, 2.92]	−0.07
Contributors exclusive to creative authorship, share	−0.61 [−3.63, 2.42]	−0.11
Contributors exclusive to creative authorship, count	−0.23 [−0.70, 0.24]	−0.29
Credited team size	−0.82 [−2.46, 0.82]	+2.51
<i>Panel C. Commuting zones</i>		
Records documenting a sample, share	−0.25 [−2.23, 1.74]	−0.11
Credited contributors per 100,000 aged 20–64	−1.59 [−6.67, 3.48]	−0.36

Note. The AKAI campaign ran during 1988, which the paper codes as pre-treatment and omits as the event-study reference year. Re-estimating with 1987 as the omitted year makes the 1988 coefficient estimable; under no anticipation it is zero. It separates from zero on 0 of 12 outcome-by-block cells. The second column expresses it as a fraction of the same outcome's 1993–2000 step, so that its size can be read against the quantity it would displace. Shares are in percentage points and counts in contributors; brackets are 95% confidence intervals under the bias-reduced estimator, clustered by commuting zone. **This is not a powerful test.** On the count of contributors exclusive to creative authorship and production the estimate is −0.250 with an interval from −0.706 to +0.205, which does not exclude a displacement of the same order as the step itself. Failing to reject no anticipation is therefore weaker evidence than the count of non-rejections suggests.

Table C.9: Pooled estimates under alternative codings of 1988

	1988 kept as pre-period	1988 rows dropped	1988 coded post-period
<i>Panel A. Records, studio and division-year effects</i>			
Creative authorship and production, share of credits	+1.57	+1.51	+1.15
Execution, share of credits	−1.34	−1.27	−0.94
Contributors exclusive to creative authorship, share	+2.11	+1.88	+1.36
Contributors exclusive to creative authorship, count	+0.44	+0.38	+0.29
Credited team size	+0.59	+0.36	+0.14
<i>Panel B. Records, commuting-zone and year effects</i>			
Creative authorship and production, share of credits	+4.14	+4.39	+4.15
Execution, share of credits	−4.16	−4.34	−4.08
Contributors exclusive to creative authorship, share	+4.56	+4.70	+4.39
Contributors exclusive to creative authorship, count	+0.68	+0.63	+0.56
Credited team size	−0.21	−0.53	−0.64
<i>Panel C. Commuting zones</i>			
Records documenting a sample, share	+2.37	+2.43	+2.30
Credited contributors per 100,000 aged 20–64	+3.87	+3.88	+3.59

Note. The pooled coefficient on $\text{Exposed} \times \text{Post}$ under the three available codings of the campaign year. The first keeps 1988's observations and assigns them to the pre-period, which is the convention used throughout the paper; the second removes them from the estimation sample; the third assigns them to the post-period. Shares are in percentage points and counts in contributors. On the count of contributors exclusive to creative authorship and production the three codings give +0.441, +0.385 and +0.291 contributors: dropping the campaign year's rows costs 13% of the estimate and coding it post-treatment costs 34%. The paper's coding is the first of the three. Because the campaign ran during 1988 the first two codings are both defensible and the third is the conservative reading; the differences between them are reported here rather than resolved.

C.7 Distance to the nearest campaign stop

The baseline design defines campaign exposure using whether a commuting zone hosted a realized 1988 campaign stop. This subsection replaces that binary measure with distance to the nearest realized stop to examine whether the response extends into nearby comparison markets.

Construction. For each continental commuting zone, I calculate the great-circle distance from its population-weighted centroid to the nearest realized 1988 campaign venue. I then interact indicators for predetermined distance bands with the post-campaign period: the commuting zone hosting the stop, 0–50 kilometres, 50–100 kilometres, 100–200 kilometres, 200–400 kilometres, and more than 400 kilometres, with the last category omitted.

The host-zone category corresponds to the baseline treatment. All eighteen exposed commuting zones contain their campaign venue, with centroid-to-venue distances between 2.6 and 34.1 kilometres, and no unexposed commuting zone is closer to a campaign venue than the furthest exposed commuting zone.

Results. Table C.10 reports the distance profiles for the main contributor-count outcome and documented sampling. For the contributor count, the host-zone estimate is +0.500 contributors. The coefficients in the non-host distance bands are small, imprecise, and show no systematic decay with distance. This pattern provides little evidence that the main organizational response extends into nearby comparison markets.

The result is similar after allowing predetermined 1980 characteristics to generate their own annual trajectories. The host-zone contributor estimate becomes +0.542, with an interval from +0.121 to +0.963.

Documented sampling behaves differently under these controls. Without them, the host-zone coefficient is +2.15 percentage points. Allowing predetermined market characteristics to generate separate trajectories reduces it to -0.04 percentage points, consistent with the sensitivity of the diffusion estimate discussed in Section 5.1.

The distance pattern is also not explained simply by market size. All eighteen exposed commuting zones lie in the top third of both the 1980 population distribution and the pre-campaign recording-activity distribution. Within that top third, the host-zone coefficient on the contributor count is +0.510 using production terciles and +0.478 using population terciles, compared with +0.500 in the full distance specification.

Distance to comparison-tour stops. A related placebo asks whether proximity to other major 1988 tours generates a similar gradient. This comparison requires care because

Table C.10: Campaign effects by distance from the nearest campaign stop

Distance band	Contributors exclusive to the category		Documented sampling (pp)	
	No controls	1980 char. $\times$ year	No controls	1980 char. $\times$ year
Commuting zone hosts a stop	+0.500* (0.235)	+0.542** (0.191)	+2.148 (1.459)	−0.041 (1.663)
0–50 km	−0.054 (0.464)	+0.116 (0.566)	+4.643 (5.215)	—
50–100 km	−0.159 (0.496)	+0.007 (0.440)	−0.482 (1.370)	−0.719 (1.421)
100–200 km	+0.187 (0.286)	+0.398 (0.227)	−0.173 (1.671)	−0.729 (1.608)
200–400 km	−0.391 (0.287)	−0.422** (0.199)	−0.959 (1.531)	−0.756 (1.432)

Note. The omitted category is more than 400 kilometres from the nearest campaign stop. Contributor counts are estimated at the record level with studio and census-division-by-year fixed effects. Documented sampling is estimated at the commuting-zone-year level with commuting-zone and census-division-by-year fixed effects. Both use all continental commuting zones. Intervals use the bias-reduced CR2 estimator with coefficient-specific Satterthwaite degrees of freedom and are clustered by commuting zone. The 0–50 kilometre category contains four commuting zones and is correspondingly imprecise. A dash indicates that the coefficient is collinear with the corresponding control block.

seventeen of the eighteen exposed commuting zones also hosted at least one comparison-tour stop, and 34 of the 169 comparison-tour venues lie inside exposed commuting zones.

I therefore report both the unscreened comparison-tour distance and versions that exclude comparison-tour venues located near realized campaign stops. Table C.11 reports the resulting estimates.

The comparison-tour distance does not produce a detectable gradient in either the unscreened or screened specifications. The result is similar under both screening radii, despite the different numbers of comparison venues retained. These estimates provide additional evidence that the campaign-distance pattern is not simply a generic feature of proximity to major arena tours.

C.8 Weighting and the equal-weighted commuting-zone estimand

The main organizational results are estimated at the record level, so local markets receive weight in proportion to the number of records they produce. This subsection shows how the estimate changes when commuting zones instead receive equal weight.

Starting from the record-level contributor-count specification and changing one element at a time, replacing studio fixed effects with commuting-zone fixed effects gives +0.682 contributors, compared with +0.690 in the corresponding studio specification. Restricting the sample to the 52 commuting zones producing at least one record in every sample year gives

Table C.11: Distance to comparison-tour stops as a placebo

Comparison-tour pool	Comparison-tour distance alone	Campaign distance, conditional
<i>Documented sampling, commuting zone (percentage points per log kilometre)</i>		
No screen	−0.556* (0.291)	−0.708** (0.281)
Comparison stops beyond 100 km of a campaign stop	−0.018 (0.282)	−0.973*** (0.312)
Comparison stops beyond 200 km of a campaign stop	+0.104 (0.277)	−1.005*** (0.351)
<i>Contributors exclusive to the category, record (per log kilometre)</i>		
No screen	+0.056 (0.128)	−0.155** (0.056)
Comparison stops beyond 100 km of a campaign stop	+0.126* (0.057)	−0.176** (0.084)
Comparison stops beyond 200 km of a campaign stop	+0.137** (0.046)	−0.121 (0.080)

Note. Distances enter in logarithms interacted with the post-campaign indicator. The screened samples exclude comparison-tour stops within the stated radius of a realized campaign stop. Fixed effects, clustering, and confidence intervals follow Table C.10.

+0.643. Collapsing those observations to commuting-zone-by-year averages and weighting each market-year by its number of records gives +0.586, with an interval from +0.126 to +1.045. Giving each commuting-zone-year equal weight instead yields +0.004, with an interval from −0.458 to +0.465.

The final two specifications use exactly the same 1,040 commuting-zone-year observations and the same 52 markets. The difference therefore comes from weighting rather than from aggregation or sample composition. The creative-authorship-and-production credit share displays the same pattern, moving from +4.37 percentage points under record weighting to −2.32 points under equal commuting-zone weighting.

These specifications use year rather than census-division-by-year effects so that each step of the comparison holds the remaining design features constant. The corresponding level of the coefficient therefore differs from the preferred estimate in the main text.

The two weighting schemes answer different questions. Record-level regressions describe the organization of the records actually produced and therefore place greater weight on markets where recording activity is concentrated. Equal-weighted commuting-zone regressions describe the average continuously active local market. The main organizational result pertains to the former estimand.

C.9 The sample window

The main analysis uses records released from 1980 through 2000. The underlying Discogs extract extends from 1965 through 2005, allowing the principal diffusion and organizational specifications to be re-estimated over a wider window.

The wider credit data are constructed using the same classification procedure as the main sample. On the overlapping 1980–2000 period, the resulting credit measures reproduce the main estimation data exactly.

Table C.12 reports the results. Over 1965–2005, the commuting-zone estimate for documented sampling is +1.99 percentage points, with an interval from +0.02 to +3.95, compared with +2.37 percentage points in the main sample. The pooled count of contributors credited exclusively in creative authorship and production is +0.390, with an interval from +0.016 to +0.764, compared with +0.441 in the main sample. The credit-share outcome remains less precisely estimated, as in the main analysis.

Figure C.5 reports the annual contributor-count estimates. Extending the sample considerably lengthens the pre-campaign period but also reduces precision in the earlier years because Discogs coverage is thinner before 1980. Under the studio specification, three of the twenty-three pre-campaign annual coefficients separate from zero; under commuting-zone fixed effects, six do. The wider window therefore provides a less stable annual pre-period than the main 1980–2000 sample, particularly under the commuting-zone specification, while leaving the pooled contributor-count result positive.

The content analysis is not extended beyond the main sample because its acoustic and lyrical representations were constructed specifically for the 1980–2000 record frame. The participation outcome is also not extended to the full 1965–2005 window because the underlying population series and predetermined baseline measures are not defined consistently across that entire period.

D Diffusion and participation

This appendix reports the additional estimates and robustness exercises underlying Section 5. It first presents the commuting-zone estimates and participation diagnostics, then examines the sensitivity of the documented-sampling result across alternative samples and specifications.

D.1 Commuting-zone estimates

Figure D.1 plots the raw annual share of records documenting sampling in each treatment arm. These series are descriptive and weight commuting zones according to the number of

Table C.12: Main estimates using a 1965–2005 sample window

	Estimate	95% interval (CR2)	<i>N</i>	Pre-years resolving
<i>Releases crediting a sample (pp)</i>				
1980–2000, as reported	+2.367	[+0.515, +4.219]	1,831	0 of 8
1980–2000, wider rule	+2.203	[+0.219, +4.186]	1,851	0 of 8
1965–2005, wider rule	+1.986	[+0.019, +3.952]	3,341	0 of 23
<i>Exclusive authorship contributors, studio</i>				
1980–2000, as reported	+0.441	[+0.060, +0.822]	70,707	0 of 8
1980–2000, wider rule	+0.447	[+0.063, +0.831]	70,733	0 of 8
1965–2005, wider rule	+0.390	[+0.016, +0.764]	102,103	3 of 23
<i>the same, commuting-zone block</i>				
1980–2000, as reported	+0.660	[+0.190, +1.129]	70,707	1 of 8
1980–2000, wider rule	+0.661	[+0.193, +1.130]	70,733	1 of 8
1965–2005, wider rule	+0.547	[−0.061, +1.156]	102,103	6 of 23
<i>Authorship share of credits, studio (pp)</i>				
1980–2000, as reported	+1.566	[−0.877, +4.009]	70,707	0 of 8
1980–2000, wider rule	+1.548	[−0.891, +3.988]	70,733	0 of 8
1965–2005, wider rule	+1.662	[−1.458, +4.781]	102,103	0 of 23
<i>the same, commuting-zone block</i>				
1980–2000, as reported	+2.652	[−0.785, +6.089]	70,707	0 of 8
1980–2000, wider rule	+2.648	[−0.788, +6.084]	70,733	0 of 8
1965–2005, wider rule	+4.408	[−1.109, +9.925]	102,103	1 of 23

Notes. The paper’s own specifications re-fitted on a wider frame. **This is an exploratory, post-hoc exercise: it is not a registered entry, it was not pre-specified, and no estimate reported anywhere else in this paper moves because of it.** The raw Discogs extract spans 1965–2005; the 1980–2000 restriction is applied when the estimation frame is built, not at the source, so the wider window requires no new data. Credits reach back with it: the role-slot table is rebuilt by the same code that builds the reported one, and on the reported frame it reproduces that table exactly. **The middle row of each block is what separates the frame rule from the window.** The reported album frame keeps exactly one release per Discogs master; restoring that rule to the wider frame leaves 70,733 albums against the reported 70,707, so the rule adds twenty-six records and only the window distinguishes the third row from the second. **The headline estimates survive the window.** Documented sampling is +1.99 percentage points [+0.02, +3.95] over 1965–2005, and the contributor count is +0.390 [+0.016, +0.764]; both intervals exclude zero. **The pre-year counts are not comparable across rows without their benchmark.** The rule applied elsewhere in this paper triggers on two of eight pre-campaign years, which corresponds to 0.4 rejections expected by chance at a nominal five per cent; the wide window has twenty-three pre years, where chance alone gives 1.2. Coverage before 1980 is thin and the annual intervals of Figure C.5 show it. The content measures of Section 7 are absent here and were not approximated: they were constructed on the reported frame and on nothing else. The rate outcomes are absent for the same kind of reason — their denominators are fixed at 1980 or begin in 1969, and none is defined across 1965–2005.

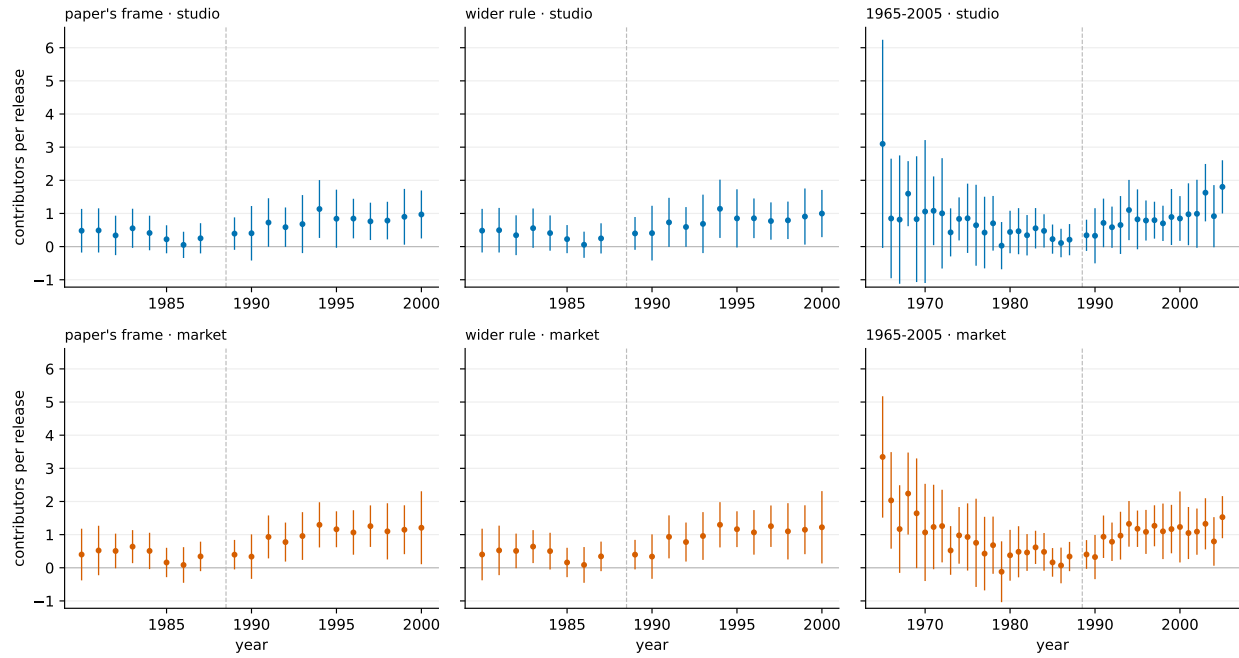


Figure C.5: Contributors credited exclusively in creative authorship and production over alternative sample windows

Note. Annual coefficients for contributors credited exclusively in creative authorship and production. The top row uses studio and census-division-by-year fixed effects; the bottom row uses commuting-zone and census-division-by-year fixed effects. Intervals use the bias-reduced CR2 estimator with coefficient-specific Satterthwaite degrees of freedom and are clustered by commuting zone. All panels share the same vertical scale. Coverage before 1980 is substantially thinner than in the main sample.

Table D.1: Commuting-zone-level estimates

Outcome	CRV1	CR2	ν	Baseline	N
<i>Panel A. Sampling adoption</i>					
Share of releases crediting a sample	+0.0237*** (0.0089)	+0.0237** (0.0090)	25.4	0.0207	1,831
Share of active studios crediting a sample	+0.0266** (0.0127)	+0.0266** (0.0129)	25.4	0.0382	1,831
<i>Panel B. Credited people</i>					
Credited people per 100k aged 20–64	+3.8750 (2.5433)	+3.8750 (2.5894)	23.4	17.8	2,604
with a sampling credit	+2.6879** (1.0564)	+2.6879** (1.0829)	23.4	0.7890	2,604
without a sampling credit	+1.1871 (1.9866)	+1.1871 (2.0153)	23.4	17.0	2,604
Market FE		Yes			
Division-by-year FE		Yes			
Clustered on		Commuting zone			

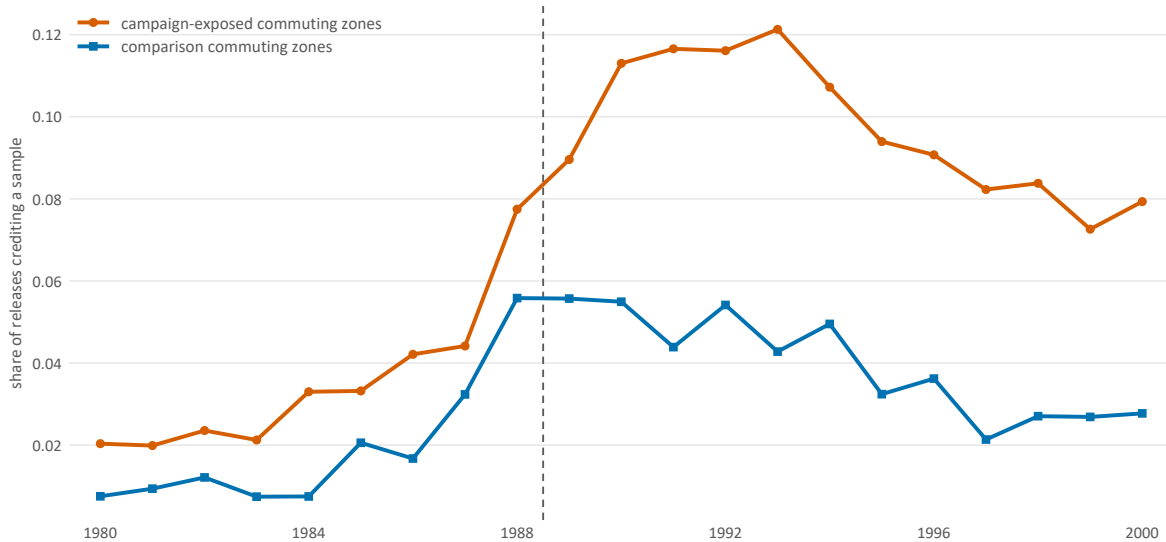
Notes. Each row reports the coefficient on $\text{Exposed}_m \times \text{Post}_t$ from a separate commuting-zone-year regression of the stated outcome, following Equation 3 in pooled form, with $\text{Post}_t = \mathbf{1}[t \geq 1989]$ and 1988 in neither window. **The two estimator columns carry the same point estimate by construction** and differ only in the standard error and therefore in the stars: CRV1 uses $G - 1$ degrees of freedom, and the bias-reduced CR2 uses the per-coefficient Satterthwaite ν reported beside it. A coefficient significant under one and not the other is the adjustment, not a disagreement. **Stars are assessed against those degrees of freedom, not against normal critical values**, which matters most in the CR2 column where ν can fall below 10. ‘Baseline’ is the mean of the outcome on the estimation sample. **The two people rows in Panel B partition the total by the person** — at least one sampling credit in the market-year against none — and sum to the first row at every market-year. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.2: Robustness of the documented-sampling estimate

Specification	Coef.	Baseline	<i>N</i>	Clusters
R1 all markets	+0.0257*** (0.0080)	0.0167	3,345	343
R1 concert markets	+0.0237*** (0.0089)	0.0207	1,831	116
R1 concert markets, two largest dropped	+0.0186** (0.0091)	0.0191	1,789	114
R2 clustered on state	+0.0237** (0.0111)	0.0207	1,831	43
R3 placebo	+0.0061 (0.0091)	0.0140	2,967	325
R4 full, by period	+0.0044 (0.0089)	0.0207	1,831	116
R4 full, by year	+0.0049 (0.0090)	0.0207	1,831	116
R5 no income, by period	+0.0105 (0.0098)	0.0207	1,831	116
R5 no income, by year	+0.0108 (0.0099)	0.0207	1,831	116
R7 reissues excluded	+0.0247*** (0.0092)	0.0209	1,825	116
R8 concert markets, 3 nearest	+0.0204 (0.0122)	0.0317	747	36
R8 concert markets, 5 nearest	+0.0171 (0.0116)	0.0323	831	40
R8 all markets, 3 nearest	+0.0165 (0.0124)	0.0328	769	37
R9 leave one out	+0.0221** (0.0094)	0.0202	1,810	115
R9 leave one out	+0.0204** (0.0087)	0.0196	1,810	115
R9 leave one out	+0.0242** (0.0093)	0.0204	1,810	115
R9 leave one out	+0.0240** (0.0092)	0.0205	1,810	115
R9 leave one out	+0.0243*** (0.0092)	0.0203	1,810	115
R9 leave one out	+0.0199** (0.0083)	0.0200	1,810	115
R9 leave one out	+0.0189** (0.0082)	0.0201	1,810	115
R9 leave one out	+0.0267*** (0.0086)	0.0207	1,810	115
R9 leave one out	+0.0233** (0.0091)	0.0204	1,810	115
R9 leave one out	+0.0242*** (0.0092)	0.0206	1,810	115
R9 leave one out	+0.0222** (0.0096)	0.0201	1,810	115
R9 leave one out	+0.0253*** (0.0093)	0.0207	1,810	115
R9 leave one out	+0.0237** (0.0091)	0.0207	1,810	115
R9 leave one out	+0.0256*** (0.0093)	0.0208	1,810	115
R9 leave one out	+0.0256*** (0.0088)	0.0207	1,810	115
R9 leave one out	+0.0247*** (0.0092)	0.0207	1,810	115
R9 leave one out	+0.0249***	0.0208	1,810	115

records they produce. They show that the divergence underlying the event-study estimates is also visible in the unadjusted data.

Figure D.1: Share of records documenting sampling, by treatment arm



Note. Raw annual shares of records documenting sampling, pooled within treatment arm. Exposed commuting zones are the 18 markets reached by the 1988 campaign; comparison commuting zones hosted another major US tour without the AKAI promotion. Because records rather than commuting zones receive equal weight, markets producing more records contribute more to each series. Table C.4 reports the corresponding values.

Figure D.2 summarizes the pooled documented-sampling estimate across the principal alternative specifications reported below.

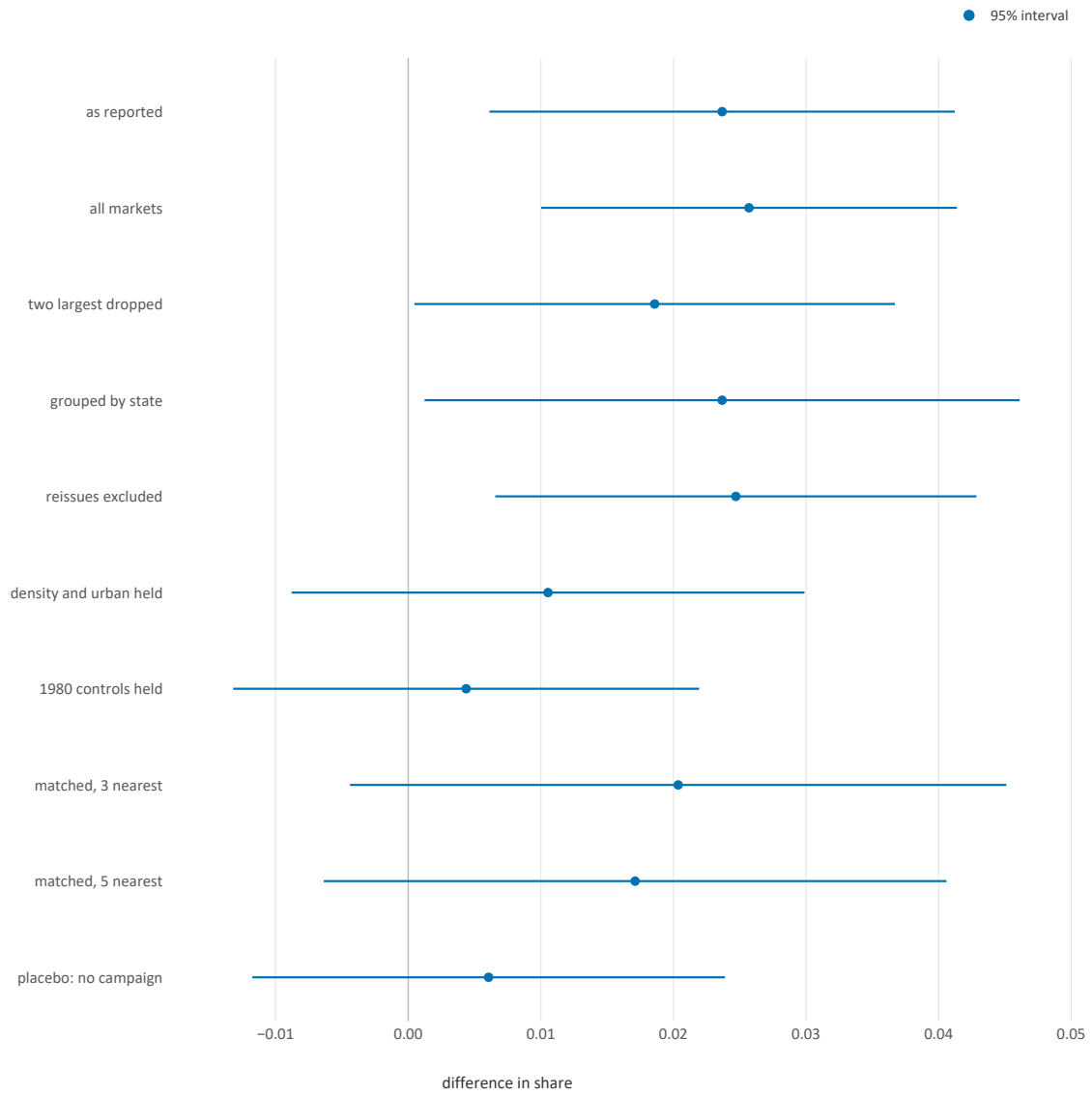
D.2 Participation

Section 5.2 finds no detectable change in participation relative to local population. This subsection reports the specification sensitivity behind that conclusion, separates the numerator from the population denominator, and decomposes the baseline movement according to whether contributors are associated with records documenting sampling.

Figure D.3 places the alternative specifications on a common scale. The baseline estimate is positive but imprecise, and its sign changes when predetermined commuting-zone characteristics are allowed to generate separate year-specific trajectories. The estimates therefore do not support a stable change in participation relative to local population.

Table D.3 reports the same specification ladder together with additional diagnostics. Panel B separates the numerator and denominator of the participation rate. Absolute contributor counts rise in the baseline specification, while working-age population itself

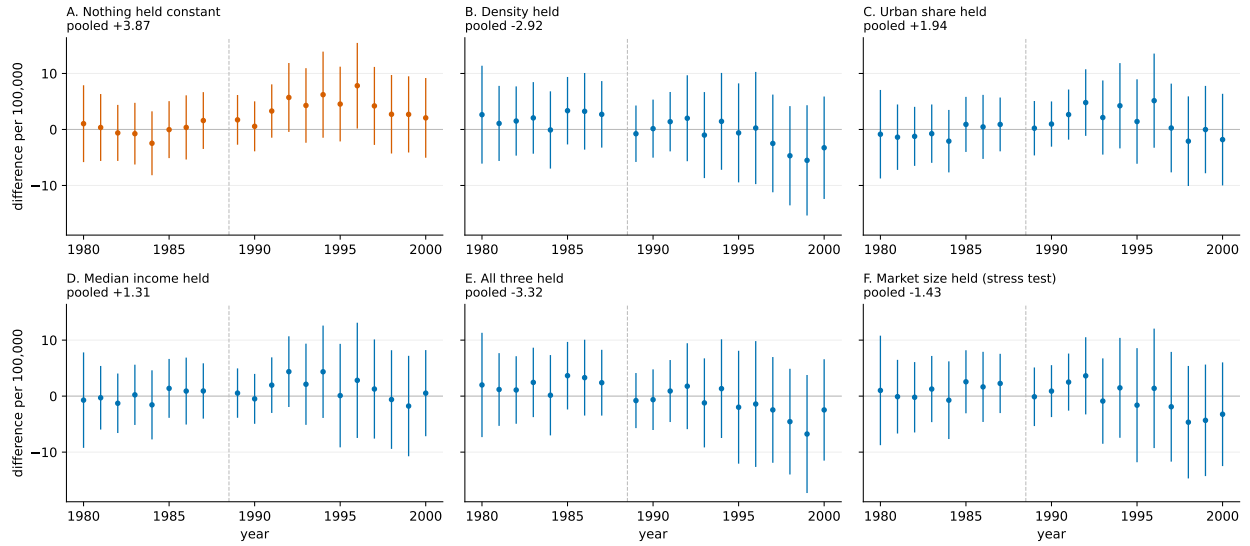
Figure D.2: Specification sensitivity of documented sampling



Note. Each row reports the pooled coefficient on $\text{Exposed} \times \text{Post}$ for the commuting-zone share of records documenting sampling under the stated specification, with its 95% CRV1 interval. Table D.2 reports the corresponding estimates.

Figure D.3: Participation under alternative specifications

Credited people per 100,000 residents aged 20-64, by year



Note. Pooled estimates for unique credited contributors per 100,000 residents aged 20–64. Predetermined controls are 1980 population density, 1980 urban share, and 1979 median income, each interacted with year indicators. OLS intervals use CR2 with coefficient-specific Satterthwaite degrees of freedom. The Poisson specifications estimate contributor counts with log working-age population as an offset and use conventional inference clustered by commuting zone because CR2 is unavailable for that estimator.

shows little differential movement. The participation result is therefore not generated by a post-campaign increase in the population denominator.

Panel C decomposes participation into contributors associated with records documenting sampling and all other contributors. Panel D reports measurement checks for the population denominator and contributor construction.

Figure D.4 shows the annual decomposition. The positive movement in the baseline aggregate is concentrated in the smaller sampling-linked component, while the much larger group of other contributors shows little corresponding movement. This decomposition is descriptive because documented sampling is itself affected by campaign exposure.

D.3 Additional evidence on documented sampling

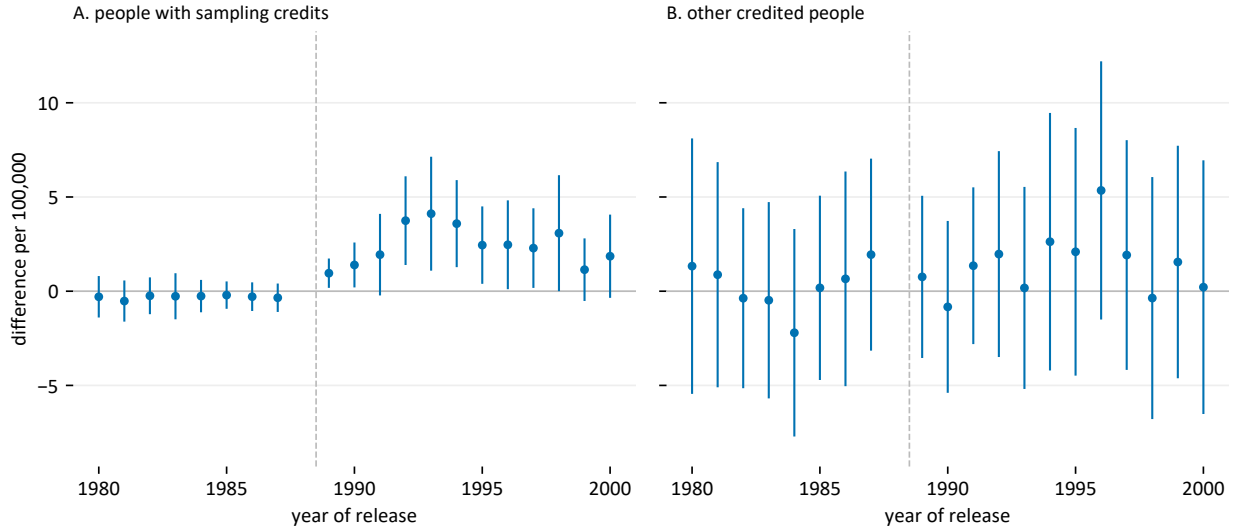
Two additional figures clarify the diffusion evidence. Figure D.5 replaces the record share with the share of active studios producing at least one record that documents sampling. Figure D.6 shows how the annual record-level path changes when predetermined market characteristics are allowed to generate separate trajectories.

Table D.3: Participation: specification sensitivity, decomposition, and diagnostics

	$\hat{\beta}$	95% interval
<i>Panel A. Credited contributors per 100k aged 20–64</i>		
No controls	+3.88	[−1.48, +9.23]
1980 density $\times$ year	−2.92	[−8.92, +3.08]
1980 urban share $\times$ year	+1.94	[−3.63, +7.51]
1979 median income $\times$ year	+1.31	[−4.99, +7.62]
All three $\times$ year	−3.32	[−9.54, +2.89]
Poisson, log-population offset, no controls	−12.9%	[−26.1%, +2.7%]
Poisson, log-population offset, all three	−9.4%	[−22.1%, +5.4%]
<i>Panel B. Numerator and denominator</i>		
Credited contributors, count, no controls	+458.5	[+68.3, +848.7]
Credited contributors, count, all three	+260.7	[−96.3, +617.6]
Log working-age population, no controls	+0.016	[−0.024, +0.057]
Log working-age population, all three	+0.001	[−0.051, +0.053]
<i>Panel C. Participation by sampling-linked status</i>		
Contributors associated with sampling	+2.72	[+0.43, +5.02]
Other credited contributors	+1.16	[−3.22, +5.54]
<i>Panel D. Measurement diagnostics</i>		
Denominator timing: annual base $\div$ 1980-frozen base, no controls		0.56
Denominator timing: annual base $\div$ 1980-frozen base, all three		0.81
Imputed denominator rows removed		14
Largest annual change from removing them		0.99
Largest annual change from the entity-cleaning rule		0.20
Correlation of the annual path before and after that rule		0.9999

Note. Commuting-zone-level regressions following Equation (3) on 124 commuting zones observed from 1980 through 2000, of which 18 are exposed. Panel A’s first five rows and Panels B and C use ordinary least squares with CR2 confidence intervals and coefficient-specific Satterthwaite degrees of freedom. Panel A’s final two rows use Poisson specifications with log working-age population as an offset and conventional inference clustered by commuting zone. Panel C partitions the baseline participation measure into contributors associated with records documenting sampling and all remaining contributors; the two components sum to total participation in each commuting-zone-year.

Figure D.4: Participation by sampling-linked status



Note. Annual coefficients from regressions of unique credited contributors per 100,000 residents aged 20–64 on campaign exposure interacted with year indicators; 1988 is the omitted reference year. Contributors are classified as sampling-linked when at least one record with which they are associated in that commuting-zone-year documents sampling. All remaining contributors form the complementary group. The two components sum to aggregate participation in every commuting-zone-year. Bars are 95% confidence intervals under CR2 with coefficient-specific Satterthwaite degrees of freedom. The decomposition is descriptive because documented sampling is itself an outcome of campaign exposure.

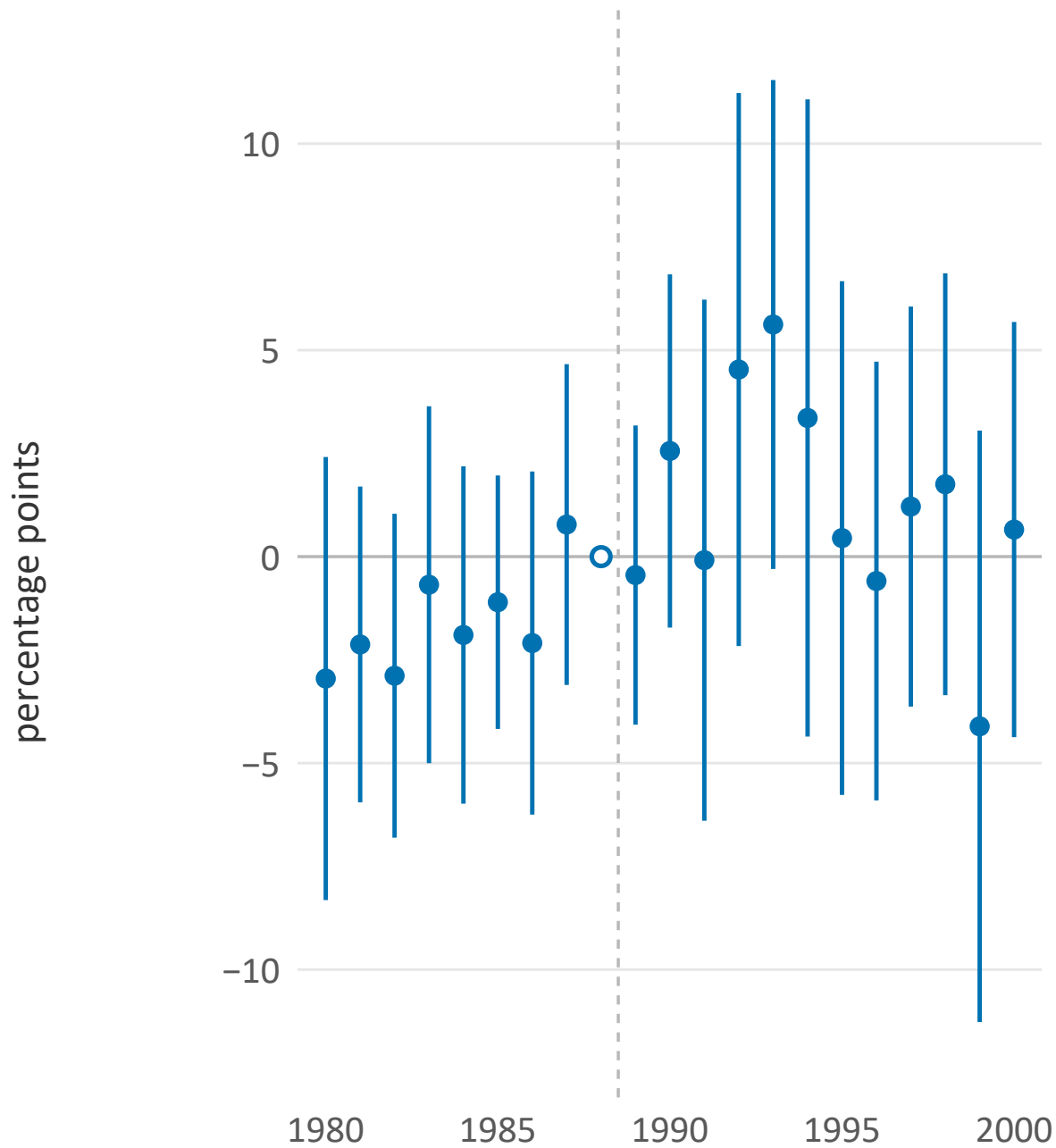
D.4 Robustness of documented sampling

This subsection reports the main sensitivity exercises for the commuting-zone documented-sampling result. Figure D.2 and Table D.2 summarize the pooled estimates, while the figures below report the corresponding annual paths where informative.

Alternative samples, weighting, and inference. Expanding the comparison pool to all continental commuting zones produces an estimate of 2.6 percentage points, with an interval from 1.0 to 4.1. Dropping the two largest commuting zones gives 1.9 points (0.05 to 3.7). Clustering by state rather than commuting zone leaves the coefficient at 2.4 points (0.12 to 4.6). Excluding reissues produces 2.5 points (0.7 to 4.3). Requiring at least five or ten records before a commuting-zone-year enters the sample, or weighting commuting-zone-years by the number of records, yields estimates between 2.4 and 3.0 percentage points.

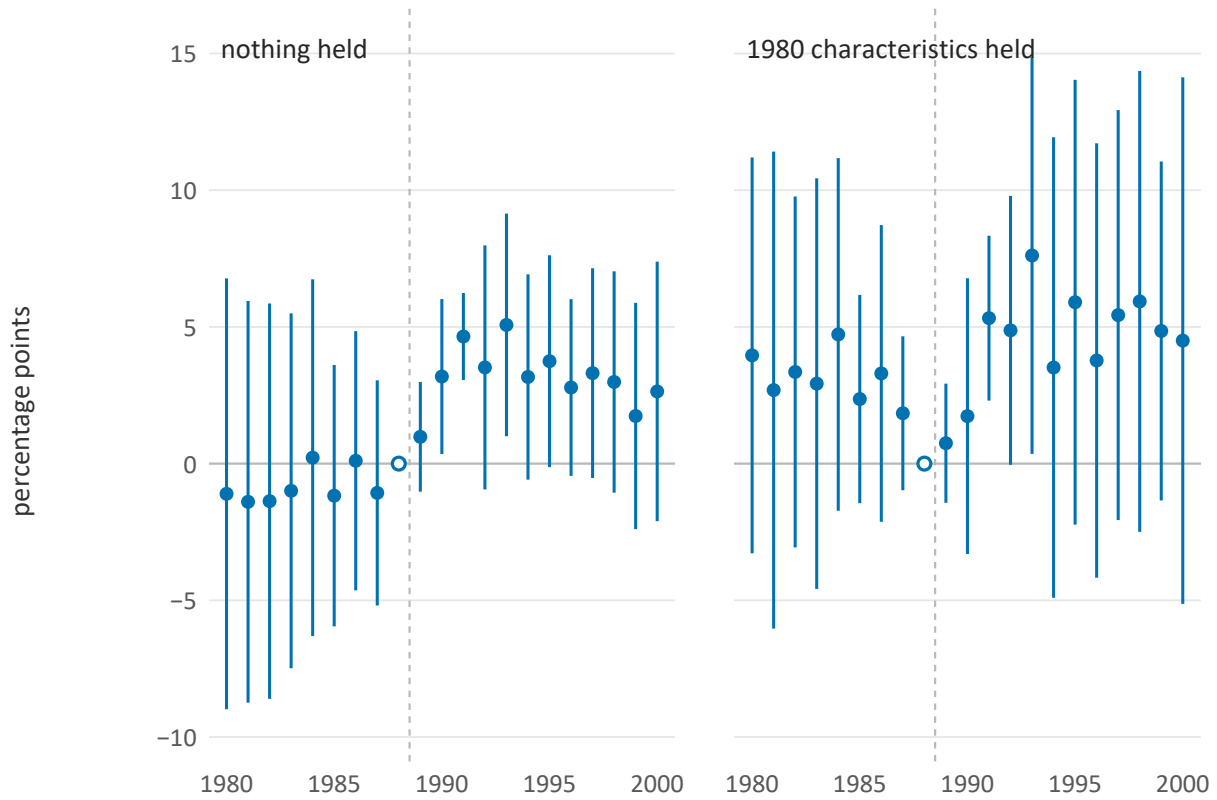
Leave-one-out. Dropping each exposed commuting zone in turn produces estimates between 1.9 and 2.7 percentage points. The result is therefore not driven by a single exposed market.

Figure D.5: Share of active studios associated with documented sampling



Note. Annual coefficients on campaign exposure for the share of active studios producing at least one record documenting sampling; 1988 is the omitted reference year. The specification includes commuting-zone and census-division-by-year fixed effects.

Figure D.6: Record-level documented sampling with predetermined characteristics



Note. Record-level event-study estimates for documented sampling without and with 1980 population density, 1980 urban share, and 1979 median income interacted with year indicators. The attenuation in the pooled coefficient reflects primarily a change in the estimated pre-campaign differential rather than disappearance of the positive post-campaign path. Panels share a common vertical scale.

Permutation inference. Randomly reassigning exposure across 9,999 draws yields $p = 0.011$. I also restrict reassignment to locations more similar to the realized itinerary. Permuting exposure within census divisions gives $p = 0.020$, while permuting within census division crossed with terciles of pre-1988 record production gives $p = 0.015$. These exercises compare the observed estimate with alternative geographic assignments and do not assume that the realized tour itinerary was randomly generated.

Placebo tours. Assigning pseudo-exposure using comparable major tours that were not accompanied by the AKAI campaign produces a pooled estimate of 0.6 percentage points, with an interval from -1.2 to 2.4 . Including both campaign exposure and the number of comparison tours in the same regression leaves the campaign coefficient at approximately 2.0 percentage points, while the comparison-tour measure shows little corresponding association.

Negative-control outcomes. I apply the same design to local employment, working-age population, and personal income. Log employment changes by -1.8 percent (-6.2 to 2.6), and working-age population shows little differential movement. Personal income has a positive pooled coefficient, but its annual path already differs throughout the pre-campaign period rather than breaking around 1988.¹⁶ These outcomes provide no evidence of a discrete broader local-economic change coinciding with campaign exposure.

Cancelled and considered tour destinations. Appendix B.4 distinguishes two documented cancelled destinations, Cincinnati and Tacoma, from Birmingham and Chapel Hill, which appear only as less well-documented possible destinations. None hosted a realized *Bad* Tour stop and none is classified as exposed.

Assigning pseudo-exposure to all four commuting zones produces a documented-sampling estimate of -0.3 percentage points (-2.5 to 1.8). Restricting the exercise to Cincinnati and Tacoma leaves only two pseudo-exposed clusters and produces a very imprecise estimate of $+1.0$ percentage points (-14.4 to 16.3). The exercise therefore provides no precise commuting-zone evidence from the documented cancellations alone.

The record-level comparison contains more observations but remains descriptive and imprecise. The documented cancellations do not reproduce the positive creative-authorship-and-production movement found in exposed markets. Given the small number and uneven size of these markets, I use this exercise as a falsification check rather than as a separate source of identification.

¹⁶Employment is the County Business Patterns series imputed by Eckert et al. (2021); income is BEA personal income; and population is the SEER working-age count, each aggregated to commuting zones.

Predetermined market characteristics. Allowing 1980 population density, 1980 urban share, and 1979 median income to generate separate year-specific trajectories reduces the pooled documented-sampling estimate from 2.37 to 0.49 percentage points (-1.29 to 2.26). Using density and urban share without income gives 1.08 points (-0.87 to 3.03).

Figure D.6 shows that this attenuation is associated primarily with a change in the estimated pre-campaign differential. The positive post-campaign path remains visible, but the adjusted specification attributes more of the exposed-comparison difference to pre-existing market characteristics. This sensitivity is why Section 5.1 uses the diffusion result primarily to establish that campaign exposure shifted documented sampling rather than placing particular weight on a single pooled magnitude.

Genre concentration. The diffusion response is concentrated in genres where sampling was already more prevalent. Dropping Hip Hop reduces the estimate to roughly $+0.7$ percentage points on a baseline of 1.2 percent. Dropping Funk/Soul and Electronic as well reduces it further to approximately zero on a baseline of 0.6 percent. This contrasts with the organizational response, which Appendix F.2 shows is not concentrated in the same genres.

D.5 Annual robustness figures

Figure D.7: Wider and narrower commuting-zone samples

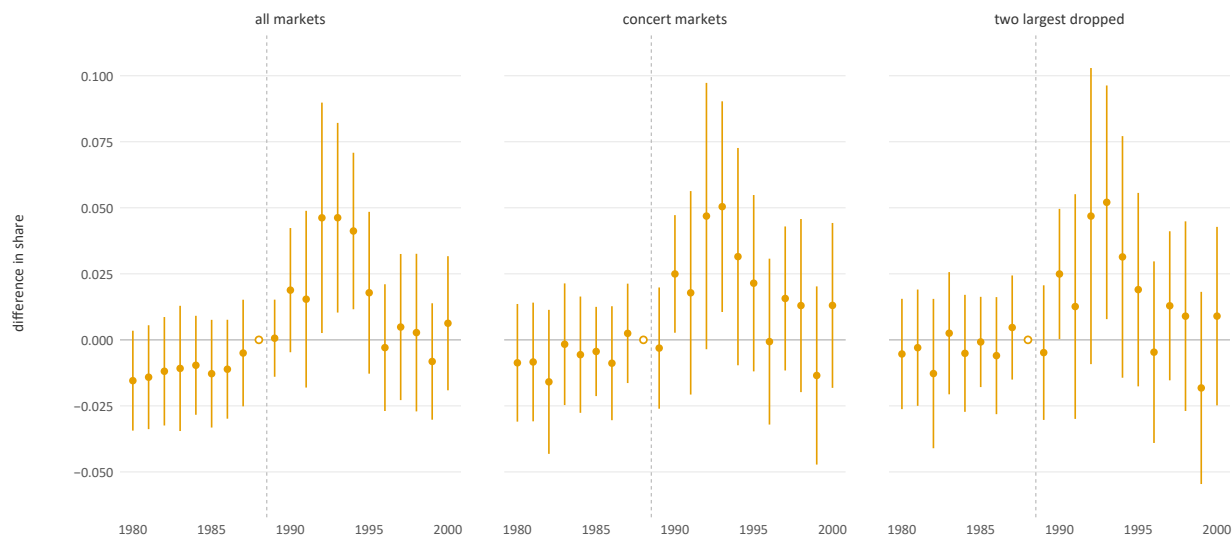
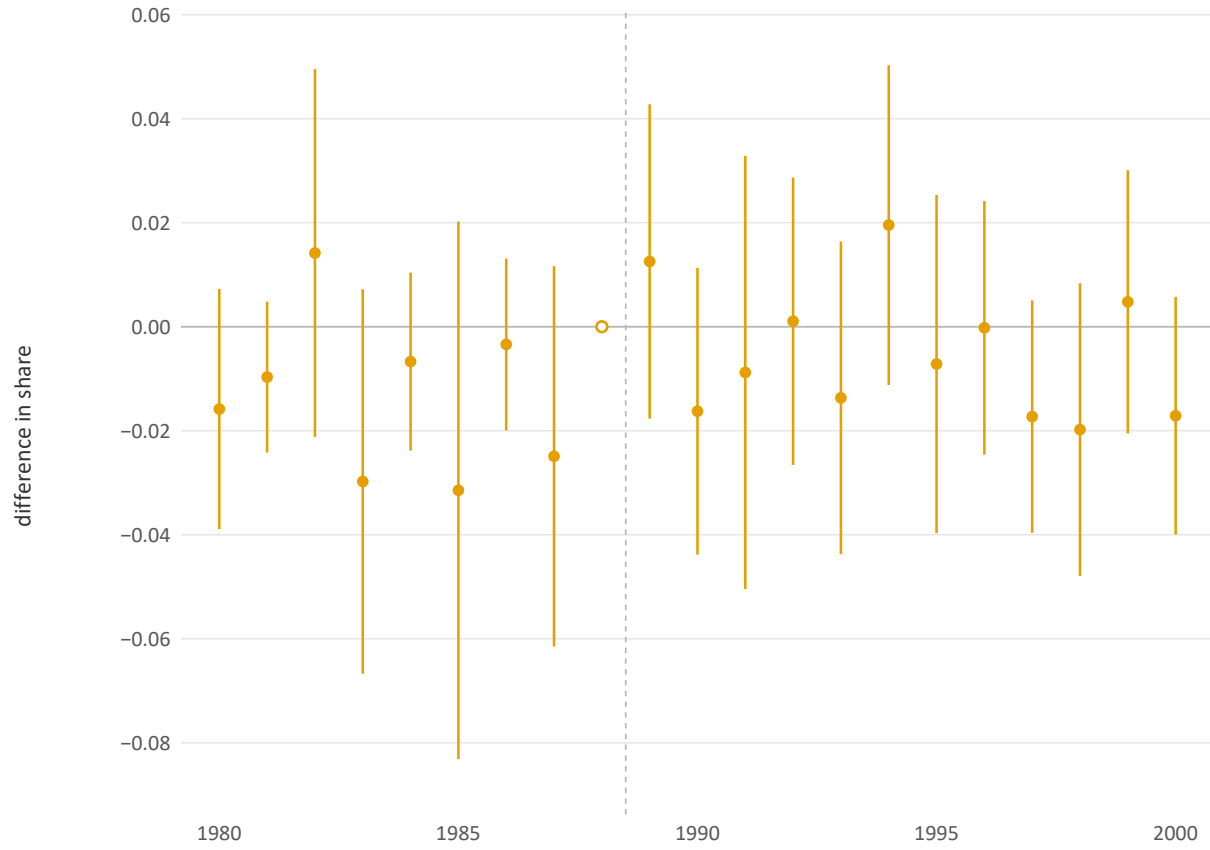


Figure D.8: Placebo tours



Note. Annual coefficients from the placebo comparison assigning exposure to commuting zones visited by other major 1988 tours, with no AKAI campaign in either group. The specification includes commuting-zone and year fixed effects; 1988 is the omitted reference year. Bars are conventional CRV1 95% confidence intervals clustered by commuting zone. The pooled placebo estimate is +0.76 percentage points $[-1.19, +2.70]$.

Figure D.9: Predetermined market characteristics

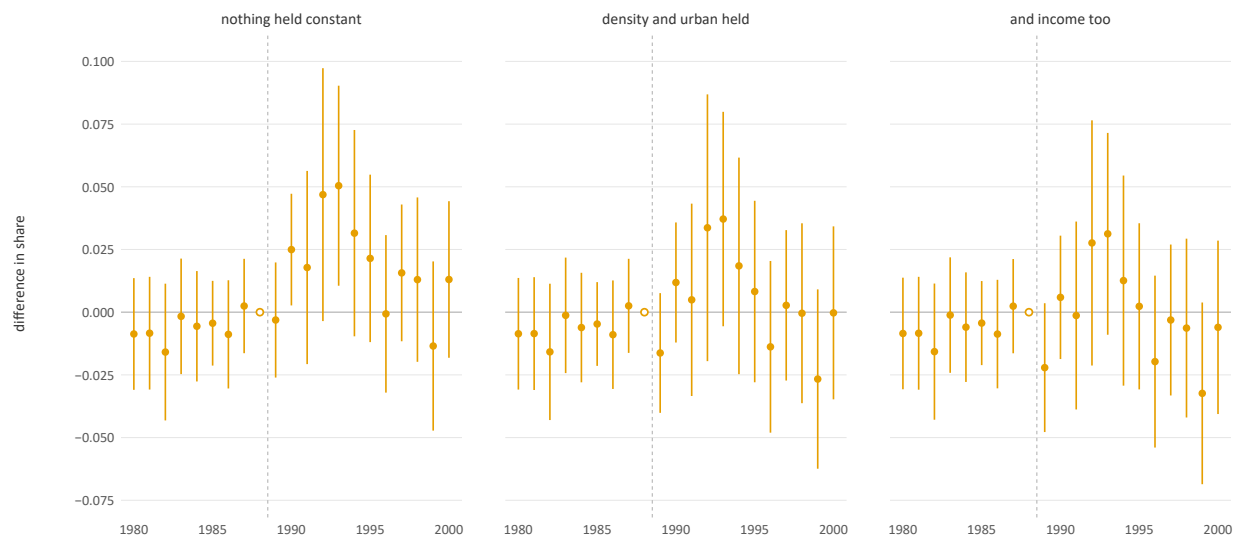


Figure D.10: Excluding reissues and re-pressed sessions

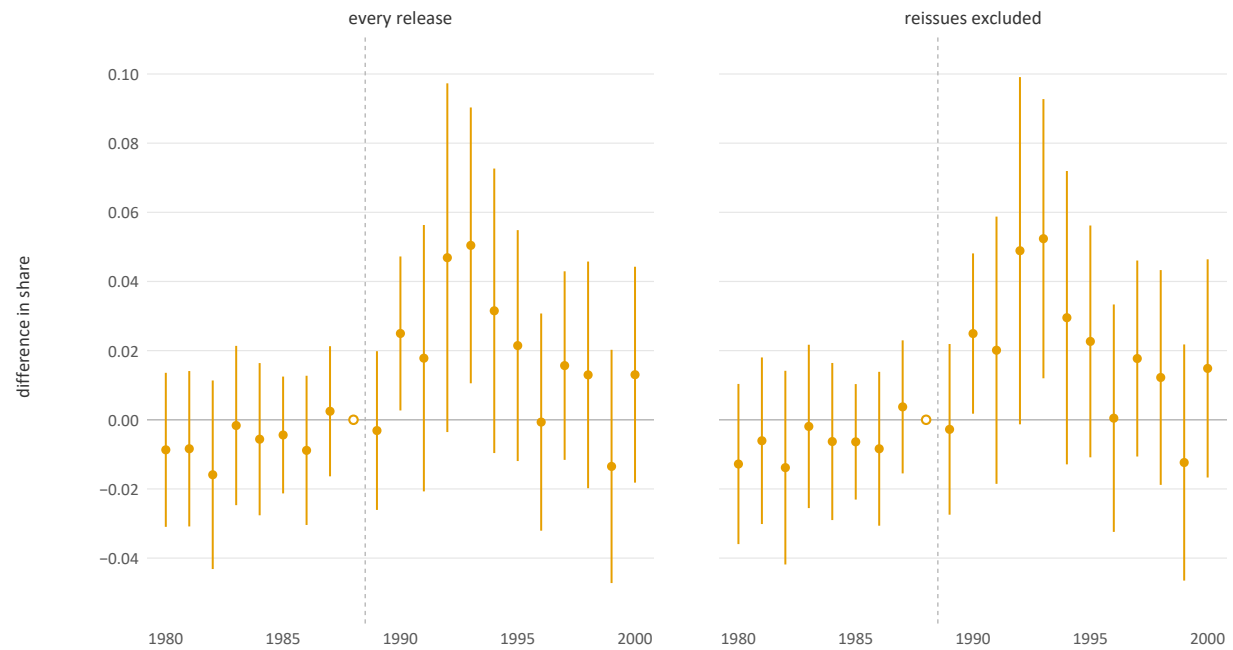
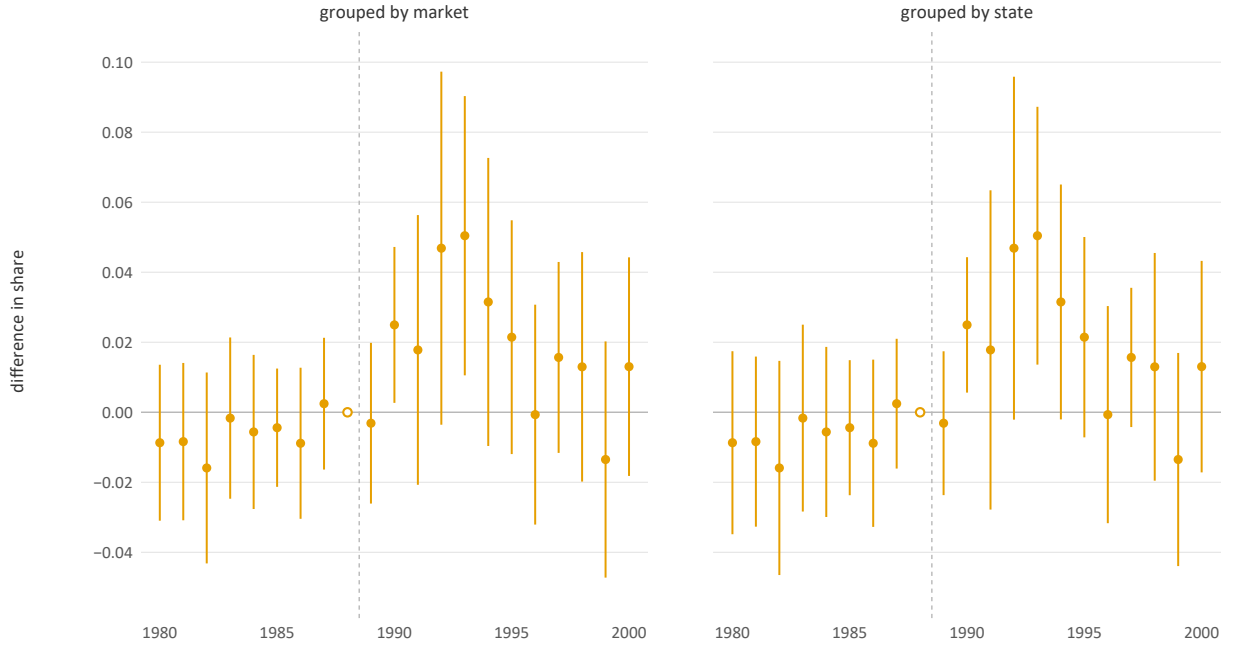


Figure D.11: Alternative clustering level



D.6 Equipment in the *Mix* directories

The preceding diffusion measures infer production practices from the records produced by studios. The *Mix* directories provide a complementary source that directly lists studio equipment. Appendix A.5 describes the construction and coverage of these data.

Figure D.15 reports the share of full-format studio listings mentioning Akai and E-mu equipment. Akai equipment appears at similar rates across treatment arms through the mid-1980s and becomes more prevalent in exposed-market listings beginning around 1988. E-mu equipment moves differently. Total sampler ownership, however, shows little corresponding differential change. The MPC60 itself is absent from listings before 1988, the year in which it was introduced.

These patterns are descriptive rather than causal estimates. Equipment information appears only in the more detailed *Mix* listing format, and the probability that a listing uses that format changes differentially after campaign exposure. The set of studios for which equipment is observable therefore changes across treatment arms and over time.

Restricting the data to the 358 studios that provide full-format listings both before and after 1988 produces an Akai difference of +0.095, with a 95% interval from -0.069 to $+0.259$, across 16 exposed commuting zones. The directories therefore provide descriptive evidence consistent with greater Akai presence in exposed markets, but they do not support

Figure D.12: Leaving one exposed commuting zone out

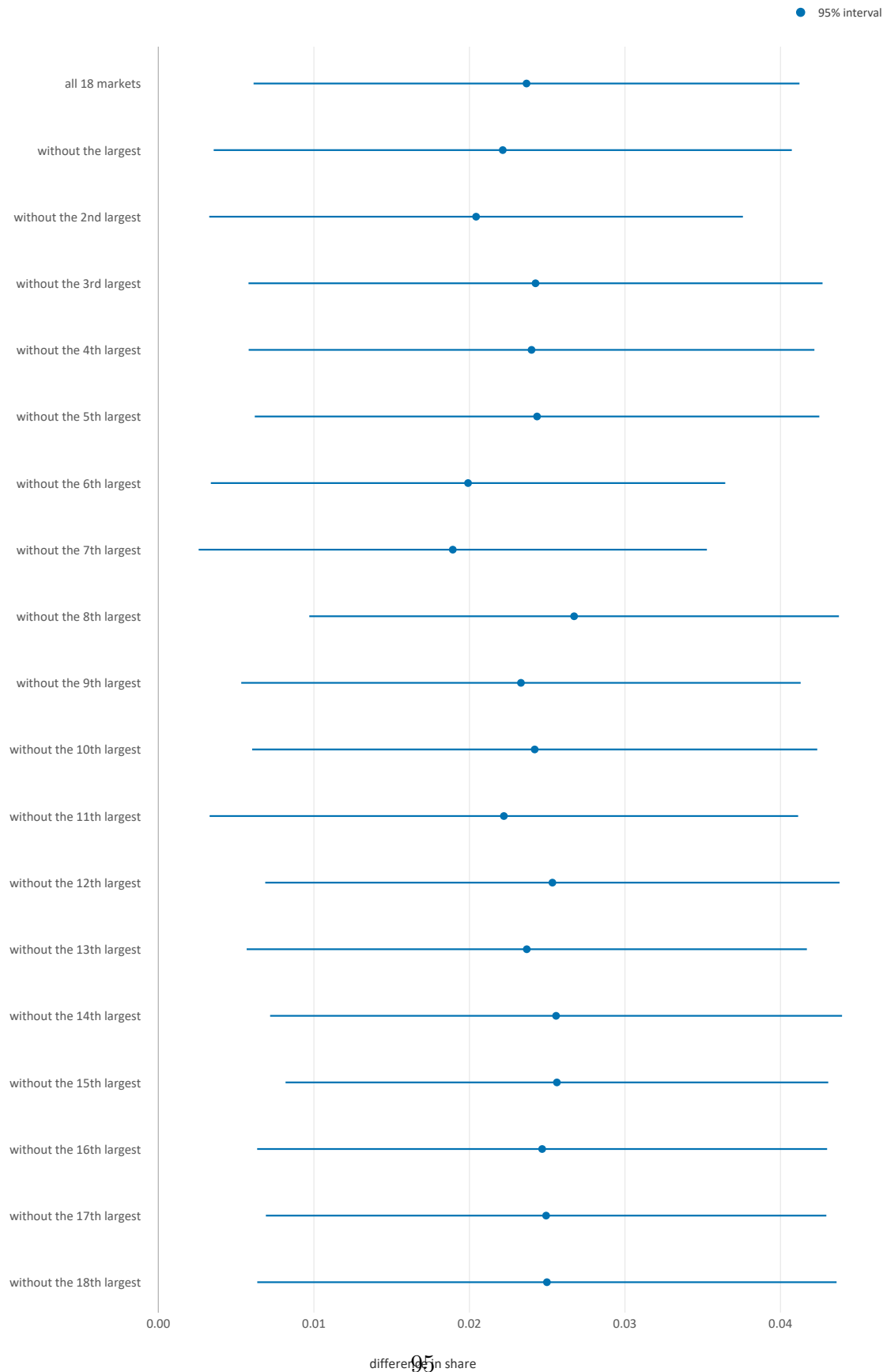


Figure D.13: Campaign exposure and comparison-tour activity

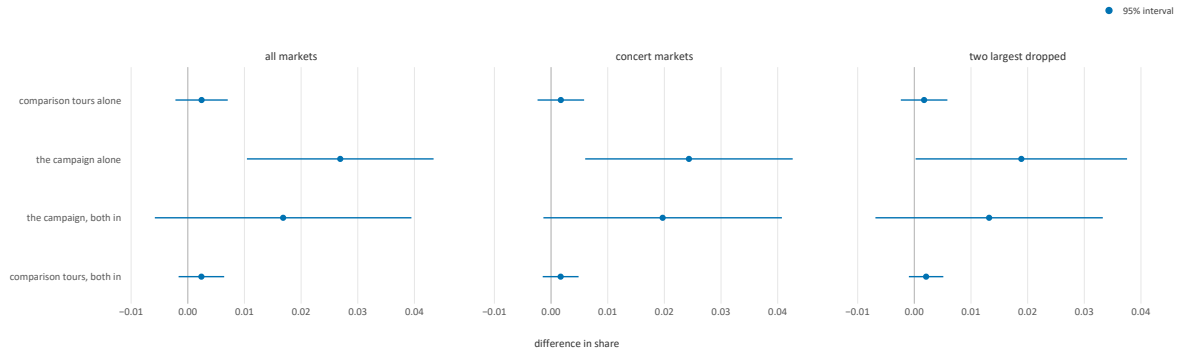
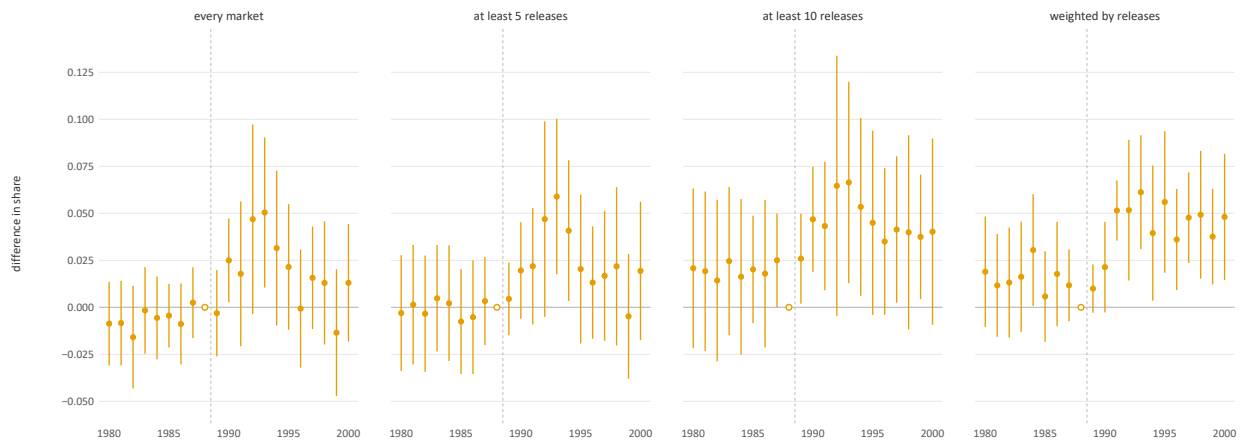
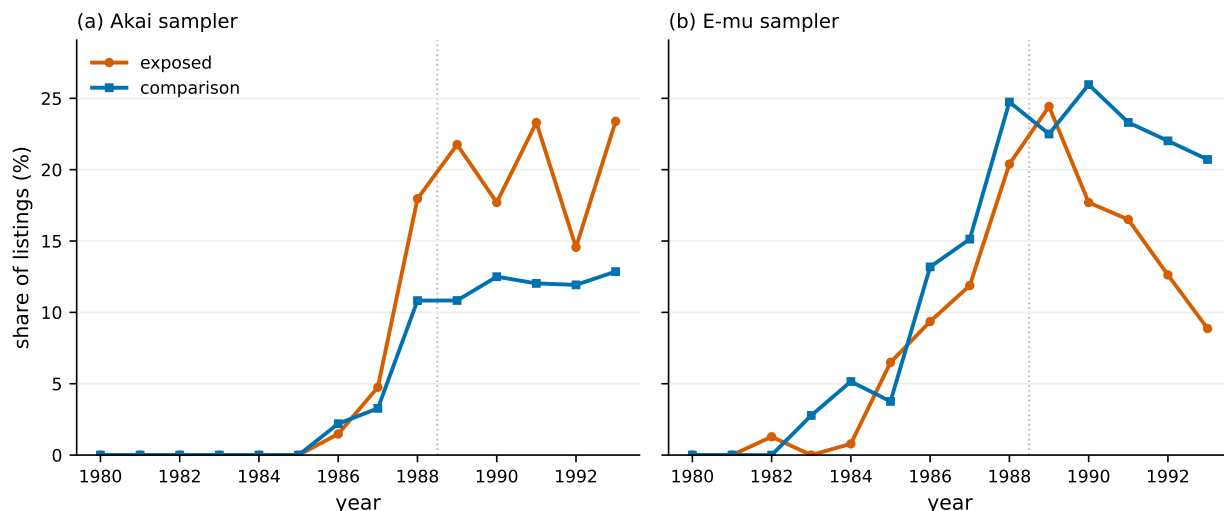


Figure D.14: Minimum record counts and record weighting



an identified estimate of MPC60 adoption.

Figure D.15: Sampler brands in studio listings, by treatment arm



Note. Share of full-format *Mix* directory listings mentioning each equipment manufacturer, by year and treatment arm. The sample contains 4,265 full-format listings out of 18,869 total listings. Equipment is unobserved, rather than absent, in short-format listings. The vertical rule marks the 1988 campaign. Because the probability of receiving a full-format listing changes differentially across treatment arms, these series are descriptive and should not be interpreted as treatment effects. Restricting to studios with full-format listings on both sides of 1988 yields an Akai difference of +0.095 [−0.069, +0.259].

E The organization of production

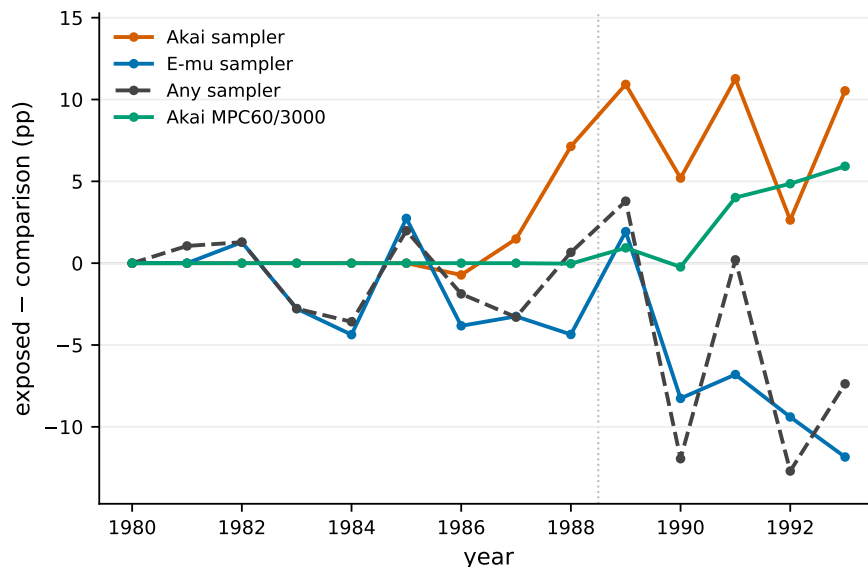
This appendix provides the additional estimates underlying Section 6. I first report the main organizational results under commuting-zone rather than studio fixed effects. I then examine the composition of creative authorship and production, alternative measures of individual breadth, alternative credit taxonomies, robustness to specification choices, and several measurement and identification checks.

E.1 The main results under commuting-zone fixed effects

The preferred specification in Section 6 holds the recording studio fixed. Equation (5) instead replaces studio fixed effects with commuting-zone fixed effects, allowing changes in the composition of active studios to contribute to the response.

Figures E.1–E.3 reproduce the main organizational figures under this alternative specification. The direction and timing of the results are similar. The shift toward creative authorship and production is somewhat larger, and the corresponding count of contributors whose functions are confined to that family rises by about 0.75 per record over 1993–2000,

Figure D.16: Equipment differences between exposed and comparison markets



Note. Difference between exposed- and comparison-market shares for the equipment measures reported in Figure D.15. Full-format *Mix* listings, 1980–1993. Because full-format coverage changes differentially across treatment arms, the series are descriptive rather than causal estimates.

compared with 0.52 in the preferred studio specification. Individual function bundles again show no detectable broadening.

The studio specification remains preferred because it compares records produced by the same recording locations before and after the campaign. The commuting-zone specification captures the broader response of local production, including changes in which studios are active.

E.2 Inside creative authorship and production

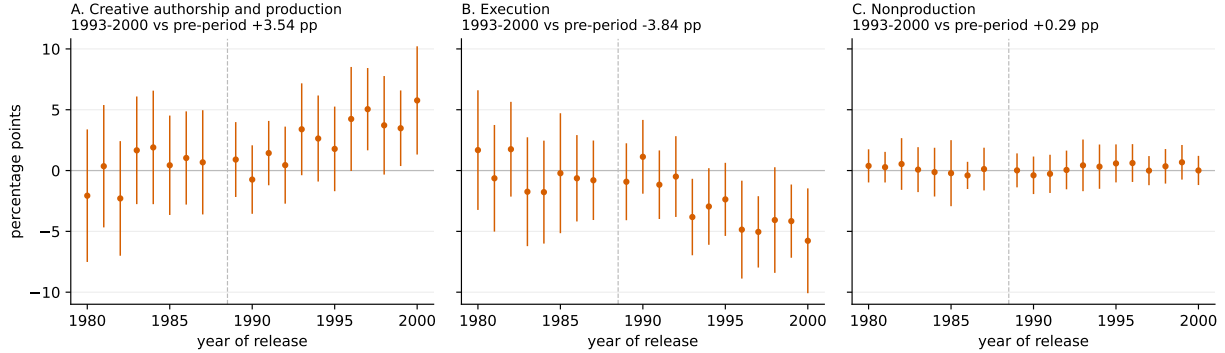
Section 6 separates creative authorship and production into authorship, consisting of writing and composition, and production and arrangement. This subsection reports a finer decomposition of the same family.

The underlying five components partition creative authorship and production exactly. Their pooled coefficients therefore sum to the coefficient for the broader family, and the same identity holds year by year in the event study.

The production component on its own is less informative because its pre-campaign path is less stable than that of the broader production-and-arrangement block. I therefore do not place weight on production-specific estimates. The block decomposition in the main text is the preferred level of aggregation.

Figure E.1: The three function families, commuting-zone specification

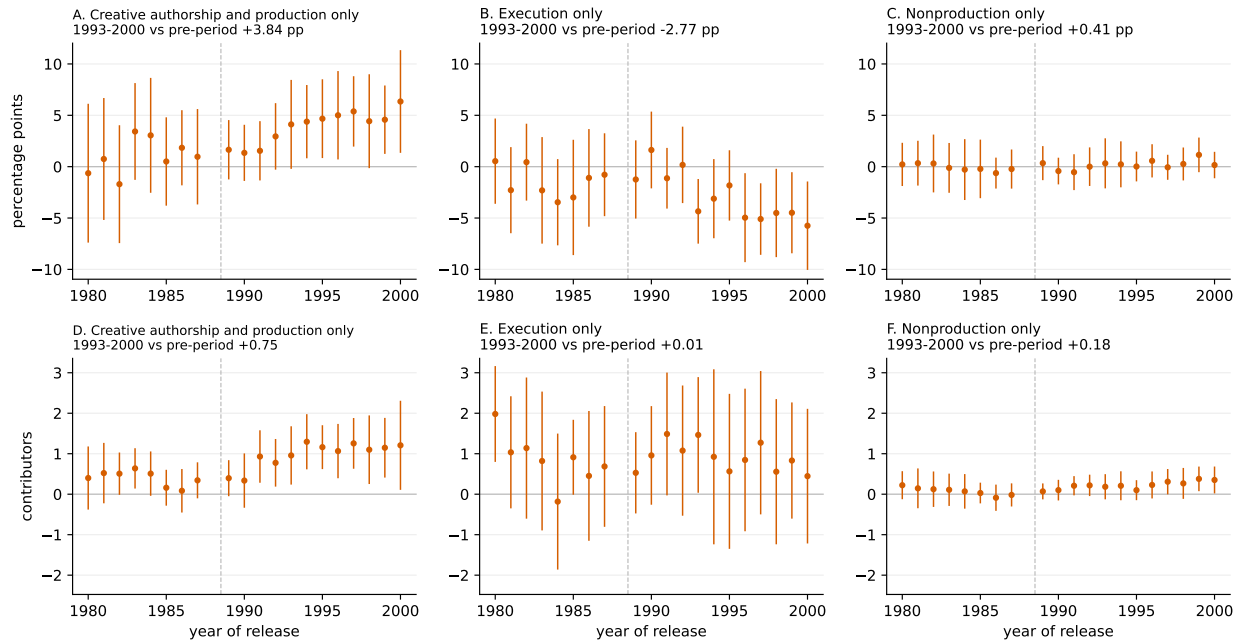
The allocation of credited work across the three families



Note. Counterpart to Figure 6 using Equation (5), with commuting-zone and census-division-by-year fixed effects. The three function families partition classified credits, so their annual coefficients sum to zero. Intervals use CR2 with coefficient-specific Satterthwaite degrees of freedom and are clustered by commuting zone. 70,707 records in 116 commuting zones, of which 18 are exposed.

Figure E.2: Team composition in shares and counts, commuting-zone specification

The same partition of the credited team, as a share of contributors and as a count



Note. Counterpart to Figure 7 using Equation (5). The count of contributors whose functions are confined to creative authorship and production rises by about 0.75 per record over 1993–2000, compared with 0.52 in the preferred studio specification. Intervals use CR2 with coefficient-specific Satterthwaite degrees of freedom and are clustered by commuting zone. 70,707 records in 116 commuting zones, of which 18 are exposed.

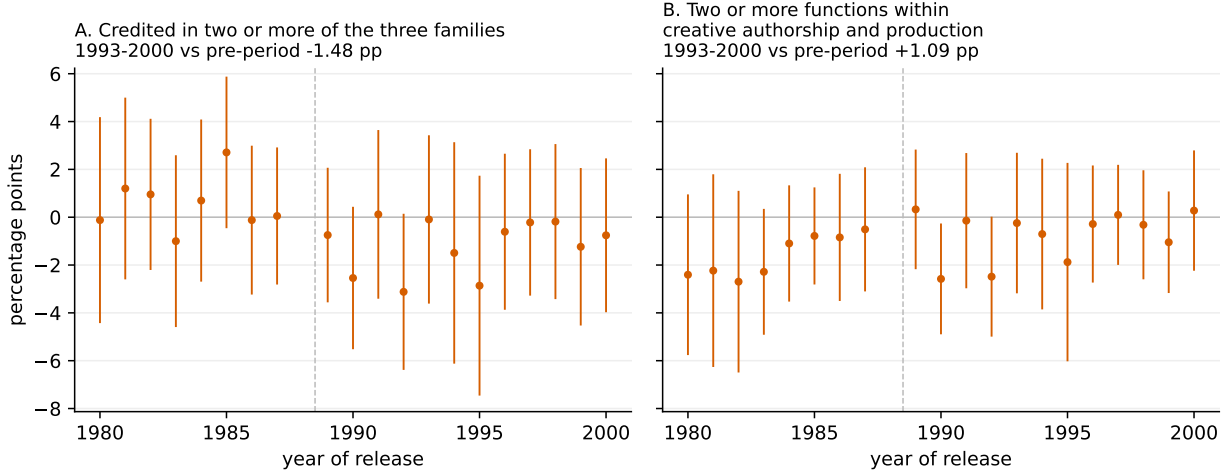
Table E.1: Creative authorship and production, decomposed

Outcome	Coef.	Baseline	1989–92	1993–2000
<i>Panel A. The two blocks</i>				
Authorship	+1.14* (0.60)	16.8	+0.78	+2.30
Production and arrangement	+0.43 (0.62)	17.5	−0.18	+1.27
<i>Panel B. The five components</i>				
Writing	+0.84 (0.64)	14.4	+0.54	+1.96
Composition	+0.30 (0.17)	2.4	+0.25	+0.34
Arrangement	+0.30 (0.23)	2.9	+0.23	+0.14
Production	+0.05 (0.60)	14.1	−0.54	+0.94
Conducting (thin)	+0.07 (0.12)	0.6	+0.13	+0.19
<i>Sum of the five</i>	+1.57	<i>equals the family coefficient exactly</i>		

Notes. All quantities are in percentage points of a record’s classified credits. Each row is the coefficient on $\text{Exposed}_{m(a)} \times \text{Post}_{t(a)}$ from a separate album-level regression following Equation 4 in pooled form; the last two columns are unweighted means of the annual coefficients over the stated years, not refits, because a refitted window can resolve where the unrestricted annual path does not. The five components of Panel B are Discogs categories that partition the family exactly, so their coefficients sum to the family’s own to machine precision, pooled and at every one of the twenty annual years; Panel A groups them into writing-and-composition and production-and-arrangement. **Inference is the bias-reduced estimator with a per-coefficient Satterthwaite correction**, so these intervals are wider than those in Table E.3, which is CRV1; the two are not comparable in width. **Conducting is 0.6 per cent of classified credits, below the thinness floor used elsewhere in this paper, and carries no claim**; it is reported because dropping it would break the partition. **Production carries adverse pre-period evidence** from the earlier proximity design, where its pre-period was rejected in two parameterisations, so no production-specific claim is made from it. $n = 70,707$ records, 116 markets, 18 exposed. 1988 is the omitted reference year.

Figure E.3: Individual breadth, commuting-zone specification

Contributors do not measurably broaden their credited functions



Note. Counterpart to Figure 8 using Equation (5). The pooled estimate for breadth across the three function families is -1.60 percentage points ($-4.27, +1.06$), while breadth within creative authorship and production is $+0.65$ points ($-0.70, +1.99$). No post-1993 annual coefficient separates from zero on either measure.

Breadth within creative authorship and production. A contributor can remain entirely within creative authorship and production while holding several distinct functions inside that family. I therefore separately measure whether contributors accumulate multiple authorship-and-production functions.

Table E.2 reports this measure at two levels of aggregation. Both estimates are small and statistically indistinguishable from zero. At the finer function level, the interval runs from approximately -1.2 to $+1.6$ percentage points around a pre-period mean of about 10 percent. The data therefore do not detect the broadening of individual function bundles that would accompany a strong generalist response.

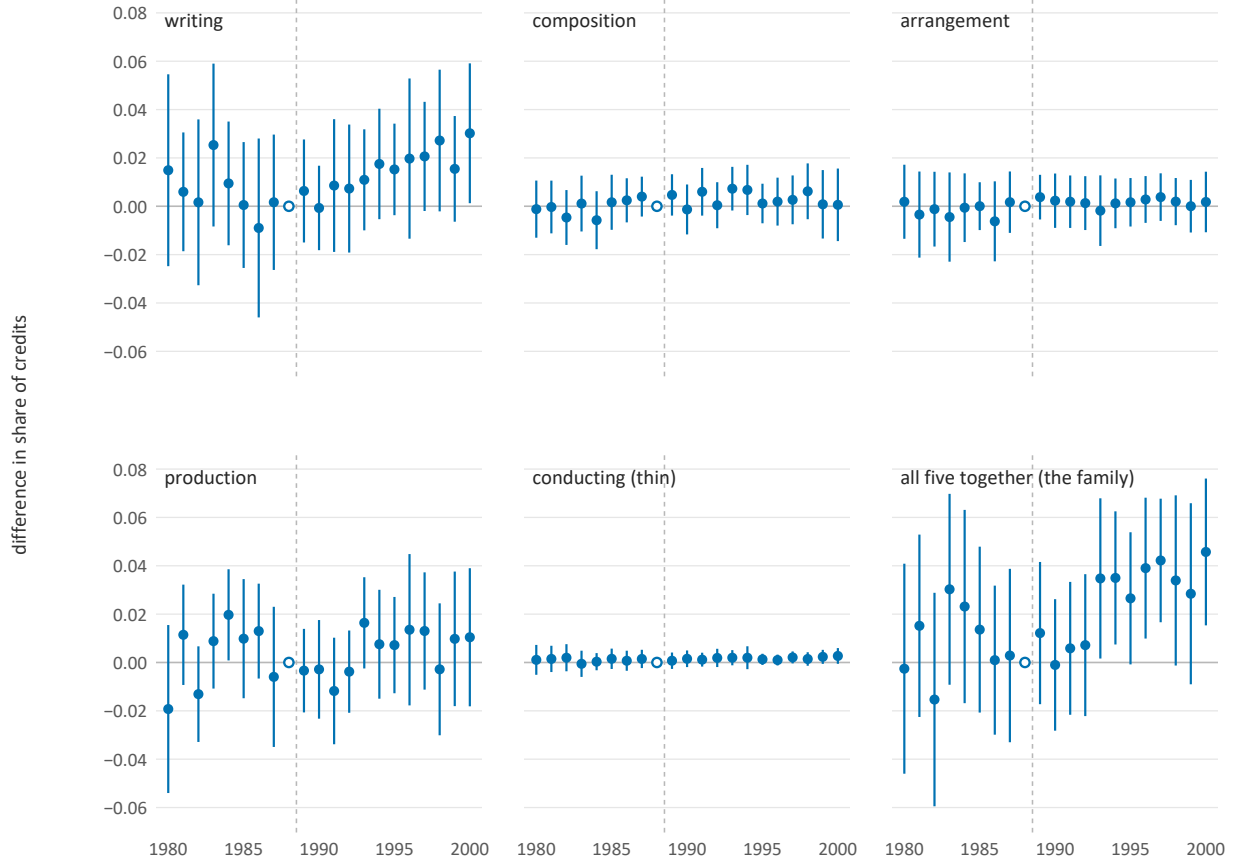
E.3 Contributor exclusivity and individual breadth

The credit-share outcomes describe how credited functions are allocated. Contributor-based measures ask a different question: whether those functions are carried by people concentrated in particular parts of production.

The share of contributors whose functions are confined to creative authorship and production rises by about 2.7 percentage points over 1993–2000. The corresponding count, reported in Section 6, rises more clearly and forms the main organizational result.

The breadth results point in the opposite direction from a generalist interpretation. The share of contributors holding multiple functions within creative authorship and production

Figure E.4: Components of creative authorship and production



Note. Annual event-study estimates for the five components of creative authorship and production, together with the broader family they partition. Conducting represents about 0.6 percent of classified credits. Specification, sample, and inference follow Figure E.6.

Table E.2: Breadth within creative authorship and production

Outcome	Coef.	Baseline	1993–2000
Holds two or more creative roles (role grain)	−0.04 (0.56)	10.3	−0.47
Holds two or more creative functions (category grain)	−0.29 (0.60)	9.4	−1.02

Notes. In percentage points. The outcome is the share of a record’s credited contributors who hold more than one distinct authorship-or-production credit on that record, at two nested grains: role counts credit strings, so “written by” and “lyrics by” are two, while category counts functions, where they are one. The denominator is every credited contributor, so a record crediting no authorship or production work contributes a genuine zero rather than leaving the sample. **This is a within-function breadth measure and is a different quantity from the specialist share of Table E.3**, which asks whether a contributor works outside the family at all: a person can work only in authorship and production while holding several functions inside it. Specification, sample and inference as Table E.1. $n = 70,707$ records, 116 markets, 18 exposed.

Table E.3: Record-level estimates under both fixed-effect specifications

		Baseline share	Pooled coefficient	1993–2000 versus pre
<i>Panel A. Studio and division-year effects</i> (pooled $\nu = 7.92$; 1993–2000 contrast $\nu = 7.67$)				
Credit lines	Creative authorship and production	34.31	+1.57 (1.06)	+2.72 (1.47)
	Execution	55.92	−1.34 (0.92)	−2.52 (1.38)
	Nonproduction	9.77	−0.23 (0.48)	−0.20 (0.67)
Exclusive contributors	Creative authorship and production	23.47	+2.11* (1.06)	+2.70 (1.49)
	Execution	46.56	−0.52 (0.93)	−1.70 (1.31)
	Nonproduction	11.40	−0.35 (0.56)	−0.39 (0.75)
	Two or more categories	18.57	−1.24 (0.75)	−0.61 (0.76)
<i>Panel B. Commuting-zone and division-year effects</i> (pooled $\nu = 10.08$; 1993–2000 contrast $\nu = 10.40$)				
Credit lines	Creative authorship and production	34.31	+2.65 (1.54)	+3.54* (1.80)
	Execution	55.92	−2.81* (1.27)	−3.84** (1.50)
	Nonproduction	9.77	+0.16 (0.42)	+0.29 (0.46)
Exclusive contributors	Creative authorship and production	23.47	+3.17 (1.89)	+3.84 (2.23)
	Execution	46.56	−1.82* (0.91)	−2.77** (1.17)
	Nonproduction	11.40	+0.26 (0.51)	+0.41 (0.57)
	Two or more categories	18.57	−1.60 (1.20)	−1.48 (1.48)

Note. Record-level regressions of organizational outcomes on campaign exposure interacted with the post-campaign period. Panel A follows Equation (4); Panel B follows Equation (5). “Credit lines” is the share of classified credits assigned to the corresponding function family. “Exclusive contributors” is the share of contributors whose functions are confined to that family. The 1993–2000 column reports a linear contrast from the annual event study comparing the mean coefficient over 1993–2000 with the mean pre-campaign coefficient. Standard errors use CR2 and are clustered by commuting zone. The contributor-exclusivity categories form a partition only when the multi-family category is included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

changes by only about 0.2 percentage points, with an interval that includes zero. The corresponding broader measure spanning multiple function families also shows no detectable increase.

Concentration within creative authorship and production. Measures such as a Herfindahl index across contributors holding creative-authorship-and-production credits are defined only for records containing at least one such credit. The probability that a record contains any credit in this family changes over time and differs across treatment arms, so conditioning on that subset would introduce a changing sample.

I therefore do not use conditional concentration measures as outcomes. An unconditional indicator for whether exactly one contributor holds all creative-authorship-and-production credits is defined for every record and changes by +1.4 percentage points ($-0.8, +3.7$) from a pre-period mean of 11.3 percent. This provides no detectable evidence of increased concentration within the family.

E.4 Authorship versus production and arrangement

Figure E.6 reports the annual decomposition of creative authorship and production into the two blocks used in the main text.

E.5 First appearances and repeat credits

A change in credit shares can arise either because additional people appear in a function family or because people already represented in that family receive additional credits. The credit data allow these two margins to be separated.

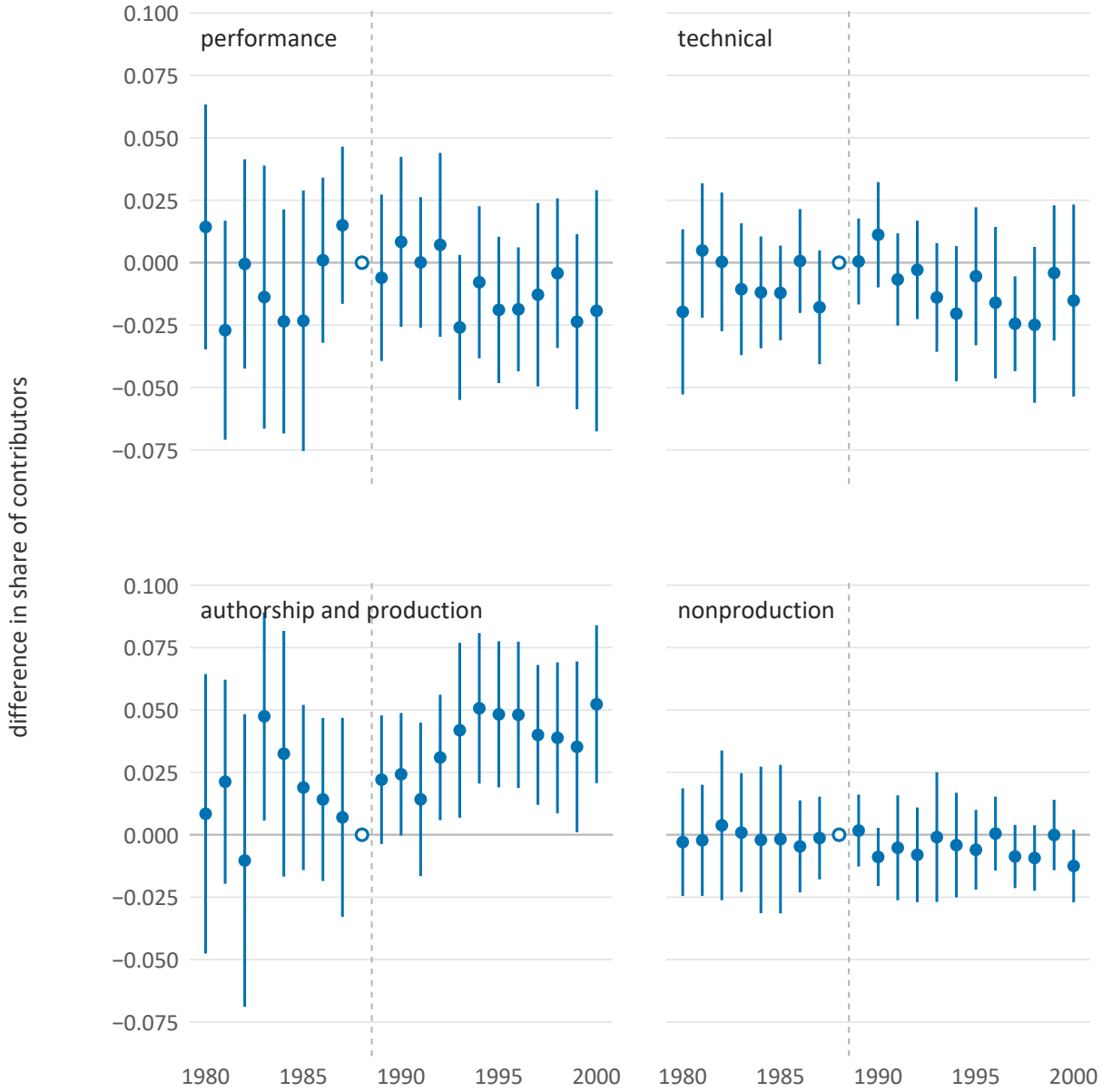
For each record and function family, every classified credit is either the contributor’s first credit in that family or an additional credit assigned to someone already represented there. Thus

$$\underbrace{s_{fa}}_{\text{family share}} = \underbrace{\phi_{fa}}_{\text{first appearances}} + \underbrace{\rho_{fa}}_{\text{repeat credits}}, \quad (\text{E.1})$$

where all three quantities are expressed as shares of record a ’s classified credits. The identity holds record by record.

Under studio fixed effects, the decline in execution is concentrated in repeat credits: repeat-credit share falls by 2.04 percentage points ($-3.51, -0.57$), while first appearances change by -0.48 points ($-2.58, +1.62$). Under commuting-zone fixed effects, both components decline, with first appearances falling by 2.66 points and repeat credits by 2.46 points.

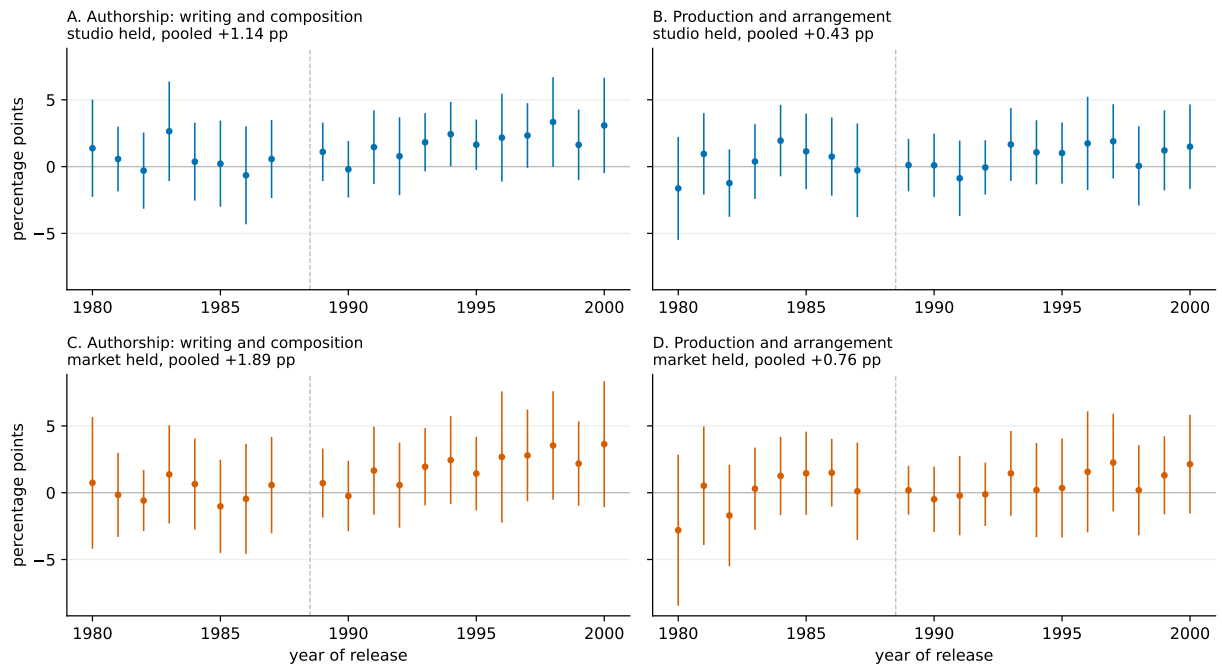
Figure E.5: Contributor exclusivity by function family



Note. Share of contributors whose credited functions on the record are confined to the indicated family. Contributors spanning more than one family are omitted from the family-specific shares and appear separately in the team-composition partition used in the main text. Specification, inference, and sample follow Figure 6.

Figure E.6: Authorship versus production and arrangement

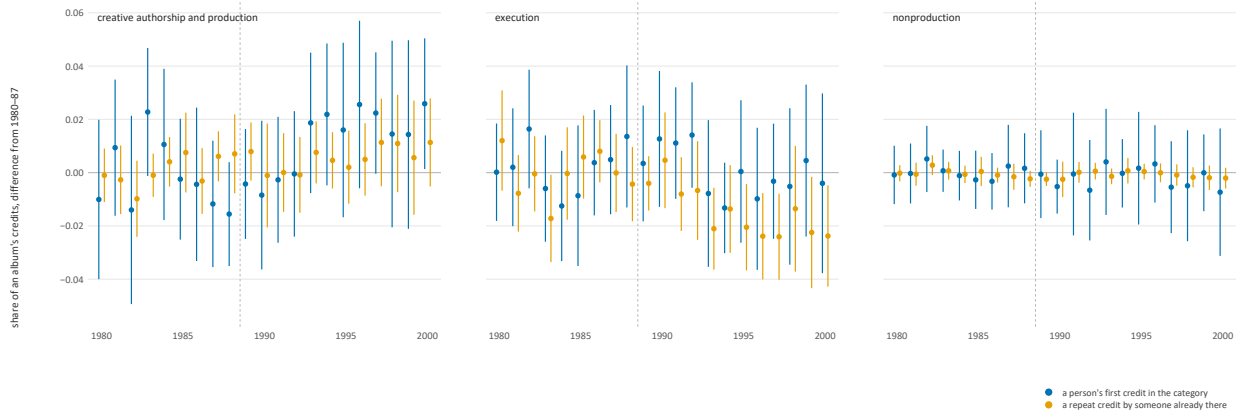
The rising category split into its two blocks, by year



Note. Creative authorship and production split into authorship and production-and-arrangement functions under both fixed-effect specifications. The two blocks partition the broader family, so their annual coefficients sum to the family's coefficient. Intervals use CR2 with coefficient-specific Satterthwaite degrees of freedom and are clustered by commuting zone.

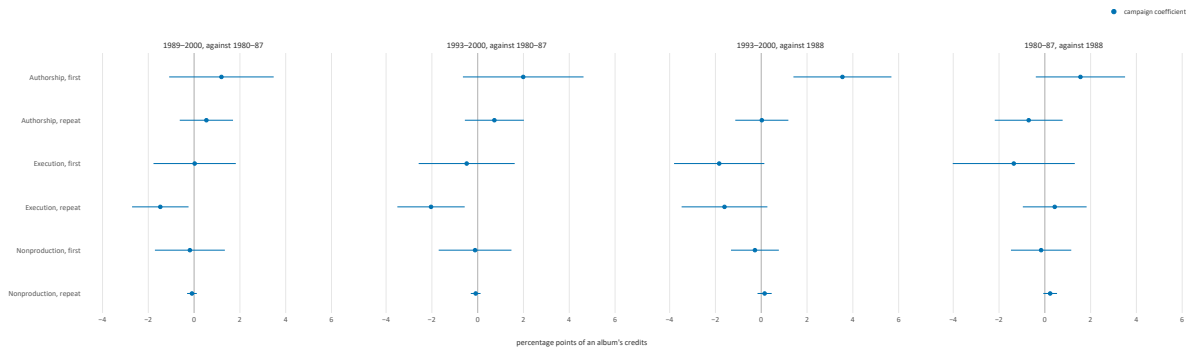
For creative authorship and production, the increase is larger in the first-appearance component than in repeat credits under both specifications, although neither component is estimated precisely enough to support a separate claim. These decompositions concern credit shares rather than distinct contributor counts.

Figure E.7: First appearances and repeat credits within each function family



Note. Each function-family share is decomposed according to Equation (E.1) into credits that are a contributor's first appearance in that family on the record and subsequent credits assigned to contributors already represented there. The two components sum exactly to the corresponding family coefficient. Both outcomes are shares of classified credits, not contributor counts. Intervals use CR2 and are clustered by commuting zone.

Figure E.8: First appearances and repeat credits under alternative comparison periods



Note. Linear contrasts from the annual estimates in Figure E.7, using either 1988 or the average pre-campaign period as the comparison. Each contrast carries its own CR2 standard error and Satterthwaite degrees of freedom.

E.6 Alternative credit taxonomies

The main analysis uses three function families: execution, creative authorship and production, and nonproduction. This subsection reports two finer partitions of the same classified credits.

The first separates execution into performance and technical functions. Both components move in the same direction, but dividing the decline across two outcomes reduces precision.

Table E.4: First appearances and repeat credits within each function family

Outcome	Baseline	Average annual difference over the window, with 95% CI			
		1989–2000 vs. pre	1993–2000 vs. pre	1993–2000 vs. 1988	1980–87 vs. 1988
<i>Panel A. A person's first credit in the category</i>					
Authorship and production	27.3	+1.19 (-1.09, +3.48)	+1.99 (-0.64, +4.62)	+3.55 (+1.41, +5.68)	+1.56 (-0.39, +3.51)
Execution	41.9	+0.02 (-1.77, +1.82)	-0.48 (-2.58, +1.62)	-1.84 (-3.81, +0.14)	-1.36 (-4.02, +1.31)
Nonproduction	8.5	-0.18 (-1.71, +1.34)	-0.11 (-1.70, +1.47)	-0.28 (-1.32, +0.77)	-0.16 (-1.48, +1.15)
<i>Panel B. A repeat credit by someone already there</i>					
Authorship and production	7.0	+0.54 (-0.63, +1.70)	+0.73 (-0.56, +2.02)	+0.02 (-1.13, +1.18)	-0.70 (-2.19, +0.78)
Execution	14.0	-1.48 (-2.71, -0.24)	-2.04 (-3.51, -0.57)	-1.60 (-3.48, +0.27)	+0.43 (-0.96, +1.83)
Nonproduction	1.2	-0.09 (-0.31, +0.12)	-0.09 (-0.30, +0.13)	+0.15 (-0.16, +0.46)	+0.23 (-0.07, +0.53)
<i>Panel C. The two together, which is the family share</i>					
Authorship and production	34.3	+1.73 (-1.25, +4.71)	+2.72 (-0.69, +6.13)	+3.57 (+1.01, +6.13)	+0.85 (-1.82, +3.53)
Execution	55.9	-1.45 (-4.14, +1.24)	-2.52 (-5.73, +0.70)	-3.44 (-5.80, -1.08)	-0.92 (-3.53, +1.69)
Nonproduction	9.8	-0.28 (-1.71, +1.15)	-0.20 (-1.76, +1.36)	-0.13 (-1.28, +1.03)	+0.07 (-1.37, +1.51)

Notes. All quantities are in percentage points of a record's classified credits. Every credit line on a record is either the first time a record holder takes in that category on that record, or a repeat by someone already credited there, so Panel A and Panel B partition the record by record and their coefficients sum to Panel C's exactly—including to the published family coefficient of Table E.1. This is reproduced to machine precision before any row here is written. Every entry is a linear contrast of the annual coefficient in Equation 4 in event-study form and carries its own bias-reduced standard error and Satterthwaite degrees of freedom; none is conditional on the window, because a refitted window can resolve where the unrestricted annual path does not. Each column is a window of years against a baseline. **The first two columns are the ones the text quotes:** both use the whole 1980–1987 pre-period as the baseline and differ only in whether the years to 1992 are included, so neither rests on 1988 being an ordinary year. The third repeats the contrast against 1988 alone and the fourth is the pre-period level, so the third and fourth columns sum to the second. That level is not conditional on 1988 a different family separates from zero, which is why both baselines are shown. Entries in bold have intervals excluding zero. **Panel A is a share of credits and not a count of people.** It rises when a larger share of a record's credits go to people appearing in that category for the first time on the record, which will happen when more distinct people work there, but it does not count them: the count version of this outcome carries how fully each record was documented, which does not follow parallel trends before 1988. No denominator here is conditional on the category, which the corresponding coverage rates would not support. All standard errors are bias-reduced and use per-coefficient Satterthwaite degrees of freedom. These outcomes were not pre-specified.

Over 1993–2000, performance changes by approximately -1.6 percentage points and technical functions by -0.9 points. Their sum reproduces the -2.5 -point decline in execution.

The second taxonomy places credits along six stages of production: writing, pre-production, session, post-production, packaging and releasing, and an unassigned residual. The largest positive movement is in writing, while the largest negative movement is in session functions. The residual category remains stable.

The alternative partitions therefore preserve the direction of the main reallocation while distributing it across more narrowly defined outcomes.

Table E.5: The same classified credits partitioned three ways

	Baseline share	Pooled coefficient	Step 1993–2000	Significant years
<i>Panel A. Three function families</i>				
Creative authorship and production	34.3	+1.57	+2.72	5 of 8
Execution	55.9	−1.34	−2.52	6 of 8
Nonproduction	9.8	−0.23	−0.20	0 of 8
<i>Panel B. Four underlying families</i>				
Creative authorship and production	34.3	+1.57	+2.72	5 of 8
Performance	38.0	−1.07	−1.57	0 of 8
Technical	18.0	−0.27	−0.94	1 of 8
Nonproduction	9.8	−0.23	−0.20	0 of 8
<i>Panel C. Six production stages</i>				
Writing	16.0	+1.08	+1.52	2 of 8
Pre-production	14.1	−0.11	+0.59	0 of 8
Session	39.9	−1.44	−2.35	1 of 8
Post-production	8.1	+0.10	−0.27	0 of 8
Packaging and releasing	12.8	+0.09	+0.16	0 of 8
Unassigned residual	9.1	+0.28	+0.35	0 of 8

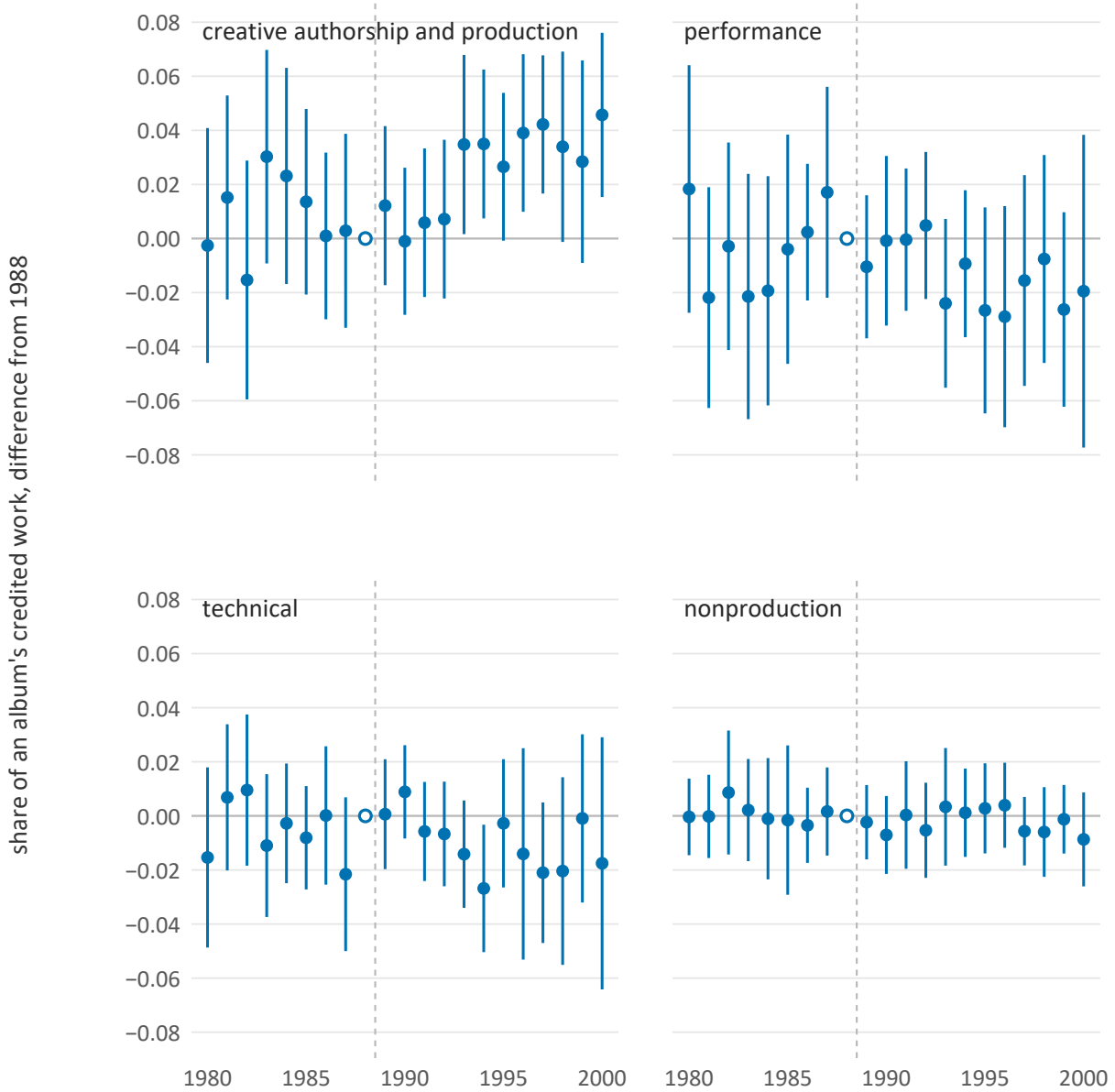
Note. All panels use the same 70,707 records, specification, and estimator; only the partition of classified credits changes. Coefficients are in percentage points. The 1993–2000 step compares the mean annual coefficient over 1993–2000 with the mean pre-campaign coefficient. “Significant years” reports the number of annual coefficients from 1993–2000 whose CR2 interval excludes zero. Within each panel, coefficients sum to zero mechanically.

E.7 Robustness of the organizational estimates

Table E.7 reports a specification ladder for the creative-authorship-and-production credit share and contributor share. The main organizational conclusion does not depend on a single exposed market, the inclusion of Hip Hop, reissues, or the entry and exit of recording studios.

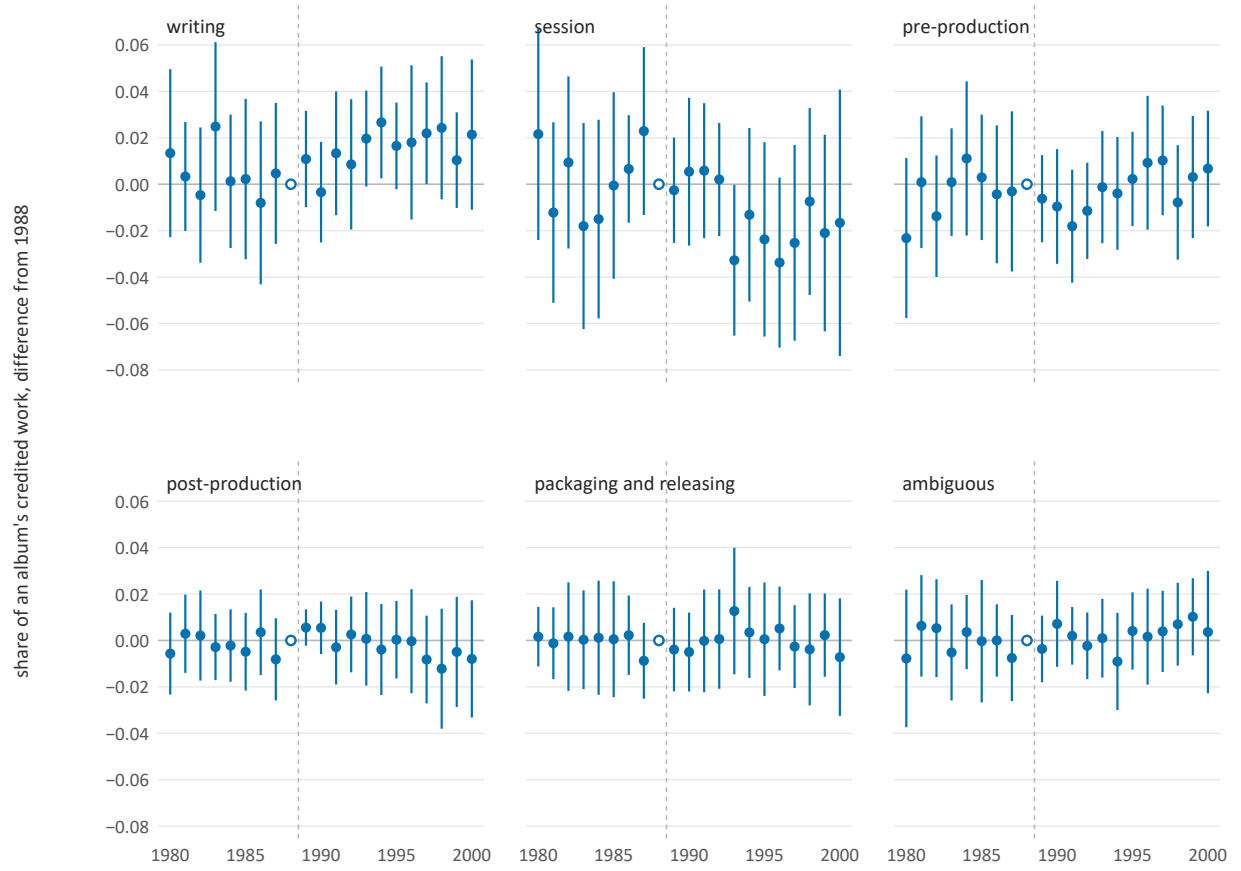
The estimates are more sensitive to specifications that absorb differential trends associated with genre composition or predetermined market characteristics. This sensitivity is one reason

Figure E.9: The four underlying credit families



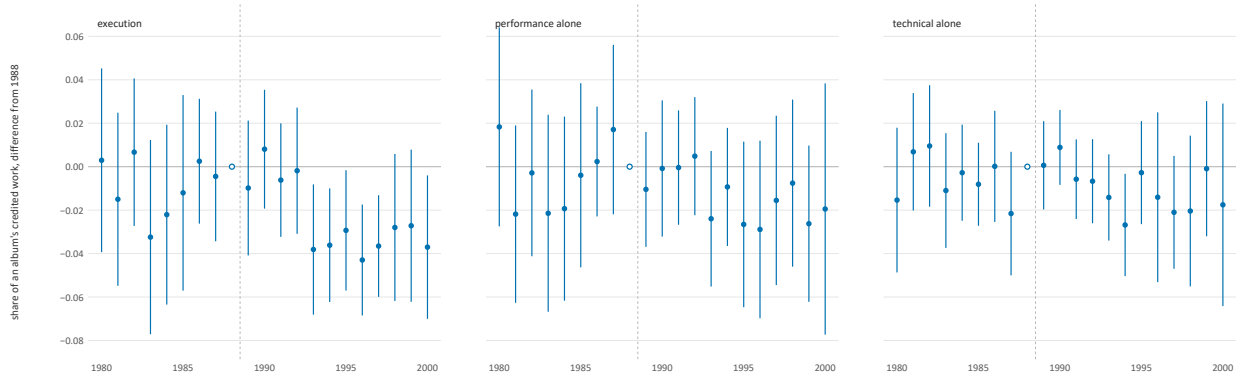
Note. The credit list partitioned into performance, technical, creative authorship and production, and nonproduction. Performance and technical are combined into execution in the main analysis. Specification, inference, and sample follow Figure 6.

Figure E.10: Six stages of production



Note. Classified credits partitioned according to stage of production. The residual category consists primarily of credits whose stage cannot be assigned uniquely. Specification, inference, and sample follow Figure 6.

Figure E.11: Execution and its performance and technical components



Note. Execution shown alongside the performance and technical families it combines. The execution coefficient equals the sum of the other two coefficients, while its confidence interval reflects their covariance.

the main text emphasizes the annual path, the contributor count, and the within-studio specification rather than relying on a single pooled credit-share estimate.

Predetermined machine-production intensity. One potential concern is that exposed markets may contain more genres or studios that were already moving toward machine-based production before the campaign. I construct a predetermined index measuring the pre-1988 prevalence of functions associated with programming, sequencing, drum machines, electronic drums, MIDI, and computers. Sampling-related labels themselves are excluded from the index.

Let High_a indicate whether record a belongs to a genre or studio above the median of this predetermined index. I estimate

$$Y_a = \beta_1(\text{Exposed}_{m(a)} \times \text{Post}_{t(a)}) + \beta_2(\text{Exposed}_{m(a)} \times \text{Post}_{t(a)} \times \text{High}_a) + \beta_3(\text{Post}_{t(a)} \times \text{High}_a) + \mu_{s(a)} + \rho_{d(a),t(a)} + \varepsilon_a \quad (\text{E.2})$$

The genre-level index is defined for the full sample, while the studio-level index is available for studios observed before 1988.

Table E.6: Predetermined machine-production intensity

	$\hat{\beta}_1$ low intensity	$\hat{\beta}_2$ differential	$\hat{\beta}_3$ post gradient	Exposure effect at exposed mean
<i>Panel A. Genre-level index, all 70,707 records</i>				
Creative authorship and production	+0.83	+1.67	+7.47	+1.45
authorship	+0.88	+0.65	+1.92	+1.12
production and arrangement	-0.05	+1.02	+5.55	+0.32
<i>Panel B. Studio-level index, 48,211 records</i>				
Creative authorship and production	+0.37	+1.86	+0.23	+1.49
authorship	+0.89	+0.13	+0.73	+0.97
production and arrangement	-0.52	+1.73	-0.50	+0.52

Note. Percentage-point coefficients from Equation (E.2). The last column evaluates the exposure effect at the mean value of the high-intensity indicator among exposed commuting zones. The two subcomponents sum exactly to creative authorship and production.

The estimated exposure effect evaluated at the exposed markets' machine-production intensity is +1.45 percentage points using the genre-level index and +1.49 using the studio-level index, close to the +1.57 pooled coefficient in the baseline specification.

The genre-level specification also reveals a large national post-1988 movement toward creative authorship and production in genres already intensive in machine-based production. This gradient is not itself a causal estimate because it compares genres that differed systematically before 1988. Importantly, its composition also differs from the campaign response. The

national genre gradient is concentrated primarily in production and arrangement, whereas the campaign effect is more heavily concentrated in authorship. Controlling for the gradient therefore attenuates the exposure estimate but does not reproduce its composition.

Table E.7: Robustness of the creative-authorship-and-production outcomes

Specification	Varies	Credit lines		Contributors	
		$\hat{\beta}$	95% interval	$\hat{\beta}$	95% interval
<i>Panel A. Studio and division-year effects</i>					
as reported	—	+1.57	[−0.31, +3.44]	+2.11	[+0.24, +3.98]
clustered on state	inference	+1.57	[−0.15, +3.29]	+2.11	[+0.44, +3.78]
county within 100 km	treatment	+1.67	[−0.20, +3.53]	+2.21	[+0.35, +4.08]
genre-by-year effects	design	+1.31	[−0.50, +3.11]	+1.70	[−0.02, +3.41]
1980 characteristics × post	controls	+3.41	[−0.21, +7.04]	+4.08	[+0.32, +7.83]
1980 characteristics × year	controls	+3.46	[−0.26, +7.17]	+4.07	[+0.22, +7.92]
Hip Hop dropped	sample	+1.56	[−0.42, +3.55]	+2.04	[+0.05, +4.03]
<i>Panel B. Alternative commuting-zone specification</i>					
baseline of this block	—	+4.39	[+0.78, +8.00]	+4.70	[+0.51, +8.90]
clustered on state	inference	+4.39	[+1.18, +7.61]	+4.70	[+1.11, +8.30]
county within 100 km	treatment	+4.42	[+0.89, +7.94]	+4.71	[+0.62, +8.80]
genre fixed effects	design	+3.51	[+0.05, +6.96]	+3.28	[−0.54, +7.10]
1980 characteristics × post	controls	−0.83	[−4.33, +2.67]	−1.08	[−5.30, +3.13]
1980 characteristics × year	controls	−0.80	[−4.35, +2.76]	−1.05	[−5.32, +3.21]
Hip Hop dropped	sample	+3.98	[+0.53, +7.42]	+4.35	[+0.28, +8.43]
<i>Panel C. Sample checks, studio specification</i>					
all commuting zones	sample	+1.93	[+0.24, +3.62]	+2.47	[+0.79, +4.15]
reissues excluded	sample	+1.51	[−0.25, +3.27]	+2.04	[+0.26, +3.81]
studios active before and after 1988	sample	+1.52	[−0.35, +3.40]	+2.06	[+0.16, +3.96]

Note. Pooled coefficients for creative authorship and production. “Credit lines” is its share of classified credits; “contributors” is the share of contributors whose functions are confined to the family. Panel A follows the preferred studio specification. Panel B additionally changes the fixed-effect structure and treatment of 1988 and should therefore be interpreted as an alternative specification rather than the direct commuting-zone counterpart to Panel A. Intervals in this table use conventional cluster-robust inference for comparability across the stored robustness specifications.

The choice of predetermined controls matters particularly under the commuting-zone specification. Table E.8 shows that 1980 population accounts for most of the difference between the alternative control sets.

E.8 Event-study robustness

Table E.7 summarizes the pooled coefficients across the main robustness specifications. Because the principal organizational result emerges only several years after the campaign, I also re-estimate the full annual event study under each specification.

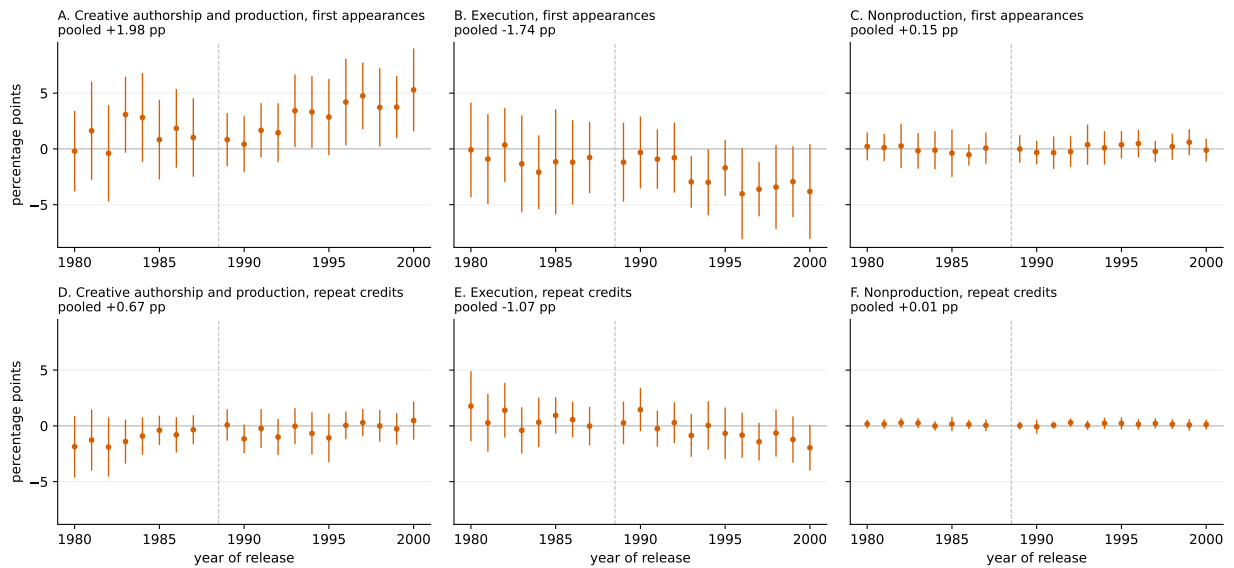
Table E.8: Sensitivity to predetermined market characteristics

Control entered	In set	Studio block		Commuting-zone block	
		$\hat{\beta}$	95% interval	$\hat{\beta}$	95% interval
No controls	—	+1.51	[−0.68, +3.69]	+4.39	[+0.78, +8.00]
1980 log density	A, B	+1.55	[−0.46, +3.57]	+2.75	[−1.62, +7.13]
1980 urban share	A, B	+1.23	[−0.91, +3.38]	+1.15	[−1.34, +3.63]
1979 median income	A	+1.90	[−0.42, +4.21]	+4.39	[−0.40, +9.18]
<i>Set A: all three</i>	A	+1.43	[−1.28, +4.13]	+1.60	[−1.26, +4.45]
1980 log population	B	+3.46	[−0.33, +7.26]	−0.75	[−4.72, +3.23]
1980 college share	B	+1.51	[−0.63, +3.66]	+4.37	[+0.91, +7.83]
1980 unemployment rate	B	+1.24	[−0.74, +3.22]	+4.37	[+0.84, +7.91]
<i>Set B: all five</i>	B	+3.90	[−0.17, +7.97]	−0.83	[−4.33, +2.67]

Note. Alternative specifications excluding 1988. Set A contains density, urban share, and median income. Set B contains population, density, urban share, college share, and unemployment. The table isolates the contribution of the individual predetermined characteristics to the sensitivity of the pooled creative-authorship-and-production share.

Figure E.12: First appearances and repeat credits, commuting-zone specification

Each category's share split into first appearances and repeat credits



Note. Counterpart to Figure E.7 using commuting-zone rather than studio fixed effects. The largest difference is in the first-appearance component of execution, which declines more under the commuting-zone specification.

Across the principal specifications, the annual paths preserve the main timing pattern: little differential movement before the campaign or during the first several post-campaign years, followed by a shift beginning around 1993. Specifications that most strongly attenuate the pooled creative-authorship-and-production coefficient, particularly genre-by-year effects and some predetermined-characteristic controls, also attenuate the later annual coefficients.

E.9 Additional record-level specifications

Table E.9: Main record-level estimates under alternative inference procedures

Outcome	Conventional	Bias-reduced	ν
<i>Panel A. Share of credit lines</i>			
Performance	−1.07 (0.89)	−1.07 (1.05)	7.9
Technical	−0.27 (0.63)	−0.27 (0.73)	7.9
Creative authorship and production	+1.57 (1.00)	+1.57 (1.06)	7.9
Nonproduction	−0.23 (0.40)	−0.23 (0.48)	7.9
<i>Panel B. Share of contributors working only in that family</i>			
Performance	−0.41 (0.75)	−0.41 (0.89)	7.9
Technical	−0.08 (0.61)	−0.08 (0.68)	7.9
Creative authorship and production	+2.11** (1.00)	+2.11* (1.06)	7.9
Nonproduction	−0.35 (0.47)	−0.35 (0.56)	7.9

Notes. In percentage points. **These are the eight four-family companion regressions** — performance, technical, creative authorship and production, and nonproduction, each in credit-share and exclusive-contributor form. **They are not the rows of Table E.3**, which merges performance and technical into execution. **Table E.3 reports the bias-reduced (CR2) intervals with coefficient-specific Satterthwaite degrees of freedom; this table reports the conventional (CRV1) companion beside them. The coefficient is identical under both by construction:** only the variance differs. The conventional estimator uses $G - 1 = 115$ degrees of freedom, the bias-reduced one the per-coefficient Satterthwaite value in the last column. At eighteen exposed clusters the conventional interval over-rejects, which is why the bias-reduced version is the one reported in the body. **Interval widths should be compared within an estimator and not across the two columns**, since the difference between them reflects the degrees-of-freedom correction and not the precision of the outcome.

Figure E.13: Creative authorship and production across specifications, studio fixed effects

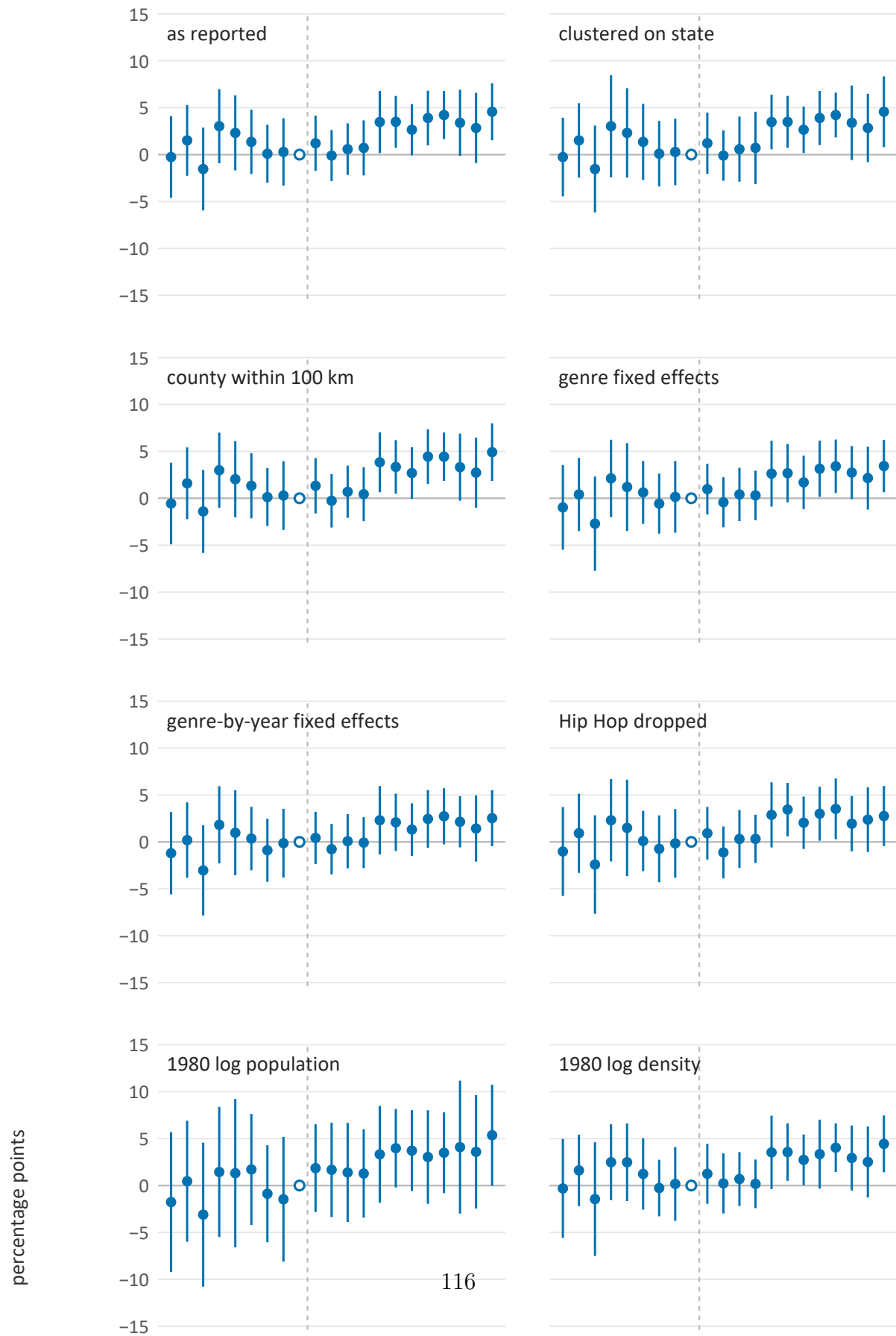


Figure E.14: Creative authorship and production across specifications, commuting-zone fixed effects

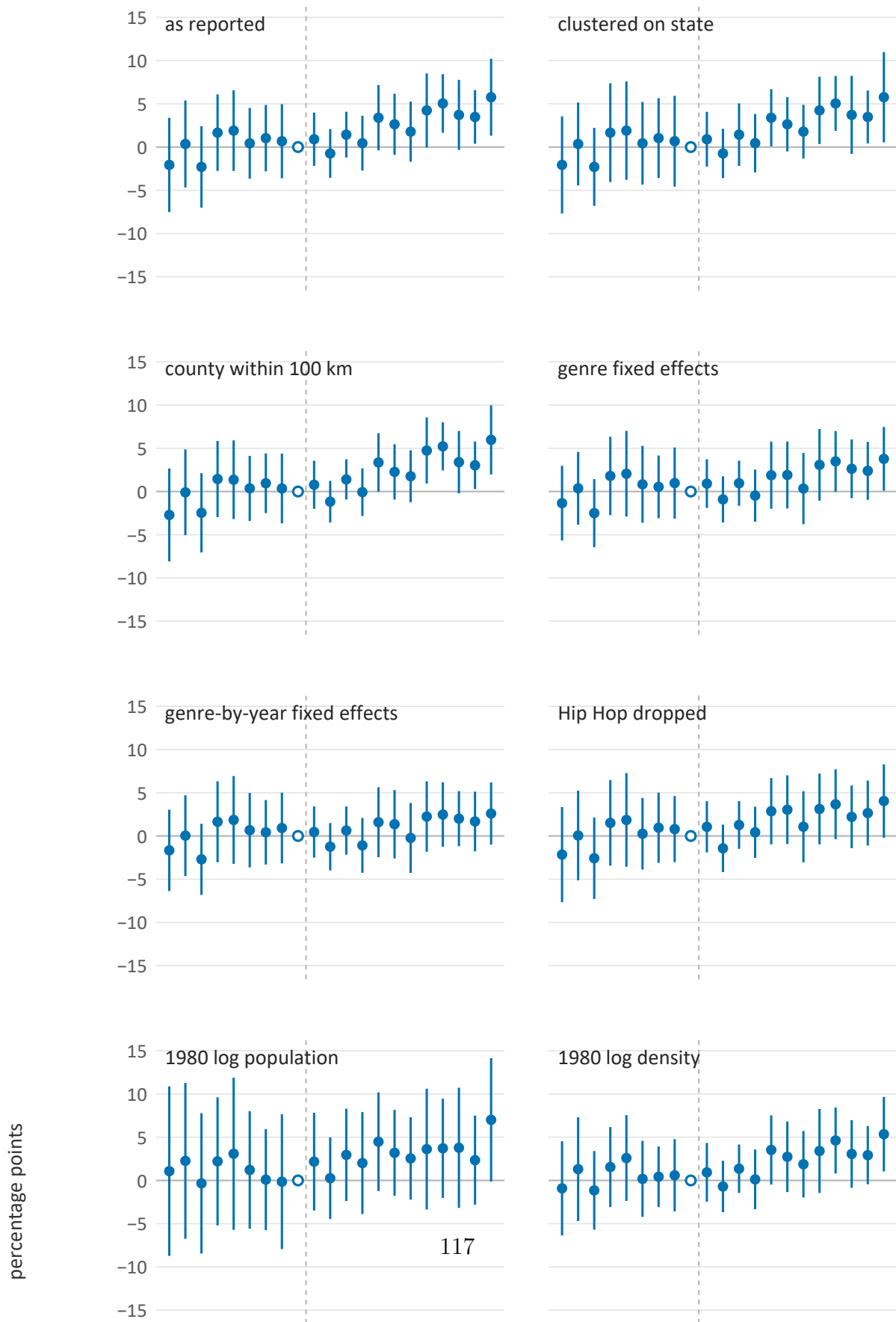


Figure E.15: Execution across specifications, studio fixed effects

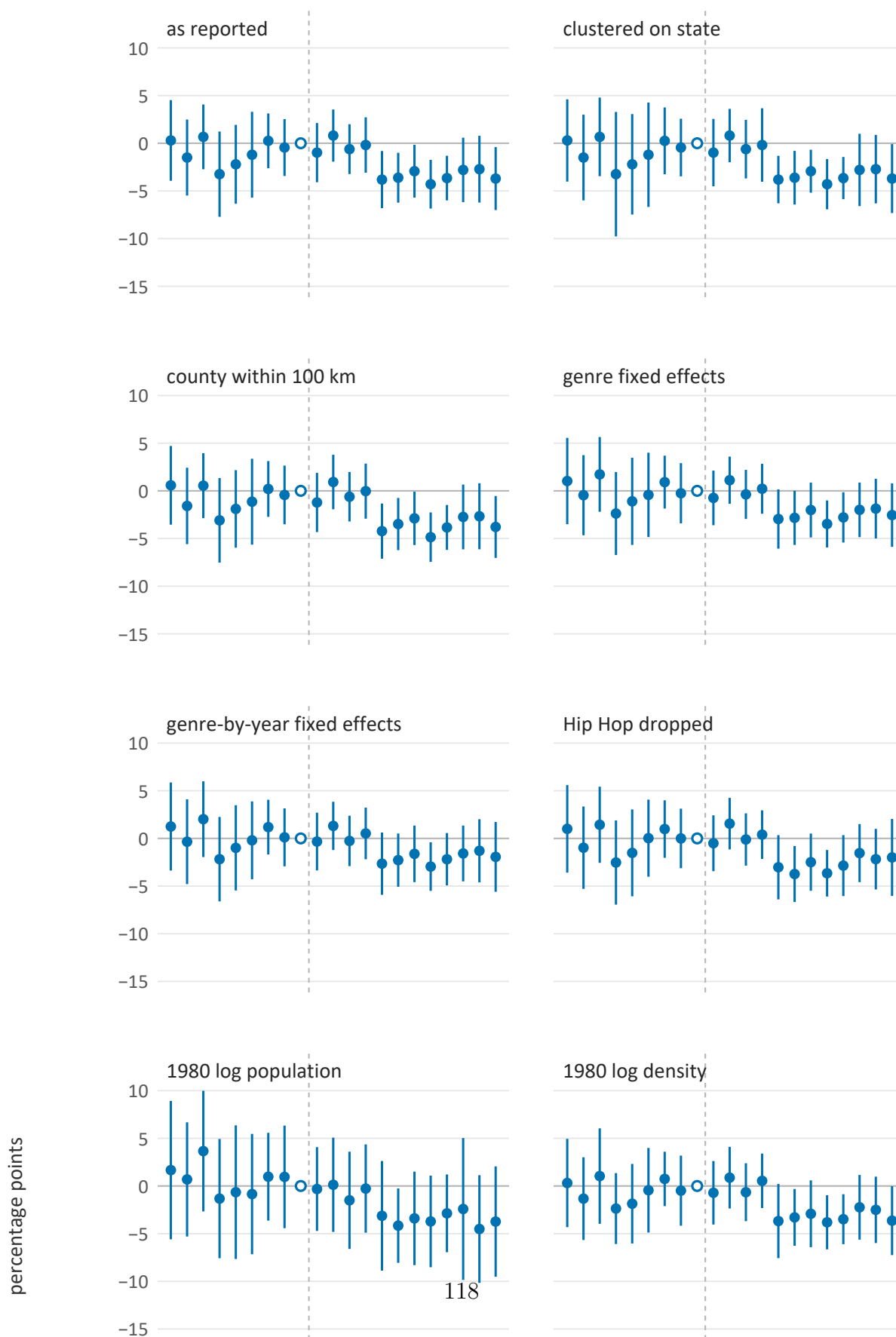


Figure E.16: Execution across specifications, commuting-zone fixed effects

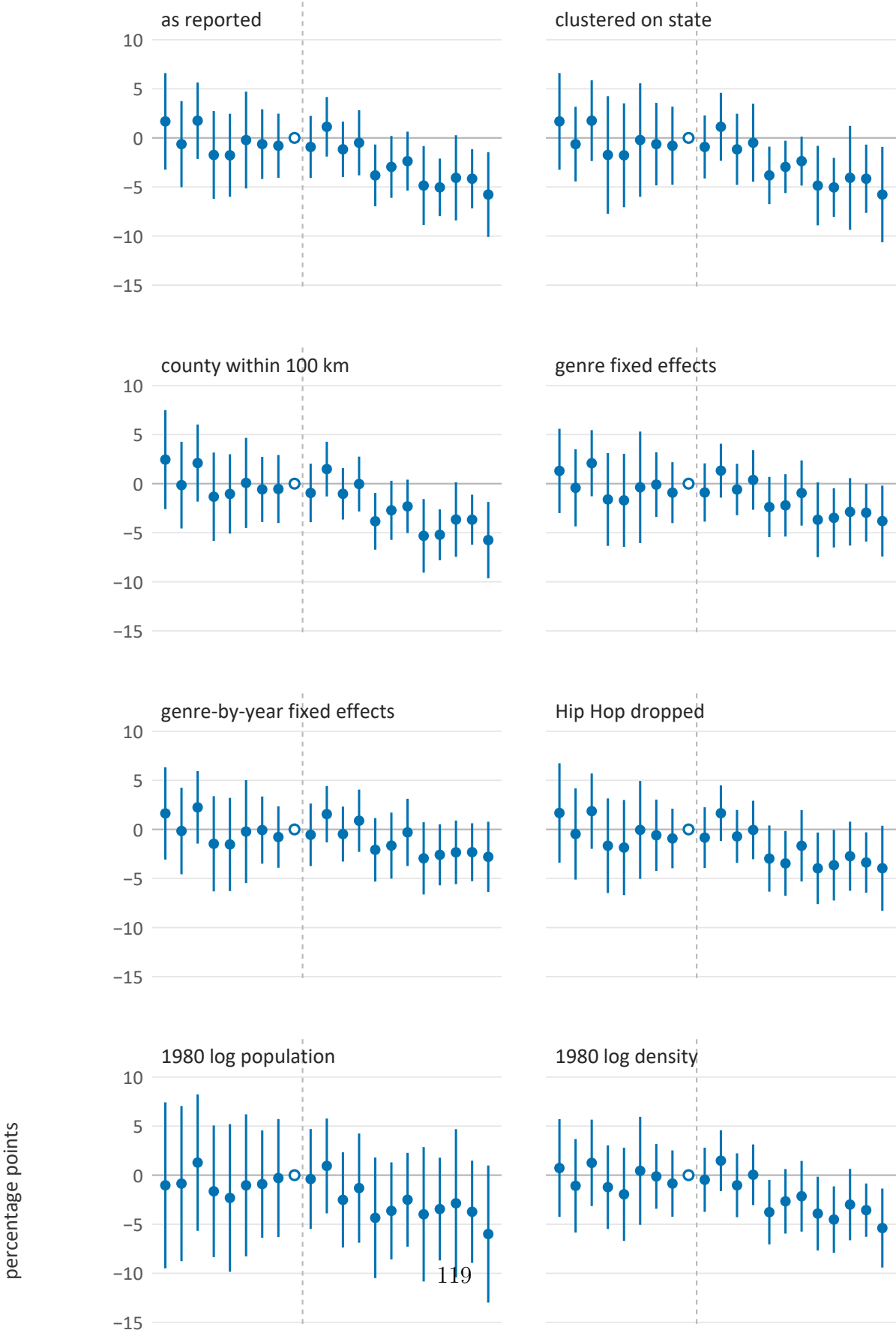


Figure E.17: Nonproduction across specifications, studio fixed effects

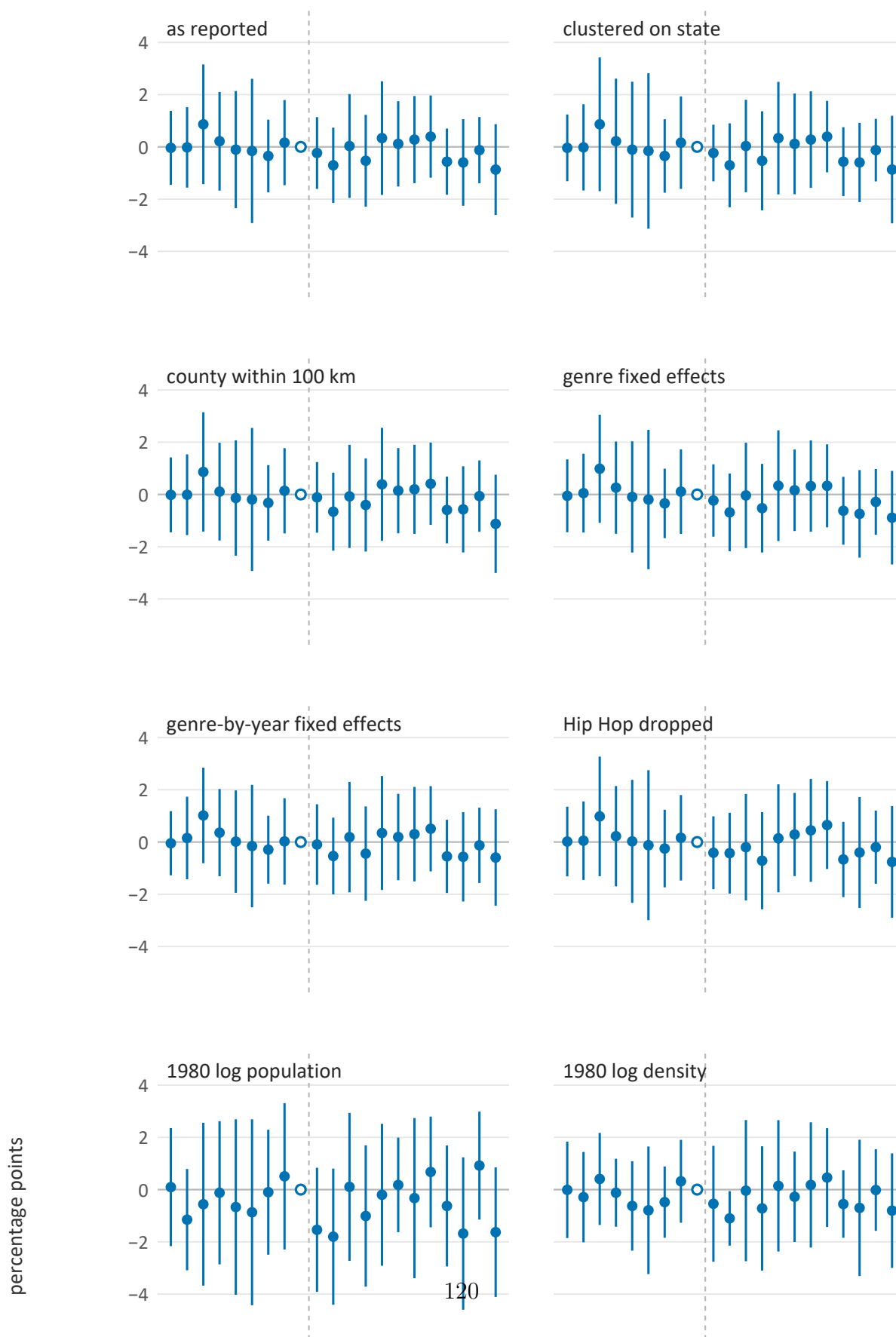


Figure E.18: Nonproduction across specifications, commuting-zone fixed effects

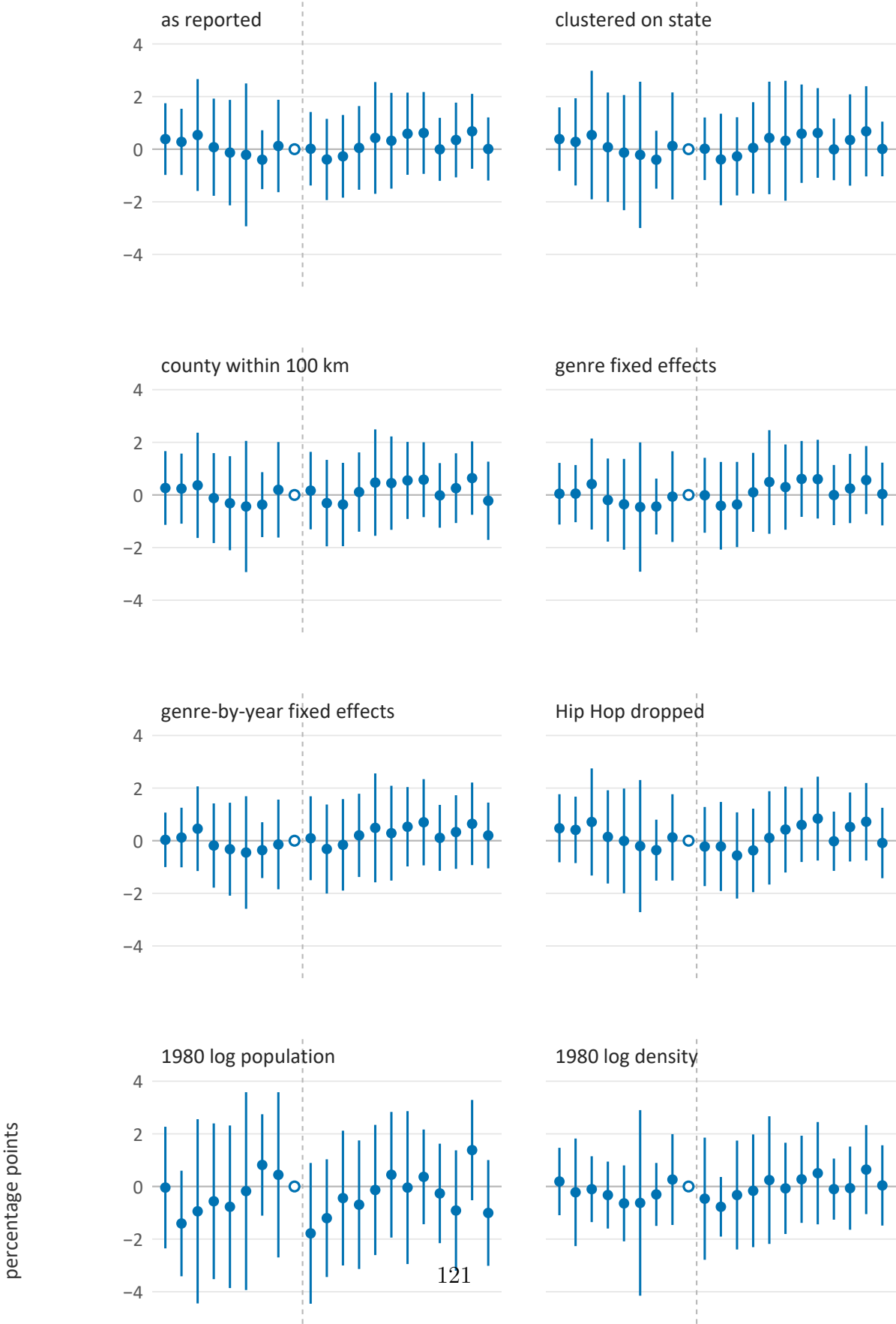


Figure E.19: Alternative commuting-zone samples

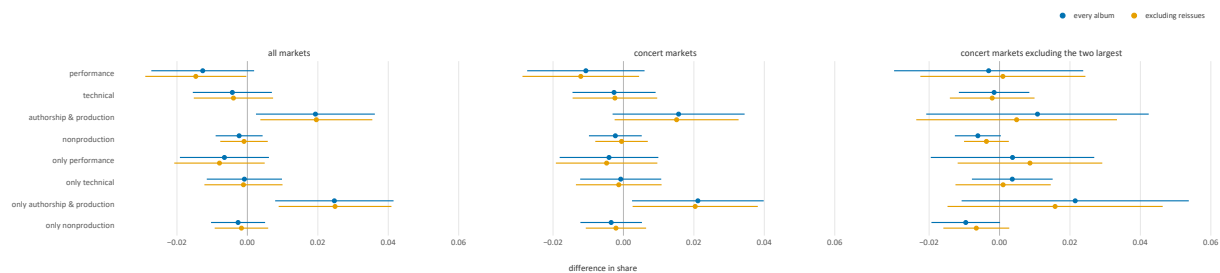


Figure E.20: Excluding reissues

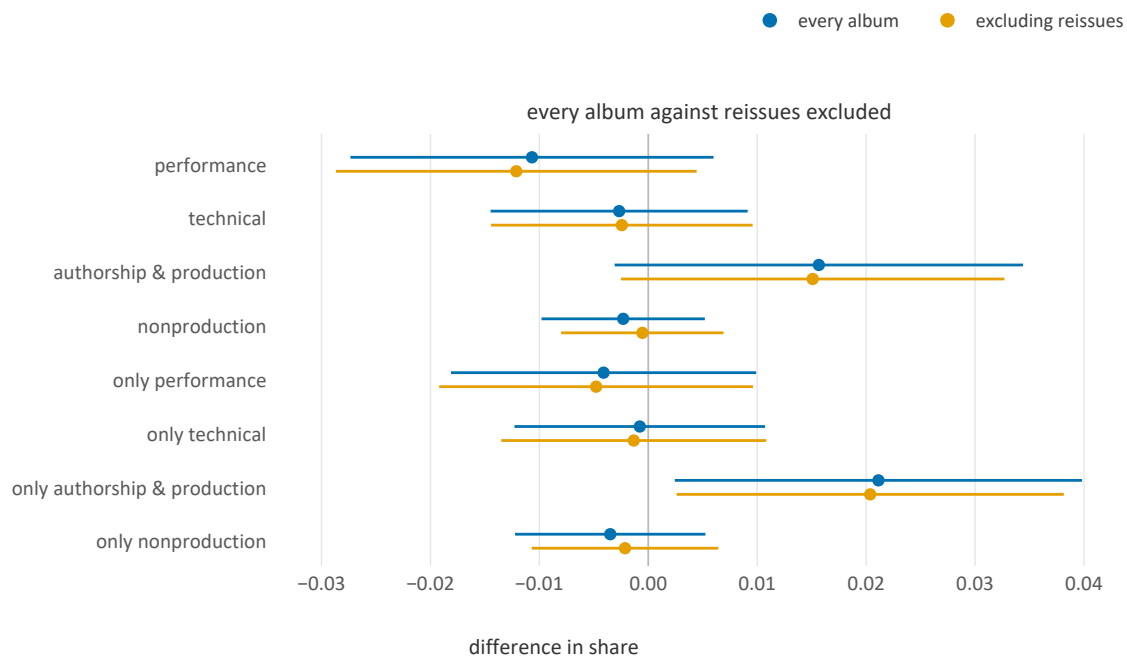


Figure E.21: Adding genre-by-year fixed effects

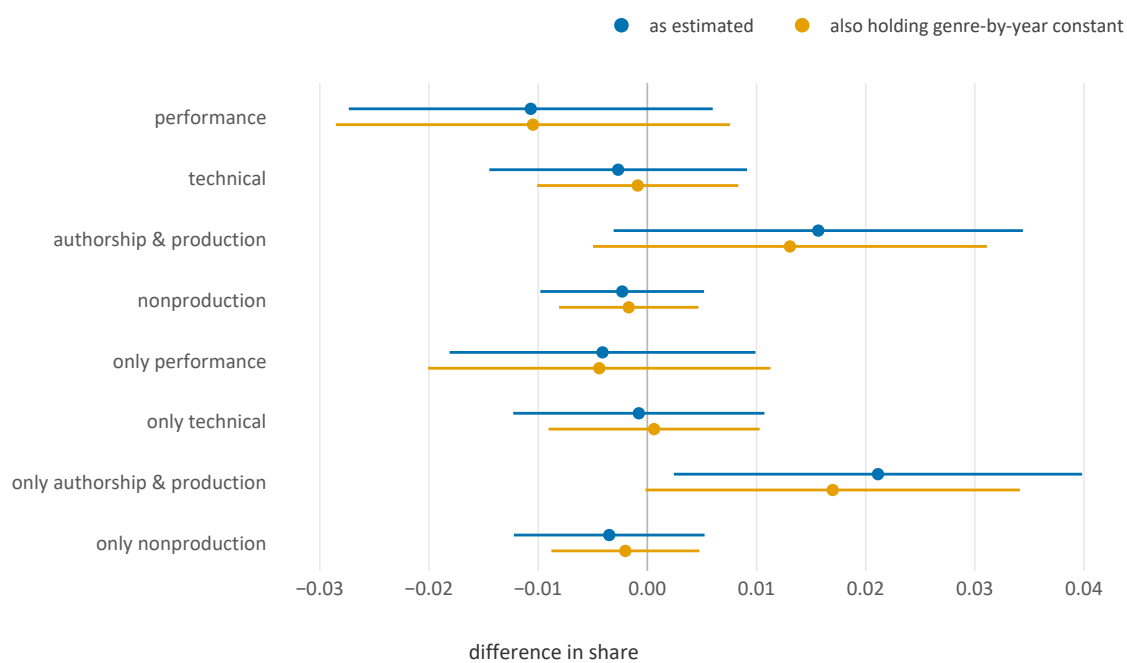


Figure E.22: Credits of the billed main artist

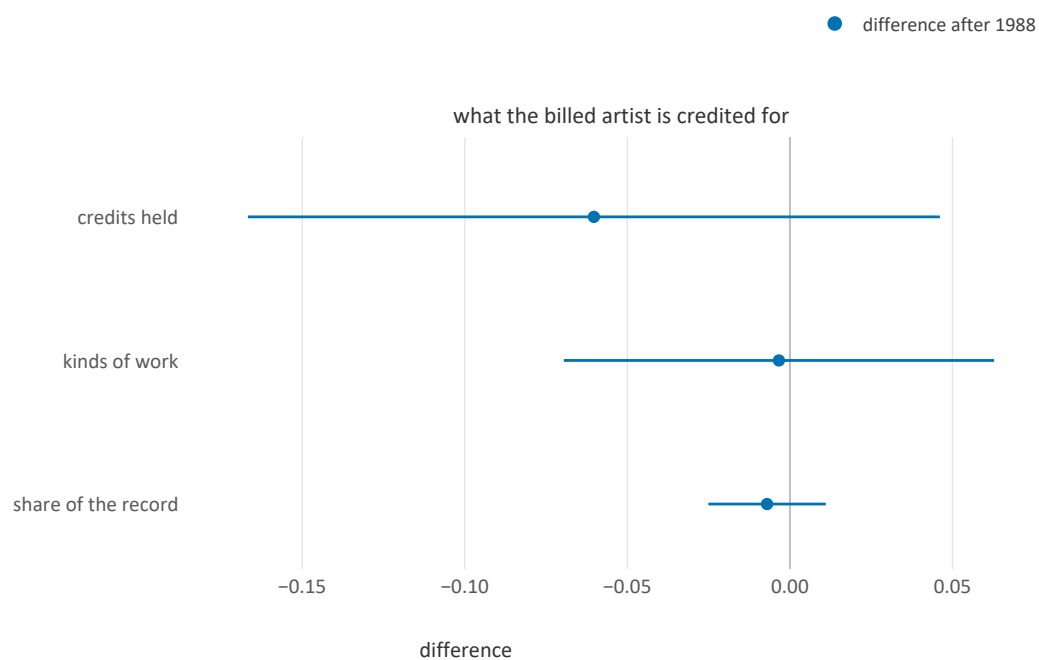
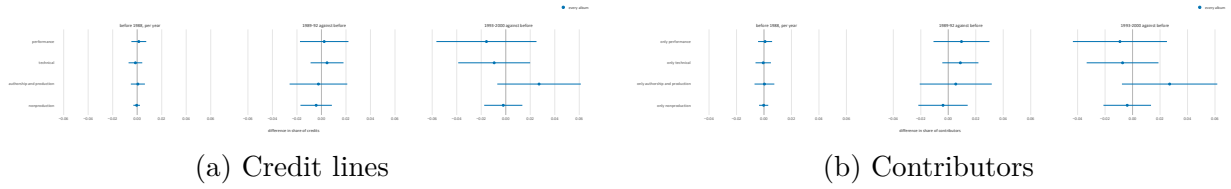


Table E.10: Pre-campaign slopes and post-campaign changes

	Coef.	ν
exititPanel A. Share of credit lines		
Performance		
Pre-period slope, per year	+0.13 (0.27)	8.6
Step, 1989–92 vs. pre	+0.22 (0.86)	8.3
Step, 1993–2000 vs. pre	−1.57 (1.76)	7.7
Technical		
Pre-period slope, per year	−0.14 (0.25)	8.6
Step, 1989–92 vs. pre	+0.45 (0.59)	8.3
Step, 1993–2000 vs. pre	−0.94 (1.26)	7.7
Creative authorship and production		
Pre-period slope, per year	+0.06 (0.25)	8.6
Step, 1989–92 vs. pre	−0.25 (1.03)	8.3
Step, 1993–2000 vs. pre	+2.72 (1.47)	7.7
Nonproduction		
Pre-period slope, per year	−0.04 (0.12)	8.6
Step, 1989–92 vs. pre	−0.43 (0.56)	8.3
Step, 1993–2000 vs. pre	−0.20 (0.67)	7.7
exititPanel B. Share of contributors working only in that family		
Performance		
Pre-period slope, per year	+0.08 (0.22)	8.6
Step, 1989–92 vs. pre	+0.96 (0.89)	8.3
Step, 1993–2000 vs. pre	−0.92 (1.47)	7.7
Technical		
Pre-period slope, per year	−0.06 (0.25)	8.6
Step, 1989–92 vs. pre	+0.88 (0.57)	8.3
Step, 1993–2000 vs. pre	−0.73 (1.12)	7.7
Creative authorship and production		
Pre-period slope, per year	+0.03 (0.32)	8.6
Step, 1989–92 vs. pre	+0.54 (1.14)	8.3
Step, 1993–2000 vs. pre	+2.70 (1.49)	7.7
Nonproduction		
Pre-period slope, per year	−0.02 (0.15)	8.6
Step, 1989–92 vs. pre	−0.38	8.3

Figure E.23: Pre-campaign slopes and post-campaign changes



Note. Linear contrasts from the annual event-study estimates: the pre-campaign slope and the post-campaign changes by function family. The left panel uses credit shares; the right uses contributor composition. Intervals use CR2.

E.10 Counts underlying the contributor shares

The main organizational result is most transparent in contributor counts. Table E.11 reports the 1993–2000 changes in the four mutually exclusive team-composition cells and total credited team size.

Table E.12 reports complementary measures that allow contributors to appear in more than one row. Under the commuting-zone specification, contributors with any creative-authorship-and-production credit rise by about 0.79 per record, while the corresponding number of credits also increases. Under studio fixed effects these overlapping measures are less precisely estimated. The mutually exclusive contributor count is therefore the most stable measure across specifications.

Table E.13 shows the arithmetic behind the increase in the share of contributors concentrated in creative authorship and production. Holding total team size at its pre-period average while increasing the number of such contributors to its later average raises their share by 3.35 percentage points. Holding their count fixed while allowing total team size to change lowers the share by 0.65 points. Together these movements approximately reproduce the estimated 2.70-point increase.

E.11 Copyright attribution

Sample clearance can add writing and composition credits for creators of sampled material even when they did not participate in producing the new record. I therefore assess how much such attribution could account for the main contributor result.

A direct matching approach is infeasible because the source recordings underlying documented samples have much poorer credit coverage than the records in the estimation sample. Of the 4,203 records documenting sampling, only a small fraction can be linked to source releases with usable writer credits.

I therefore implement a mechanical removal exercise. For records documenting sampling from 1992 onward, I remove potentially attribution-related writing and composition credits

Table E.11: Credited team composition

Contributors per album	Coef.	Pre-period mean	<i>N</i>
<i>Panel A. Studio and division-year effects ($\nu = 7.67$)</i>			
Creative authorship and production only	+0.516** (0.155)	3.48	70,707
Execution only	+0.057 (0.288)	7.72	70,707
Nonproduction only	+0.001 (0.077)	1.77	70,707
Two or more categories	−0.121 (0.130)	2.40	70,707
<i>Total credited contributors</i>	+0.452 (0.434)	15.39	70,707
<i>Panel B. Commuting-zone and division-year effects ($\nu = 10.40$)</i>			
Creative authorship and production only	+0.753** (0.259)	3.48	70,707
Execution only	+0.008 (0.311)	7.72	70,707
Nonproduction only	+0.179* (0.083)	1.77	70,707
Two or more categories	+0.039 (0.106)	2.40	70,707
<i>Total credited contributors</i>	+0.979* (0.536)	15.39	70,707
Census division-by-year effects	Yes		
Clustering	Commuting zone (116 zones, 18 exposed)		

Note. Each row reports the change in the number of credited contributors per record, comparing the mean annual coefficient over 1993–2000 with the mean pre-campaign coefficient. The four mutually exclusive cells partition the credited team: execution only, creative authorship and production only, nonproduction only, and contributors spanning two or more families. Their coefficients therefore sum to the change in total credited team size. Standard errors use CR2 and are clustered by commuting zone. Credits identify attributed functions rather than hours worked, authority, or labor input. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.12: Overlapping counts of contributors and classified credits

Per album	Coef.	Pre-period mean	N
<i>Panel A. Studio and division-year effects ($\nu = 7.67$)</i>			
Contributors with any creative authorship and production credit	+0.368 (0.238)	5.83	70,707
Creative authorship and production credit lines	+0.513 (0.321)	7.56	70,707
Contributors with any execution credit	−0.080 (0.371)	10.07	70,707
Execution credit lines	−0.750 (0.695)	14.25	70,707
Contributors credited in both [†]	−0.163 (0.136)	2.29	70,707
<i>Panel B. Commuting-zone and division-year effects ($\nu = 10.40$)</i>			
Contributors with any creative authorship and production credit	+0.793** (0.266)	5.83	70,707
Creative authorship and production credit lines	+1.079** (0.375)	7.56	70,707
Contributors with any execution credit	+0.016 (0.317)	10.07	70,707
Execution credit lines	−0.195 (0.428)	14.25	70,707
Contributors credited in both [†]	+0.009 (0.109)	2.29	70,707
Census division-by-year effects	Yes		
Clustering	Commuting zone (116 zones, 18 exposed)		

Note. Unlike Table E.11, rows in this table can overlap and therefore do not form a partition. Specification and inference follow Table E.11. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.13: Accounting for the increase in the contributor share

	Percentage points
Holding total contributors fixed, contributors working exclusively in creative authorship and production rise	+3.35
Holding that number fixed, total contributors rise	−0.65
Combined	+2.61
Estimated 1993–2000 change in the share	+2.70

Note. Arithmetic decomposition of the change in the share of contributors whose functions are confined to creative authorship and production. This is an accounting exercise rather than an additional estimate.

and reconstruct the organizational outcomes. Table E.14 reports increasingly broad removals.

Removing one, two, or three writing credits per sampled source explains between 2.5 and 6.1 percent of the observed increase in the share of contributors confined to creative authorship and production. Removing all eligible writing and composition credits from sampled records after 1991 explains approximately 17.1 percent of the increase. Copyright attribution can therefore account for part of the measured response, but not most of it.

Table E.14: Removing potentially attribution-related writing credits

Removal scenario	Credit share	Share of contributors
1 author per sampled source	1.2%	2.5%
2 authors per sampled source	2.9%	4.6%
3 authors per sampled source	4.4%	6.1%
Extreme: all removable authorship credits	19.3%	17.1%

Note. Share of the observed 1993–2000 organizational change accounted for after removing the stated writing and composition credits from records documenting sampling and released from 1992 onward. Studio and census-division-by-year fixed effects.

E.12 Reassigning campaign exposure

As an additional sensitivity exercise, I compare the observed organizational estimate with estimates obtained after assigning exposure to alternative sets of eighteen commuting zones.

I consider three reassignment schemes: unrestricted reassignment, reassignment holding the number of exposed markets fixed within census division, and reassignment holding it fixed within census division and terciles of pre-campaign recording activity. The last scheme compares the observed treatment assignment with alternatives that are more similar geographically and in pre-campaign production.

For the share of contributors confined to creative authorship and production, the corresponding two-sided reassignment p -value is 0.183 under the most restrictive scheme. For the count of those contributors, the corresponding value is 0.042. The unrestricted and census-division-only schemes yield larger values for the count.

Because the tour itinerary was not randomly assigned, these exercises are best interpreted as comparisons with alternative geographically and economically similar exposure assignments rather than as randomization inference.

E.13 Sensitivity to violations of parallel trends

Following [Rambachan and Roth \(2023\)](#), I assess how sensitive the organizational results are to departures from parallel trends related to the differential movements observed before the

Table E.15: Reassigning campaign exposure to alternative commuting zones

Reassignment scheme	Share exclusive to the category	Contributors exclusive to the category
Any 18 commuting zones	0.254	0.148
Within census division	0.288	0.150
Within census division and third of pre-1988 production	0.183	0.042
<i>The same procedure on two outcomes that should not move</i>		
Contributors with any execution credit	—	0.852
Total credited contributors	—	0.474

Note. Two-sided reassignment p -values based on 9,999 alternative assignments. Each draw assigns exposure to eighteen of the 116 commuting zones, subject to the stratification rule indicated in the row. Studio and census-division-by-year fixed effects throughout. The exercise compares the observed estimate with other plausible assignments and does not assume that the realized tour itinerary was random.

campaign.

Under $\Delta^{RM}(\bar{M})$, post-campaign deviations from parallel trends are bounded relative to the largest adjacent change observed in the pre-period. Table E.16 reports the resulting robust confidence sets.

The contributor count is the most stable organizational outcome under this exercise. The 1993–2000 average remains above zero through $\bar{M} = 1$ under both studio and commuting-zone fixed effects. Under the studio specification, the corresponding robust confidence set is $[+0.05, +1.82]$ at $\bar{M} = 1$. The credit-share outcomes become inconclusive under substantially smaller departures.

This provides an additional reason to place greater weight on the contributor count than on the credit-share measures. The exercise concerns the post-campaign average relative to the event-study normalization rather than the pre-versus-post contrast reported in the main text, so the point estimates are not numerically identical. Appendix C.5 describes the relationship between these quantities.

E.14 The 1991 ruling

A separate descriptive exercise asks whether the credit changes following the 1991 *Grand Upright* ruling resemble the pattern that sample-clearance attribution would generate.

Because realized sampling after 1991 is itself endogenous to clearance costs, I define exposure using whether a record’s studio had documented sampling before 1991. I estimate

$$Y_a = \sum_{k \neq 1991} \beta_k (T_a \times \mathbf{1}\{t(a) = k\}) + \mu_{s(a)} + \rho_{t(a)} + \varepsilon_a, \quad (\text{E.3})$$

Table E.16: Robust confidence sets under bounded violations of parallel trends

		Robust confidence set, $\Delta^{RM}(\bar{M})$			
	$\bar{M} = 0$	0.25	0.5	1	1.5
<i>Panel A. Records, studio and division-year effects</i>					
Creative authorship and production, share of credits	[1.36, 5.78]	[0.34, 6.88]	[-1.29, 8.55]	[-5.03, 12.34]	[-9.00, 16.31]
Execution, share of credits	[-5.48, -1.41]	[-6.46, -0.56]	[-7.92, 0.85]	[-11.29, 4.18]	[-14.82, 7.71]
Contributors exclusive to creative authorship, share	[2.37, 6.50]	[0.74, 7.74]	[-1.54, 9.76]	[-6.35, 14.44]	[-11.32, 19.33]
Contributors exclusive to creative authorship, count	[0.38, 1.35]	[0.32, 1.45]	[0.25, 1.55]	[0.05, 1.82]	[-0.20, 2.11]
Credited team size	[0.34, 3.18]	[-0.04, 3.70]	[-0.60, 4.34]	[-1.87, 5.77]	[-3.21, 7.22]
<i>Panel B. Records, commuting-zone and year effects</i>					
Creative authorship and production, share of credits	[-0.22, 7.34]	[-0.66, 8.15]	[-1.27, 9.07]	[-3.21, 11.55]	[-5.68, 14.33]
Execution, share of credits	[-7.46, -0.88]	[-7.96, -0.44]	[-8.64, 0.10]	[-10.79, 2.05]	[-13.21, 4.47]
Contributors exclusive to creative authorship, share	[0.91, 8.68]	[0.28, 9.33]	[-0.52, 10.20]	[-2.90, 12.66]	[-5.68, 15.52]
Contributors exclusive to creative authorship, count	[0.61, 1.86]	[0.52, 1.93]	[0.41, 2.02]	[0.08, 2.32]	[-0.30, 2.70]
Credited team size	[0.08, 4.04]	[-0.51, 4.38]	[-1.27, 4.75]	[-3.05, 5.68]	[-4.91, 6.97]
<i>Panel C. Commuting zones</i>					
Records documenting a sample, share	[-0.93, 4.20]	[-1.18, 4.58]	[-1.44, 5.00]	[-2.49, 6.31]	[-3.90, 7.82]
Credited contributors per 100,000 aged 20–64	[-1.58, 10.21]	[-2.11, 10.90]	[-2.59, 11.63]	[-3.79, 13.67]	[-5.36, 16.20]

Note. Identified sets for the 1993–2000 average under bounded failures of parallel trends, following [Rambachan and Roth \(2023\)](#) allows the post-1988 violation to be at most $\bar{M}$ times the largest movement observed between adjacent pre-period years; $\bar{M}$ parallel trends and returns the conventional interval. **The quantity bounded here is the 1993–2000 average measure in 1988, not the step against the pre-period average that Section 6.2 reports;** the two are not interchangeable and estimates differ accordingly. The procedure is applied to that average as a single post-treatment parameter rather than to annual coefficients, which is stated because it is a choice: the full annual problem did not solve. Shares are in percentage points, counts in contributors. The count of contributors exclusive to creative authorship and production remains distinguishable from $\bar{M} = 1.00$ on a point estimate of +0.86 contributors; the credit-line and exclusivity shares do not survive $\bar{M} = 0.5$, and the commuting zone diffusion estimate does not survive the smallest violation considered.

where T_a indicates that the studio had documented sampling before 1991. A more demanding specification replaces year effects with genre-by-year effects.

If sample clearance were the principal mechanism, the clearest response should appear in writing and composition credits rather than production and arrangement. Table E.17 shows that both move.

Several features make this exercise unsuitable as a causal design for the ruling itself. The annual differences do not display a discrete break at 1991, the pre-period is not stable, and the composition of records produced by early-sampling studios changes substantially over time. I therefore do not interpret the coefficients as causal effects of the ruling.

The pattern is nevertheless informative about attribution. Production and arrangement rise alongside authorship; much of the authorship movement occurs in repeat credits rather than first appearances; and the composition differs from the campaign effect, which is substantially more authorship-weighted. These facts are difficult to reconcile with mechanical writer attribution accounting for most of the organizational response, although they do not imply that clearance had no effect.

Table E.17: Credit changes around the 1991 ruling

Outcome	Studio and year effects		Studio and genre-by-year	
	$\hat{\beta}$	95% interval	$\hat{\beta}$	95% interval
Authorship credit lines	+0.78	[+0.39, +1.18]	+0.37	[+0.03, +0.71]
Production and arrangement	+0.48	[+0.19, +0.77]	+0.40	[+0.15, +0.65]
Execution credit lines	−0.89	[−1.93, +0.14]	−0.24	[−1.24, +0.76]
Nonproduction	+0.07	[−0.09, +0.23]	+0.01	[−0.15, +0.17]
Total credit lines	+0.44	[−1.04, +1.92]	+0.54	[−0.89, +1.97]
Authorship, first appearances	+0.15	[+0.02, +0.28]	+0.06	[−0.06, +0.19]
Authorship, repeat credits	+0.63	[+0.30, +0.97]	+0.31	[+0.03, +0.59]

Note. Pooled 1992–2000 coefficients from Equation (E.3). Treatment indicates that the record’s studio documented sampling before 1991. Standard errors are clustered by studio.

E.15 Following contributors over time

The record-level analysis asks how functions are allocated within the production unit. A complementary question is whether the same contributors become more narrowly specialized over time.

I follow contributors observed before 1988 and assign each to the commuting zone in which they recorded most often during the pre-campaign period. Exposure is therefore fixed before the campaign.

Table E.18 shows little detectable change in the number of function families or distinct functions held by these contributors, in their probability of being credited within a single family, or in their subsequent participation. The results therefore provide no evidence of a broad within-person narrowing or broadening among incumbents.

E.16 Release format

Short-form records contain substantially fewer classified credits than long-form records, and their prevalence rises over the sample period. This can mechanically affect share outcomes even when contributor counts move similarly.

Table E.20 therefore reports the principal organizational outcomes separately for long- and short-form records, while Table E.19 tests whether the campaign response differs across formats.

The contributor-count result is similar across the two subsamples. Over 1993–2000, the estimate is +0.486 contributors among long-form records and +0.586 among short-form records, compared with +0.516 in the full sample. By contrast, the credit-share response is substantially larger among short-form records.

Table E.18: Contributors observed before 1988, followed over time

	Pre-period	1993–2000
Credited on any record in the year	0.1503	+0.0050 (0.0059)
Records credited per year	0.2835	−0.0042 (0.0271)
Number of broad categories held on a record	1.1495	−0.0098 (0.0142)
Number of distinct credited functions on a record	1.5304	−0.0380 (0.0412)
Working exclusively in creative authorship and production on a record	0.2553	+0.0146* (0.0084)
Spanning execution and creative authorship and production on a record	0.1385	−0.0101 (0.0128)

Note. Contributor and year fixed effects. Exposure is assigned using the contributor’s principal pre-1988 commuting zone. Standard errors are clustered by that commuting zone. Conventional cluster-robust intervals are reported because CR2 could not be formed with the high-dimensional contributor fixed effects.

This difference is consistent with the denominator problem discussed in Appendix A.8. Short-form records contain an average of 15.5 classified credits, compared with 28.8 for long-form records. A given change in contributor composition therefore generates a larger movement in credit shares on a shorter credit list.

The interaction estimates for contributor counts are small and imprecise across specifications, while the corresponding interactions for credit shares are consistently positive and considerably larger. The format exercise therefore reinforces the interpretation that contributor counts provide the cleaner measure of the organizational response.

F Additional evidence on the nature of the technology

This appendix provides additional evidence on how the technological episode studied in the paper should be interpreted. I first examine whether the organizational response is better described by new machine-specific functions, augmentation of incumbent contributors, or changes in the allocation of existing functions. I then ask whether the reorganization is concentrated in parts of production that were already intensive in sampling or other machine-based production practices. Finally, I report the descriptive characteristics of artists associated with sampled records.

Table E.19: Campaign exposure interacted with release format

	Studio block		Commuting zone	
	Pooled	Step	Pooled	Step
<i>Panel A. Contributors confined to creative authorship and production</i>				
β_1 , long-form	+0.588	+0.754	+0.762	+0.821
β_2 , short-form difference	-0.289	-0.531	-0.113	+0.065
95% interval	(-1.07, +0.49)	(-1.63, +0.57)	(-0.75, +0.52)	(-1.02, +1.15)
<i>Panel B. Share of classified credits in creative authorship and production</i>				
β_1 , long-form	-0.54	-0.49	-0.64	-0.73
β_2 , short-form difference	+3.84	+6.45	+6.00	+8.06
95% interval	(+0.40, +7.29)	(-1.20, +14.11)	(-0.25, +12.24)	(-0.33, +16.44)

Note. Panel A is measured in contributors and Panel B in percentage points of classified credits. Each specification includes the interaction of campaign exposure, the post-campaign period, and a short-form indicator, together with the corresponding lower-order terms. Intervals use CR2 and are clustered by commuting zone.

Table E.20: Main organizational outcomes by release format

	All records	Long-form only	Short-form only
<i>Panel A. Contributors confined to creative authorship and production</i>			
Step, 1993–2000 vs. pre	+0.516** (0.155)	+0.486** (0.176)	+0.586 (0.412)
Pooled, Post = 1[$t \geq 1989$]	+0.441** (0.165)	+0.327* (0.147)	+0.619 (0.418)
+ genre-by-year effects	+0.310 (0.178)	+0.279 (0.172)	+0.455 (0.372)
Pre-period mean	3.41	4.01	2.02
<i>Panel B. Share of classified credits in creative authorship and production</i>			
Step, 1993–2000 vs. pre	+2.718 (1.467)	+1.083 (0.903)	+3.660 (2.847)
Pooled, Post = 1[$t \geq 1989$]	+1.566 (1.058)	-0.011 (0.731)	+3.384 (1.894)
+ genre-by-year effects	+1.306 (1.053)	-0.455 (0.684)	+2.870 (1.611)
Pre-period mean (%)	36.45	30.97	49.09
Records	70,707	43,996	26,711

Note. Panel A is measured in contributors and Panel B in percentage points of classified credits. Long-form records carry an *Album*, *LP*, or *Mini-Album* descriptor. All specifications include studio and census-division-by-year fixed effects. Standard errors use CR2 and are clustered by commuting zone.

F.1 New functions, augmentation, and reallocation

One way technological change can reorganize production is by creating new functions associated directly with the technology. Another is by augmenting incumbent workers so that the same people perform more work. The evidence in this setting points instead toward a reallocation of existing production functions.

Table F.1: Additional evidence on the technological mechanism

Mechanism	Prediction in this setting	Evidence
New function creation	Functions named for the machine or its operation become quantitatively important	Little evidence. Machine-specific credits remain extremely rare and show little aggregate movement
Labor augmentation	Contributors active before 1988 subsequently hold more credited work	Little evidence. Incumbents do not receive more credits per year after campaign exposure
Automation and reallocation	Functions associated with execution lose relative weight as other production functions become more important	Consistent. The allocation of credits shifts away from execution and toward creative authorship and production
Expertise leveling	Production becomes disproportionately associated with artists outside traditional channels of specialist training	Consistent descriptively. Artists associated with sampled records are more likely to be self-taught and less likely to report formal musical training

Note. The first three rows provide complementary evidence on how the technological episode affected production. The final row is descriptive because documented sampling and artist characteristics are not experimentally assigned. The mechanisms are not mutually exclusive.

Machine-specific functions remain small. If the principal organizational response operated through the creation of new functions associated directly with the sampler, credits naming programming, sequencing, sampling, drum machines, or related machine operations should become quantitatively important.

They do not. Taken together, these functions account for roughly four classified credits per thousand. Their aggregate share changes by approximately one hundredth of a percentage point, with an interval that includes zero. Individual machine-specific labels are even smaller. The one label that clearly increases is *samples*, but its baseline incidence is only a few credits per hundred thousand.

The organizational response documented in Section 6 is therefore much larger than the emergence of explicitly machine-named functions. The main change occurs within function categories that predate the sampler rather than through the addition of a quantitatively important new category of work.

Incumbent contributors are not measurably augmented. A second possibility is that the technology increased the amount of credited work performed by musicians already active before the campaign. I examine contributors credited before 1988 and compare their subsequent credited activity across exposed and comparison markets.

The annual estimates do not indicate an increase. If anything, pre-1988 contributors associated with exposed markets receive slightly fewer credits per year after the campaign. The pooled estimate is approximately -0.02 credits per year, with an interval from -0.07 to $+0.03$. Their probability of appearing on a record in a given year changes by only about one tenth of a percentage point.

These estimates provide little evidence that the organizational response operated through incumbent contributors performing more credited work. They are instead consistent with the record-level evidence that adjustment occurs through the composition of the production team.

F.2 Where the organizational response is concentrated

The MPC60 combined sampling with rhythm programming and sequencing. I therefore ask whether the organizational response is concentrated in parts of recorded-music production that were already intensive in these practices before the campaign.

I begin with documented sampling. Hip Hop and Electronic had the highest pre-campaign sampling rates, at 36.5 and 5.0 percent respectively, compared with 2.6 percent or less in every other genre. I classify these two genres as sampling-prone using only pre-1988 information.

The differential response in the creative-authorship-and-production share between sampling-prone and other genres is -1.4 percentage points, with an interval from -3.6 to $+0.8$. The corresponding interaction therefore provides no evidence that the credit-share response is larger in genres that sampled more before the campaign.

A split based on a record's own documented sampling status produces a similar pattern, but this comparison conditions on a post-treatment outcome and is descriptive. A split based on the studio's pre-campaign sampling history points toward a larger response in previously sampling studios, although the differential is imprecisely estimated.

Sampling captures only one part of the technology. Documented sampling is an incomplete measure of the production practices the MPC60 could affect. The sampler also provided sequencing, rhythm programming, and drum-machine functions that had a substantially broader pre-existing user base.

I therefore construct a second predetermined measure using the incidence of programming, sequencing, drum-machine, electronic-drum, MIDI, and computer-related credits during

Table F.2: Machine-specific functions and incumbent contributors

Role	Coef.	Baseline
<i>Panel A. Roles named for the machine</i>		
machine slots	+0.00001 (0.00041)	0.00426
programmed by	+0.00055 (0.00037)	0.00295
sequenced by	−0.00009 (0.00010)	0.00044
sequencer	−0.00007 (0.00006)	0.00006
sampler	−0.00027* (0.00016)	0.00054
samples	+0.00003 (0.00003)	0.00003
drum machine	−0.00016* (0.00009)	0.00021
midi controller	+0.00002 (0.00003)	0.00003
<i>Panel B. Roles that predate it</i>		
predating slots	+0.0082 (0.0085)	0.38035
written by	+0.0084 (0.0059)	0.11996
producer	−0.0037 (0.0054)	0.10777
engineer	−0.0002 (0.0024)	0.07290
arranged by	+0.0014 (0.0023)	0.02367
composed by	+0.0049*** (0.0014)	0.01528
mixed by	−0.0025 (0.0030)	0.04075
<i>Panel C. Named for the machine, inside the growing family</i>		
drum programming	−0.00039 (0.00025)	0.00162
<i>Panel D. Incumbents credited before 1988</i>		
credits per year	−0.0249 (0.0237)	0.434
credited at all in year	+0.0000 (0.0043)	0.150
Panels A–C	Studio FE, division-by-year FE	
Panel D	Contributor FE, division-by-year FE	
Clustered on	Commuting zone	

Notes. Panels A–C: each row is the share of a record’s classified credits held by that role, from a separate album-level regression following Equation 4. Panel D: the unit is a contributor-year on a balanced 1980–2000 panel of everyone credited before 1988, a year with no credit is a zero, and the outcomes are credits held in the year and an indicator for any credit. ‘Drum programming’ is named for the machine AND assigned to creative authorship and production by the role catalogue, so it is reported alone and folded into neither group: a role that is both cannot adjudicate between them. Standard errors in parentheses; **inference is CRV1 with $G-1$ degrees of freedom, so the classification block retains the conventional estimator and its standard errors are**

Table F.3: Machine-specific functions relative to their pre-period means

Role	Baseline share	Change (% of baseline)
machine slots	0.0043	+0.3 (9.7)
programmed by	0.0029	+18.7 (12.7)
sequenced by	0.0004	—
sequencer	0.0001	—
sampler	0.0005	—
samples	0.0000	—
drum machine	0.0002	—
midi controller	0.0000	—
drum programming	0.0016	−24.4 (15.7)

Note. Coefficients from the machine-specific-function estimates in Table F.2, expressed relative to the corresponding pre-period mean. No coefficient is re-estimated. Ratios are omitted for functions whose baseline incidence is too small for proportional changes to be economically interpretable.

1980–1987. Sampling-specific labels are excluded. Appendix E.7 describes the construction in detail.

Using this broader machine-production measure changes the sign of the interaction. Records in machine-intensive genres show an approximately +1.7 percentage-point larger creative-authorship-and-production response, with an interval from roughly −0.2 to +3.6. The corresponding sampling-based interaction is negative. Neither interaction is estimated precisely enough to distinguish it from zero.

The comparison illustrates why documented sampling should not be interpreted as a complete measure of the production changes associated with the sampler. Different predetermined measures of technological intensity capture different parts of the device’s capabilities.

The contributor count shows stronger heterogeneity. The heterogeneity is more pronounced for the main contributor-count outcome. Using the genre-level machine-production index, the differential response in contributors whose functions are confined to creative authorship and production is approximately +0.68 contributors, with an interval from +0.16 to +1.20 under the studio specification.

Related authorship counts move in the same direction. Authorship credit lines increase by approximately 1.83, and contributors holding any authorship credit by approximately 1.04, in the high-machine-intensity group relative to the low-intensity group.

I interpret these results cautiously. The corresponding continuous moderator produces a smaller estimate, and the differential does not consistently separate from zero across alternative fixed-effect specifications and moderator definitions. The broader pattern nevertheless suggests that the count response may be stronger in parts of production that were already more intensive in programming, sequencing, and related machine-based practices.

Machine-credit measures are useful only as predetermined proxies. Machine-related credit labels predict documented sampling before the campaign. Among records released from 1980 through 1987, records containing a machine-related credit document sampling at 5.2 percent, compared with 2.2 percent among records without one. The difference remains positive after accounting for genre-by-year effects.

Their usefulness deteriorates after the campaign, however. Machine-named credits rise sharply during the 1980s, peak near 1988, and subsequently decline even as documented sampling remains elevated. I therefore use these measures only as predetermined indicators of production practices before the campaign, not as post-campaign measures of technology adoption.

Taken together, these exercises do not establish that the organizational response is confined to a particular technological subgroup. They show instead that conclusions about heterogeneity depend on which aspect of the sampler is used to define technological intensity. Sampling alone provides a narrower measure than the broader collection of programming, sequencing, and rhythm-production capabilities incorporated into the sampler.

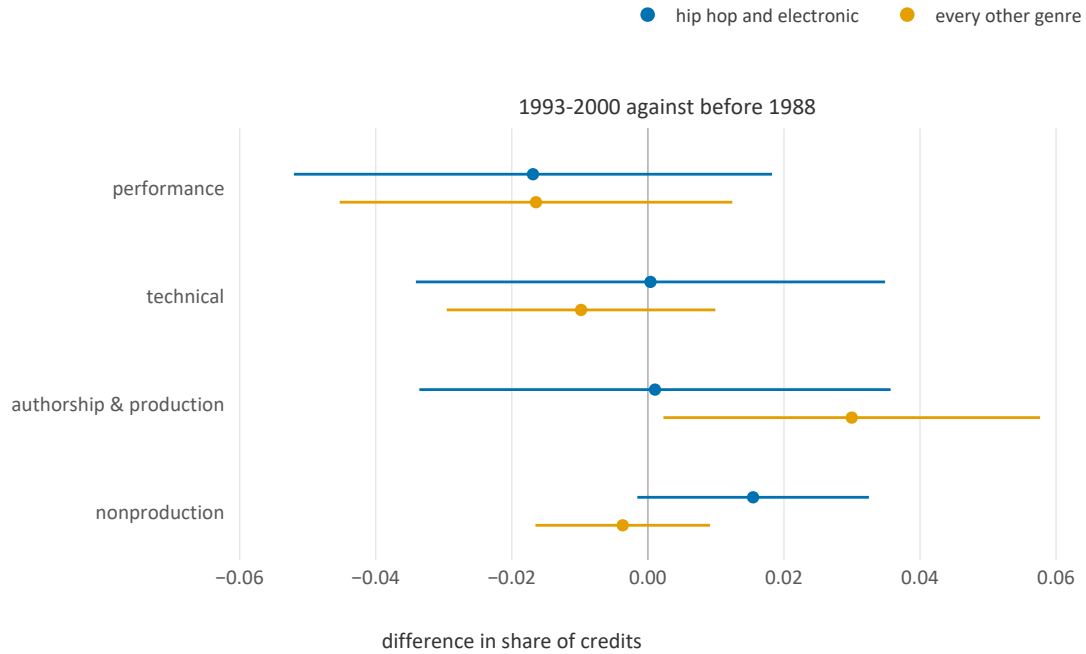
F.3 Artist backgrounds and documented sampling

The characteristics of artists associated with sampled records provide complementary descriptive evidence on the expertise margin. Because documented sampling is chosen rather than assigned, none of the comparisons in this subsection has a causal interpretation.

Table F.4 compares characteristics of main artists associated with records that document sampling with those of other main artists, before and after 1988. The clearest differences after the campaign concern musical training. Artists associated with sampled records are substantially more likely to be described as self-taught and less likely to report formal musical training or conservatory education.

These differences are consistent with the expertise-leveling characterization of the technological episode. Sampling allowed musical ideas to be realized without relying on the same combination of instrumental and technical training required in more conventional production processes. The comparison does not establish that sampling caused these artists to enter production or that training itself determined adoption.

Figure F.1: Sampling-prone genres and other genres



Note. Changes from the pre-campaign period to 1993–2000 estimated separately for sampling-prone genres (Hip Hop and Electronic) and all other genres. Sampling propensity is defined using pre-1988 records. Studio and census-division-by-year fixed effects. Intervals use conventional cluster-robust inference clustered by commuting zone.

The remaining characteristics provide context rather than a separate mechanism. Artists associated with sampled records are also younger on average and more likely to have biographies indicating technical interests. Coverage differs across characteristics, so every comparison is interpreted conditional on the available biographical information.

G Documented sampling and the organization of production

This appendix examines how records that document sampling differ from records that do not. The analysis is descriptive throughout. Documented sampling is itself an outcome of campaign exposure and of other choices made during production, so conditioning on it does not identify the causal effect of sampling.

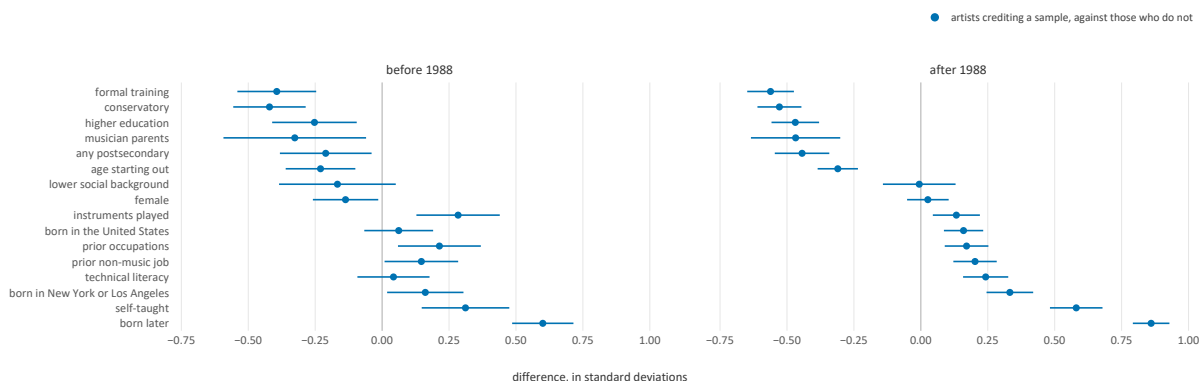
The purpose of the exercise is narrower. The main design establishes separately that campaign exposure increased documented sampling and changed the division of labor. Here I ask whether the records that actually document sampling display a corresponding difference in credited production functions.

Table F.4: Artist characteristics associated with documented sampling

Characteristic	Gap, 1980–87	Gap, 1989–2000	Change	95% CI
born later	+7.2640	+12.6590	+5.3950	[+3.6832, +7.1067]
age starting out	−1.2798	−1.6960	−0.4162	[−1.2488, +0.4164]
instruments played	+0.5592	+0.2185	−0.3407	[−0.6799, −0.0014]
self-taught	+0.1541	+0.2789	+0.1247	[+0.0311, +0.2184]
any postsecondary	−0.1028	−0.2124	−0.1097	[−0.2064, −0.0129]
higher education	−0.1100	−0.2005	−0.0905	[−0.1691, −0.0120]
lower social background	−0.0815	−0.0029	+0.0786	[−0.0470, +0.2042]
formal training	−0.1694	−0.2418	−0.0724	[−0.1460, +0.0012]
born in New York or Los Angeles	+0.0611	+0.1298	+0.0687	[+0.0048, +0.1325]
prior occupations	+0.1933	+0.1260	−0.0672	[−0.2194, +0.0850]
technical literacy	+0.0113	+0.0738	+0.0625	[+0.0183, +0.1066]
musician parents	−0.1531	−0.2152	−0.0621	[−0.2088, +0.0846]
female	−0.0454	+0.0096	+0.0550	[+0.0055, +0.1045]
conservatory	−0.1566	−0.2043	−0.0476	[−0.1071, +0.0119]
born in the United States	+0.0231	+0.0612	+0.0381	[−0.0174, +0.0937]
prior non-music job	+0.0561	+0.0760	+0.0199	[−0.0408, +0.0807]

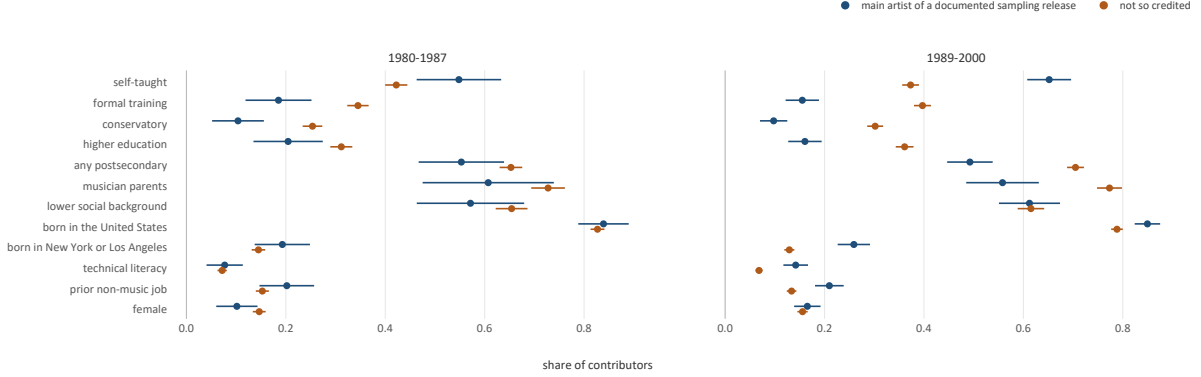
Notes. Each row is the gap between main artists who credit a sample and those who do not, in each window, and the change in that gap across 1988, ranked by the size of the change. **Nothing in this table is a treatment effect:** adoption is a choice, the two groups differ in everything, and no row is adjusted for anything. **This is the one table in the paper that reports an interval rather than a standard error and stars,** and the reason is that there is no test here to star. The interval on the change is naive — it adds four sampling variances as though the cells were independent, and the two windows share people — so it summarises spread rather than testing anything. Awards are computed and excluded: an artist wins one because of the work, so they describe a consequence of adopting rather than who adopted.

Figure F.2: Artist characteristics associated with documented sampling



Note. Standardized differences between main artists associated with records documenting sampling and other main artists during 1989–2000. Each difference is divided by the pooled standard deviation of the corresponding characteristic. Bars are 95% intervals rescaled by the same standard deviation. Descriptive throughout. Coverage varies across biographical characteristics and is reported in Table F.4.

Figure F.3: Artist differences before and after the campaign



Note. Difference in means between main artists associated with records documenting sampling and other main artists, calculated separately for 1980–1987 and 1989–2000. Bars are 95% intervals. The comparison is descriptive because documented sampling is chosen and no other characteristics are held fixed.

G.1 Documented sampling and credited functions

I compare records that document sampling with records that do not, holding fixed the recording environment and contemporaneous geographic conditions. For record a , I estimate

$$Y_a = \sum_t \beta_t (\text{HasSample}_a \times \mathbf{1}\{t(a) = t\}) + \alpha_{s(a)} + \rho_{d(a),t(a)} + \varepsilon_a, \quad (\text{G.1})$$

where HasSample_a indicates that at least one track on record a documents a sample. The specification is estimated on the 70,707 records containing at least one classified credit, of which 4,203 document sampling.

Because the annual interactions span every year, each β_t is the sampled-minus-non-sampled difference in that year. Figure G.1 re-expresses these annual differences relative to their 1988 value to show how the gap evolves over time.

I also summarize the change in the gap using

$$Y_a = \gamma \text{HasSample}_a + \delta (\text{HasSample}_a \times \text{Post}_{t(a)}) + \alpha_{s(a)} + \rho_{d(a),t(a)} + \varepsilon_a. \quad (\text{G.2})$$

Here γ describes the average pre-campaign difference between records that document sampling and those that do not, while δ describes how that difference changes after 1988. Because documented sampling is not assigned, neither coefficient has a causal interpretation.

I report both the studio and commuting-zone fixed-effect specifications used elsewhere in the paper. The two answer different descriptive questions: the studio specification compares records produced within the same recording locations, while the commuting-zone specification allows differences across studios within the same local market to contribute.

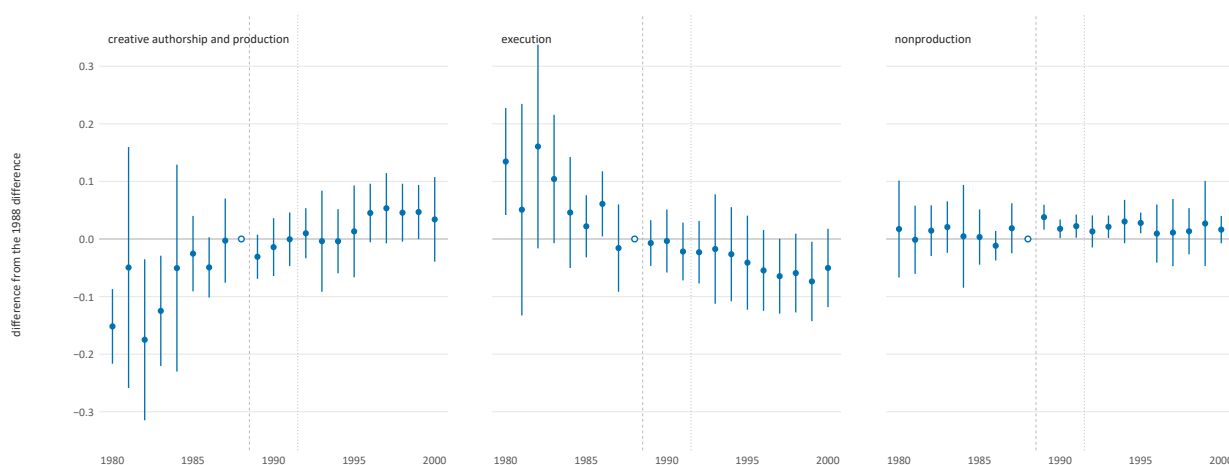
The sampled-record gap moves in the same broad direction as the main organizational result. Under the commuting-zone specification, the post-campaign difference between sampled and non-sampled records widens toward creative authorship and production and away from execution. The creative-authorship-and-production gap increases by about 7.7 percentage points, while execution falls by about 9.0 points. The corresponding change in nonproduction is much smaller.

Under studio fixed effects, the coefficients have the same signs but are smaller and less precisely estimated. The creative-authorship-and-production difference rises by about 4.6 percentage points and execution falls by about 6.0 points. The two specifications therefore agree on the broad direction of the sampled-record contrast, although not on which individual coefficients are estimated precisely.

The difference predates the campaign. The annual estimates show that sampled and non-sampled records were already diverging before 1988. Under the commuting-zone specification, several pre-campaign annual differences in creative authorship and production and execution are already distinguishable from their 1988 values. The studio specification shows a milder version of the same pattern.

This timing rules out interpreting the post-1988 widening as a break caused by the campaign. The annual coefficients instead describe an evolving difference between two endogenously selected types of record.

Figure G.1: DESCRIPTIVE: differences between records documenting sampling and other records, by year



Note. Annual differences between records documenting sampling and records that do not, expressed relative to the corresponding difference in 1988. Outcomes are shares of classified credits assigned to the three function families. The specification uses commuting-zone and census-division-by-year fixed effects. Documented sampling is endogenous, so the coefficients are descriptive and should not be interpreted as treatment effects. Bars are CR2 95% confidence intervals clustered by commuting zone.

The descriptive comparison therefore connects the two main reduced-form findings only in a limited sense. Campaign exposure increased documented sampling; campaign exposure also changed the organization of production; and records documenting sampling are disproportionately organized in the same broad direction. But the sampled-record difference predates the campaign and cannot identify whether sampling caused the organizational change, whether changing production practices generated both outcomes, or whether both reflect other differences across records.

G.2 Pooled and annual comparisons

Table G.1 reports the pooled sampled-minus-non-sampled differences and the change in those differences after 1988. Table G.2 reports the annual coefficients from Equation (G.1) in levels.

Table G.1: DESCRIPTIVE: records documenting sampling and other records, pooled

Outcome	Mean		Difference	Excl. 1988
	Sampled	Not		Difference
Performance	20.3	39.1	−6.38*** (0.42)	−6.19
Technical	22.6	17.7	+0.21 (0.30)	+0.01
Creative authorship and production	46.5	33.5	+4.63*** (0.43)	+4.62
Nonproduction	10.6	9.7	+1.54** (0.34)	+1.56
<i>The two blocks of the third family</i>				
Authorship	21.9	16.5	+3.53*** (0.34)	+3.63
Production and arrangement	24.7	17.0	+1.09* (0.42)	+0.99

Notes. In percentage points. Each row is the coefficient on HasSample_a from an album-level regression with studio and census-division-by-year fixed effects, standard errors clustered on the commuting zone, bias-reduced with a per-coefficient Satterthwaite correction. **Descriptive: whether a record documents a sample is an outcome of the design, not an assigned treatment, so no coefficient here is the effect of sampling on anything.** The four families partition every classified credit and their differences sum to zero; the two blocks partition the third family and sum to it. The final column re-estimates with the campaign year’s rows dropped. No genre composition is held fixed; Table G.3 reports the conditional versions. $n = 70,707$ records, of which 4,203 document a sample; 116 markets, 18 exposed. Standard errors in parentheses; stars are assessed against the Satterthwaite degrees of freedom rather than normal critical values, and **mark a difference between two kinds of record, not an effect.** *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table G.2: DESCRIPTIVE: records documenting sampling and other records, by year

Outcome	Mean annual difference			Years whose interval excludes zero
	1980–87	1989–93	1995–2000	
Performance	−0.50	−6.47	−7.81	1987–2000
Technical	+0.19	+0.56	−1.00	1983, 1987
Creative authorship and production	+0.15	+3.67	+7.52	1988, 1990–1992, 1994–2000
Nonproduction	+0.15	+2.24	+1.29	1989, 1994–1995
<i>The two blocks of the third family</i>				
Authorship	+0.67	+1.52	+7.36	1995–2000
Production and arrangement	−0.52	+2.15	+0.16	1990–1992

Notes. In percentage points. Each annual coefficient is the sampled-minus-non-sampled difference in that year, from the specification of Table G.1 with the single term replaced by twenty-one annual terms. **There is no reference year and these are not event studies:** the annual terms sum to HasSample_a , which the fixed effects do not absorb, so every year is estimated directly and no year is normalised to zero. **Descriptive throughout;** the regressor is an outcome. The 1980–87 coefficients are not a pre-period test, because no treatment date enters this specification. Window means are unweighted averages of the annual coefficients; 1988 and 1994 fall outside all three windows and are estimated all the same. These are the coefficients in LEVEL form; where a figure draws them measured against 1988 instead, that is a reparametrisation of this same fit and changes no coefficient. Intervals are the bias-reduced ones.

G.3 Conditioning on genre

Genre accounts for part of the observed difference between sampled and non-sampled records. Table G.3 therefore adds Discogs’ primary genre, first as a fixed effect and then interacted with year.

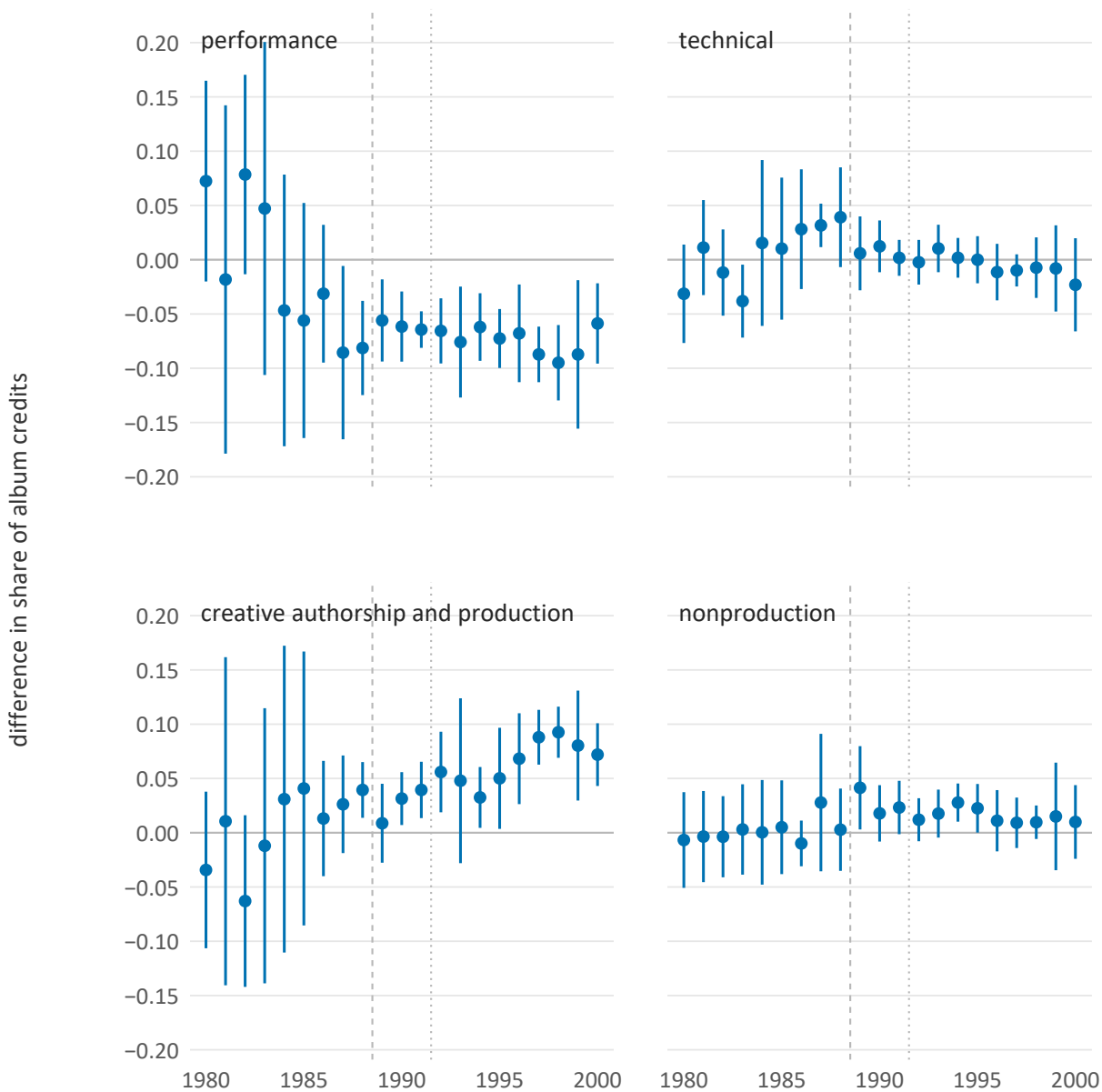
Conditioning on genre attenuates the aggregate association substantially and can change its sign under some specifications. I treat this as an alternative conditional comparison rather than as a correction. Discogs assigns a single primary genre to records that may span several styles, and contemporaneous genre may itself be affected by the production practices associated with sampling.

The sensitivity therefore shows that part of the sampled-record difference reflects which genres document sampling, while leaving open how much arises within genres.

G.4 Alternative function partitions

Figure G.2 repeats the annual comparison using the four underlying credit families, separating performance from technical functions. The direction of the reallocation is similar to the three-family presentation used in the main text, but the movement is divided across more outcomes.

Figure G.2: DESCRIPTIVE: sampled and non-sampled records across four function families



Note. Annual differences between records documenting sampling and other records using the four underlying function families: performance, technical, creative authorship and production, and nonproduction. All twenty-one annual coefficients are estimated directly. The comparison is descriptive because documented sampling is endogenous.

Table G.3: DESCRIPTIVE: conditioning on primary genre

Outcome	No genre control (Table G.1)	Genre effects	Genre $\times$ year effects
Performance	−6.38	−0.05	−0.09
Technical	+0.21	−1.11	−1.16
Creative authorship and production	+4.63	−0.84	−0.56
Nonproduction	+1.54	+2.01	+1.81
<i>The two blocks of the third family</i>			
Authorship	+3.53	+1.36	+1.88
Production and arrangement	+1.09	−2.20	−2.44

Notes. In percentage points. The specification of Table G.1 with Discogs’ primary-genre field added as a further fixed effect, alone and interacted with year. **This is an alternative conditional comparison and not a correction of the first column.** Discogs assigns one primary genre to records that routinely span several, and genre may itself respond to the technology, so the conditional comparison is a different quantity rather than a cleaner version of the same one; the paper’s descriptive object is the unconditional trajectory. The coefficients are not invariant to which genre is omitted in level terms but the reported contrasts are, and re-estimating with a different omitted genre reproduces every entry. **Descriptive throughout;** the regressor is an outcome. Same sample, estimator and clustering as Table G.1.

G.5 Studio fixed effects

Figures G.3 and G.4 reproduce the pooled and annual sampled-record comparisons holding the recording studio fixed.

The signs of the creative-authorship-and-production and execution differences agree with the commuting-zone specification, but the magnitudes are smaller and the intervals wider. The annual estimates again show that sampled and non-sampled records began to diverge before 1988.

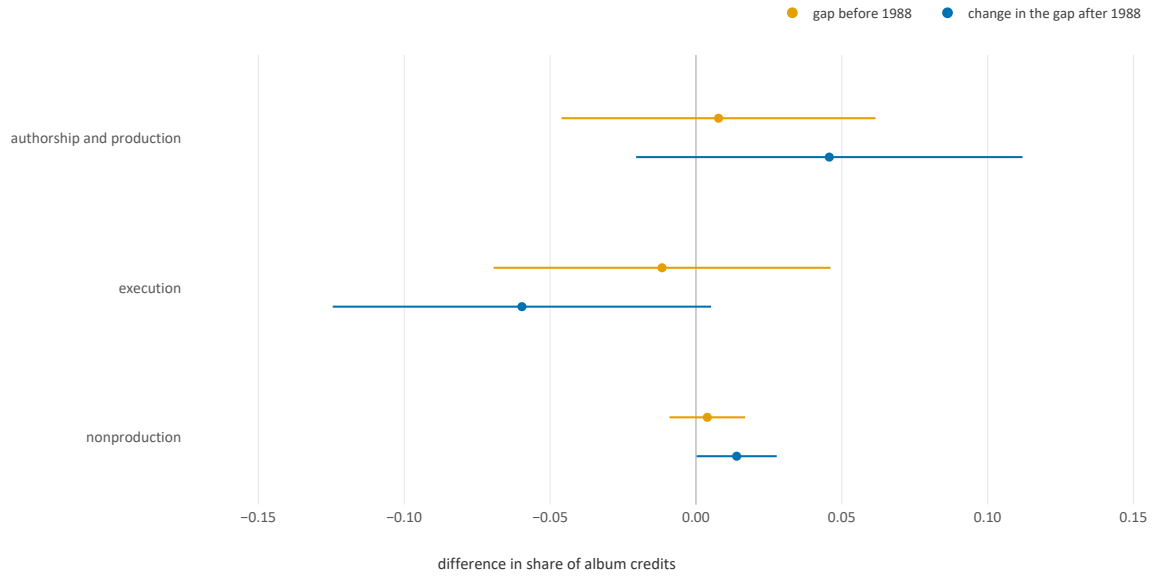
G.6 The pooled change in the sampled-record gap

Figure G.5 summarizes the pre-campaign sampled-minus-non-sampled difference and its post-1988 change under the commuting-zone specification.

G.7 Local sampling intensity

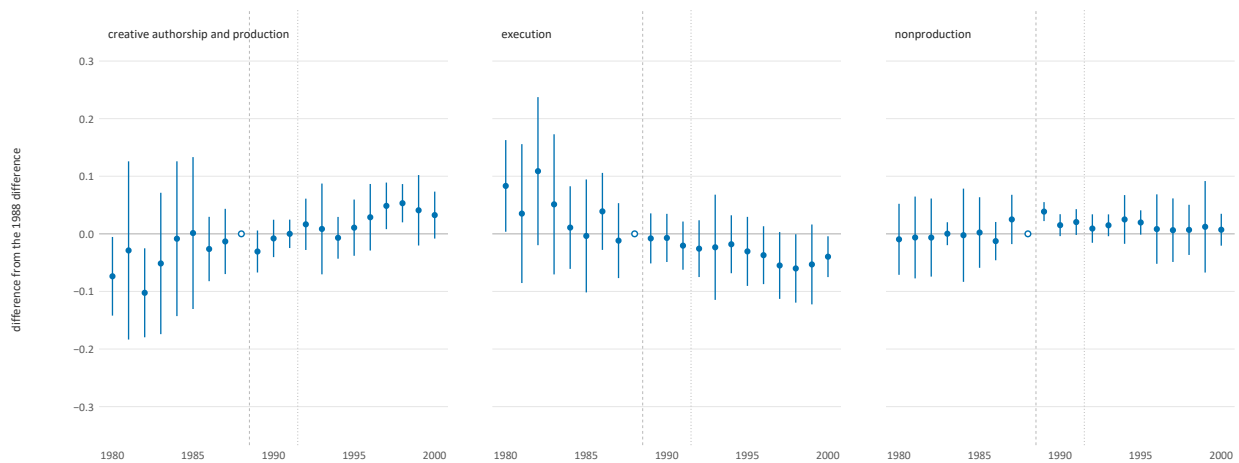
As an alternative to a record’s own sampling status, I construct the sampling rate of its commuting zone while excluding records produced in the focal record’s studio. This measure asks whether records produced in markets where other studios document more sampling also display a different credit composition.

Figure G.3: DESCRIPTIVE: sampled-record differences before and after 1988, studio fixed effects



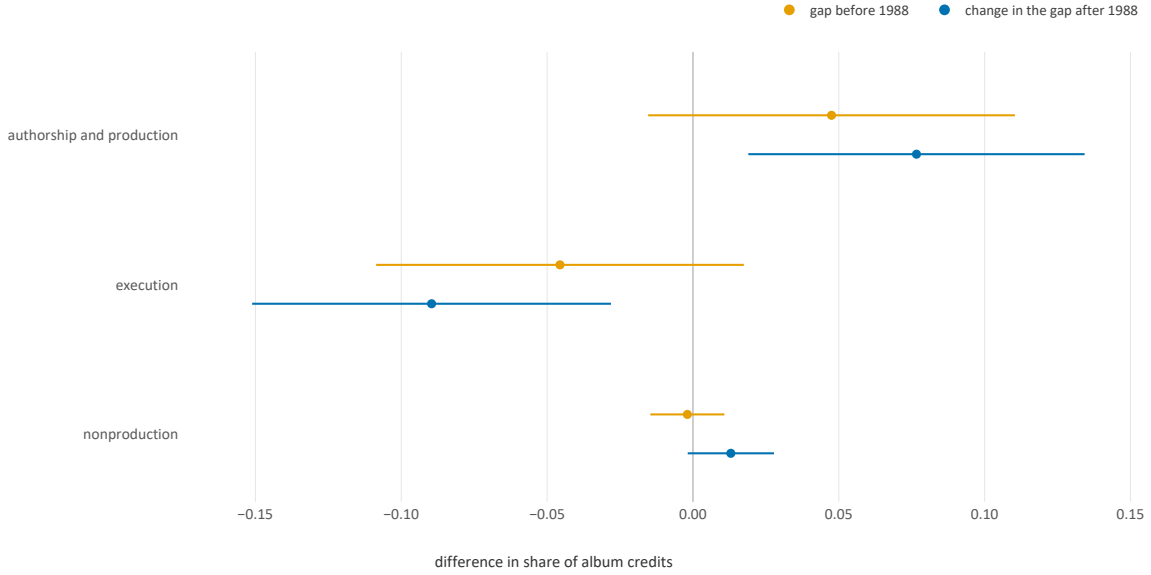
Note. Pooled differences between records documenting sampling and other records using studio and census-division-by-year fixed effects. The first coefficient reports the pre-campaign difference and the second the change after 1988. Documented sampling is endogenous, so the comparison is descriptive.

Figure G.4: DESCRIPTIVE: sampled-record differences by year, studio fixed effects



Note. Annual counterpart to Figure G.3, expressed relative to the 1988 sampled-minus-non-sampled difference. The pre-campaign estimates show that the two types of record were already diverging before the campaign.

Figure G.5: DESCRIPTIVE: sampled-record differences before and after 1988



Note. Coefficients γ and δ from Equation (G.2). The first reports the average difference between records documenting sampling and other records before the campaign; the second reports the change in that difference after 1988. Commuting-zone and census-division-by-year fixed effects. Documented sampling is endogenous, and the annual estimates in Figure G.1 show that the divergence between the two groups predates the campaign.

The resulting estimates are small and imprecise. None of the pooled coefficients separates from zero, and only a small number of annual coefficients do. Because the measure is undefined in commuting-zone-years without other active studios, it also reduces the usable sample. I therefore do not place weight on this specification.

Taken together, the descriptive evidence points in one direction but does not identify a mechanism. Records documenting sampling are more heavily oriented toward creative authorship and production and less toward execution, broadly matching the organizational response to campaign exposure. That difference, however, was already emerging before 1988 and is sensitive to genre composition. The appendix therefore provides descriptive consistency between sampling and the organizational results, not a causal link between them.

H Product characteristics and content measures

This appendix provides additional detail on the acoustic and lyrical measures used in Section 7. I first describe the two representations of record content and their coverage, then document the correction applied to the lyrical measure, alternative fixed-effect and comparison-set choices, a second audio representation, and the descriptive content profile of records documenting sampling.

H.1 Measuring distance from recent output

Section 7 measures how far a record lies from what the industry had recently produced using two representations of the finished record: acoustic characteristics and lyrics. These outcomes were developed as exploratory measures and were not pre-specified.

Acoustic characteristics. The acoustic representation uses eleven characteristics available from Spotify Echo Nest: danceability, energy, speechiness, acousticness, instrumentality, liveness, valence, tempo, mode, key, and time signature, following the dimensions used by [Askin and Mauskopf \(2017\)](#). Loudness is excluded because it is particularly sensitive to mastering.

I standardize the characteristics and represent each record by the equal-weight average of its measured tracks. For each record, I then calculate one minus its weighted mean cosine similarity to records released in the preceding year. Comparison records receive greater weight when their genre is closer to the focal record’s genre, where genre proximity is measured using the similarity of genre centroids in the preceding year.

The comparison set is therefore the recent catalogue rather than only the record’s own genre. The genre weights are relatively smooth: similarities between distinct genres range from 0.689 to 0.998, with a mean of 0.942. The resulting measure correlates 0.974 with an alternative that weights every prior-year record equally. Appendix H.5 reports a more restrictive measure that compares each record only with prior-year records in its own genre.

The acoustic measure is available for 30,865 records across 105 commuting zones, including all 18 exposed commuting zones.

Lyrics. The lyrical representation uses sentence-transformer embeddings of each track’s full lyrics. I represent a record by the equal-weight average of its track embeddings and calculate distance from recent lyrical output using cosine similarity. Because the resulting measure is mechanically related to lyric length, I correct it using the procedure described in Appendix H.2.

The lyrical measure is available for 14,642 records across 95 commuting zones, including all 18 exposed commuting zones.

Coverage. The content measures cover subsets of the full record sample, so I examine whether campaign exposure predicts whether a record is observed in the content data. Figure H.2 reports difference-in-differences estimates for coverage itself. I do not detect a differential change in coverage for the principal acoustic or lyrical samples. For the lyrical

analysis, adding observed artist characteristics changes the estimated content coefficient by at most 0.013 pre-period standard deviations.

H.2 Lyric length and the lyrical measure

The raw embedding-based lyrical-distance measure is strongly related to lyric length. Its correlation with log token count is -0.333 , and the relationship remains similar within years. More importantly, representing the same lyric using different truncation lengths changes the measured distance substantially. Using 64 rather than 510 tokens moves the raw measure by approximately 2.28 standard deviations even though the underlying lyric is unchanged.

I therefore correct the lyrical representation for this mechanical length effect. The correction is estimated entirely from within-lyric truncation experiments. Each lyric is embedded repeatedly at different token counts, and the systematic change in the embedding-based distance measure is estimated as a function of token count. The resulting adjustment is then applied to each lyric at its observed length.

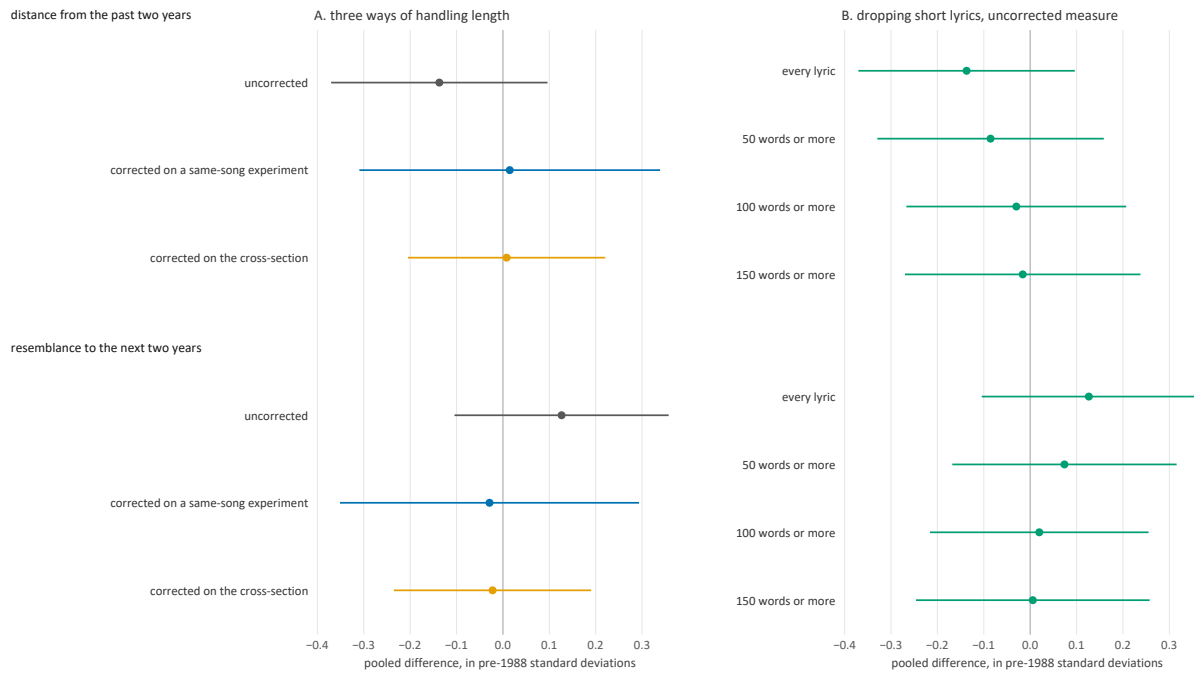
This approach differs from residualizing the outcome on lyric length in the observed cross-section. Cross-sectional residualization mechanically removes any real variation correlated with length. The truncation-based correction instead estimates the measurement distortion using repeated representations of the same underlying text.

After correction, the correlation between measured lyrical distance and log token count falls from -0.333 to $+0.035$, while the corrected measure retains a correlation of 0.923 with the original measure.

The estimated bias function changes smoothly around the model’s 510-token chunk boundary. I therefore find no evidence that the correction is generated by a discrete tokenization threshold. Above 510 tokens, where the truncation experiment becomes less stable, I hold the estimated correction constant. This affects approximately 10.8 percent of the lyrical corpus.

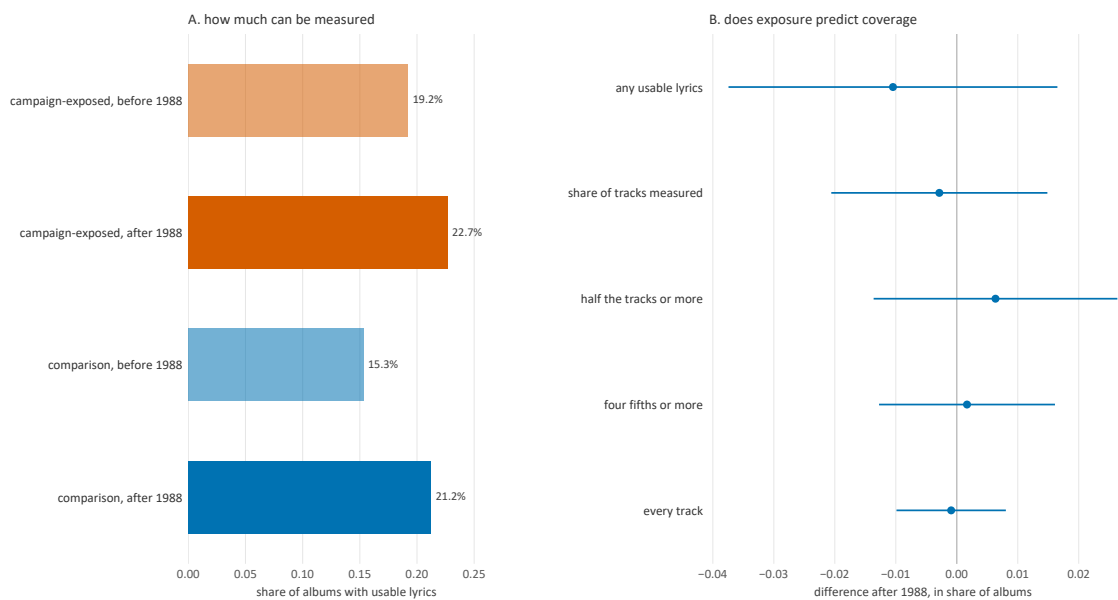
The procedure relies on the approximation that a naturally shorter lyric behaves like a truncation of a longer lyric of the same length. The truncation experiments indicate that this approximation is imperfect, particularly near 510 tokens. I therefore report the uncorrected estimate as a sensitivity check. Without the correction, the pooled lyrical coefficient is -0.137 pre-period standard deviations; with the correction it is $+0.026$. Neither provides evidence of a detectable lyrical response, but the sign change illustrates why the measurement correction matters.

Figure H.1: The lyric-length correction



Note. Lyrical-distance estimates before and after correcting for the relationship between embedding-based distance and lyric length. The figure also reports the correlation between each candidate measure and log token count. The correction is estimated from within-lyric truncation experiments rather than from the cross-section of records.

Figure H.2: Coverage of the lyrical content measure



Note. Difference-in-differences estimates for whether a record is observed in the lyrical content sample, using the same exposure design as the main content analysis. I do not detect a differential change in lyrical coverage following campaign exposure.

H.3 Studio and commuting-zone fixed effects

The main content specification uses commuting-zone rather than studio fixed effects. The economic object is the position of records produced in a local music market relative to recent industry output, and musical influence can move across studios through artists, producers, performers, and local scenes.

The acoustic result is not sensitive to this choice. Using commuting-zone and census-division-by-year fixed effects, the pooled acoustic estimate is +0.167 pre-period standard deviations, with an interval from +0.062 to +0.273. Replacing commuting-zone fixed effects with studio fixed effects gives +0.164, with an interval from +0.048 to +0.280, on the same 30,865 records.

The content result therefore does not depend on whether the specification holds the local market or the individual recording studio fixed.

H.4 No single acoustic characteristic carries the result

The acoustic-distance measure summarizes movement in an eleven-dimensional space. I therefore examine whether the aggregate response is driven by any individual acoustic characteristic.

For each characteristic, I calculate its absolute deviation from the corresponding commuting zone’s pre-campaign mean and apply the same empirical design. None of the eleven individual characteristics produces a detectable coefficient. On the same 29,141 records, the aggregate acoustic-distance measure is +0.192 pre-period standard deviations, with an interval from +0.075 to +0.310.

The result therefore reflects movement in the joint acoustic position of records rather than an obvious shift in a single measured dimension. This pattern is consistent with several characteristics changing jointly in ways that are individually difficult to detect, but the design does not identify the specific combination responsible for the aggregate movement.

H.5 Distance from the recent catalogue versus distance within genre

The main acoustic measure compares a record with the preceding year’s catalogue while weighting comparison records by genre proximity. Because the genre weights remain relatively close to one, this measure is best interpreted as distance from the industry’s recent output rather than distance from the record’s own genre.

I therefore construct a separate within-genre measure that compares each record only

with records in its own genre released during the preceding year. The within-genre and catalogue-wide measures correlate 0.62, indicating that they capture related but distinct dimensions of product position.

Table H.1: Alternative comparison sets for acoustic distance

Comparison set	Commuting-zone effects		Studio effects	
	Estimate	Years resolving	Estimate	Years resolving
Genre-weighted, whole prior year	+0.167	8 of 12	+0.164	8 of 12
Unweighted, whole prior year	+0.133	2 of 12	+0.133	2 of 12
Own genre only	+0.023	0 of 12	+0.013	0 of 12

Note. Pooled estimates in pre-period standard deviations on the same 30,865 records across 105 commuting zones, including 18 exposed commuting zones. “Years resolving” reports the number of post-campaign annual confidence intervals excluding zero. No pre-campaign annual coefficient separates from zero.

The within-genre estimate is close to zero and no post-campaign annual coefficient separates from zero. Its confidence interval is sufficiently wide that this null alone does not establish that the within-genre and catalogue-wide responses differ.

Because both measures are observed on the same records, I can test the difference directly. After standardizing each by its own pre-period dispersion, the difference between catalogue-wide and within-genre distance is +0.145 standard deviations under commuting-zone fixed effects, with an interval from +0.034 to +0.255, and +0.151 under studio fixed effects, with an interval from +0.015 to +0.287.

The content result is therefore specifically a repositioning relative to the industry’s recent catalogue. I do not detect a comparable increase in distance from a record’s own genre.

The within-genre result is not driven by a small comparison set. The median record has 1,936 same-genre prior-year peers, and even the first percentile has 191. Requiring at least 30 prior-year same-genre records drops only 15 observations and leaves the estimate essentially unchanged. A second construction based on distance from the own-genre centroid also produces a small estimate, +0.033 pre-period standard deviations.

H.6 A second audio representation

The main acoustic measure uses eleven engineered characteristics. I construct a second representation using MusicFlamingo audio embeddings to examine whether the result depends on that representation.

The comparison is made on the subset of records for which both representations are available. On 26,819 common records, the eleven-characteristic measure produces an estimate of +0.177 pre-period standard deviations, with an interval from +0.036 to +0.317. The

corresponding MusicFlamingo measure produces an estimate of +0.126, with an interval from -0.259 to $+0.511$.

The point estimates therefore have the same sign and broadly similar magnitude, but the embedding estimate is substantially less precise. The second representation neither confirms nor contradicts the main acoustic result.

To separate differences in representation from differences in comparison sets, I also apply the same genre-weighted prior-year construction used for the main acoustic measure to the MusicFlamingo embeddings. On the 26,816 records where all measures are available, the resulting coefficient is +0.152 pre-period standard deviations, with an interval from -0.228 to $+0.531$. The original two-year embedding-based version gives +0.126 on essentially the same sample.

The two MusicFlamingo constructions correlate 0.979, while their correlation with the engineered-characteristic measure remains much lower. The principal difference between the two acoustic approaches therefore comes from the representation of the recording rather than from the definition of the recent comparison set.

I also report an uncentred version of the embedding representation that was constructed alongside the primary embedding. Its pooled coefficient is +0.191 pre-period standard deviations, with an interval from -0.059 to $+0.441$. Its annual path is more positive under commuting-zone fixed effects but substantially weaker under studio fixed effects. I therefore treat it as a sensitivity representation rather than as independent confirmation of the main result.

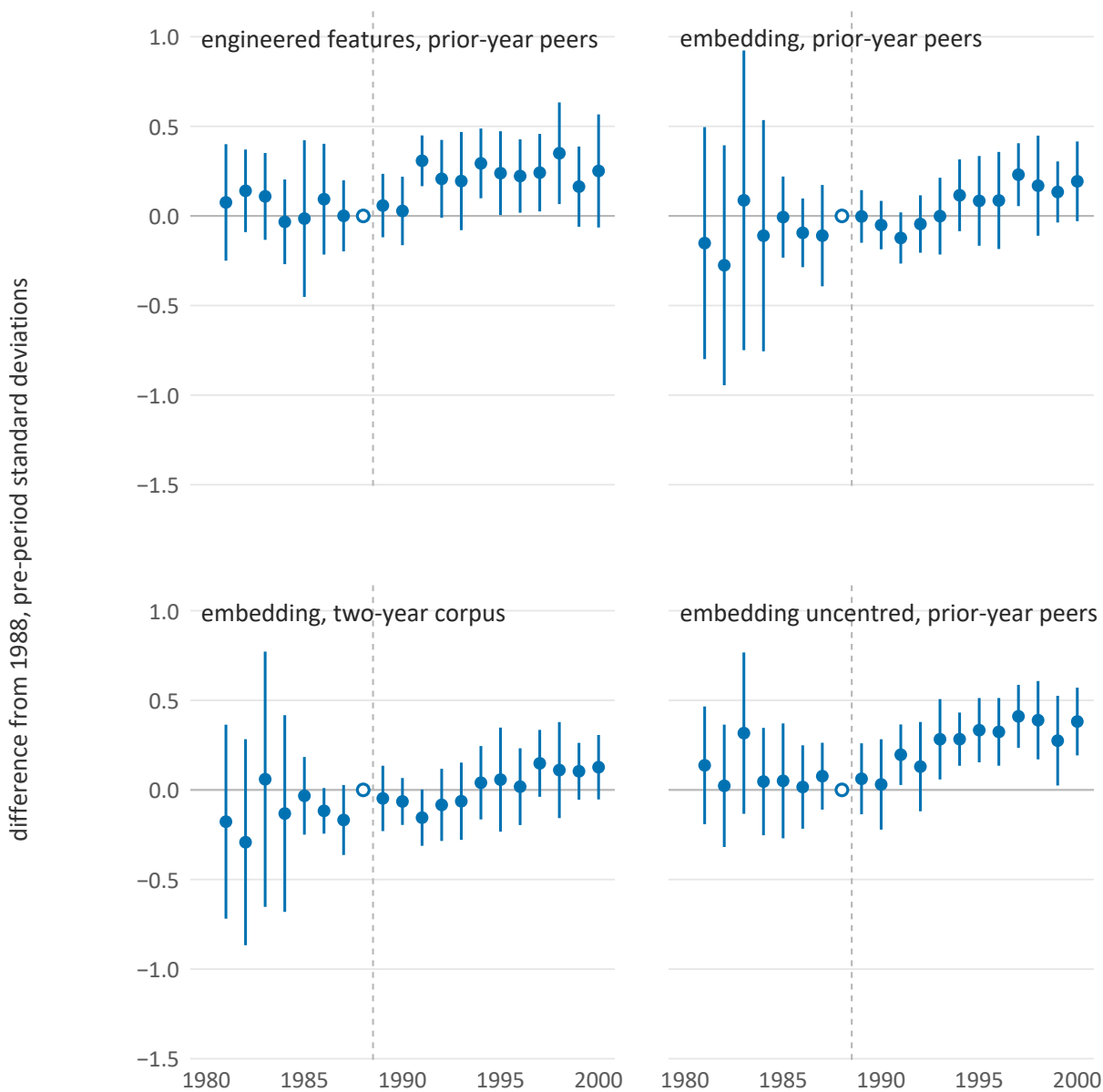
Audio coverage also warrants caution. Availability of the MusicFlamingo representation changes differentially across treatment arms around the campaign, and the underlying acoustic sample itself exhibits some differential coverage. This is one reason the product evidence is interpreted as an additional margin of adjustment rather than as the paper’s central organizational result.

H.7 The content signature of records documenting sampling

Section 7 also reports a descriptive comparison between records that document sampling and records that do not. This comparison asks what sampled records look and sound like, not what campaign exposure causally changed.

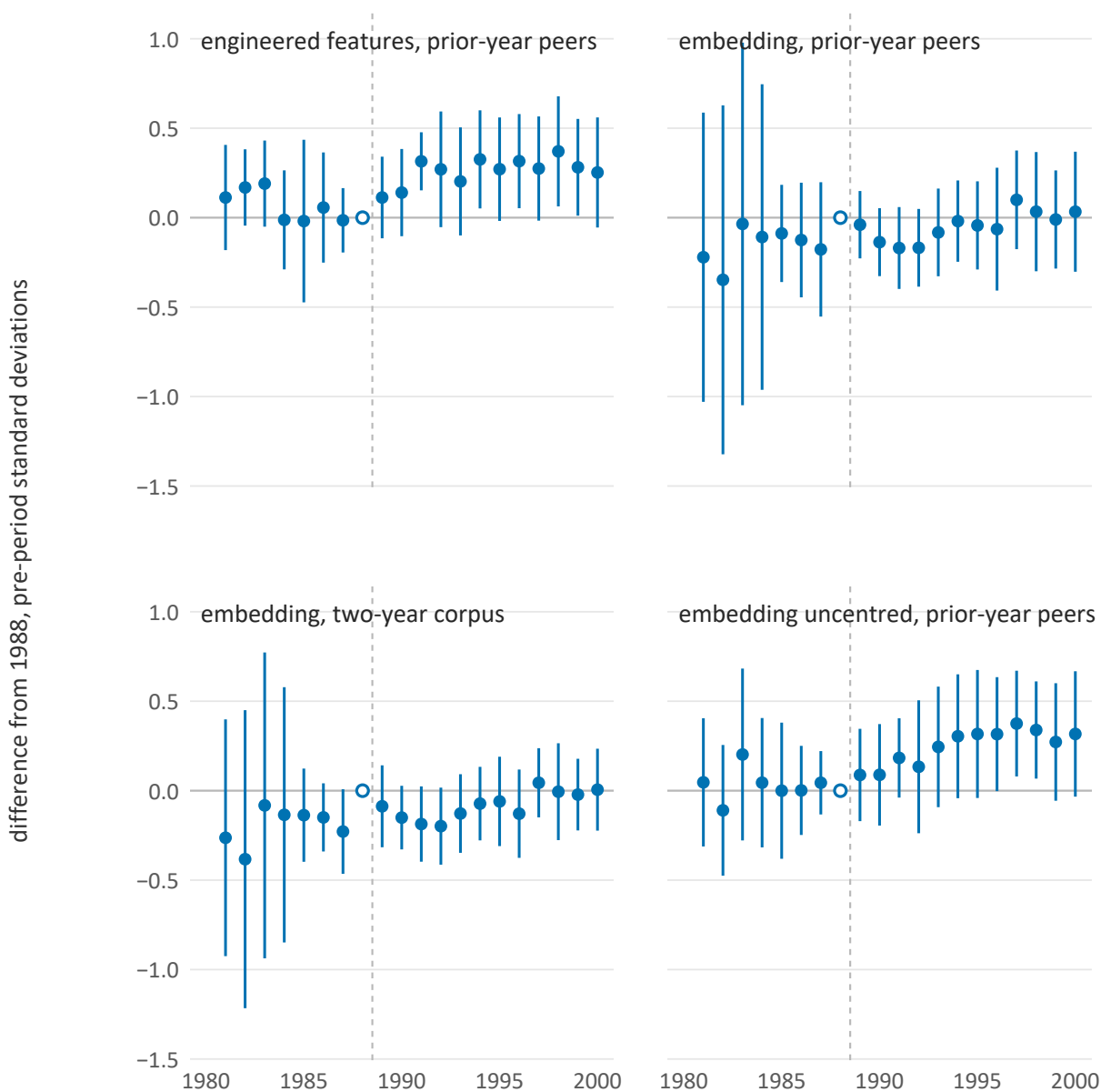
Figure 11 reports differences in acoustic and lyrical characteristics under two specifications. The first holds the recording studio and census division by year fixed. The second additionally conditions on broad genre. Comparing the two shows how much of the cross-sectional content signature associated with documented sampling is accounted for by genre composition.

Figure H.3: Alternative audio representations, commuting-zone specification



Note. Annual estimates for alternative measures of distance from recent audio output on the common sample of 26,816 records. All outcomes are expressed in pre-period standard deviations. The first panel uses the eleven engineered acoustic characteristics. The remaining panels use alternative constructions based on MusicFlamingo embeddings. Commuting-zone and census-division-by-year fixed effects.

Figure H.4: Alternative audio representations, studio specification



Note. Counterpart to Figure H.3 using studio rather than commuting-zone fixed effects. All panels use the same records and share a common vertical scale.

All estimates in this figure use the same inference procedure, with CR2 intervals and clustering by commuting zone, so that the acoustic and lyrical comparisons can be read on a common statistical basis.

The comparison is descriptive throughout. Documented sampling is chosen, is itself affected by campaign exposure, and is not used as an assigned treatment. The resulting differences therefore characterize records associated with sampling rather than identify the causal effect of sampling on record content.